47. **Solution:** At a distance 'r' from the axis the electric field, using gauss law is equal to $E = \frac{pr}{2t_o}$

Answer: B

48. **Solution**: The total Resistance between 'a' and 'b' is calculated to be $R_T = \frac{6}{11}R$. Hence the power between terminal 'a' and 'b' is

$P_{ab} = I^2R_T = \frac{6}{11}I^2R$. The power dissipated between 'b' and 'c' is

$I^2R = P_{bc}$

Then the ratio $\frac{P_{ab}}{P_{bc}} = \frac{\left(\frac{6}{11}I^2R\right)}{(I^2R)} = \frac{6}{11}$

Answer: D

49. **Solution**: To lift the wire vertically the weight of the wire should be equal to the magnetic force on the wire:- Hence:-

$F_g = F_m$

$\rightarrow mg = BIl$

$\rightarrow B = \frac{mg}{lI} = \left(\frac{m}{l}\right)\left(\frac{g}{I}\right)$

$\rightarrow B = \left(0.05\,^{kg}/_{m}\right)\left(\frac{10N}{2AKg}\right) = 0.25T,\ East$

Answer: B

50. **Solution**:- The separation distance between the conductors is a = 0.01m the current through each conductor is 10,000A. Then the magnetic force between the conductors per unit length one on the other is given by:

$\frac{F}{l} = \frac{2kI_1I_2}{a}$, but $I_1 = I_2 = 10^4A$ & $K = 10^{-7}\ \frac{TM}{A}$

After substitution you will get that $\frac{F}{l} = 2000\,^{N}/_{m}$

Answer: D

1. *C*. For a body that is moving along a circular path with uniform circular motion, the acceleration of the body has a constant magnitude
2. *B*. The lines of electric field have direction which move away from the positive charge to the negative charge. And not the reverse.
3. *B* momentum is a vector quantity
4. *C* The motion of the object is uniformly accelerated motion. To get the time required we can use the formula

$S = V_i t + \frac{1}{2}at^2$

$\rightarrow 90 = (15m/s)t + \frac{1}{2}(10m/s^2)\,t^2$

$\rightarrow$ This is a quadratic equation of the time t

$\rightarrow 5t^2 + 15t + 90 = 0$

$\rightarrow t^2 + 3t - 18 = 0$

Using the quadratic formula, you can get the answer to be $t = 3$ sec

5. A The zeroth law of thermodynamics states that if two bodies are separately in thermal equilibrium with a third body then they must be in thermal equilibrium with each other
6. D The direction of angular velocity is determined by the RHR and it is perpendicular to the plane of rotation and not tangent to the trajectory.
7. C Action and reaction forces act on two different bodies.
8. A Work done by a constant force is calculated by the formula

$W = F \times s \quad F = (4x)i \,\&\, s = x = 5m$

$= [(4x)\,i]\,[x]$

$= (20\,N)(5m)$

$W = 100J$

9. B. A Wave front is the locus of all points of a travelling wave that are in phase
10. A No electric current will be induced in a circuit if the cross-sectional area of the wire changes
11. C By the low of conservation of linear momentum

Momentum before collision = momentum after collision

$\rightarrow (V)(m) + 0 = (m + m)V_f$

Telegram: https://t.me/ExtremeSeries

→ $V \times m = 2mV_f$

→ $V_f = \frac{V}{2} = 0.5\,V$

12. **A.** The block decelerates up the inclined plane . The net force acting on the block down the block inclined plane is equal to the sum of mg $\sin\theta$ and F_f=15N by newton's 2nd law:-

$Fn = ma$

→ $Mg\sin\theta + 15N = ma$

→ $(4kg)(10m/s2)\sin 30) + 15N = (4kg)\,a$

→ $\vec{a} = 8.75\ m/s^2$

Finally to get the distance the block moves up before stopping can be obtained by:-

$$S = \frac{V_f^2 - V_i^2}{2(a)} \quad V_f = 0 \;\&\; a = \text{negative}$$

$$S = \frac{64m^2/s^2}{(2\times 8.75m/s^2)} = 3.66m$$

13. **D.** The sum of the forces 1N and 7 N can not be equal to 5 N in any way.

14. **A.** In a quasi – static process of gas expansion the work done is found using the following relation which applies an integral calculation

$$W = \int_{v_1}^{v_2} pdv = \int_{V_1}^{V_2} (dv)dv$$

$$= \alpha\frac{1}{2}V^2]_{1m^3}^{2m^3}$$

$$W = 5\times 10^{+5}\text{N/m}^5 \times \frac{3}{2}\,m^6 = 0.75\,mJ$$

15. **B.** After the ball is released from the hand of the thrower it will be under the control of the earth's gravity hence its acceleration will be equal to g

16. **C** When the block is sliding down the inclined plane the net force on it is equal to mg $\sin\theta$ - F_f. By newton's 2nd law its acceleration is

$F_{net} = m\,a$

→ $Mg\sin\theta - F_f = ma$

→ $Mg\sin\theta - mg\cos\theta\mu_k = ma$

→ $a = g\sin\theta\,(1-\mu_K)$ .....(1) $\sin\theta = \cos\theta$, for θ 45^0

When the block is projected up, the net force on the block should be mgsin $\theta + F_f$ By newton's 2nd law the upward acceleration is

$F_{net} = ma$

→ $Mgsin\,\theta + mg\cos\theta\mu_k = ma$

→ $a = g\sin\theta\,(1+\mu_k)$ .... (2)

Dividing (1) by (2) gives us the following

$$\frac{mgsin\,\theta(1-\mu_k)}{mg\,sin\theta(1+\mu_k)} = \frac{1-\mu_k}{1-\mu_k}$$

17. **B.** When a small mass m is suspended from one end of a vertical string the string becomes inclined to the vertical. The other three choices A, C, & D are not correct

18. **D.** If A+ B –C = 0

→ (3i -2j) + (-i -4j) –C =0

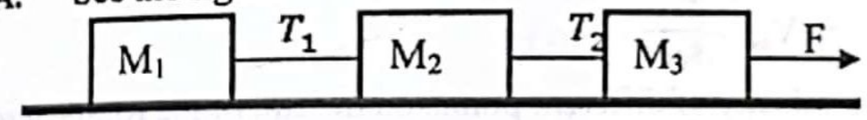
→ (2i – 6j) – C = 0

→ $\vec{C} = 2i - 6j$

Then the unit vector inthe direction of C is

$$\hat{C} = \frac{C}{|C|} = \frac{2i-6j}{2\sqrt{10}} = \frac{1\,i}{\sqrt{10}} - \frac{3}{\sqrt{10}}\,j$$

19. **A.** See the figure



The system of the three blocks acceleration in the direction of the force F.

There fore by newton's 2nd law you can write

→ $F - T_2 = m_3 a$. .... (1)

→ $T_2 - T_1 = m_2 a$.... (2)

→ $T_1 = m_1 a$ ..... (3)

→ $F - (m_2 a + T_1) = m_3 a$, substitute for T_2

→ $F - (\frac{m_2}{m_1}T_1 + T_1) = \frac{m_3}{m_1}T_1$

→ $F = T_1(\frac{m_3}{m_1} + \frac{m_2}{m_1} + 1)$

→ $F = T_1(\frac{m_1+m_2+m_3}{m_1})$

→ $T_1 = \frac{m_1 F}{m_1+m_2+m_3}$

20. **D.** If C is a vector along x axis Then $C = C_x i$

Telegram: https://t.me/ExtremeSeries

Then B + C = (-1 +C_x) i – 4j + 3k

⇨ A (B +C) = 3(-1+C_x) +8 +3 = 0

⇨ -3 +3C_x +8+3 =0

⇨ =C_x = -8/3i

21. **D**. For a rigid body to be in a complete equilibrium of both rotational and translational the sum of all torques must be zero and the sum of all forces must be zero on the body

22. **B** To calculate the inductance of an LC circuit we use the relation

$$L = \frac{1}{4\pi^2 f^2 C} = \frac{1}{4\pi^2 (120\,Hz)^2 (8\mu F)} = 0.22\,H$$

23. **A**. The bullet at point 2 on the trajectory moves both up ward and horizontally to the right . its horizontal motion has no acceleration. Its vertical velocity component is decreasing and this means its acceleration is down ward (decelerating)

24. **C** To get the average speed you can use the Formula **Average speed** $= \frac{\text{total distance travells}}{\text{total time taken}}$

In the first 5 seconds, $V_{av} = \frac{25m}{5sec} = 5\frac{m}{s}$

From t = 6 sect t = 10 sec

→ $V_{av} = \frac{45m}{5sec} = 9m/s$

25. **C**. The velocity of different points on the rim of the rolling disc is different. Accordingly the velocity of the upper most point on the disk is twice the velocity of the center of the disk

26. **B** For two interfering light waves on a screen to form fringes, after passing through two narrow slits we have the relation ship:-

$$\lambda = \frac{nd}{R}$$

Where $\lambda \equiv$ wave length of the light

d ≡ distance between slits = 0.2 mm

R ≡ disance from slit to screen = 2m

n ≡ distance between from fringes =?

Since V = λ f we have $\lambda = \frac{v}{f}$

Then → $\frac{v}{f} = \frac{nd}{R}$

→ $n = \frac{VR}{df} = \frac{(3\times10^8 m/s \times 2m)}{(0.2\times10^{-3}m \times 6\times10^{14}Hz)}$

→ $n = 5mm$

27. **B** The angular momentum of a particle is given by the relation:-

$L = Prsin\theta$, where.

'P' is the inner momentum and r is the position vector and θ is the angle between p and r. If p and r are parallel, this means $\theta = 0$ And $sin\theta = sin0^0$. Then the angular momentum in this case becomes gero.

28. **D** The velocity of bos relative to the car driver can be obtained as

$V_{BC} = V_B - V_C$

= (-5i – 4j) m/s – (3i – 2j) m/s

= [(-5- 3)i + (-4 + 2) j]m/s

V_{BC} = (-8i – 2j) m/s

29. **C**. To solve this problem use the equation of continuity That is

(V_1 A_1)small = (V_2 A_2) large pipe

Pipe

$V_1 = (\frac{A_2}{A_1}) V_2$

$V_1 = (\frac{50cm^2}{10Cm^2}) 12m/s = 60m/s$

30. **A**. The parallel capacitor with a plate separation d and with out a dielectric is given as: C= $\frac{E_0 A}{d}$

According to the question the new capacitance C_n is the sum of the capacitors C_a-and C_m connected in series.

$C_a \equiv$ capacitance of capacitor with air gap & distance $\frac{2}{3}$ d

$C_m \equiv$ capacitance of capacitor with dielectric material & distance d/

To get the new capacitance C_n, add C_a & C_m in series mode

That is $\frac{1}{c_a} = \frac{1}{c_n} + \frac{1}{c_m}$

→ $C_n = \frac{3kE_0A}{(2K+1)d}$

→ $C_n = (\frac{3k}{2k+1})$ C

31. **D**. The acceleration of a simple harmonic dscillator is given by

→ $F = ma = Kx$

→ $a = \frac{kx}{m}$

Telegram: https://t.me/ExtremeSeries

As you can see $\vec{a}$ is dependent as the value of X that is the displacement of the oscillator from its equilibrium position .
Hence $\vec{a}$ becomes minimum when velocity is maximum
at x = 0

32. **D.** The size of the ohmic conductor (length or thickness) the nature of the material & also the temperature can affect the resistance of the conductor. But voltage applied has no direct influence on resistance of the conductor

33. **C** If the spring is compressed by x = 1m for a force of F = 100N, then its spring constant K must be 100N/m. When the mass slides down the plane and stops after compressing the spring, the mass will transfer all of its GPE to the SPE of the spring.
That is you can write GPE = SPE

$$\rightarrow mgh = \tfrac{1}{2} k x^2$$

$$\rightarrow \quad h = 2m$$

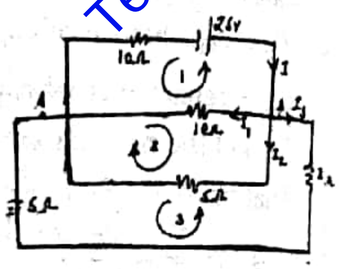
If you get the height of the inclined plane, you can get the distance the mass slided down the plane using the sine function.

That is $\rightarrow \sin 30 = \frac{h}{s}$, S ≡distance slided by the mass

$$S = \frac{h}{\sin 30} = \frac{2m}{0.5} = 4m$$

34. **B.** If the intensity of light shining on a metal surface increases, the number of photoelectrons that are emitted by photoelectric effect will increase

35. **A.** Consider the circuit below. Assign current directions through the branches from the source as shown.
Use kirchhoff's junction and loop rules



→ At the right junction
$I = I_1 + I_2 + I_3$ .....(1)
→ From loop (1)
$-25 + 10I - 10I_1 = 0 \quad \rightarrow \quad -5 = 2I + 2I_1$
.....(2)
→ From loop (2)
$-10I_1 + 5I_2 = 0 \rightarrow 2I_1 = I_2$ ..... (3)
→ From loop (3)
$20I_3 - 5I_3 + 5I_2 \rightarrow 5I_3 = I_2$ ..... (4)
The required quantities are I_3 and the P.d between points A and B

By combining the equations that we get from kirchhoff's rules, you can get the quantities to be $I_3 = 0.227$ A and the P.d = 5.68 v

36. **D.** The circuit given is a wheat stone bridge circuit. In this circuit for the voltage between B and D to be zero, the ratio of the resistors on the left should be equal to the ratio of the resistors on the right. First you need to combine the resistors in the same branch
Then

$$\frac{x+6}{12} = \frac{0.5}{0.5}$$

$$\rightarrow x + 6 = 12$$

$$\rightarrow x = 6\Omega$$

37. **A.** Note that the water pipe has similar cross sectional area from the river to the sub-city. The height difference is
$h_2 - h_1 = 1532 = \Delta h$
If you apply Burnoulli's equation it will be reduced to the form
$P = \rho g \Delta h$
$P_{min} = 10^3 kg/m^3 \times 10 N/kg \times 1532\ m = 15.32 Mpa$

38. **A.** The phase angle in an alternating circuit becomes zero if the circuit contains only a resistor. This is because the voltage and the current in such a circuit are in phase

39. **D** As the swithch is closed to the left capacitance C_1 has original charge 'q_0' given by:-

$$q_0 = C_1 V_0$$

Note that C_2 & C_3 are connected in series & so their charge is the same.

That is $q_2 = q_3$

We can write for the new charge on C_1 as:-

$$q_1 = q_0 - q_2$$

You know also from the type of connection the potential at C_1 is given as:-

$$\rightarrow V_1 = V_2 + V_3$$

$$\rightarrow \frac{q_1}{c_1} = \frac{q_2}{c_2} + \frac{q_2}{c_3}$$

$$\frac{q_1}{c_1} = q_2\left(\frac{c_2 + c_3}{c_2 c_3}\right)$$

Telegram: https://t.me/Extremeseries

$$\rightarrow q_1 = q_2 C_1 \left(\frac{C_2+C_3}{C_2 C_3}\right)$$

Then by making some substitutions, it can be shown that:-

$$\rightarrow q_1 = \left(\frac{C_1C_2+C_1C_3}{C_1C_2+C_1C_3+C_2C_3}\right) C_1 V_0$$

40. If the coordinate, x & y (positions) of the particle at any time t are given then the function for the acceleration of the particle can be obtained by taking the second derivation of the position functions that is

$$\overrightarrow{a_x} = \frac{d^2x}{dt^2} = \frac{d^2(9t+5t^2)}{dt^2} = 10\text{m/s}^2 \;\&$$

$$\overrightarrow{a_y} = \frac{d^2y}{dt^2} = \frac{d^2(5t+8)}{dt^2} = 0$$

Hence the acceleration is 10m/s^2 in the positive X axis which is constant.

41. The second law of thermodynamics does not imply that heat will flow from colder body to a hotter one as in the refrigerator with out work being done on the refrigerator (system)

42. A. The key phenomenon that demonstrates the particle nature of light is the photoelectric effect where electrons are ejected from a metal surface when light strikes the metal

43. B. By the law of conservation of angular momentum we have (angular momentum) $_{before}$ = angular momentum) $_{after}$

$$\rightarrow I_1\,\omega_1 + I_2\,\omega_2 = (I_1 + I_2)\,\omega$$

$$\rightarrow \omega = \frac{I_1\omega_1 + I_2\omega_2}{I_1+I_2}$$

44. **D** For a simply harmonic motion the maximum velocity is calculated using the formula

$$V_m = \sqrt{\frac{k}{m}}\,A$$

The period of vibration is also given by $T = 2\pi\sqrt{\frac{m}{k}}$

By combination we get the relation:-

$$V_m = \left(\frac{2\pi}{T}\right)_A = 18.84 \text{ cm/sec}$$

45. **C.** much more current passes through a filament lamp when it is turned on during which the filament has low temperature and low

Telegram: https://t.me/Extremeseries

resistance. After some time the temperature increases and the resistance also increases

46. C The impedance of a capacitor is given by

$$X_c = \frac{1}{2\pi fc} = 19.9\ ohm$$

$$X_c = \frac{1}{2\pi\times10^3\ Hz\times8\times10^{-6}\ F}$$

47. A. If 25 w maximum power is delivered to each resistor, the voltage at each resistor will be 25v, by the formula $P = \frac{V^2}{R}$. Since the two parallel resistors take the same voltage the total maximum voltage that can be applied between A &B is 50 V .

48. A What follows from the kinetic theory of gases is that the pressure of a gas is proportional to the average of square of the molecular speed of particles.

49. By neuton's Law of universal gravitation, the attraction of the earth on the satellite is inversely proportional to the square of the separation distance between the earth and the satellite .

50. A for the ladder not to slip it has to fulfill both conditions of equilibrium Look at the forces that are acting on the ladder

From the first condition

$$\rightarrow F_N = w = 50N\ \&$$

$$\rightarrow H = F_f = F_N\mu = 25N$$

From the 2nd condition

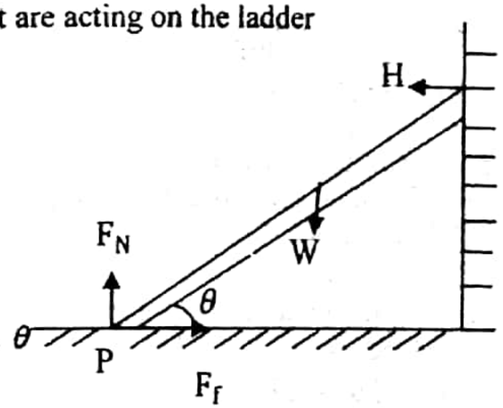
Take point 'P' as a pivot

$$\rightarrow \quad H l \sin\theta = \frac{Wl\cos\theta}{2}$$

$$\rightarrow \quad 50N \sin\theta = 50N \cos\theta$$

$$\rightarrow \quad \text{If } \sin\theta = \cos\theta,$$

than 'θ' = 45^0(minimum angle)



SOLUTION TO PHYSICS PROBLEMS OF 2004

CODE 24

1. If an ion is acceleratedthrough an electric potential difference, the electric field does work on the ion.
   By the principle of work energy theorem the work done on the ion is equal to the change in KE of the ion.
   That is:-$V_q = \Delta KE$

$$\longrightarrow q = \frac{\Delta KE}{V}$$

$$\longrightarrow q = \frac{7.37 \times 10^{-17} J}{115V} = 6.4 \times 10^{-19} c$$

   Answer - A

2. Electric force between the protons can be calculated as follows:-

$$F = \frac{kq1q2}{r2} \quad , q1 = q2 = p$$

$$= \frac{kp2}{r2}$$

$= 9 \times 10^{9N}\, m^2N \times (1.6 \times 10\text{-}^{19}c)^2$
C^2
$(2 \times 10^{-15}m)^2$
F= 57.6N
Answer C

3. Since the intensity of a wave is proportional to the square of the amplitude, then if amplitude is doubled, the intensity will increase by a factor of 4.
   Answer:- C

4. The temperature of a gas is due to the Kinetic energy of the molecules of the gas
   Answer B

5. If A+ B= 2i, then

$$\longrightarrow (3i - 2j) + (-i - aj) = 2i$$
$$(3i - i) + (2i - aj) = 2i$$
$$(3 - 1)i - (2 + a)j = 2i$$
$$2i - (2 + a)j = 2i$$

   This means:- -2 –a =0
   a = -2
   Answer C

Telegram:https://t.me/Extremeseries

6. In a purely inductive circuit the current takes a cosine function while the voltage is sine function. Hence the current lags the voltage by 90^0 or $\pi/2$
   Answer C

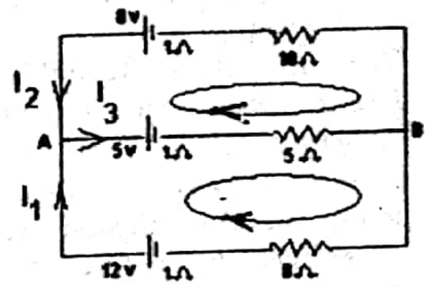
7. The SI unit of electric field is N/C
   Answer D

8. Initial KE is $KE_1 = \frac{1}{2} mv_1^2$
   Final KE is $KE_2 = \frac{1}{2} m(2V_1)^2$, $V_2 = 2V_1$
   $KE_2 = 1/2m4V_1^2$
   $= (1/2\, mv_1^2)4$
   $KE_2 = 4KE_1$

9. Answer B
   The projectile at its maximum height, has a vertical acceleration of $10m/s^2$ which is equal to acceleration due to gravity.

10. Answer B
    The sum and difference of two non-zero vectors can be equal in magnitude if the two vectors are acting at right angles with each other. They should be perpendicular to each other.

11. AnswerD
    This problem can be solved by applying Kirchhoff's junction and loop rules as follows. Look at the circuit.



    -From the junction rule we have;- i
    $-I_1 + I_2 = I_3$
    From loop rule:-
    .upper loop:-
    $5I_3 + I_3 + 5v - 8v + I_2 + 10I_2 = 0$
    $6I_3 + 11I_2 = 3v$ ii
    .lower loop:-
    $-8I_1 - I_1 + 12v - 5v - I_3 - 5I_3 = 0$
    $-9I_1 - 6I_3 = -7v$ iii
    Solving equations number i, ii and iii simultaneously, will give the current through the 5 to be 0.0.48A
    Answer:-C

12. If a cell supplies 0.9A through a 2 Ω resistor we can write:-emf=0.9(r +2 Ω) and again if the same sell supplies 0.3A through a 7 Ω visitor we can write:- Emf = 0.3A (r + 7Ω).
    Solving the above two equations simultaneously, we get r= 0.5Ω
    Answer A

SOLUTION TO PHYSICS PROBLEMS OF 2004

CODE 24

1. If an ion is acceleratedthrough an electric potential difference, the electric field does work on the ion.
   By the principle of work energy theorem the work done on the ion is equal to the change in KE of the ion.
   That is:-$V_q = \Delta KE$

$$\longrightarrow q=\frac{\Delta KE}{V}$$

$$\longrightarrow q=\frac{7.37x10^{-17}J}{115V}=6.4x10^{-19}c$$

   Answer - A

2. Electric force between the protons can be calculated as follows:-

$$F=\frac{kq1q2}{r2} \quad , q1=q2=p$$

$$=\frac{kp2}{r2}$$

$= 9 \times 10^{9N}\,\underline{m^2}N \times (1.6 \times 10\text{-}^{19}c)^2$
C^2
$(2 \times 10^{-15}m)^2$
F= 57.6N
Answer C

3. Since the intensity of a wave is proportional to the square of the amplitude, then if amplitude is doubled, the intensity will increase by a factor of 4.
   Answer:- C

4. The temperature of a gas is due to the Kinetic energy of the molecules of the gas
   Answer B

5. If A+ B= 2i, then

$$\longrightarrow (3i-2j)+(-i-aj)=2i$$
$$(3i-i)+(2i-aj)=2i$$
$$(3-1)i-(2+a)j=2i$$
$$2i-(2+a)j=2i$$

   This means:- $-2-a=0$
   a = -2
   Answer C

6. In a purely inductive circuit the current takes a cosine function while the voltage is sine function. Hence the current lags the voltage by 90^0 or $\pi/2$
   Answer C

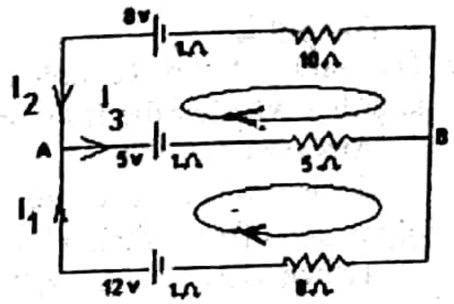
7. The SI unit of electric field is N/C
   Answer D

8. Initial KE is $KE_1 = \frac{1}{2} mv_1^2$
   Final KE is $KE_2 = \frac{1}{2} m(2V_1)^2$, $V_2 = 2V_1$
   $KE_2 = 1/2m4V_1^2$
   $=(1/2\ mv_1^2)4$
   $KE_2 = 4KE_1$
   Answer B

9. The projectile at its maximum height, has a vertical acceleration of $10m/s^2$ which is equal to acceleration due to gravity.
   Answer B

10. The sum and difference of two non-zero vectors can be equal in magnitude if the two vectors are acting at right angles with each other. They should be perpendicular to each other.
    AnswerD

11. This problem can be solved by applying Kirchhoff's junction and loop rules as follows. Look at the circuit.



    -From the junction rule we have;-
    $-I_1 + I_2 = I_3$ i
    From loop rule:-
    .upper loop:-
    $5I_3 + I_3 + 5v - 8v + I_2 + 10I_2 = 0$
    $6I_3 + 11I_2 = 3v$ ii
    .lower loop:-
    $-8I_1 - I_1 + 12v - 5v - I_3 - 5I_3 = 0$
    $-9I_1 - 6I_3 = -7v$ iii
    Solving equations number i, ii and iii simultaneously, will give the current through the 5 to be 0.0.48A
    Answer:-C

12. If a cell supplies 0.9A through a 2 Ω resistor we can write:-emf=0.9(r +2 Ω) and again if the same sell supplies 0.3A through a 7 Ω visitor we can write:- Emf = 0.3A (r + 7Ω).
    Solving the above two equations simultaneously, we get r= 0.5Ω
    Answer A

Telegram: https://t.me/ExtremeSeries

13. Refraction of light is responsible for the colors seen on a thin oil film on the water surface

AnswerD

14. To get the velocity we need to get the wave length of the sound first. For the one end opened pipe given $\ell = \lambda/4$ $\lambda = 4\ell = 2.72m$

→ Then using the wave equation $V = \lambda f$ we get

$V = \lambda f = (2.72m)(110Hz)$

$V = 299.2m/s$

Answer.C

15. For a spring mass system the KE at the equilibrium position is equal to the PE at the extreme points when displacements is maximum

That means:- $\frac{1}{2} mv^2_{max} = \frac{1}{2} KA^2$, A = amplitude

$$A = \sqrt{\frac{m}{k}} V_{max}$$

$$A = \sqrt{\frac{1kg}{100N/m}} 3m/\sec = 0.3m$$

Answer.B

16. The curve that is drawn between two points on a thermodynamics p-v diagram represents the thermodynamics process that is taking place.It may be expansion or compression of a gas adiabatically or isothermally.

Answer.B

17. The value of the Reynolds number is a number having no unit and describes whether the flow of the fluid is stream lined or turbulent.

Answer.C

18. By the law of conservation of linear momentum, the total momentum before and offer collision are equal

Hence:- (P_T) before $= (P_T)$after

$m_1V = (m_1 + m_2)V_c$, V_c= common velocity

$$Vc = \frac{m_1 V}{m_1 + m_2}$$

Answer.C

19. In the first experiment the car had Vi = 0.5m/s V_f=0 and s=1m

For this we can write:- $s = \frac{V_f^2 - V_i^2}{2a}$

$$a = \frac{0 - (0.25m^2/s^2)}{2m}$$

$$a = -1/8 m/s^2$$

In the 2nd experiment the car has Vi= 1m/s and V_f = om/s but we need the distance taking the same deceleration, we have

$$s = \frac{Vf^2 - Vi^2}{2a}$$

$$S = 0 - (m/s) \quad \frac{=4m}{-1/4 m/s^2}$$

Answer .D

20. For the first car you can write:- $d = \frac{1}{2} a_1 t^2$ and $30m/s = a_1 t$ and you get $d = 15t$

For the 2nd car you can write:- $x = \frac{1}{2} a_2 (9t^2)$ and $30m/s = a_2(3t)$ and get $x = 45t$. Solving $d = 15t$ and $x = 45t$ simultaneously you will get the distance the 2nd car covered to be $x = 3d$.

Answer .A

21. The total energy of a particle executing a simple Harmonic motion is give by the relations total energy $= \frac{1}{2} KA^2$

This means the energy is directly proportional to the square of the amplitude

Answer .C

22. When the ball is a submerged there will be three forces acting on it as shown here. F_{ap} is the applied force we need to get. As the ball is completely submerged it is in equilibrium under the three forces. That means

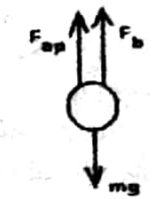
$Fap + mg = Fb$

$Fapp = F_b - mg$, but $F_b = d_w V_b g$

$= d_w V_b . g - d_b V_b g$

$= V_b . g(d_w - d_b)$, $V_b = 4/3\ \pi r^3$

$Fapp = 0.258N$ /after substitution/



Answer. B

23. The pressure due to the water at the bottom of the cylinder is given by:- $P = \rho gh$

Hence:- 5000 pa= 10^3kg/m3 x 10N/kg x h

h=0.5m

If water is replaced by ethyl alcohol then the new pressure is given by

$P = \rho gh$

$= 800kg/m^3 \times 10N/k \times 0.5m$

$P = 4000\ pa$

Answer .A

Telegram: https://t.me/Extremeseries

24. When climbing up a ladder that leans against a wall, it is safer and essential for the ground at the bottom of the ladder to be having move fiction or be rough to keep the ladder from sliding.

Answer. B

25. The wheel has w_0 =o and w_f= 12rad/s^2 and t=3sec. To get its angular acceleration and the angular displacement, we use the formulae:-

$\alpha = \frac{wf-wi}{t}$ = 4rad/s^2 and

$\theta = \frac{1}{2}\alpha t^2$= 18rad

Answer .A

26. The block is given an initial velocity V_i = 8m/s up the inclined place and will come to rest, V_f =0. Two forces are acting of the block down the plane and cause it to decelerate.

That is:- F =ma

$$a = F/m = \frac{mg\sin\theta + F_f}{M}$$

$$a = -35/4\ m/s^2$$

To get the distance moved by the block up the inclined place you can use

$$S = \frac{V_f^2 - Vi^2}{2a}$$

$$= \frac{-(8m/s)^2}{-(35/2)m/s^2}$$

S= 3.66m

Answer .A

27. To solve this problem we can use the forming $X = V_i t + \frac{1}{2} at^2$, for constant acceleration. The given quantities are:- Vi= 12m/s and the distance 'x' is equal to 8cm = 0.08m and the time is t =2sec

Then:- $x = Vi\ t + \frac{1}{2} at^2$

$$0.08m = 12m/s \times 2sec + \frac{1}{2} a(2sec)^2$$

$$a = \frac{-23.92m/s^2}{2} = -11.96m/s$$

$$a = -12m/s^2$$

Answer .A

28. The impedance of a circuit containing inductor(L) resistor (R) and capacitor (C) connected in series is given by the relation.

$$Z = \sqrt{R^2 + (X_L^2 - X_C^2)}$$

But $R = 5\Omega$, $X_L = 2\pi fL = 2.512\Omega$ and $X_C = \frac{1}{2\pi fc} = 15.9\Omega$

After substitution you will get:- $Z = 14.3\Omega$

No answer is given

29. The main reason for using AC to transmit electricity than Dc is tominimize the energy loss. This can be done by minimizing the current through the wire and increasing the voltage using transformer. Dc cannot be used for transformer.

Answer .B

30. The ,magnitude of the magnetic force on the elections is calculated using the formula

$F = Bvq\cos 60$,

$= (5 \times 10^{-5}T)(1.4 \times 10^7\ m/s)(1.6 \times 10^{-19}c)(0.866)$

$F = 9.6992 \times 10^{-17}N$

$F \cong 9.7 \times 10^{-17}N$

Answer .B

31. The loudness L_2 = 60db is greater than the loudness L_1=30db by 30. Hence we can write:-

$\rightarrow db_2 - db_1 = 10\log(I_2/I_1)$

$30db = 10\log(I_2/I_1)$

I_2/I_1 = antilog(3) = 1000

Answer:- D

32. At any given instant there is a wide range of particle speeds and therefore a wide range of molecular kinetic energies, however, the average kinetic energy of all the gas particles is proportional to the absolute temperature. There for we cannot say that the distribution of the velocities of the particles is uniform.

Answer: - D

33. Four forces are acting on the ladder and the ladder is in equilibrium. Hence we can apply the first and the second conditions of equilibrium as follows.

Look at the figure below.

Vertically:- N_g = weight of ladder = 200N ---------- i

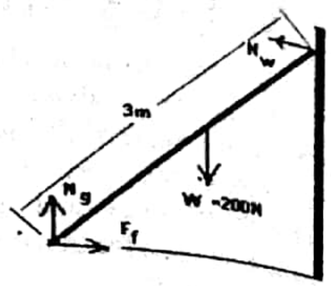
Horizontally:-$N_w = F_f = N_g\mu$ ---------- ii

Next take the top of the ladder as a pivot and apply the second condition as follows.

Telegram: https://t.me/ExtremeSeries

Note that W and F_f rotate the ladder anti clock wise and N_g rotates the ladder clockwise.

Clock wise torque =anti clock wise torque



- $N_g x 3m x \sin 30^0 = F_f 3\cos 30^0 + W1.5\sin\sin 30^0$
- $300 = 2.598F_f + 150$
- $150 = 519.6\mu$
- $\mu = 0.289$
- Then, since $N_w = F_f = N_g\mu = 200Nx.288=57.6$

Answer;- A

34. The moment of inertia of a circular loop(a hoop) is mr2. But the moments of inertia for solid disk, solid cylinder and solid sphere are ½ mr^2, $\frac{1}{2}mr^2$ and $\frac{2}{5}mr^2$respectively. Hence the circular loop has the largest moment of inertia

Answer .C

35. This a type of collision in which we can apply the law of conservation of momentum.

$\rightarrow Ep_i = Ep_f$

$0 = m_1v_1 + m_2v_2$, m_1= 0.4kg & m_2 =4kg

Since KE_2= 100J = $\frac{1}{2}m_2v^2$

$V2 = \sqrt{50}$ m/s

$-m_1v_1 = m_2v_2$

$V1 = \frac{-m2}{m1}v_2 = -10\sqrt{50}$ m/s

Then KE of smaller mass mi is given by

$KE_1 = ½\ mv_1^2 = ½ x(0.4kg)(10\sqrt{50m/s})^2$

KE_1 = 1000J

Answer: - C

36. For a circuit containing RLC, the frequency of the damped oscillation is obtained using the relation

$$w = \sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}$$

If you substitute the given values for R,C& L in this equation you will get W=15.72 x10^3 Hz

Answer .B

37. The formula for include voltage in one of the coils is:

$Emf = -M\frac{dI}{dt}$, m= 10^{-4}H

$= (-10^{-4}H)\ [d(10xAsin(1000t))]$

$dt = (-10^{-4}H)(10A)(1000)(\text{Cos } 1000t)$

But the peak voltage is at cos(1000t)= -1

Hence Emf=$(-10^{-4}H)(10^4A)(-1)$

Emf= 1V ,

Answer .D

38. This emf product in the wire is called motional emf. If is given by the equation

emf =BVL, 'B' is the magnetic field around the wire.

'B' is given by:- $B = \frac{\mu oI}{2\pi r}$

Substituting 'B'in the above Emf, you will get

$$-emf = \frac{\mu_o vil}{2\Pi r}$$

Answer A

39. The resistance between the ends of the rod is the sum of the resistances of the two materials.

Hence:- $R_T = R_1 + R_2$

- But, $R_1 = \frac{\rho_1 l_1}{A}$ and $R_2 = \frac{\rho_2 l_2}{A}$
- After substitution we will get
- R_T = 378Ω

Answer: - A

40. You have two formulae for capacitance given by:-

$C = \frac{Q}{V}$ and $C = Eo\frac{A}{d}$

If you combine then we have:- $d = \frac{EoAV}{Q}$

$= \frac{EoV}{(Q/A)A}$, Qis charge density

$d = \frac{8.85 \times 10^{-12}\ F/m \times 15ov}{300 \times 10^{-6}\ C/M^2}$ $\quad d = \underline{4.425\mu m}$

Answer .D

41. In any electric circuit simple or complex the maximum power transfer will take, pace when the load resistance 'R' is equal to the internal resistance of the battery or power source

Answer .B

Telegram: https://t.me/ExtremeSeries

44. Look at the diagrammatical representation of the problem
There are three forces T, Fe and mg on the changed object. The body is suspended at an angle 'Q' means, it is in equilibrium. Hence by applying the 1st condition of equilbrivmwe can proceed as follows:-

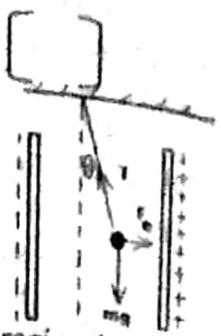
$\Sigma F_x = 0 \Rightarrow Fe = Eq = T\sin\theta$ .....(1)

$\Sigma F_y = 0 \Rightarrow mg = T\cos\theta$ .....(2)

Combine(1) & (2) $E = \frac{mg\sin\theta}{q} \Rightarrow \frac{}{\cos\theta}$

$E = \frac{mg\tan\theta}{Q}$

Since:- $V = Ed = \frac{mgd\tan\theta}{q}$



Answer .A

43. The absence of a particular color of light in the region is due to the interference of light of that color in that region.
Answer;- C

. The intensity of sound (I)and the distance (r) form the source are related by the relation:- $\frac{I_1}{I_2} = \frac{r_2^2}{r_1^2}$

Then if $r_2 = 3r_1$ then by substituting in the above equation you will get that:-
$I_2 = I_1/9$
Answer .D

45. When the water flows through the hose, it is known that the rate of mass flow that is m/t is constant.
Hence you can write:- At two different points with different cross sections:-
- $m_1/t = m_2/t$ But t is cancelled and we have
- $m_1 = m_2$
- density (V_1) = density (V_2)
- $A_1v_1 = A_2v_2$
- $V_2 = \frac{A_1}{A_2}V_2 = 31.6\text{m/sec}$

Answer:- B

46. When the block is released, the spring potential energy of the spring will be changed to the gravitational potential energy of the block;
Hence:- $1/2Kx^2 = mgh$
$h = \frac{kx^2}{2mg} = 10.2m$
Answer;- A

47. As you can see the wheat stone bridge circuit in not balanced. Nd if the current that passes through the system is 2A, the source voltage must be 10v.
Then you can write:-
- $V_A = 2A \times 2\Omega = 4v$
- $V_B = 2A \times 3\Omega = 4v$
- Then $V_B - V_A = 6v - 4v = -2v$
- Answer:- A

48. To convert a galvanometer in to an ammeter we need to connect a very small resistance resistor (shunt) in parallel to it.
Answer .B

49. The circuit represents wheat stone bridge circuit for capacitors. We need to assume that the circuit to be balanced bridge circuit.
In this case points C and D will be at the same potential. This implies that the middle capacitance can be ignored or has no effect.
Then:-Adding C_1 and C_2 at the top inseries= 5F
Adding C_3 and C_4 at the bottom in series = 5F
Then the remaining branch becomes parallel and so we get the equivalen capacitance to be 10F.
Answer:-A

50. In the young's double slit experiment if we use white light instead of monochromatic light, then only the central fringe become white but the others are colored.
Answer .D

Telegram: https://t.me/ExtremeSeries

1. **solution:** photons are stream of particles emmited from an electromagnetic wave and they are quantum of light.
   Answer: D
2. **Solution:** the time constant of the circuit τ is given by
   $$\tau=RC=10^{6}\Omega x5x10^{-6}F=5\sec onds$$
   The charge stored in the capacitor after the switch is closed is obtained from
   $$Q=Q_{\circ}(1-e^{\frac{t}{RC}})=C\varepsilon=5\mu Fx30V=150\mu C$$
   Answer: A
3. **Solution:** According to Newton's third law the force exerted on water by the fins d of the fish is action and the reaction is directed westward.
   Answer: C
4. **Solution:** since both molecules have the ame temperature implies that they also have the same molecular kinetic energy. There fore, the speed of the molecule is given by
   $$v=\sqrt{\frac{2KE}{m}}$$ and the lighter molecule will have more speed than the heavier molecule. Therefore, choieces A, B, and D are not true.
   Answer: No answer
5. **Solution:** they disagree in the kinetic energy of the ball if one of them is in motion relative to the ball because the speed of the ball for the person at rest is different from the speed of the ball observed by the person in motion because of the relative speed of the ball.
   Answer: D
6. **solution:** $\vec{A}$ has opposite in direction with $\vec{B}$ if and only if c is negative.
   Answer: B
7. **Solution:** According to Faraday's law of electromagnetic induction the emf in a coil depends on
   i. the number of turns of the coil,
   ii. the speed of the approaching magnet, and
   iii. the magnetic moment of the magnet.
   Answer: A
8. **Solution:** The frequency of the wave is calculated from
   $$f=\frac{v}{\lambda}=\frac{50^{m}/_{s}}{25000m}=2x10^{-3}HZ$$

Answer: A

9. Given; the frequency of the wave is f=0.33 HZ. Required T=?
   Solution: the period of thse wave is related to its frequency as
   $$T=\frac{1}{f}=\frac{1}{0.33HZ}=3\sec onds$$
   Answer: D
10. **Solution:** Unit vectors are used to describe the position of an object relative to another object.
    Answer: A
11. **Solution:** In a reversible process, every state along the path is an equilibrium state, so the system can return to its initial condition by going along the same path in the reverse direction.
    Answe: A
12. **Solution:** from the givev informations in the question it is clear that the electric potential at A and B are the same(A and B also lie on an equipotential line). When the charge is moved along an equipotential lie the work done is always zero because the p.d. between A and B is zero.
    Answer: A
13. Given; C_{air} =1 Pf, C_{wax} = 2 pF and $d_1=2d_o$ required k=?
    Solution: in the absence ofa dielectric the capacitance of a paraleel-plate capacitor is given by
    $$C_o=\frac{\varepsilon_o A}{d}=1pF.....1$$
    When a dielectic made of wax is inserted between the plates its capacitance becomes
    $$C_{wax}=\frac{k\varepsilon_o A}{d_1}=\frac{k\varepsilon_o A}{2d}=2pF.....2$$
    Dividing eq(2) by eq(1) gives k=4
    Answer: B
14. **Solution:** A transformer is a device used to step-up or step-down voltage. In a step-up transformer voltage increases due to large number of turns of coils in the secondary.
    Answer: C
15. **solution:** The current flowing across a single bulb rating at 220 V, 500 W is
    $$I_1=\frac{P}{V}=\frac{500W}{220V}=\frac{25}{11}A$$

Telegram: https://t.me/ExtremeSeries

If the power line is protected by a 15 A fuse, then the number of bulbs it accommodates equals

$$No\ of\ bulbs = \frac{I}{I_1} = \frac{15\,A}{\frac{25}{11}A} = 6$$

Answer: C

16.solution: the value of $\vec{B}X\vec{A} = (5i - 3j + k)X(8i + j - 2k) = 5i + 18j + 29k$

Answer; D

17. Solution: when the particle is brought to rest its velocity becomes zero. this means the derivative of position with respect to time becomes zero.

$$\frac{dX(t)}{dt} = \frac{d(25t - 3t^2)}{dt} = 25 - 9t^2 = 0$$

Solving the quadratic equation gives $t = \frac{5}{3}s = 1.67\,s$

Answer; B

18. Solution: First we have to determine the deceleration of the car from

$$a = \frac{v^2 - u^2}{2s} = \frac{0 - (30\,m/s)^2}{2(150\,m)} = -3\,m/s^2$$

The aerage force of friction that bring the car to a stop is

f=ma =(2000 kg)(3 m/s^2)=6000 N

Answer: B

19. Solution: The moment of inertia of a wheel is I = mr^2 for a group of identical masses it is given by

$$I_{tot} = 4mr^2 = 4(3\,kg)(0.75\,m)^2 = 6.75\,kgm^2$$

Answer: C

20.solution: In a refrigerator and heat pumps heat energy is ejected from the cold reservoir to kitchen in homes.If the room is thermally insulated the temperature of the room remains the same according to the second law of thermodynamics.

Answer: D

21. solution: the change in the length of aluminum is obtained from

$$\Delta L = L_o \alpha \Delta T = 3\,m\,x\,2.31/(K)x100\,K = 6.93x10^{-3}\,m$$

$$\%\,change\ in\ length = \frac{\Delta L}{L}x100\% = \frac{6.93\,x\,10^{-3}\,m}{3\,m}x100\% = 0.23\%$$

Answer: D

22.Given:

$P_{abs} = 1.81x10^5\,Pa, P_{atm} = 1.01x10^5\,Pa, \rho_{oil} = 800\,kg/m^3$ and $g = 10\,m/s^2$

Reqyired h=?

Solution: the depth of oil is obtained from

$$P_{abs} = P_{atm} + \rho_{oil}gh,$$

$$h = \frac{P_{abs} - P_{atm}}{\rho g} = \frac{80000\,Pa}{800\,kg/m^3\,x10\,m/s^2} = 10\,m$$

Answer: C

23.solution: the angular momentum of the particle is calculated from

$L = m(\vec{r}X\vec{v}) = 1.5kg(1.5i + 2.2j)mX(4.2i - 3.6j)\,m/s$

1.5kg x(-7.32 k)m^2/s = - 22 k kg m^2/s

Answer: A

24. Given; $\vec{F}_1 = 100\,N\,at\,30°, \vec{F}_2 = 80\,N\,at\,60°, and\ m = 2\,kg$

Required a=?

Solution: the given forces are at right angles to each other and the magnitude of their resultant is obtained from Pythagoras theorem as follows

$$F_R = \sqrt{F^2{}_1 + F_2{}^2} = \sqrt{(100\,N)^2 + (80\,N)^2} = 128\,N$$

The acceleration of the object is obtained from Newton's second law

$$a = \frac{F_R}{m} = \frac{128\,N}{2\,kg} = 64\,m/s^2$$

Answer: B

25. Given; u = 20 m/s, v=0, and a= - 5 m/s^2 required S=?

Solution: The distance covered by the car before it comes to a stop is

$$s = \frac{v^2 - u^2}{2a} = \frac{0 - (20\,m/s)^2}{2(-5\,m/s^2)} = 40\,m$$

Answer: C

26.Solution: According to classical theory prediction as the wavelength approaches zero, the amount of energy being radiated should increase.

Telegram: https://t.me/ExtremeSeries

The total radiation emmited(the area under the curve) increases with increasing temperature. Since from the given curve

$\lambda_1 > \lambda_2 > \lambda_3 > \lambda_4 \Rightarrow T_1 > T_2 > T_3 > T_4$. Therefore, choice A is not true

Answer: A

27. **Solution:** When a long straight current carrying wire held parallel to a rectangular loop two magnetic forces at a perpendicular distance exert equal force in opposite direction and these forces are couples that produce turning or rotational motion due to the net torque.

Answer: D

28. **Solution:** Among the given alternatives the only device that uses the principle of electromagnetic induction is the Ac generator while the others use a Dc current for their operation.

Answer: A

29. Given; $E = 5x10^5$ N/C, $q=1.6x10^{-19}$ C Required; F =?

Solution: The electrostatic force acting on the electon is given by

$$F_{ele} = qE = 1.6x10^{-19}\,C\,x5x10^5\,N/C = 8x10^{-16}\,N$$

Answer: C

30. **solutiion:** the equivalent resistance of the 6 Ω and the 12 Ω resistors connected in series are each 18 Ω along path ABC and along path ADC and this resistors are in parallel to the source. Therefore the total resistance of the circuit becomes 9 Ω according to the rule of parallel connection of resistors.

From this informations the total current of the circuit becomes

$$I = \frac{V}{R_T} = \frac{6V}{9\Omega} = \frac{2}{3}A$$

The current across each branch equals $I_1 = I_2 = \frac{1}{3}A$.

The Potential difference betweenpoints A and B = V_{BA}= $I_1R_1 = \frac{1}{3}Ax6\Omega = 2V$

The Potential difference between points A and D= V_{DA}= $I_2R_2 = \frac{1}{3}Ax12\Omega = 4V$

The potential difference between points B and D is

$V_{BD}=V_D - V_B = 4\ V - 2\ V = 2\ V$

Answer: B

31. **Solution:** The universal gravitational mass of the body on earth and Jupiter are respectively,

$$F_E = \frac{GM_Em}{r^2_E} \text{ and } F_E = \frac{GM_Jm}{r^2_J}$$

Given that $M_J = 381M_E$, and $W_J=2.4W_E$, and taking the ratio F_E to F_J you will get

$$\frac{F_E}{F_J} = \frac{\frac{GM_Em}{r^2_E}}{\frac{GM_Jm}{r^2_J}} = \frac{M_E}{M_J}(\frac{r_J}{r_E})^2$$

$$\frac{318}{2.4} = (\frac{r_J}{r_E})^2 \Rightarrow r_J = 11.5r_E$$

Answer: D

32. **solution:** According to the law of conservation of mechanical energy principle the gravitational potential energy at the top of the incline equals to the sum of translational kinetic energy and rotational kinetic energy of the ball.

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

The moment of inertia of the ball is $I = \frac{2}{5}mr^2$ and the linear speed is related to the angular speed by v=rώ. Substituting this in the equation gives

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}(\frac{2}{5}mr^2)\omega^2 = \frac{1}{2}mv^2 + \frac{1}{5}mv^2 = \frac{7}{10}mv^2$$

Solving for v we get $v = \sqrt{\frac{10}{7}gh}$

Answer: A

33. **Solution:** Let us applying both conditions of equilibrium to the problem

From the first condition: $\Sigma F_Y = 0 \Rightarrow V = W = mg = 200\,N = N_g$

From the second condition: $\Sigma\tau_p = 0$

$$\Sigma\tau_{clock} = \Sigma\tau_{anti} \Rightarrow 200\ N(1.5\cos 30^o)m = N_w(3\sin 30^o)m$$

N_w=57.7 N

Answer: C

34. **solulution:** When theelevator accends with an acceleration 'a'the force exerted on the fluid is given by

F=m(a + g)=ρV(a + g). substituting V=AΔh gives

F= ρAΔh(a + g)

Telegram: https://t.me/Extremeseries

The pressure exerted on the fluid is obtained from

$$P = \frac{F}{A} = \frac{\rho A \Delta h(a+g)}{A} = \rho \Delta h(a+g)$$

Answer: B

35. **Solution:** The kinrtic energy of a particle is related to the momentum of the particle bythe relation

$$E_{K1} = \sqrt{\frac{P^2}{2m_1}} \; and \; E_{K2} = \sqrt{\frac{P^2}{2m_2}}$$

Since the particles have the same momenta, then

$$\frac{E_{K1}}{E_{K2}} = \sqrt{\frac{m_2}{m_1}}$$

From the above relation we conclude that $E_{K1}=E_{K2}$ if and only if $m_1=m_2$

Answer: C

36. **solution:** The fringe width is related to the slit distance(d), the distance between the slit and the screen(D), and the wavelength by

$$\omega = \frac{\lambda D}{d} \quad if \; D_2 = \frac{D_1}{2} \; and \; d_2 = \frac{d_1}{2}, then$$

$$\frac{\omega_2}{\omega_1} = \frac{\frac{\lambda D_2}{d_2}}{\frac{\lambda D_1}{d_1}} = \frac{D_2 d_1}{D_1 d_2} = 1 \Rightarrow \omega_1 = \omega_2$$

Answer: B

37. **Solution:** Given that $C_1=4\ \mu F, C_2=8\ \mu F$ and $C_T = \frac{24}{11}\mu F$

Required: C_3=?

Since all the capacitors are connected in series, we apply rule of series connection to the problem to determine the capacitance of the unknown one.

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} = \frac{1}{4\mu F} + \frac{1}{8\mu F} + \frac{1}{C_3} = \frac{11}{24\mu F}$$

$$\frac{2C_3 + C_3 + 8}{8\mu F(C_3)} = \frac{11}{24\mu F} \Rightarrow 72\mu F(C_3) + 192\mu F^2 = 88\mu F(C_3)$$

$16(C_3)=192\ \mu F$

$C_3=12\ \mu F$

Answer: A

38 **Solution:** When a charged particle is set into motion in a region of magnetic field along a circular path the magnetic force exerted on it is supplied by the centripetal force of the charge.

$$F_m = F_c \Rightarrow qvB = \frac{mv^2}{r} \qquad q = \frac{mv}{Br}$$

Answer: D

39.**Solution:** The general equation of an alternating current as a function of time is expressed as

$I(t) = A\sin(2\pi ft)$, where A is amplitude and f is the frequency.

Equating this with the given equation we get

A=10.0 A and f=50 HZ

Answer: C

40.Given: R=45 Ω, Z=75 Ω, ΔV_{rms}=210 V. Required: P_{ave}=?

Solution: The total impedance of the RLC circuit is obtained from

$$Z = \sqrt{R^2 + z^2} = \sqrt{(45\Omega)^2 + (75\Omega)^2} = 93\Omega$$

The average power delivered to the circuit is

$$P_{ave} = \frac{\Delta V_{rms}{}^2}{2Z} = \frac{(210V)^2}{186\Omega} = 237.6\ W$$

Answer: No answer

41.**Solution:** the thickness of the film 't' is calculated from the relation

$$t = \frac{\lambda}{4n}$$

Since there is blue light reflected from the surface therefore, the thickness of the film remains the same.

Answer: A

42.**solution:** The energy stored across the plates of a parallel plate capacitor with a dielectric material inserted between the plates is calculated from

$$U = \frac{1}{2}(k\varepsilon_o AdE^2)$$

From the above formula we see that the energy stored depends(or affected) on plate area, distance between the plates and on the value of the dielectric constant, but when charge decreases energy stored also decreases instead of increasing according to the equation $U = \frac{Q^2}{2C}$.

Telegram: https://t.me/ExtremeSeries

Answer: C

43. **solution:** Let the power output produced when 240 V source is used be

$P_{o1} = \frac{V^2_{o1}}{R}$ and the power output when using the 220 V source be

$P_{o2} = \frac{V^2_{o2}}{R}$.

The percentage increase of power output is calculated from

$$\%increase = \frac{P_{o1} - P_{o2}}{P_{o1}} = \frac{V^2_{o1} - V^2_{o2}}{V^2_{o1}} = \frac{(240)^2 - (220)^2}{(240)^2} x100\% = 19\%$$

Answer: C

44. **Solution:** The defecting magnetic force on this section of the wire is

$$\vec{F}_m = I(\vec{L}X\vec{B}) = 2.4A(0.75\vec{i}\ X1.6\vec{k})Tm = (-2.88\vec{k})N$$

Answer: D

45. **Solution:** The voltage drop

Across path BC is $V_1 = 8R$, and

The voltage drop along path DE is also $V_2 = V_1 = 8R$, because they are in parallel. From rules of parallel connection the current along paths BC and DE join at the junction point to give the current across the resistors at F and at G since these resistors are in series. Therefore the current becomes 8 A.

Answer: B

46. **Solution:** During the refraction a wave both speed and wavelength are altered while frequency of the wave remains the same.

Answer: C

47. **Solution:** The pressure of a fluid in motion is expressed by the relation

$$P = \frac{1}{2}(\rho v^2) = \frac{1}{2}(\frac{m}{V})v^2$$

If m1=2m, and v1=1/2 v, then

$$P_1 = \frac{1}{2}(\frac{2m}{V})\frac{v^2}{4} = \frac{1}{4}(\rho v^2) = \frac{P}{2}$$

Answer: C

48. **Solution:** The speed of water through the nozzle is calculated from continuity equation

$$A_1 v_2 = A_2 v_2 \ but\ A_1 v_1 = 0.012 m^3/s = volume\ flux$$

$$A_2 = \frac{\pi d^2_2}{4} = \frac{3.14(2.2x10^{-2}cm)^2}{4} = 3.8x10^{-4}m^2$$

$$v_2 = \frac{0.012 m^3/s}{3.8x10^{-4}m^2} = 31.6 m/s$$

Answer: C

49. **Solution:** The collision between the masses is inelastic and applies the law of conservation of momentum to the problem gives

$m_1 u = (m_1 + m_2)v \Rightarrow 10g(u) = (1010g)v \Rightarrow u = 101v....1$

Applying the conservation of energy principle after impact gives

Kinetic energy at the bottom=potential energy at the top

$\frac{1}{2}mv^2 = mgh \Rightarrow v = \sqrt{2gh} = \sqrt{2x10x0.45}m/s = 3m/s.....2$

Substituting eq(1) in eq(2) gives

u=102x3 m/s=303 m/s

Answer: A

50. **solution:** Let Vre=$3x10^3$ m/s and Vfr=$5x10^3$ m/s

After fuel is ejected and the mass of the rocket is halved the speed of the rocket relative to earth becomes

V FR = V're(after mass is halved) + Vre(before fuel is ejected)

$5x10^5$ m/s= V're + $3x10^3$ m/s

The speed of the rocket after its mass is reduced to half is $2x10^3$ m/s.

Answer: D

Telegram: https://t.me/ExtremeSeries

SOLUTION TO ETHIOPIAN UNIVERSITY ENTRANCE EXAMINATION (EUEE) PHYSICS 2006/2014

1. Solution:- Given $r_2=2r_1$

Let r_1 be the charges between two point charges Q_1 and Q_2 and the force be F_1.

$$F_1 = \frac{kQ_1Q_2}{r_1^2}$$

Q_1 ——— r_1 ——— Q_2

When the distance between the charges is doubled, then the new force becomes

$$F_2 = \frac{kQ_1Q_2}{r_2^2}$$

Q_1 ——— $r_2 = 2r_1$ ——— Q_2

Taking the ratio of the forces gives

$$\frac{F_1}{F_2} = \frac{\frac{kQ_1Q_2}{r_1^2}}{\frac{kQ_1Q_2}{r_2^2}} = \left(\frac{r_2}{r_1}\right)^2 = \left(\frac{2r_1}{r_1}\right)^2 = 4$$

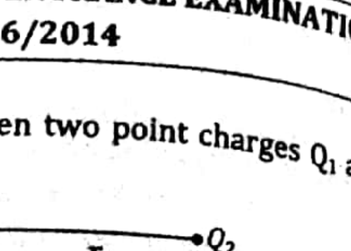
$$4F_2 = F_1 \text{ and } F_2 = \frac{1}{4}F_1$$

When the distance between the charges doubled the force becomes $\frac{1}{4}$ as much the original.

Answer:- C

2. Solution:- The slope of a velocity time graph represents the acceleration.

$$Slope = \frac{\Delta v}{\Delta t} = \frac{v-v}{t} = a$$



Answer:- B

3. Solution:- Given $q = 2.5 \times 10^6 C$

Work done in moving the charge from A to B in a region of an electric field is

$W = \Delta u = 6.3 \times 10^{-4} J$

Required:- The potential difference V between A and E.

The p.d between points A and B is given by

$$V = \frac{\Delta u}{q} = \frac{6.3 \times 10^{-4} J}{2.5 \times 10^{-6} C} = \frac{63}{25} \times 10^2 \frac{J}{C}$$

$$V = 2.52 \times 10^{-2} \, volts$$

Answer:-B

4. Solution:- In a periodic transverse wave a point of the string vibrates or moves in simple harmonic motion with the same angular frequency as the wall.

Answer:- C

5. Solution:- Given $|\vec{D} + \vec{E}| = 3\,|\vec{D} - \vec{E}|$

Required:- The angle θ between $\vec{D}$ and $\vec{E}$

The magnitude of $|\vec{D} + \vec{E}|$ and $\vec{D} - \vec{E}$ is obtained from cosine law. Stated as follows.

$|\vec{D} + \vec{E}| = D^2 + E^2 + 2DEcos\theta$

$|\vec{D} - \vec{E}| = D^2 + E^2 - 2DEcos\theta$

$|\vec{D} + \vec{E}| > 3|\vec{D} + \vec{E}|$ *implies*

$D^2 + E^2 + 2DEcos\theta > 3(D^2 + E^2 - 2DEcos\theta)$

$D^2 + E^2 + 2DEcos\theta = 3D^2 + 3E^2 - 6DEcos\theta$

$2D^2 - 2E^2 = -8DEcos\theta$

$D^2 + E^2 = 4DEcos\theta$

Of D=E, then

$2D^2 = 4D^2 cos\theta$

$cos\theta = \frac{1}{2} \Longrightarrow \theta = 60^0$

Note:- The two vectors must have the same magnitude and the angle between the vectors must be 60^0.

Answer:- D

6. Solution:- The pressure of a fluid kept in a container depends on the density of the fluid and the depth of the fluid. It is independent on the shape of the container and the volume of the fluid.

Answer:- B

Telegram: https://t.me/ExtremeSeries

7. Solution:- The capacitance of an air gap parallel plate capacitor separated by a distance d_1 is given by $C_1 = \frac{EoA}{d_1}$ and the energy stored in the capacitors.

$$u_1 = \frac{1}{2}C_1V_1^2 = \frac{1}{2}\left(\frac{EoA}{d_1}\right)(E_1d_1)^2 = \frac{1}{2}EoAd_1E_1^2$$

$$V_1 = E_1d_1 = V_2 = V$$

When the distance between the plates is increased to d_2

$$C_2 = \frac{EoA}{d_2} \text{ and } U_2 = \frac{1}{2}C_2V_2^2$$

$$U_2 = \frac{1}{2}\frac{EoA}{d_2}(E_2d_2)^2 = \frac{1}{2}EoAE_2^2d_2$$

From $V_1 = V_2$ we have $\frac{E_1}{E_2} = \frac{d_1}{d_2} > 1 \Rightarrow E_1 > E_2$

$$E_1 = \frac{d_2}{d_1}$$

$$U_1 = \frac{1}{2}EoAd_1\left(\frac{d_2^2}{d_1^2}\right)E_2 = \frac{1}{2}EoA\left(\frac{d_2^2}{d_1}\right)$$

Taking the ratio U_1 to U_2 gives

$$\frac{U_1}{U_2} = \frac{\frac{1}{2}EoAE_2^2\frac{d_2^2}{d_1}}{\frac{1}{2}EoAE_2^2d_2} = \frac{d_2}{d_1} > 1$$

$$U_1 > U_2$$

- The electric field between the plates decrease above increase plate separation.
- The capacitance of the parallel plates capacitor also decrease.
- The potential difference between the plates remains the same.

Answer:- No answer

8. Solution:- For a particle moving in a circular path the centripetal acceleration is always perpendicular to the tangential velocity (Velocity vector) of the particle.

Answer:- C

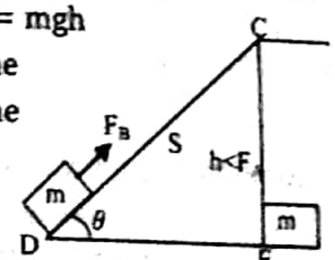
9. Solution:- According to Faraday's law of electromagnetic induction we can induce current in the inner loop by varying the

current in the outer loop of wire. We all such a phenomenon self induction. The current induced in the inner loop always opposes the change in current in the outer loop according to Lenz's law.

Answer:- A

10. Solution:- In lifting the box by A from E to C vertically up in air increase the PE of the box by $\Delta V_A = mgh$

When the second man (B) drags the box along a ramp from D to C the increase in the PE of the box is



$$\Delta V_B = mgsin\theta(S) \text{ but } \frac{h}{s} = sin\theta \Rightarrow h = sin\theta$$

- A and B change the PE by the same amount.

Answer:- A

11. Solution:- Given $U_1 = U_2 = U, \theta_1 = 30^0 \text{ and } \theta_2 = 60^0$ the time taken by the projectiles to hit the ground are respectively, given by

$$t_1 = \frac{2U_1sin\theta_1}{g} = \frac{2Usin30^0}{g}$$

$$t_2 = \frac{2U_2sin\theta_2}{g} = \frac{2Usin60^0}{g}$$

$$\frac{t_1}{t_2} = \frac{sin30^0}{sin60^0} = \frac{\frac{1}{2}}{\frac{\sqrt{3}}{2}} = \frac{1}{\sqrt{3}}$$

$t_2 = \sqrt{3}t_1$ the projectile filed at angle of incline $\theta_1 = 30^0$ will reach (hit) the ground first than the one field at angle of 60^0.

Answer:-B

12. Solution:- The velocity of the planet at A and B are given by

$$V_a = \frac{2\pi r_a}{T_a} = \frac{2\pi\left(\frac{r_b}{8}\right)}{t} \qquad t_a = t_b = t$$

Telegram: https://t.me/ExtremeSeries

$$V_b = \frac{2\pi r_b}{T_a} = \frac{2\pi r_b}{t}$$

$$\frac{V_a}{V_b} = \frac{\frac{2\pi r_b}{8t}}{\frac{2\pi r_b}{t}} = \frac{1}{8}$$

The Velocity of the planet at point B is

$$V_b = \frac{1}{8}V_a = \frac{1}{8}\left(60{,}000\frac{m}{s}\right) = 7{,}500\ m/s$$

Answer:- A

13. Solution:- Given current flow =I=25mA

time taken=t=3hr=$3 \times 3600sec$

The total charge delivered by the battery is given by

$q = It = 25 \times 10^{-3}A \times 10{,}800\ sec$

$= 270\ A.S = 270C$

Answer:- D

14. Solution:- When a cup of coffee at 80^0C is placed on a table whose temperature is 20^0C heat is transferred to the table and as a result the thermal energy of the coffee decreases and the molecules move more slowly.

Answer:- A

15. Solution:- Given $\vec{A} = 2\hat{\imath} + \hat{\jmath} + \hat{k}$ and $\vec{B} = \hat{\imath} - 4\hat{\jmath} + \lambda$. If $\vec{A}$ is perpendicular to $\vec{B}$, then

$\vec{A}.\vec{B} = 0$

$\vec{A}.\vec{B} = AxBx + AyBy + A_2B_2 = 0$

$= 2(1) + 1(-4) + 1(\lambda) = 0$

$= 2 - 4 + \lambda = 0$

$\lambda = 2$

Answer:- C

16. Solution:- The mass flow rate in the hose with diameter 16cm to fill a 10 liter bucket in 20 seconds is given by

$$\eta_1 = \frac{\Delta m_1}{\Delta t_1} = \frac{\Delta(\rho_1 V_1)}{\Delta t_1} = \rho_1\frac{\Delta V_1}{\Delta t_1} = \frac{\rho_1 A_1 d_1}{\Delta t_1}$$

When using a hose with diameter $d_2 = 8cm$ the mass flow rate is given by

$$\eta_2 = \frac{\Delta m_2}{\Delta t_2} = \frac{\Delta(\rho_2 V_2)}{\Delta t_2} = \frac{\rho_2 A_2 d_2}{\Delta t_2}$$

The mass flow rate is the same using 16cm diameter hose and 8cm hose. Thus

$$\frac{\rho_1 A_1 S_1}{\Delta t_1} = \frac{\rho_2 A_2 S_2}{\Delta t_2}$$

$$\frac{\pi d_1^2}{4(20sec)}(S_1) = \frac{\pi d_2^2}{4t_2}S_2$$

$Since\ S_1 = S_2$ then

$$\frac{4}{20sec} = \frac{1}{t_2} \Rightarrow t_2 = 5sec$$

From Bernoulli's continuity equation

$A_1V_1 = A_2V_2$

$\frac{\pi d_1^2}{4}V_1 = \frac{\pi d_2^2}{4}V_2$ $since\ d_1 = 2d_2$ we have

$V_1(d_2^2) = V_2 d_2^2$

$V_2 = 4V_1$

Since the volume flux is the same using both hoses so a 10 liter bucket can be filled in 10 seconds when using an 8cm diameter hose.

Answer:-B

17. Solution:- The number of complete cycles performed by the spring are 20 and the time taken is 36 second. The frequency of the vibrating spring is:-

$$frequency = \frac{No\ of\ complete\ cycle}{time\ taken}$$

$$= \frac{20cycles}{36sec} = \frac{5C0}{9sec} \approx 0.55Hz$$

$$Period\ of\ oscillation = \frac{time\ taken}{No\ of\ cycles} = \frac{36se}{20cy} = 1.8\ sec$$

Answer:- No answer

18. Solution:- Given $R = 4\Omega, L = 27mH, C = 380\mu F, f = 50H$ the inductive reactance X_L of the circuit is

$X_L = 2\pi fL = 2(3.14)(50Hz)(2.7 \times 10^{-2}H)$

$= 3.14 \times 2.7\Omega$

Telegram: https://t.me/Extremeseries

$= 8.4\Omega$

The capacitive reactance of the circuit is

$$X_C = \frac{1}{2\pi fc} = \frac{1}{2(3.14)(50Hz)(380\times10^{-6}F)}$$

$$X_L - X_C = 8\Omega - 8.4\Omega = 0.4\Omega$$

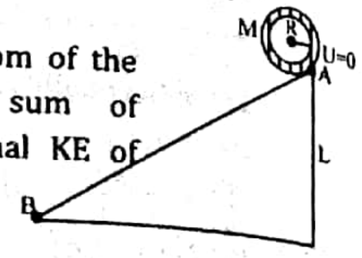
The total ienipedance (Z) of the circuit is given by

$$Z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{(4\Omega) + (-0.4\Omega)^2} \approx 4\Omega$$

Answer:-B

19. Solution:- The total ME of the hoop at the top (point A) is

$$E_{MA} = Mgh = PE_A$$



The total ME at the bottom of the incline plane is the sum of translational and rotational KE of the hoop. Thus,

$$E_{MB} = \frac{1}{2}mv^2 + \frac{1}{2}Iw^2$$

The moment of inertia of a hoop is $I = Mr^2$, therefore;

$$E_{MB} = \frac{1}{2}mv^2 + \frac{1}{2}MRw^2$$

Applying the law of conservation of ME to the problem gives

$$MRw^2 = mgh$$

$$w^2 = \frac{gh}{R^2} \Rightarrow w = \frac{\sqrt{gh}}{R}$$

The angular momentum of the hoop about its center of mass is given by

$$L = Iw = MR^2\left(\frac{\sqrt{gh}}{R}\right) = MR\sqrt{gh}$$

Answer:- D

20. Solution:- Given :- $A = 1cm^2 = 10^{-4}m^2, \ell = 2km = 2{,}000m,$

$\rho = 2\times10^{-8}\Omega m$

Required:- R=?

The resistance of the copper cable is given by

$$R = \frac{\rho\ell}{A} = \frac{2\times10^{-8}\Omega m \times 2{,}000m}{10^{-4}m^2}$$

$= 4\times10^{7}\Omega = 0.4\Omega$

Answer:- D

21. Solution:- Given m=500kg, F=W=mg=5000N

$A=0.2cm^2 = 2\times10^{-5}m^2$

$L = 3m, Y = 2\times10^{11}Pa$

Required:- ΔL?

The Young's modulus of the steel wire is

$$Y = \frac{Stress}{Strain} = \frac{\frac{F}{A}}{\frac{\Delta L}{L}}$$

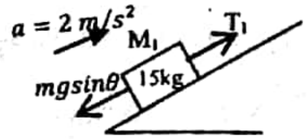
$$\Delta L = \frac{FL}{AY} = \frac{5{,}000N\times3m}{2\times10^{-5}m^2\times2\times10^{11}N/m^2}$$

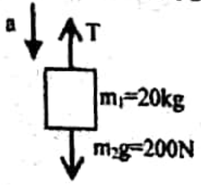
$$= \frac{15\times10^{3}\times10^{-5}}{4\times10^{11}}m$$

$$\Delta L = 15\times10^{-3}m = 3.75\times10^{-3}m = 3.75mm$$

Answer:-A

22. Solution:- Applying Newton's 2nd law to the system of blocks let us determine the size of the tension force on each strings.





Net force=m_1a on m_1

$$T_1 - mgsin\theta = m_1a$$

$$T_1 - 150N(0.6) = 15kg\left(2\frac{m}{s^2}\right)$$

$$T_1 - 90N = 30N$$

$$T_1 = 120N$$

Net force on= m_2a on m_2

$$200N - T_2 = 20kg\times2\,m/s^2$$

$$T_2 = 200N - 40N$$

$$T_2 = 200N - 40N$$

$$T_2 = 160N$$

The turning effect (torque) on the pulley to produce rotation equals to

$$T_{net} = (T_2 - T_1)R = I\alpha = I\left(\frac{a}{R}\right)$$

$$(160N - 120N)(R) = \frac{I\left(2\frac{m}{s^2}\right)}{2\times10^{-1}m}$$

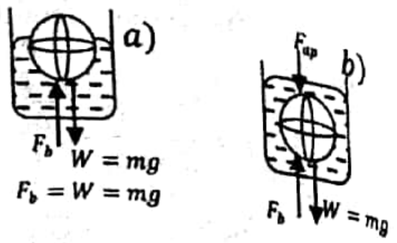
Telegram: https://t.me/ExtremeSeries

$$I = \frac{40N \times (2 \times 10^{-1})^2}{2\frac{m}{s^2}} = \frac{1.6Nm}{2\frac{m}{s^2}} = 0.8kgm^2$$

Answer:-A

23. Solution:- Given d=6cm, $\rho_b = 0.4\frac{kg}{m^3}$, $g = 10\frac{m}{s^2}$, $\rho_w = 1\frac{g}{cm^3}$.

The tennis ball floats in water when it is completely submerged in water when adown ward force is applied on it as shown in the figure. (b)



The up thrust of the water (buoyant force) is given by $F_b = \rho_w g$.

From the figure in part b

$F_{ap} + W = F_b = \rho_w V_s g \qquad V_{id} = V_s = \frac{4}{3}\pi r^3$

$F_{ap} + \rho_s V_s g = \rho_w V_s g$

$F_{ap} = (\rho_w - \rho_s)V_s g$

$= \left(1\frac{g}{cm^3} - 0.4\frac{g}{cm^3}\right)\left(10\frac{m}{s^2}\right)\left(\frac{4}{3}\pi(27 \times 10^{-6}m^3)\right)$

$= 0.6\ \frac{g}{cm^3} \times 10\frac{m}{s^2} \times \frac{4}{3} \times \frac{22}{7} \times 27 \times 10^{-6}m^3$

$= 600\frac{kg}{m^3} \times 10\frac{m}{s^2} \times \frac{88}{7} \times 9 \times 10^{-6}m^3$

$= 678.85 \times 10^3 \times 10^{-6}N$

$F_{ap} = 0.67885N \approx 0.68N$ *down ward*

Answer:- A

24. Solution:- Given d=0.03mm, D=4.8m fringe width=w=4.5cm

Required=λ =?

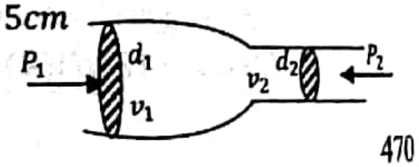
The wavelength of monochromatic light is obtained from

$$\lambda = \frac{wd}{d} = \frac{4.5\times10^{-2}m \times 4.8m}{3\times10^{-5}m} = 7{,}200m$$

Answer:- No answer

25. Solution:- Given:- $d_1 = 10cm,\ d_2 = 5cm$

$P_1 = 80{,}000pa$



$P_2 = 6{,}000pa$

From Bernoulli's is continuity equation we have

$A_1V_1 = A_2V_2 \Longrightarrow \frac{\pi d_1^2}{4}V_1 = \frac{\pi d_2^2}{4}V_2$

$(10cm)^2V_1 = (5cm)^2V_2 \Longrightarrow V_2 = 4V_1$

From Bernoulli's energy equation we also have

$P_1 + \frac{1}{2}\rho V_1^2 = P_2 + \frac{1}{2}\rho V_2^2$

$\frac{1}{2}\rho(V_2^2 - V_1^2) = P_1 - P_2$

$500\frac{kg}{m^3} \times 15V_1^2 = 74{,}000pa$

$15V_1^2 = \frac{74{,}000pa}{500\frac{kg}{m^3}}$

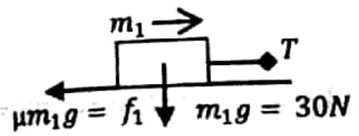
$V_1^2 = \frac{\frac{740m^2}{s^2}}{75} \approx 9.86\frac{m^2}{s^2}$

$V_1 = \sqrt{9.86}\frac{m}{s} \approx 3.15\frac{m}{s}$

$V_2 = 4V_1 = 4\left(3.15\frac{m}{s}\right) = 12.6\frac{m}{s}$

Answer:- C

26. Solution:- Given:- $m_1 = 3kg, = 5kg, \mu = 0.3, \theta = 53^0$ let us apply Newton's 2nd law to each block.

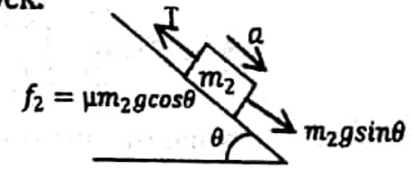


The opposing friction force between M_1 & the surface is

$f_1 = \mu m_1 g = 0.3 \times 30N = 9N$

$F_{net}(m_1) = T - f_1 = m_1 a$

$t - 9N = 3kg(a) - - - (1)$



The opposing friction force between M_1 & the surface is

$f_2 = \mu m_2 g\cos\theta$

$= 0.3 \times 50N \times 0.6 = 9N$

$50N(0.8) - 9N - T = 5kg(a)$

$31N - T = 5kg(a) - -(2)$

Solving (1) and (2)simultaneously gives

Telegram: https://t.me/ExtremeSeries

$$T - 9N = 3kg(a)$$
$$+\ 31N - T = 5kg\ (a)$$
$$22N = 8kg(a) \Rightarrow a = 2.75\frac{m}{s^2}$$

$Substituting\ a = 2.75\frac{m}{s^2}\ in\ (1) gives$

$$T - 9N = 3kg\left(2.75\frac{m}{s^2}\right) = 8.25N$$
$$T = 17.25\ N$$

Answer:- B

27. Solution:- Given m=3kg, w=mg=30N

$\theta_1 = 53^0\ and\ \theta_2 = 37^0$

If the given system is in equilibrium, then the tension in each string is calculated from the following equations.

$$T_1 = \frac{mgcos\,\theta_2}{\sin(\theta_1 + \theta_2)}\ but\ \sin(\theta_1 + \theta_2) = \sin 90^0 = 1$$

$$Thus, T_1 = 3kg \times 10\frac{m}{s^2}\cos 37^0$$
$$= 30N(0.8) = 24N$$
$$T_2 = mgcos\,\theta_1 = 30N\cos 53^0 = 30N(0.6) = 18N$$

Answer: D

28. Solution:- The frequency of the spring mass system when a block of mass m is hanged is given by

$$f_1 = f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

When this mass is replaced by a second mass m_2=2m the frequency of oscillation is

$$f_2 = f = \frac{1}{2\pi}\sqrt{\frac{k}{2m}}$$

$$\frac{f_1}{f_2} = \frac{f}{f_2} = \frac{\frac{1}{2\pi}\sqrt{\frac{k}{m}}}{\frac{1}{2\pi}\sqrt{\frac{k}{2m}}} = \sqrt{2}$$

$$\frac{f}{f_2} = \sqrt{2} \Rightarrow f_2 = \frac{f_1}{\sqrt{2}} = \frac{1}{\sqrt{2}}f$$

Telegram: https://t.me/ExtremeSeries

Answer:- A

29. Solution:- Given $u = 20\frac{m}{s}$, $x = 36m$, $\theta = 37^0$ the time taken by the ball to hit the cross bar is obtained from $x = V_{ox}t \Rightarrow$

$36m = 20\frac{m}{s}, \theta = 37^0(t)$

$$t = \frac{36m}{15\frac{m}{s}} = 2.25\ s$$

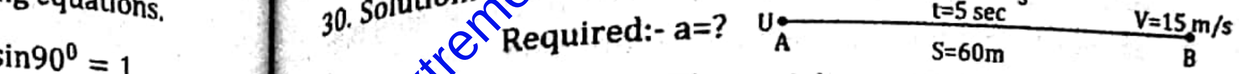
The height reached by the ball is given by

$$h = V_{oy}t - \frac{1}{2}gt^2 = 20\frac{m}{s}sin37^0(2.258) - 5\frac{m}{s^2}(2.25\ s)^2$$
$$= 27m - 25.3m \approx 2.85m\ when\ g = 9.8\frac{m}{s^2}$$

Answer:-B

30. Solution:- Given $S_{AB} = 60m, t = 5\ \&, V = 15\frac{m}{s}$

Required:- a=?



The initial speed (u) is obtained from

$$S = \left(u + \frac{V}{2}\right)t \Rightarrow u = \frac{2s}{t} - V$$

$$\frac{2 \times 60m}{5s} - 15\frac{m}{s} = 24\frac{m}{s} - 15\frac{m}{s} = 9\frac{m}{s}$$

The acceleration of the body is calculated from

$$a = \frac{V - u}{t} = \frac{15\frac{m}{s} - 9\frac{m}{s}}{5\ sec} = \frac{6\frac{m}{s}}{5\ s} = 1.2\frac{m}{s^2}$$

Answer:-D

31. Solution:- A heat sink in a machine is used for the purpose of absorbing heat and for this reason it must have high specific heat capacity and low conductivity.

Answer:- A

32. Solution:- Given $V_p = 6V, N_p = 50, N_s = 100$

Required:- V_s =?

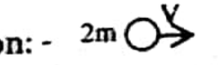
The voltage in the secondary is obtained by using the transformer equation given by

$$\frac{V_p}{V_s} = \frac{N_p}{N_s} \Rightarrow V_s = \frac{N_s}{N_p}(V_p) = \frac{100}{50}(6V) = 12V$$

Answer:- B

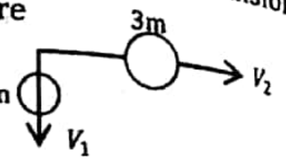
33. Solution:- For a reversible isothermal process the change in the entropy of a body is zero

Answer:- C

34. Solution: - 

The collision is elastic before collision

The total momentum before collision in the x – direction is



$P_{bef}(x) = m_1u_1 + m_2u_2 = 2mu$ 2m right +3mu left

$= -mv\ left = mv\ right$

$P_{bef}(y) = 0$

After collision particle 2m is moving in the y direction and particle 3m is moving in the x direction. Thus,

$P_{after}(x) = 2m(0) + 3m(u_1) = 3mu_2$

$P_{after}(y) = -2mu_1\hat{j} + 3m(0) = -2mu_1\hat{j}$

Applying the law of conservation of momentum along both axes gives.

$P_{bef}(x) = P_{aft}(x)$

$$mv\ left = 3mV_2V_2 = \frac{u}{3}left = 0.33u\ left$$

From the law of conservation of KE we also have

$E_{ki}(2m) + E_{ki}(3m) = E_{kf}(2m) + E_{kf}(3m)$

$$E_{ki}(2m) + \frac{3}{2}mu^2 = 0 + \frac{1}{6}mu^2$$

$$E_{ki}(2m) = \frac{1}{6}mu^2 - \frac{3}{2}mu^2 = -\frac{4}{3}mu^2 = 1.33mu^2$$

Answer:-B

35. Solution:- Given:- $\lambda = 0.1m$ distance between points=D=3.2m For two coherent sources constructive interference results when the path difference (distance) or when the points are at a distance of

$d = (2m + 1)\lambda = (2 + 1)(0.1m) = 0.3m\ for\ m = 1$

Answer:- A

36. Solution:- Given N=50, $\theta = 37^0, B_1 = 200mT$

$B_2 = 600mT, \Delta t = 4s$

The charge in flux ($\Delta\emptyset$) is given by

$$\Delta\emptyset = \Delta BAcos37^0$$

The induced emf in the coil is obtained from

$$\varepsilon = \frac{N\Delta\emptyset}{\Delta t} = \frac{N\Delta BAcos37^0}{\Delta t} = \frac{N\Delta Bs^2cos37^0}{\Delta t}$$

$$160 \times 10^{-3}V = \frac{50(0.4T)(s^2)(0.8)}{4s}$$

$$s^2 = \frac{16 \times 10^{-2}Vs}{4T} = 4 \times 10^{-2}\mu^2$$

$S = 2 \times 10^{-1}m = 0.2m$

The total length of the coil (L) is given by

$$L = 50(4S) = 50(4)(0.2m) = 40m$$

Answer:- No answer

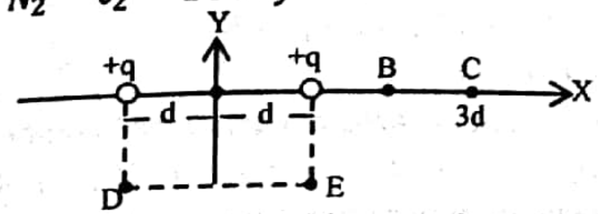
37. Solution:- Given:- $N_0 = 224, t_1 = 3.5\ days, t_2 = 14\ days$ the half life of sample is given by

$$N_1 = \frac{N_0}{2}e^{t_1\frac{1}{2}}\ or\ t_1 = \frac{N_1}{2}ln2$$

and $N_2 = \frac{N_0}{2}e^{t_2}\ or\ t_2 = \frac{N_2}{2}ln2$

$$\frac{N_1}{N_2} = \frac{t_1}{t_2} = \frac{3.5days}{14\ days} = \frac{1}{4}$$

Answer:- B

38. Solution:- 

The net electric field strength at point A is

$$\overrightarrow{E_A} = \overrightarrow{E_1} + \overrightarrow{E_2} = \frac{kq}{d^2}i + \left(-\frac{kq}{d^2}\hat{i}\right) = 0$$

The net electric field strength at point B is

$$\overrightarrow{E_B} = \overrightarrow{E_1} + \overrightarrow{E_2} = \frac{kq}{(3d)^2}i + \frac{kq}{d^2}\hat{i} = \frac{10}{9}\frac{kq}{d^2}\hat{i}$$

The net electric field strength at point C is

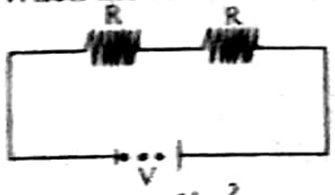
Telegram: https://t.me/ExtremeSeries

$$\overrightarrow{E_C} = \overrightarrow{E_1} + \overrightarrow{E_2} = \frac{kq}{(4d)^2} i + \frac{kq}{2d^2} \hat{i} = \frac{5}{16} \frac{kq}{d^2} \hat{i}$$

In the same way the net electric field strength at points D and E is also non zero.

Answer:- A

39. Solution:- When the resistors are connected in series



$P_S = 8W \quad I_S = \frac{V}{2R}$

$R_T = 2R$

$$P_S = \left(\frac{V}{2R}\right)^2 (2R) = \frac{V^2}{4R^2} \times 2R = \frac{V^2}{2R} = 8W$$

When these same resistors are connected in parallel

$$R_T = \frac{R}{2} \text{ and } I_P = \frac{V}{R_T} = \frac{2V}{R}$$

The power delivered for parallel connection is

$$P_P = I_P^2 R_T = \left(\frac{2V}{R}\right)^2 \times \frac{R}{2} = \frac{4V^2}{R^2} \times \frac{R}{2} = \frac{2V^2}{R}$$

$$\frac{P_S}{P_P} = \frac{\frac{V^2}{2R}}{\frac{2V^2}{R}} = \frac{1}{4} \Rightarrow \frac{8W}{P_P} = 32W$$

Answer:- D

40. Solution:- $\ell_B = 2\ell_A, R_B = 2R_A$ let $R_A = R$

Required:- R_B =?

The cross sectional area of each wire is respectively,

$A_A = \pi r_A^2 = \pi R_A^2$ and $A_B = \pi r_B^2 = \pi (2R_A)^2 = 4\pi R_A^2$

The resistance of wire B is given by

$$R_B = \frac{\rho \ell_B}{A_B} = \frac{\rho \times 2\ell_A}{4\pi R_A^2} = \frac{1}{2}\left(\frac{\rho\, 2\ell_A}{4\pi R_A^2}\right) = \frac{1}{2} R_A = \frac{1}{2} R$$

Answer:-C

41. Solution:- Given:- $\beta_1 = 120dB,\ S_1 = 4m$

$\beta_2 = 100dB, S_2 = ?\quad I_0 = 10^{-12} \frac{W}{m^2}$

The intensity level for each sound is obtained from

$$\beta = 10 \log \frac{I_1}{I_0} dB$$

For the first sound $\beta_1 = 10 \log \frac{I_1}{I_0} dB$

$$120\text{dB} = 10 \log \frac{I_1}{10^{-12} \frac{W}{m^2}} dB$$

$$12 = \log_{10} \frac{I_1}{10^{-12} \frac{W}{m^2}}$$

$$10^{12} = \frac{I_1}{10^{-12} \frac{W}{m^2}} \Rightarrow I_1 = 1 \frac{W}{m^2}$$

For the second sound

$$\beta_2 = 10 \log_{10} \frac{I_2}{I_0} dB$$

$$10 = \log_{10} \frac{I_1}{10^{-12} \frac{W}{m^2}}$$

$$10 = \log_{10} \frac{I_2}{I_0} \Rightarrow \frac{I_2}{I_0} = 10^{10}$$

$$I_2 = 10^{10}(I_0) = 10^{10}\left(10^{-12} \frac{W}{m^2}\right) = 10^2 \frac{W}{m^2}$$

From the definition of intensity of sound we have

$$I_1 = \frac{P_1}{S_1^2} \text{ and } I_2 = \frac{P_2}{S_2^2} \text{ since } P_1 = P_2$$

$$\frac{I_1}{I_2} = \left(\frac{S_2}{S_1}\right)^2 \Rightarrow \frac{1 \frac{W}{m^2}}{10^2 \frac{W}{m^2}} = \left(\frac{S_2}{4m}\right)^2$$

$$\left(\frac{S_2}{4m}\right)^2 = 10^2 \Rightarrow \frac{S_2}{4m} = 10 \Rightarrow S_2 = 10(4m) = 40m$$

Answer:- C

42. Solution:- At a temperature $T_1 = 300k$, the KE of the gas particles is given by

$$E_{K1} = \frac{3}{2} kT_1 = \frac{1}{2} m \overline{v_1^2} \qquad v_r^2 = 2v_1^2$$

The KE at a temperature T_2 is E_{K2} and is given by

Telegram: https://t.me/ExtremeSeries

$$E_{K2} = \frac{3}{2}KT_2 = \frac{1}{2}m\bar{v}_2^2$$

$$\frac{E_{K1}}{E_{K2}} = \frac{T_1}{T_2} = \frac{\frac{1}{2}m\bar{v}_1^2}{\frac{1}{2}m\bar{v}_2^2} = \frac{\bar{v}_1^2}{\bar{v}_2^2} = \frac{\bar{v}_1^2}{4\bar{v}_1^2} = \frac{1}{4}$$

$$\frac{300K}{T_2} = \frac{1}{4} \Rightarrow T_2 = 4(300K)1{,}200K$$

Answer:- D

43. Solution:- Given $n = 1.25, \lambda = 600nm$ the minimum thickness (t) the soap bubble that results constructive interference is obtained from

$$t = \left(n - \frac{1}{2}\right)\lambda = (1.25 - 0.5)(600nm)$$

$$= 0.75 \times 600nm \quad = 450nm$$

Answer:-D

44. Solution:- Given $\vec{B} = (1.6\hat{k})\text{T}$ points in the Z – axis

$\vec{\ell} = (0.75\hat{\imath})\text{m}$ lies on the x – axis

$I = 2.4$

The magnetic deflecting force on this section of the wire is calculated from

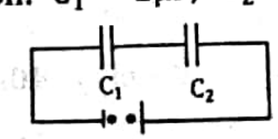
$$\vec{F} = I(\vec{\ell} \times \vec{B}) = 2.4A\,(0.75m\hat{\imath} \times 1.6\hat{k}\text{T})$$

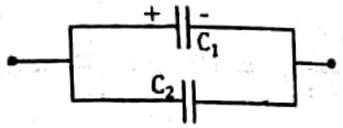
$$= (2.4A \times 0.75m \times 1.6\text{T})(\hat{\imath} \times \hat{k})$$

$= 2.88N\hat{\imath} = 2.88N$ in the +ve y direction

Answer:-B

45. Solution:- $C_1 = 2\mu\text{F}, \quad C_2 = 3\mu\text{F}, \quad \text{V} = 120\text{V}$





After the battery is removed

$$\frac{1}{c_{eg}} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{2\mu\text{F}} + \frac{1}{3\mu\text{F}} = \frac{5}{6\mu\text{F}}$$

$$c_{eg} = \frac{6}{5}\mu\text{F} = 1.2\mu\text{F}$$

$$Q_T = c_{eg} \times V = 1.2\mu\text{F} \times 120\text{V}$$

$$Q_T = 144\mu\text{C} = Q_1 = Q_2$$

Telegram: https://t.me/ExtremeSeries

when the capacitors are connected in parallel $V_1^1 = V_2^1$

$$\frac{q_1^1}{C_1} = \frac{q_2^1}{C_2} \Rightarrow \frac{q_1^1}{2\mu\text{F}} = \frac{q_2^1}{3\mu\text{F}}$$

$$q_2^1 = 1.5q_1^1$$

The total charge of the circuit when the capacitors are connected in parallel is

$$q^1 = q_1^1 + q_2^1 = q_1^1 + 1.5q_1^1 = 2.5q_1^1$$

When the capacitors are joined with same polarity

$$q_2^1 - q_1^1 = q_1 + q_2 = 144\mu\text{C} + 144\mu\text{C} = 288\mu\text{C}$$

$$1.5q_1^1 - q_1^1 = 288\mu\text{C}$$

$$0.5q_1^1 = 1.5(576\mu\text{C}) = 576\mu\text{C}$$

$$q_2^1 = 1.5q_1^1 = 1.5(576\mu\text{C}) = 846\mu\text{C}$$

Answer:-A

46. Solution :- Original volume of glass at $0^0C = V_0 = 1{,}000cm^3$

Final temperature of glass flask=$T_f = 100^0C$

Change in volume of glass flask=$\Delta V_g = 15.2cm^3$

From volume expansion we have

$$\Delta V_{ap} = V_0 \propto_{ap} \Delta Tg$$

$$15.2cm^3 = 10^3cm^3 \times \propto_{ap} (100^0C)$$

$$\propto_{ap} = \frac{15.2}{10^5{}^0C} = 1.52 \times 10^{4/}{}^0C$$

Using $\propto_{real} = 1.82 \times 10^{4/}{}^0C$ *and* $\propto_{ap} = 1.52 \times 10^{4/}{}^0C$

$$\propto_{glass} = \propto_{real} - \propto_{ap} = 1.82 \times 10^{4/}{}^0C - 1.52 \times 10^{4/}{}^0C$$

$$= 0.3 \times 10^{4/}{}^0C = 3 \times 10^{5/}{}^0C$$

Answer:-C

47. Solution:- Given thermal efficiency= $\eta = 65\%, T_C = 20^0C$

Required:- T_h

Thermal efficiency=$\eta = \frac{T_h - T_C}{T_h}$

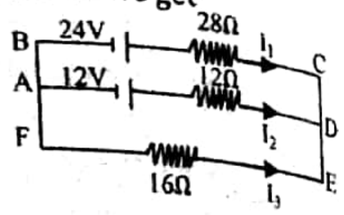
$$0.65 = 1 - \frac{T_C}{T_h} = 1 - \frac{20^0C}{T_h}$$

$$\frac{20^0C}{T_h} = 1 - 0.65 = 0.35$$

$$T_h = \frac{20\,^0C}{0.35} = \frac{2{,}000^0C}{35} = 57^0C$$

Answer:-A

48. Solution:- Applying Kirchhoff's point rule circuit we get



$I_3 = I_1 + I_2 - - - (1)$

For loop ABCDA

$24v - 28I_1 + 12I_2 - 12V = 0$

$12v - 28\Omega I_1 + 12\Omega I_2 = 0$

$3I_2 - 7I_1 = -3 - - - (2)$

For loop ADEFA

$12v - 12\Omega I_2 - 16\Omega I_3 = 0$

$3I_2 + 4I_3 = 3 - - - (3)$

Substituting (1) in (3) gives

$3I_2 + 4I_1 + 4I_2 = 3 \Rightarrow 7I_2 + 4I_1 = 3 - - - (4)$

Solving (2) and (4) simultaneously gives

$$3I_2 - 7I_1 = -3 \times 4$$
$$7I_2 + 4I_1 = 3 \times 7$$
$$12I_2 - 28I_1 = -12$$
$$49I_2 + 28I_1 = 21$$
$$61I_2 = 9A$$

$$I_2 = \frac{9}{61}A$$

Using (1) we get

$$I_3 = I_1 + I_2 = \frac{9}{61}A + \frac{30}{61}A = \frac{39}{61}A \approx 0.5A$$

Answer:-C

49. Solution:- $\frac{x}{5m} = cos53^0$

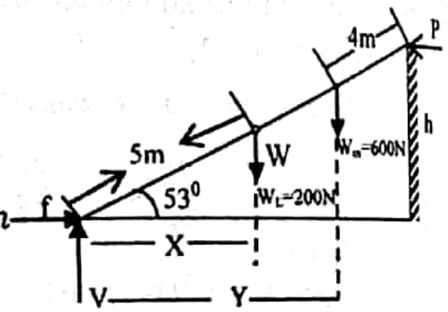
$x = 0.5(0.6) = 0.3m$

$\frac{y}{6m} = cos53^0$

$y = 6(0.6)m = 3.6m$

$\frac{h}{10m} = sin53^0 = 0.8$

$h = 8m$



From the first condition of equilibrium

$$\sum Fx = 0 \Rightarrow f - P = 0 \Rightarrow f = P$$

$$\sum Fy = 0 \Rightarrow v - W_l - W_m = 0 \Rightarrow V = W_L + W_m$$
$$= 200N + 600N = 800N$$

From the second condition of equilibrium we have

$$\sum T_A = 0$$

W_L and W_m produce a clock wise torque

$$\sum T_{clock} = W_L(x) + W_m(y)$$
$$= 200N(3)m + 600N(3.6m)$$
$$= 600Nm + 2160Nm = 2760Nm$$

$$\sum T_{anti} = P(h) = P(8m)$$

$$\sum T_{anti} = \sum T_{clock}$$

$$8m(P) = 2760Nm$$

$$P = \frac{2760Nm}{8m} = 345N$$

$$f = P = \mu mg = 345N$$

$$\mu(800N) = 345N \Rightarrow \mu = \frac{345N}{800N} = 0.4313$$

Answer:- B

50. Solution:- Given $m = 1.5kg, k = 200\frac{N}{m}$ $x = 2cm$ $f = 10N$. The ME at the extreme position equals to the elastic PE.

$$E_{MA} = \frac{1}{2}kx^2 = \frac{1}{2}\left(200\frac{N}{m}\right)(2 \times 10^{-2}m) = 10^{-2}\frac{N}{m}$$
$$= 0.04J$$

Loss of $ME = EMo - EMA = W_f = f(x)$

$$0.8kgv^2 = 0.24J$$

$$v^2 = \frac{0.24J}{0.8kg} = \frac{2.4}{8}\frac{m^2}{s^2} = 0.3\frac{m^2}{s^2}$$

$$V = \sqrt{0.3}\frac{m}{s} = 0.55\frac{m}{s} = 55\frac{cm}{s}$$

Answer:-D

Telegram: https://t.me/ExtremeSeries

CHEMISTRY 2001/2009

Physical Constants

a) Gas constant (R)= $8.314 Jmol^{-1}K^{-1}$= 0.0821 L-atm $mol^{-1}K^{-1}$
b) Avogadro's number (N_A) =$6.023 \times 10^{23} mol^{-1}$
c) Faraday's Constant (F) = $9.6485309 \times 10^4 mol^{-1}$

SI Units and Conversion Factors

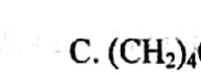
d) 1ton = 907.185kg
e) 1 metric ton = 1000kg
f) $1Å = 10^{-10}m$
g) 1L-atm = 101.3J

Atomic Numbers (Z) and Atomic Weights (A)

Element	H	He	Li	B	C	N	O	F	Ne	Na	S	Cl	Ca
Z	1	2	3	5	6	7	8	9	10	11	16	17	20
A	1.0	4.0	6.9	10.8	12.0	14.0	16.0	19.0	20.2	22.98	32..1	35.5	40.1

Element	Cr	Mn	Fe	Cu	Zn	Ag	Cd	Au	Hg
Z	24	25	27	29	30	47	48	79	80
A	51.9	54.9	55.9	63.5	65.4	107.9	112.4	197.0	200.6

1. Which of the following is most likely to be miscible in carbon tetrachloride, CCl_4?
   A. Na_2CO_3 B. CH_3F C. $(CH_2)_4O_2$ D. I_2
2. If we add a catalyst to a chemical reaction that is at equilibrium, then a catalyst increases the rate at which equilibrium is achieved,
   A. but it does not change the composition of the equilibrium
   B. And it will change the composition of the equilibrium
   C. And will shift the equilibrium in the direction that produces heat
   D. And will shift the equilibrium in the direction that occupies a larger volume.
3. Which of the following is not a synthetic fiber?
   A. Acrylic B. Nylon C. Polyester D. Silk
4. Which of the following is a polyprotic acid
   A. HClO B. HBrO C. HIO D. $H_2C_2O_4$
5. If a gas absorbs 50kJ of heat and does 25kJ of work, then what will be the change in the internal energy (ΔE) of the gas?
   A. 75kJ B. -75kJ C. 25kJ D. 0kJ
6. Which substance is reduced in the reaction,
   $C(s) + 2N_2O(g) \rightarrow CO_2(g) + 2N_2(g)$?
   A. C B. N_2O C. CO_2 D. N_2

7. A covalent bond is unlikely to exist in which of the following substances?
   A. H_2 B. SeH_2 C. SiF_4 D. CaO
8. A reaction reaches a state of equilibrium only when:
   A. the reactants and the products are reacting
   B. the concentration of the reactants and products become equal
   C. the products react together at the same rate at which they are formed
   D. all the reactants and the products are in the same state of matter
9. Which of the following species is not amphoteric?
   A. SO_4^{2-} B. HS^- C. HSO_4^- D. HCO_3^-
10. Which one of the following is the major use of carbon dioxide?
    A. Manufacture of washing soda
    B. Manufacture of baking soda
    C. Refrigeration
    D. Production of carbonated beverages
11. What is the maximum number of electrons in an atom that can have the quantum $n = 5$, $m_l = +1$?
    A. 8 B. 16 C. 32 D. 64
12. Which one of the following Lewis structures is correct for the triiodide ion (I_3^-)?
    A. $[I - I - I]^-$ B.  C. D. $[I = I = I]^-$
13. At 373K, $K_p = 5 \times 10^{-5}$ atm for the reaction
    $COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$
    For equal pressures of the reactant and products ($P(CO) = P(COCl_2) = P(Cl_2) = 1atm$) placed in a flask which of the following is true at equilibrium?
    A. The total pressure of the system must remain at 3 atm
    B. $P(CO)$ and $P(Cl_2)$ must decrease from their initial values
    C. $P(CO)$ and $P(Cl_2)$ must increase from their initial values
    D. $P(COCl_2)$ will remain constant
14. If the equilibrium constant for the reaction
    $2NO(g) \rightleftharpoons N_2(g) + O_2(g)$
    Is $K_c = 3.6 \times 10^3$ at 275K, then what will be the value of K_c for the reaction
    $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$?
    A. 2.70×10^{-4} B. 2.70×10^4 C. 12.96×10^4 D. 3.60×10^3
15. Which of the following conjugate acid- base pair is correctly listed for the reaction below?
    $H_2O + (CH_3)_3N \rightleftharpoons (CH_3)_3NH^+ + OH^-$

Telegram: https://t.me/ExtremeSeries

A. H_2O, $(CH_3)_3N$ C. H_2O,$(CH_3)_3NH^+$
B. $(CH_3)_3NH^+$, OH^- D. $(CH_3)_3N$, $(CH_3)_3NH^+$

16. Which is the most commercially important halogen?
A. Bromine B. Chlorine C. Fluorine D. Iodine

17. Which of the following does not involve in an increase in the entropy of the system?
A. Boiling B. Freezing C. Melting D. Sublimation

18. How many moles of MnO_2 are produced in a basic solution when MnO_4^- (permanganate ion) and CN^- (cyanide) react to form MnO_2 and CNO^- according to the following unbalanced reaction?
$MnO_4^-(aq) + CN^-(aq) \longrightarrow MnO_2(s) + CNO^-(aq)$
A. 1 B. 2 C. 3 D. 5

19. If the diameter of the chromium (Cr) atom is about 2.4 Å, then how many chromium atoms would have to be lined up to span 1.0cm?
A. 2.4×10^7 B. 4.2×10^7 C. 2.4×10^8 D. 4.2×10^8

20. A commercial bleaching solution ("berekina") contains 5 mass percent (mass%) sodium hypochlorite, NaOCl. What is the mass of NaOCl in a bottle containing 2500g of bleaching solution?
A. 50g B. 100g C. 125g D. 150g

21. What is the percent of dissociation in 1.0M solution of HF containing $[H^+] = 2.7 \times 10^{-2}$?
A. 2.7% B. 3.7% C. 27% D. 37%

22. When 1 mole of methane (CH_4) is burned at constant pressure, 890 kJ of energy is released as heat. What will be the value of ΔH for the process in which 10g sample of methane is burned at constant pressure?
A. -890kJ B. -556kJ C. +556kJ D. +890kJ

23. The empirical formula of an organic compound containing carbon and hydrogen is CH_2. The mass of one liter of this organic gas is exactly equal to that of one liter of N_2. What is the molecular formula of the organic gas?
A. C_2H_4 B. C_3H_6 C. C_4H_8 D. C_6H_{12}

24. Which of the following is determined by the Millikan's oil drop experiment?
A. Charge on the electron C. Mass of electron
B. e/m ratio for an electron D. Velocity of electron

25. What is the normality of a 2.3M sulphuric acid solution?
A. 1.15N B. 2.3N C. 4.6N D. 6.9N

26. Which of the following is the correct expression of the rate reaction, $2N_2O_5 \rightleftharpoons 4NO_2+O_2$, in terms of NO_2?
A. $\Delta[NO_2]/\Delta t$ B. $-\Delta[NO_2]/\Delta t$ C. $\frac{1}{4}\Delta[NO_2]/\Delta t$ D. $-\frac{1}{4}\Delta[NO_2]/\Delta t$

27. For the reaction,
$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$
K_p is equals to:
A. K_c B. K_cRT C. $K_c(RT)^2$ D. $K_c(RT)^{-1}$

28. In which of the following the yield of the product in the reaction,
$A_2(g) + 2B(g) \rightleftharpoons C(g)$ + heat,
would be higher?
A. High temperature and high pressure
B. High temperature and low pressure
C. Low temperature and high pressure
D. Low temperature and low pressure

29. Which of the following monosaccharides circulate in the bloodstream and supplies the body with a source of energy?
A. Fructose C. Sucrose
B. Glucose D. Starch

30. Which of the following ions act as both a Bronsted acid and base in water?
A. HCO_3^- B. SO_4^{2-} C. NO_3^- D. PO_4^{3-}

31. Which of the following acids is the strongest?
A. $CH_3COOH(K_a = 1.8 \times 10^{-5})$
B. $HCOOH(K_a = 1.8 \times 10^{-4})$
C. $HCN(K_a = 7.2 \times 10^{-10})$
D. C_6H_5COOH $(K_a = 6.3 \times 10^{-5})$

32. How many mL of 1M H_2SO_4 is required to neutralize 10mL of 1.0M NaOH solution?
A. 2.5 B. 5.0 C. 10.0 D. 20.0

33. How much work, in kJ, is done when a gas expands from 5.0L to 15.0L under a constant pressure of 2.00atm?
A. -20.0kJ B. -2.03kJ C. +2.03kJ D.+20.0kJ

34. Which alkaline earth metal shares diagonal relationship with aluminium?
A. Ba B. Be C. Ca D. Mg

35. Which is the metal ion in the porphyrin of heme?
A. Calcium B. Iron C. Molybdenum D. Magnesium

36. Which are the number of moles and the mass of a copper sample containing 5.00×10^{20} atoms?

Telegram: https://t.me/ExtremeSeries

A. 3.8×10^{-4}mol Cu and 5.2×10^{-2}g Cu
B. 8.3×10^{-4}mol Cu and 5.2×10^{-2}g Cu
C. 5.2×10^{-2}mol Cu and 8.3×10^{-4}g Cu
D. 5.2×10^{-2}mol Cu and 3.8×10^{-4}g Cu

37. Given the following reaction, what mass of gaseous carbon dioxide can be absorbed by 1kg of lithium hydroxide?
$2LiOH(s) + CO_2(g) \longrightarrow Li_2CO_3(s) + H_2O(l)$
A. 920g B. 1840g C. 2760g D. 3680g

38. If a sample containing 36g NH_3 is reacted with 180g of CuO, according to the following reaction, then what is the limiting reactant and how many grams of N_2 will be formed?
$2NH_3(g) + 3CuO(s) \longrightarrow N_2(g) + 3Cu(s) + 3H_2O(g)$
A. NH_3; $10.6gN_2$ C. CuO; $22.3gN_2$
B. NH_3; $22.3gN_2$ D. CuO; $10.6gN_2$

39. Which of the following statements is not correct?
A. Fats are esters of glycerol and the fatty acids
B. Fats produce more energy per gram than either proteins or carbohydrates
C. Fats are insoluble in water, which permits their storage in the body
D. True vegetarians, who do not eat meat, fish, eggs, or dairy products have zero cholesterol in their bodies.

40. In the fermentation of sugar in an enzymatic solution, which is initially 0.2M, the concentration of the sugar is reduced to 0.1M in 10hours and to 0.05M in 20 hours. What is the order of the reaction and the rate constant?
A. First order $k = 1.92 \times 10^{-5}s^{-1}$
B. First order $k = 3.85 \times 10^{-5}s^{-1}$
C. Second order $k = 1.38 \times 10^{-4}M^{-1}s^{-1}$
D. Second order $k = 2.72 \times 10^{-4}M^{-1}s^{-1}$

41. If A $Cr^{3+}(aq)$ solution is electrolyzed using a current of 15.A, then What mass of chromium, Cr(s), is plated out after 3 days?
A. 400g B. 700g C. 800g D. 2100g

42. Which of the following has the greatest number of atoms?
A. 4g of hydrogen C. 40g of calcium
B. 24g of ozone D. 127g of iodine

43. How many spectral lines are emitted from a hydrogen atom excited to the state designated by the principal quantum number, n = 3?
A. 1 B. 2 C. 3 D. 4

Telegram: https://t.me/ExtremeSeries

44. In which of the following period of the Periodic Table is an element with atomic number 20 placed?
A. 1 B. 2 C. 3 D. 4

45. What is the hybridization of sulfur atom in SF_6?
A. sp^2 B. sp^3 C. sp^3d D. sp^3d^2

46. The reaction, $2O_3 \rightleftharpoons 3O_2$, proceeds through the mechanism given below:
$O_3 \rightleftharpoons O_2 + O$ fast
$O_3 + O \rightarrow 2O_2$ slow
What would be the rate law expression for the reaction?
A. Rate = $k[O_3]^2[O_2]^{-1}$ C. Rate = $k[O_3][O_2]$
B. Rate = $k[O_3]^2[O_2]$ D. Rate = $k[O_3]^2$

47. Which of the following substances undergoes hydrolysis in aqueous solution?
A. HCl B. $HC_2H_3O_2$ C. NH_4OH D. NH_4Cl

48. The pH of 0.1M solution of a weak acid is 3.0. What is the value of the ionization constant for the acid?
A. 10^{-7} B. 10^{-5} C. 10^{-3} D. 0.1

49. Consider the following half – reactions:
$Fe(OH)_2(s) + 2e^- \rightarrow Fe(s) + 2OH^-(aq)$ $E^0 = -0.88V$
$NiO_2(s) + 2H_2O(l) + 2e^- \rightarrow Ni(OH)_2(s) + 2OH^-(aq)$ $E^0 = +0.49V$
What is the standard cell potential for a voltaic cell using these electrodes?
A. -1.37V B. -0.39V C. 0.49V D. 1.37V

50. Which of the hydrogen halide has the highest enthalpy of vaporization?
A. HI B. HBr C. HCl D. HF

Booklet Code: 28

1. What is the standard entropy change for the reaction:
   $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$
   Given the following data: (S^0 of $SO_2(g)$ = 248.1J/(mol.K); S^0 of $O_2(g)$ = 205.0J/(mol.k); S^0 of $SO_3(g)$ = 256.6J/(mol.k)
   A. -188.0J/K B. +17.0J/K C. -196.5 J/K D. +60.1J/K
2. How many moles are there in 159g of alanine, $C_3H_7NO_2$?
   A. 0.560 B. 0.992 C. 1.78 D. 3.31
3. Which one of the following is NOT true of metallic bonding?
   A. It gives rise to excellent electrical conductivity
   B. Electrons are fee to move throughout the structure
   C. The strength of metallic bonds increases down a group.
   D. The strength of metallic bonding affects the boiling point of metals.
4. Synthetic fibers are:
   A. naturally produced.
   B. Manufactured through the use of chemical substances.
   C. Produced from animal hair fibers.
   D. Produced from plant substances.
5. An amphiprotic species is a molecule or ion that can _____.
   A. Accept protons. C. both accept and donate protons
   B. donate more than one protons D. be formed into a double ion.
6. Which one is true for a triprotic acid, such as phosphoric acid, H_3PO_4?
   A. $K_{a1} > K_{a2} > K_{a3}$ C. $K_{a1} > K_{a2} = K_{a3}$
   B. $K_{a3} > K_{a2} > K_{a1}$ D. $K_{a1} = K_{a2} = K_{a3}$
7. Which group of elements are the most reactive of all the metallic elements?
   A. alkali metals C. Transition metals
   B. Alkaline earth metals D. Group 2B metals
8. All of these are characteristics of most ionic compounds in the solid phase EXCEPT,
   A. high melting point. C. high electrical conductivity.
   B. solubility in water. D. insolubility in organic solvents.
9. Consider the following:
   $2NO(g) + Cl_2 \quad 2NOCl(g)\ \Delta H = -78.38kJ$
   What conditions of temperature and pressure will produce the highest yield of NOCl at equilibrium?
   A. High temperature High pressure.
   B. High temperature Low pressure.
   C. Low temperature High pressure.
   D. Low temperature Low pressure.
10. In the process known as 'roasting', a(n) _____ in chemically converted to a(n) _____.
    A. carbonate/oxide C. sulfide/oxide
    B. oxide/sulfate D. phosphate/ phosphide
11. Which of the following statement is NOT true according to 1st law of thermodynamics?
    A. Energy can neither be created nor destroyed although it can be changed from one form to another.
    B. The total energy of the universe is constant.
    C. The change in the internal energy of a closed system is equal to the sum of heat absorbed by the system and work done by the system.
    D. The change in internal energy of a system is zero in all the processes.
12. How many chloride ions are in 1.0 mole of $CaCl_2$?
    A. $1.20 \times 10^{23} Cl^-$ ions C. $1.81 \times 10^{24} Cl^-$ ions
    B. $6.02 \times 10^{23} Cl^-$ ions D. $1.20 \times 10^{24} Cl^-$ ions
13. Which of the following types of solutions are possible?
    I. solid dissolved in a liquid III. gas dissolved in a gas
    II. gas dissolved in a liquid IV. Solid dissolved in a solid
    A. I and II B. I, II, III & IV C. I, II and IV D. I
14. What is the normality of 1.0 M solution of Na_2CO_3?
    A. 1N B. 0.5N C. 2N D. 3N
15. What is the type of monomer unit in a natural protein polymer?
    A. Amino acids C. α - amino acids
    B. Essential amino acids D. Nucleic acids
16. Which of the following substances could be amphiprotic?
    I. NO_3^- II. H_2O III. NH_3 D. HCO_3^-
17. Which one of the following does NOT form hydroxide ions when placed in water?
    A. Ionic hydrides C. Nonmetal oxides
    B. Ionic metal oxides D. Ionic nitrides
18. Which of the following is true for a 0.10M solution of a weak base, HB?
    A. $[H^+] = 0.10M$ C. $[HB] > [H^+]$
    B. $[B^-] = 0.10M$ D. pH = 1.0
19. What is the bond order of O_2+?

Telegram: https://t.me/ExtremeSeries

A. 2 B. 2.5 C. 1.5 D. 3

20. Which of the following is characteristic of an antibonding molecular orbital?
    A. It is a molecular orbital with a high probability of finding electrons parallel to thee region between bonded atoms.
    B. It has no electrons.
    C. It is a molecular orbital with a high probability of finding electrons away from the region between bonded atoms.
    D. It is a molecular orbital with a high probability of finding electrons in the region between bonded atoms.
21. What is the molarity of a solution obtained by dissolving 0.01 moles of NaCl is 500mL of solution?
    A. 0.01M B. 0.005M C. 0.02M D. 0.10M
22. Which of the following does NOT affect the rate of a chemical reaction?
    A. Enthalpy of the reaction C. Surface area
    B. Concentration of reactants D. Temperature
23. Which of the following is NOT a valid expression for the rate of the reaction given below?
    $4NH_3 + 7O_2 \rightarrow 4NO_2 + 6H_2O$
    A. $-\frac{1}{7}\frac{\Delta[O_2]}{\Delta t}$ B. $\frac{1}{4}\frac{\Delta[NO_2]}{\Delta t}$ C. $-\frac{1}{6}\frac{\Delta[H_2O]}{\Delta t}$ D. $-\frac{1}{4}\frac{\Delta[NH_2]}{\Delta t}$
24. Each of the choices below gives a reaction and the corresponding rate law. Of these choices, which one could be an elementary process or individual step in a chemical reaction?
    A. $2A \rightarrow P$ rate = k[A]
    B. $A + B \rightarrow P$ rate = k[A][B]
    C. $A + 2B \rightarrow P$ rate = $k[A]^2$
    D. $A + B + C \rightarrow P$ rate = k[A][C]
25. Consider the reaction in which nitric oxide is oxidized to nitrogen dioxide:
    $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$
    For which the rate law is rate = $k[NO]^2[O_2]$. If this reaction takes place in a sealed vessel and the partial pressure of nitric oxide is doubled, what effect would this have on the rate of reaction?
    A. The reaction rate would increase by a factor of four.
    B. The reaction rate would increase by a factor of three.
    C. The reaction rate would increase by a factor of eight.
    D. The reaction rate would increase by a factor of two.

Telegram: https://t.me/Extremeseries

26. Which one of the following will change the value of an equilibrium constant?
    A. Changing the temperature.
    B. Adding other substances that do not react with any of the species involved in the equilibrium.
    C. Varying the initial concentrations of reactants.
    D. Varying the initial concentration of products.
27. The equilibrium constant for reaction (1) is K. what is the equilibrium constant for reaction (2)?
    $SO_2(g) + \frac{1}{2}O_2(g) = SO_3(g)$ (1)
    $2SO_3(g) = 2SO_2(g) + O_2(g)$ (2)
    A. K^2 B. 2K C. ½K D. $1/K^2$
28. Which of the following are carbohydrates?
    I. nucleic acids IV. aldoses
    II. polyhydroxyketones V. triglycerides
    III. disaccharides
    A. II, III and IV B. I and V C. I, III and V D. III and IV
29. Which of the following is true concerning this acid-base reaction?
    $HS^- + CH_3Cl \rightarrow CH_3SH + Cl^-$
    A. CH_3Cl is a Bronsted – Lowry acid.
    B. HS^- is a Bronsted – Lowery acid.
    C. HS^- is a Lewis acid.
    D. HS^- is a Lewis base.
30. For the stepwise dissociation of aqueous H_3PO_4, which of the following is NOT a conjugate acid-base pair?
    A. HPO_4^{2-} and PO_4^{3-} C. $H_2PO_4^-$ and HPO_4^{2-}
    B. H_3PO_4 and $H_2PO_4^-$ D. H_2PO_4 and PO_4^{3-}
31. Which solution below has the highest concentration of hydroxide ions?
    A. pH = 3.21 B. pH = 12.59 C. pH = 7.93D. pH=9.82
32. The standard reduction potential of Zn^{2+}/Zn is (-0.76) V and that of Cu^{2+}/Cu is +0.34V. What would be the electromotive force (emf) of the cell constructed between these two electrodes?
    A. 1.10V B. 0.42V C. -1.10V D. -0.42V
33. At constant T and P, which one of the following statements about the enthalpy (ΔH) and internal energy (ΔU) changes is correct for the reaction?
    $CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$
    A. $\Delta H = \Delta U$ B. $\Delta H < \Delta U$ C. $\Delta H > \Delta U$

D. ΔH and ΔU are independent of physical states of the reactants

34. Which one of the following conditions regarding a chemical process ensures its spontaneity at all temperatures?
A. $\Delta H < 0, \Delta S < 0$ C. $\Delta H < 0, \Delta S > 0$
B. $\Delta H > 0, \Delta S < 0$ D. $\Delta H > 0, \Delta S > 0$

35. Why are silicate minerals NOT commonly used as sources of metals?
A. They are rare minerals.
B. They usually do not contain important metals.
C. They are difficult to reduce and concentrate.
D. They are only found at excessive depths in the oceans.

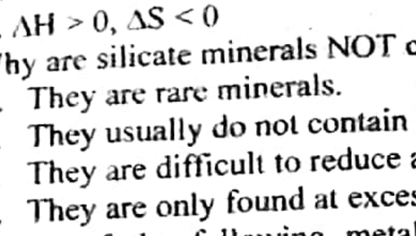
36. Which of the following metals is the best conductor of heat and electricity?
A. Copper B. Silver C. Gold D. Tungsten

37. HNO_3 is used as:
I. a hydrogenating agent.
II. an acid.
III. an oxidizing agent.
IV. an ammoniating agent.
A. II and III B. I and II C. I, II, III and IV D. II, III and IV

38. Which equation represents an oxidation – reduction reaction?
A. $H_2SO_4 + 2NH_3 \rightarrow (NH_4)_2SO_4$
B. $H_2SO_4 + Na_2CO_3 \rightarrow Na_2SO_4 + H_2O + CO_2$
C. $2K_2CrO_4 + H_2SO_4 \rightarrow K_2Cr_2O_7 + K_2SO_4 + H_2O$
D. $2H_2SO_4 + Cu \rightarrow CuSO_4 + 2H_2O + SO_2$

39. Which set of quantum numbers (n, l, m_l; m_s) is NOT permitted by the rules of quantum mechanics?
A. 1, 0, 0, ½ B. 2, 1, -1, -½ C. 3, 3, 1, -½ D. 4, 3, 2, ½

40. What can you conclude from the figure below?



A. Hund's rule has been violated.
B. The pauli exclusion principle has been violated.
C. The Aufbau principle has been violated.
D. This is a valid orbital diagram.

41. Which set contains only covalently bonded molecules?
A. BCl_3, $SiCl_4$, PCl_3 C. I_2, H_2S, NaI
B. NH_4Br, N_2H_4, HBr D. Al, O_3, As_4

42. How many grams of $Ca(OH)_2$ are contained in 1500mL of 0.0250M $Ca(OH)_2$ solution?
A. 1.85g B. 2.78g C. 3.17g D. 4.25g

43. The conventional equilibrium constant expression (K_c) for the system as described by the equation:
$2ICl(s) \quad I_2(s) + Cl_2(g)$ is:
A. $[Cl_2]$ B. $[Cl_2]/[ICl]^2$ C. $[I_2][Cl_2]/[ICl]^2$ D. $[I_2][Cl_2]/2[ICl]$

44. What is the conjugate base of HSO_4^-?
A. H^+ B. H_2SO_4 C. OH^- D. SO_4^{2-}

45. An electrochemical cell constructed for the reaction:
$Cu^{2+}(aq) + M(s) \rightarrow Cu(s) + M^{2+}(aq)$
has an $E^0_{cell} = 0.75V$. The standard reduction E^0_{red} potential for $Cu^{2+}(aq)$ is 0.34V. What is the standard reduction potential for $M^{2+}(aq)$?
A. -1.09V B. -0.410V C. 0.410V D. 1.09V

46. Which of the following compounds would be expected to have the highest melting point?
A. BaF_2 B. $BaCl_2$ C. $BaBr_2$ D. BaI_2

47. Which of the following is true about chlorofluorocarbons?
A. React directly with stratospheric ozone to destroy it.
B. Interact with UV energy and become free radicals which destroy ozone.
C. Become free radicals that react with oxygen to create ozone.
D. React with free radicals to remove carbon dioxide.

48. Which metal can be found as the free element?
A. Cr B. Fe C. Mn D. Pt

49. The enthalpy change for which reaction represents the standard enthalpy of formation for hydrogen cyanide, HCN?
A. $H(g) + C(graphite) + N(g) \rightarrow HCN(g)$
B. $½H_2(g) + C(graphite) + ½N_2(g) \rightarrow HCN(g)$
C. $HCN(g) \rightarrow ½H_2(g) + C(graphite) + ½N_2(g)$
D. $H_2(g) + 2C(graphite) + N_2(g) \rightarrow 2HCN(g)$

50. Which one of the compounds below is most likely to be ionic?
A. CCl_4 B. NO_2 C. $ScCl_3$ D. ClO_2

51. What is the correct order when the substances O_2, H_2O, OF_2, and H_2O_2 are arranged in order of increasing oxidation number for oxygen?
A. O_2, H_2O, OF_2, H_2O_2 C. H_2O_2, O_2, H_2O, OF_2
B. H_2O, H_2O_2, O_2, OF_2 D. OF_2, O_2, H_2O_2, H_2O

52. A monoatomic ion that has 20 protons and a +2 charge

Telegram: https://t.me/ExtremeSeries

A. has 16 protons.
B. has the symbol Ar^{2+}
C. has 18 neutrons.
D. is isoelectronic with Ar.

53. What is the molality of a solution that contains 51.2g of naphthalene, $C_{10}H_8$, in 500mL of carbon tetrachloride? Given the density of CCl_4 = 1.60g/mL.
A. 0.25m B. 0.50m C. 0.75m D. 0.84m

54. A homogeneous liquid reaction mixture is often heated to increase the rate of reaction. This is best explained by the fact that raising the temperature:
A. increases the heat of reaction.
B. decrease the energy of activation.
C. increases the vapor pressure of the liquid.
D. increases the average kinetic energy of the reactants.

55. Which of the following titrations will have an equivalence point at a pH > 7.00?
A. Weak acid with strong base.
B. Strong acid with a strong base.
C. Weak acid with a weak base.
D. Strong acid with a weak base

56. A 0.10M solution of a weak acid, HX, is 0.059% ionized. What is the dissociation constant for the acid?
A. 3.8×10^{-9} B. 3.5×10^{-8} C. 6.5×10^{-7} D. 4.2×10^{-6}

57. Consider the standard voltaic (or galvanic) cell: Fe, Fe^{2+} versus Au, Au^{3+}. Which answer identifies the cathode and gives the E^0_{red} for the cell? Given $E^0(Au^{3+}/Au) = 1.50V$, $E^0(Fe^{2+}/Fe) = -0.44V$.
A. Fe, -0.44V B. Au, 1.06V C. Au, 1.94V D. Fe, 1.94V

58. What is the enthalpy for the reaction,
$C_2H_6(g) + 7/2O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l)$?
Given the following information:
$2C(s) + 3H_2(g) \rightarrow C_2H_6(g)$ $\Delta H_f^0 = -84.7kJ.mol^{-1}$
$C(s) + O_2(g) \rightarrow CO_2(g)$ $\Delta H_f^0 = -393.5\ kJ.mol^{-1}$
$H_2(g) + ½O_2(g) \rightarrow H_2O(l)$ $\Delta H_f^0 = -285.8\ kJ.mol^{-1}$
A. -764kJ B. -1560kJ C. -1664kJ D. -3120kJ

59. How many atoms are present in 64g of SO_2?
A. 6.02×10^{23} C. $3 \times 6.02 \times 10^{23}$
B. $2 \times 6.02 \times 10^{23}$ D. $4 \times 6.02 \times 10^{23}$

60. What is the correct order of the atomic radius in the elements below?
A. Mg < Al < Cl C. Mg > Al > Cl
B. Al < Mg < Cl D. Cl > Al > Mg

61. The outermost electron (s) of which of the following experiences is the greatest effective nuclear charge?
A. K B. Ca C. Br D. P

62. Which of the following is NOT probable shape of a molecule?
A. A tetrahedral $GeCl_4$ C. A trigonal bipyramidal PCl_5
B. A bent HCN D. A T-shaped BrF_3

63. Four elements A, B, C, D have atomic numbers - 1, Z, Z + 1 and Z + 2, respectively. If Z is 9, then the bond between which pair of elements will be ionic?
A. A and C B. C and D C. B and D D. B and C

64. Which of the following statements is INCORRECT for the reaction represented by the figure below?
A. Y represents the transition state of the reaction.
B. The enthalpy of the reaction is given by b – a.
C. The forward reaction is endothermic.
D. b is the activation energy of the forward reaction.

65. Given the initial – rate data below for the reaction:
$aA + bB \rightarrow$ Products,
what is the correct rate law?

Experiment	Concentration of A(M)	Concentration of B (M)	Intial Rate M/s
1	0.100	0.100	4.0×10^{-5}
2	0.200	0.100	1.6×10^{-4}
3	0.100	0.200	8.0×10^{-5}

A. Rate = $k[A][B]$ C. Rate = $k[A]^2[B]^2$
B. Rate = $k[A]^2[B]$ D. Rate = $k[A][B]^2$

66. Which of the following compounds is the strongest base?
A. NH_2Cl B. NH_3 C. D.

67. For preparing a buffer solution of pH 6.0 by mixing sodium acetate and acetic acid, what would be the ratio of the concentration of salt and acid ($K_a = 10^{-5}$)?
A. 1:10 B. 10:1 C. 100:1 D. 1:100

68. Given the following data at 25°C:
$2NOCl(g) \rightarrow 2NO(g) + Cl_2(g)$ $\Delta H^0 = +75kJ$
$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$ $\Delta H^0 = -113.05kJ$
$2NO_2(g) \rightarrow N_2O_4(g)$ $\Delta H^0 = -58.03kJ$
Calculate ΔH^0 at 25°C for the reaction:
$N_2O_4(g) + Cl_2(g) \rightarrow 2NOCl(g) + O_2(g)$
A. +96.08kJ B. +299kJ C. 186.8kJ D. 85.52kJ

Telegram: https://t.me/Extremesms

69. How many moles of $KBrO_3$ are required to prepare 0.0700 moles of Br_2 according to the reaction:
$KBrO_3 + KBr + HNO_3 \rightarrow KNO_3 + Br_2 + H_2O$
A. 0.0233 B. 0.0732 C. 0.210 D. 0.220

70. Apossible meehanism for the reaction, $2A+B \rightarrow C + D$, is:
1. $A + A \quad A_2$ fast, equilibrium
2. $A_2 + A \rightarrow A_3$ slow
3. $A_3 + B \rightarrow A + C + D$ fast
According to the mechanism, the rate law will be:
A. Rate = $k[A]$
B. Rate = $k[A]^2$
C. Rate = $k[A]^3$
D. Rate = $k[A]^2[B]$

71. What is the $[H_3O^+]$ in a 0.050M solution of $Ba(OH)_2$?
A. 1.0×10^{-13}M
B. 5.0×10^{-10}M
C. 1.0×10^{-5}M
D. 5.0×10^{-2}M

72. Which species has the largest F – A – F bond angle where A is the central atom?
A. BF_3 B. CF_4 C. NF_3 D. OF_2

73. When the atom Li, Be, B, Na are arranged in order of increasing atomic radius, which is the correct order?
A. Li, Be, B, Na
B. Li, Na, B, Be
C. Na, Li, Be, B
D. B, Be, Li, Na

74. One of the steps in the manufacture of nitric acid is the oxidation of ammonia shown in this equation:
$4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g)$.
If gaseous water appears at a rate of 0.025 $mol.min^{-1}$, at what rate does ammonia disappear?
A. 0.0040$mol.min^{-1}$
B. 0.017$mol.min^{-1}$
C. 0.038$mol.min^{-1}$
D. 0.150$mol.min^{-1}$

75. What is the half-life of the irreversible first order reaction, $A \rightarrow B$, if 75% of A is converted to B in 60 minutes?
A. 30 minutes B. 45 minutes C. 60 minutes D. 80 minutes

76. At a certain temperature for the reaction:
$2NO(g) + O_2(g) \quad 2NO_2(g)$
The equilibrium concentration are: $[NO] = 0.52M$; $[O_2] = 0.24M$; $[NO_2] = 0.18M$.
What is the value of K_c at this temperature?
A. 0.063 B. 0.50 C. 1.4 D. 2.0

77. A 25.00 mL sample of 0.1050M H_2SO_4 is titrated with a NaOH solution of unknown concentration. The phenolphthalein endpoint was reached

when 17.23mL of the NaOH solution had been added. What is the concentration of the NaOH?
A. 0.07617M B. 0.1447M C. 0.1524M D. 0.3047M

78. During the electrolysis of an aqueous solution of $AgNO_3$, what would happen to the mass of silver metal deposited if the current is doubled and the electrolysis time is decreased to ½ of its initial value?
A. It would stay the same.
B. It would increase to twice its initial value.
C. It would decrease to ¼ of its initial value
D. It would decrease to ½ of its initial value.

79. What is the balanced equation for the cell notation of the following galvanic cell?
$Pb(s)|Pb^{2+}(aq) \| Cl_2(g) | Cl^-(aq) | C(s)$
A. $Pb(s) + 2Cl^-(aq) \rightarrow Pb^{2+}(aq) + Cl_2(g)$
B. $Pb(s) + Cl_2(g) \rightarrow Pb^{2+}(aq) + 2Cl^-(aq)$
C. $Pb^{2+}(aq) + 2Cl^-(aq) \rightarrow Pb(s) + Cl_2(g)$
D. $Pb^{2+}(aq) + 2Cl^-(aq) \rightarrow PbCl_2(s)$

80. Nitrogen – fixing bacteria convert:
A. nitrate to nitrogen gas.
B. nitrogen gas to nitrate.
C. nitrate to ammonia.
D. ammonia to nitrate.

Telegram: https://t.me/ExtremeSeries

Directions : Each of the following questions is followed by four possible alternatives. Read each question and carefully blacken only the letter of your choice on the separate answer sheet provided

Physical constants

A. Gas constant, R = 8.314J mol^{-1} K^{-1} = 0.0821 L- atom mol^{-1} K^{-1}

B. Avogadro's number, $N_A = 6.023 \times 10^{23}$ mol^{-1}

A. Plank's constant, h = 2.9979×10^{8}Js

D.Speed of light, c = 2.9979×10^{8} ms^{-1}

E. Faraday's constant (F) = 9.6485309×10^{4} C mol^{-1}

SI Units and Conversion Factors

A. 1ton = 907.185kg

B. 1 metric ton = 1000 kg

C. 1 A = 10^{-10}m

D. 1 L – atm = 101.3

ATOMIC NUMBERS (Z) AND ATOMIC WEIGHTS (A)

Element	H	He	Li	B	C	N	O	F	Ne	Na	P	S	Cl	Ca
Z	1	2	3	5	6	7	8	9	10	11	15	16	17	20
A	1.0	4.0	6.9	10.8	12.0	14.0	16.0	19.0	20.2	23.0	31.0	32.0	35.5	

Element	Ti	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ag	Cd		Hg
Z	22	24	25	26	27	28	29	30	47	48	79	80
A	47.	52.0	54.	55.	58.	58.	63.5	65.4	107.9		197.0	200.6

1. What percent is gold in 18 karat gold?
   A. 18 C.25
   B. 89 D. 75
2. Which one of the following is a colligative property of solvent in a solution?
   A. Lowering of vapor pressure
   B. Osmotic pressure
   C. Boiling point
   D. Change in entropy
3. Which one of the following is the criterion for a spontaneous change using the thermodynamic parameter entropy (S)?
   A, $\Delta S_{system} < 0$ C. $\Delta S_{universe} \leq 0$

B. $\Delta S_{system} \leq \Delta S_{surr}$ D. $\Delta S_{universe} > 0$

4. Which element corresponds to the electron configuration [Ne] $3s^2$ $3p^1$?1065+2
   A. Mg B. Al C. Si D, Ga
5. Which bond between the following groups of elements will be the most polar?
   A. S-F B.S-Cl C. S- Br D. S- I
6. What is the unit for the rate of a chemical reaction if the concentration is expressed in moles per liter and the time in seconds?
   A, $Lmol^{-1}s^{-1}$ B. $molL^{-1}s^{-1}$ C, s^{-1} D. $moll^{-1}$ s^{-2}
7. What is the most important type of solute- solvent interaction in a solution of $Ca Cl_2$ in water?
   A. Dipole- dipole C. Hydrogen bonding
   B. Dispersion forces D. Ion- dipole
8. A bleaching solution contains 5.2 mass percent sodium hypochlorite, NaOCl. What is the mass of Na OCl in a bottle containing 2500g of bleaching solution?
   A. 90 B. 130 C. 480 D. 520
9. Which of the following is NOT a state function?
   A. Work C. Internal energy
   B. Enthalpy D. Pressure
10. Which of the following metals has the highest positive oxidation state?
    A. Ti B. Cr C. Mn D. Cu
11. Which of the following is NOT a natural polymer?
    A. Proteins B. Starch C. Cellulose D. Teflon
12. The coefficients required to correctly balance the reaction below are
    $Al(NO_3)_3 + Na_2S \longrightarrow Al_2S_3 + NaNO_3$
    A. 2, 1, 3, 2 B. 2, 3, 1, 6 C. 1, 1, 1, 1 D. 4, 6, 3, 2
13. One mole of $(NH_4)_2HPO_4$ contains_____moles of hydrogen atoms.
    A.one B. five C. six D. nine
14. A 500 g toothpaste sample has 0.2 g fluoride concentration, what is the concentration of fluoride in terms of PPm?
    A. 200 B. 250 C. 400 D. 1000
15. Which solution below has the highest concentration of hydroxide ion?
    A. PH = 3.21 C. PH = 9.82
    B. PH = 7.93 D. PH = 12.59
16. What substance is reduced in the following reaction?

Telegram: https://t.me/ExtremeSeries

$Cr_2O_7^{2-} + 6S_2O_3^{2-}\ 14 \rightarrow 2Cr^{3+} + 3S_4O_6^{2-} + 7H_2O$

A. $Cr_2O_7^{2-}$ B. $6\ S_2O_3^{2-}$ C. H^+ D. Cr^{3+}

17. Which one of the following pairs CANNOT be mixed together to form a buffer solution ?
A. NH_3 and NH_4Cl C.Rb OH and HBr
B. $NaC_2H_3O_2$ and $HC_2H_3O_2$ D. H_3PO_4 and KH_2PO_4

18. Which one of the following aqueous solutions will be basic?
A. Na Cl C. Na_2CO_3
B. Na_2SO_4 D. $Fe\ Cl_3$

19. Consider the reaction:
$CH_4(g) + 4Cl_2(g) \rightarrow CCl_4(g) + 4\ HCl(g),\ \Delta H = -434kJ$
What energy change occurs when 1.2 moles of methane reacts?
A. 5.2×10^5 J are released
B. 5.2×10^5J are absorbed
C. 3.4×10^5J are released
D. 3.6×10^5J are absorbed

20. When electricity is passed through aqueous solution of aluminum chloride, 13.5 g of Al are deposited. What must be the number of Faradays of electricity passed?
A. 2.0 B. 1.5 C. 1.0 D. 0.5

21. Which of the following is the anodic reaction in the discharge of a lead- acid storage battery?
A. $Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4(s) + 2e^-$
B. $Pb\ SO_4(s) + 2e^- \rightarrow Pb(s) + SO_4^{2-}(aq)$
C. $PbO_2(s) + SO_4^{2-}(aq) + 4H^+\ 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$
D. $PbSO_4(s) + 2H_2O(l) \rightarrow PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^-$

22. With which one of the following is gold complexed to recover it from its ore?
A. NO_3^- B. CO_3^{2-} C. SO_4^{2-} D. CN^-

23. For the reaction, $3O_2(g) \rightarrow 2O_3(g)$; $\Delta H = 284$ KJ. What are the signs of ΔG and ΔS respectively?
A. positive and negative C. negative and negative
B. positive and positive D. negative and positive

24. Which of the following is NOT true about anode of a voltaic cell?
A. Reduction occurs at it.
B. It is assigned as the negative pole
C. Electrons flow from it

D. Cations in solution move away from the surface of the electrode as the redox reaction occurs.

25. What is the orbital hybridization of the oxygen atom in water?
A. sp B. sp^2 C. sp^3 D. sp^3d

26. Within a specified period, and increase in atomic number is usually accompanied by
A. An increase in atomic radius and an increase in electronegativity.
B. A decrease in atomic radius and an increase in electronegativity.
C. An increase in atomic radius and a decrease in electronegativity
D. A decrease in atomic radius and a decrease in electronegativity.

27. Which of the following sets of quantum numbers is not allowed in the hydrogen atom?
A. $n = 2, l = 1, m_l = -1$ C. $n = 8, l = 7, m_l = -6$
B. $n = 3, l = 0, m_l = 0$ D. $n = 2, l = 0, m_l = 2$

28. NF_3, BF_3, and ClF_3 all have molecular formulae of the type XF_3. Which molecule(s) has (have) trigonal pyramidal molecular geometry?
A. NF_3 and BF_3 C. NF_3 and ClF_3
B. BF_3 and ClF_3 D. NF_3

29. Which of the following molecules **CANNOT** be formed?
A. NCl_3 B. OF C. ONF_3 D. PCl_3

30. Of Br_2, Ne, HCl, and N_2, is likely to have the largest intermolecular dispersion forces?
A. Br_2 B. Ne C. HCl D. N_2

31. Thiosulfate ion is oxidized by iodine according to the following reaction
$2S_2O_3^{2-}(aq) + I_2(aq) \rightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$
If in a certain experiment, 0.006 mol of $S_2O_3^{2-}$ is consumed in 1.0 L of solution each second. At what rates are $S_4O_6^{2-}$ and I^- produced in this solution?
A. 0.012 $molL^{-1}S^{-1}$ ($S_4O_6^{2-}$) and 0.006 $molL^{-1}S^{-1}$ (I^-)
B. 0.006 $molL^{-1}S^{-1}$ ($S_4O_6^{2-}$) and 0.006 $molL^{-1}S^{-1}$ (I^-)
C. 0.003$molL^{-1}S^{-1}$) $S_4O_6^{2-}$) and 0.006 $molL^{-1}S^{-1}$ (I^-)
D. 0.012 $molL^{-1}S^{-1}$) $S_4O_6^{2-}$) and 0.012 $molL^{-1}S^{-1}$ (I^-)

32. A particular reaction was found to depend on the concentration of the hydrogen ion, $[H^+]$. The initial rate changed as a function of $[H^+]$ as follows:

Telegram: https://t.me/Extremeseries

$[H^+]$ (M)	0.005	0.01	0.02
Initial rate (M/s)	3.4×10^{-7}	1.7×10^{-7}	0.85×10^{-7}

What is the order of the reaction In $[H^+]$?

A. -1 B. 1 C. 2 D. 0

33. The conversion of ozone into O_2 proceeds in two elementary steps as follows

$O_3(g) \rightarrow O_2(g) + O(g)$

$O_3(g) + O(g) \rightarrow 2\,O_2(g)$

What is the equation for the overall reaction?

A. $2\,O_3(g) \rightarrow 3O_2(g)$

B. $2O_3(g) + O(g) \rightarrow 3O_2(g) + O(g)$

C. $O_3(g) + O(g) \rightarrow O_2(g) + 2\,O(g)$

D. $2O_3(g) + O(g) \rightarrow 3O_2(g)$

34. Which of the following factors does NOT affect the rate of a chemical reaction that is zero order?

A. Concentration of reactants C. Temperature

B. Surface area of reactants D. Presence of a catalyst

35. How would you prepare 0.90 L of 1.2×10^{-2} M NaCl aqueous solution?

A. Weigh 0.012g of NaCl, dissolve in a small amount of water and dilute to 0. 90L

B. Weigh 0.63 g of Na Cl, dissolve in a small amount of water and dilute to 0.90L

C. Weigh 1.2 g of NaCl, dissolve in a small amount of water and dilute to 0.90L

D. Weigh 1.4 g of NaCl, dissolve in a small amount of water and dilute to 0. 90L

36. Which of the following solutes will produce the largest increases in boiling point upon addition to 1kg of water?

A. 1 mol of $Co(NO_3)_2$

B. 2 mol of KCl

C. 3 mol of ethylene glycol($C_2H_6O_2$)

D. 1 mole of $CaCl_2$

37. What is the correct order of increasing acid strength in the following species: HBrO, $HBrO_2$, $HBrO_3$, and $HClO_3$?

A. $HBrO < HBrO_2 < HBrO_3 < HClO_3$

B. $HBrO < HBrO_2 < HClO_3 < HBrO_3$

C. $HClO_3 < HBrO_3 < HBrO_2 < HBrO$

D. $HBrO_3 < HBrO_2 < HBrO < HClO_3$

38. A 50.0 mL sample of 0.50 M acetic acid (CH_3COOH, $Ka = 1.8 \times 10^{-5}$) is titrated with 0.150 M NaOH solution. What is the PH of the solution when no NaOH is added?

A. 1.6 B. 2.28 C. 2.58 D. 3.17

39. The enthalpy change for the conversion of white phosphorus to red phosphorus is – 17.7 KJ mol^{-1} . Which of the following statement is true?

A. One mole of red phosphorus will give off more heat when burn in air than one mole of white phosphorus.

B. One mole of white phosphorus will give off more heat when burned in air than one mole of red phosphorus.

C. The heat generated by both types is the same

D. The enthalpy change for the conversion of red phosphorus to white phosphorus will be – 35.4 KJ mol-1

40. Which of the following is Not a spontaneous process?

A. The rusting of iron

B. The dissolution of salt in water

C. A clear solution becomes a uniform color after a few drops of dye are added

D. Cleaning a room

41. What are the expected products of the electrolysis of an aqueous solution of $CuCl_2$?

A. Cu and H_2 at the cathode and Cl_2 and O_2 at the anode

B. Cu at the cathode and Cl_2 at the anode

C. Cu and H_2 at the cathode and only Cl_2 at the anode

D. Cu at the cathode and Cl_2 and O_2 at the anode.

42. Which is the second most abundant element in the earth's crust?

A. Aluminum B. Silicon C. Oxygen D. Iron

43. The reaction of a dialcohol with diprotic carboxylic acid is an example of

A. a condensation polymerization

B. a copolymer

C. an addition polymerization

D. an thermo set polymer

44. Which of the following is NOT a carbohydrate?

A. Glucose B. Cellulose C. Glycine D. Starch

45. The exothermic step in the dissolution of a salt in water is

A. separation of the salt into ions in gaseous state

Telegram: https://t.me/ExtremeSeries

B. separation of the water molecules
C. hydration(salvation) of the ions from the salt
D. separation of the salt into ions in gaseous state and separation of the water

46. Which of the following is NOT correct?
A. Addition of a strongly solvated solute decreases the solubility of a gas in liquid
B. for solid dissolved in liquid that is exothermic, increase in temperature increases the solubility
B. The solubility of a substance is its concentration in the saturated solution
D. For solution to occur solvent- solute attractions must overcome solute- solute and solvent-solvent attractions

47. How many milligrams are in 0.2 decigrams?
A. 20 milligrams
B. 2000 milligrams
C. 0.002 milligrams
D. 0.00002 milligrams

48. How many grams of Ca CO_3 will be formed by passing excess CO_2 gas through 100 mL of 0.25 M Ca (OH) $_2$ solution?
A. 10 grams B. 2 grams C. 2.5 grams D. 5 grams

49. Which one of the following electromagnetic radiation has the longest wavelength?
A. gamma rays
B. microwaves
C. radio waves
D. x- rays

50. What are the possible values of the azimuthal quantum number for an electron in the 4f subshell?
A. 0,1,2,3 B. 1, 2, 3 C. 3 D. 4

51. Which one of the following is the correct electronic configuration of Ni?
A. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$
B. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9 4s^1$
C. 1s22s22p63s23p64s^{2}4p^{6}5s2
D. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2 4p^2$

52. What type of hybridization is used by carbon in $C_2 H_2$ (acetylene) ?
A. sp B. sp^3 C. sp^2 D. sp^2d

53. Which of the following rate relationships is valid for the reaction 2 NO_2 (g) $\rightarrow$ 2 NO (g) + O_2 (g)?
A. $\frac{\Delta[NO_2]}{\Delta t} = \frac{\Delta[NO]}{\Delta t}$
B. $-\frac{\Delta[NO_2]}{\Delta t} = \frac{\Delta[O_2]}{\Delta t}$
C.- $\frac{1}{2}\frac{\Delta[NO_2]}{\Delta t} = \frac{\Delta[O_2]}{\Delta t}$
D. $\frac{\Delta[O_2]}{\Delta t} = \frac{\Delta[NO]}{\Delta t}$

54. Which of the following statement is NOT correct regarding equilibrium constant? The value of the equilibrium constant does not depend upon the _____
A. intial concentration of the reactants
B. initial concentration of the products
C. nature of the reactants
D. presence of a catalyst

55. Consider the reaction PCl_5 (g) $\rightleftarrows$ PCl_3 (g) + Cl_2 (g)
In a closed container at equilibrium at a constant temperature. What will be the effect of adding more PCl_5 on the equilibrium concentration of Cl_2 (g) ? It
A. Decreases
B. Cannot be predicated without the value of K_p
C. Remains unaffected
D. Increases

56. What is the mole fraction of a solute in one molal aqueous solution?
A. 0.009 B. 0.018 C. 0.027 D. 0.036

57. Which of the following aqueous solutions will have the lowest freezing point? 0.1 molal solution of _____.
A. urea
B. acetic acid
C. sodium chloride
D. calcium chloride

58. When the equation for the following reaction in basic solution is balanced, what is the sum of the coefficients?
$MnO_2 + HO_2^- \rightarrow MnO_4$
A. 11 B. 31 C. 14 D. 6

59. The standard emf for the cell reaction, $Zn + Cu^{2+} \rightarrow Cu + Zn^{2+}$, is 1.1 V. What would the emf for the cell reaction be when 0.1M Cu^{2+} and 0.1 M Zn^{2+} solutions are used at 298 K?
A. 1.10 V B. 0.10V C. -1.10V D. -0.110

60. Which one of the following is a polyamide molecule?
A. Terylene B. Rayon C. Nylon D. Polystyrene

61. Which metal atoms can form ionic bonds by losing electons from both the outermost and next to outermost principal energy levels?
A. Fe B. Mg C. Pb D. Ca

62. The energy required to remove an electron from an iron atom amounts to 7.21×10^{-19} J. What is the maximum wavelength of light that can perform this?
A. 250 nm
B. 276nm
C. 289.nm
D. 300nm

Telegram: https://t.me/ExtremeSeries

63. What is the molecular geometry of O_3?
A. Trigonal planar
B. Bent
C. Trigonal pyramidal
D. Tetrahedral

64. At 295 K, a 3.0 L flask contains 7 mol HI, 4 mol H_2, and 0.6 mol I_2 in equilibrium. What is the value of K at 295 K ?

$$H_2\,(g) + I_2\,(g) \rightarrow 2\,HI\,(g)$$

A. 10.2 B. 20.4 C. 30.4 D. 61.2

65. The equilibrium constant is 0.09 at 295 K for the reacrtion

$H_2O\,(g) + Cl_2O(g) \rightarrow 2HOCl\,(g)$ For which of the following sets of conditions is the system at equilibrium?

A. A 1.0 L flask contains 1.0 mol of HOCl, 0.10 mol of Cl_2O, and 0.1 mol of H_2O
B. A 2.0 L flask contains 0.084 mole of HOCl, 0.080 mole of Cl_2O, and 0.98 mol of H_2O
C. A 3.0 L flask contains 0.25 mol of HOCl, 0.0010 mol of Cl_2O, and 0.56 mol of H_2O
D. A 4.0 L flask contains 0.25 mol of HOCl, 0.0010 mol of Cl_2O, and 0.56 mol of H_2O

66. 1000 g of a 55% (mass percentage) nitric acid solution is to be diluted to 20% (mass percentage) metric acid solution. How many liters of water should be added to the starting solution?
A. 0.45 B. 1.55 C. 2.30 D. 2.75

67. When pure methanol is mixed with water, the solution becomes warm. Is this an ideal solution?
A. Yes, for ideal solution $\Delta H_{soln} = 0$
B. No, for ideal solution $\Delta H_{soln} = 0$
C. Yes, for ideal solution $\Delta H_{soln} > 0$
D. No, for ideal solution $\Delta H_{soln} < 0$

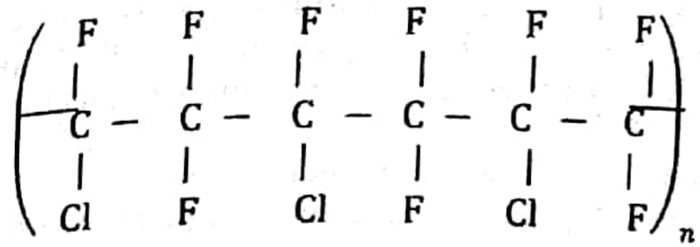
68. What is the percentage of HF molecules ionized in a 0. 10 M HF solution($Ka = 6.8 \times 10^{-4}$) ?
A. 2 B. 5 C. 8 D. 10

69. How does the titration of a strong acid with a strong base differ from titration of a weak acid a strong base with respect to the quantity of base required to reach the required to reach the equivalence point?
A. The quantity of base required to reach the equivalence point is the same in both titrations.

B. The quantity of base required to reach the equivalence point for the titration of a strong acid with a strong base is higher than for the titration of a weak acid with a strong base.
C. The quantity of base required to reach the equivalence point for the titration of a strong acid with a base is lower than for the titration of a weak acid with a strong base.
D. The quantity of base required to reach the equivalence point for the titration of a strong acid with a strong base is slightly higher than for the titration of a weak acid with a strong base.

70. Which of the following reaction is a redox reaction?
A. $3\,Se\,(s) + 4HNO_3\,(aq \rightarrow 3Se\,O_2\,(s) + 4\,NO\,(g) + H_2O\,(l)$
B. $Ca\,Cl_2\,(aq) + 2\,AgNO_3\,(aq) \rightarrow Ca\,(NO_3)_2\,(aq) + 2\,AgCl(s)$
C. $BaCO_3\,(s) \rightarrow BaO\,(s) + CO_2\,(g)$
D. $3Mg(OH)_2\,(s) + 2NH_3\,(g) \rightarrow Mf_3N_2(s) + 6H_2O(l)$

71. A polymer has the following structure



What is the monomer for this poymer?
A. $CF\,Cl = CF_2$ B. $CF_3\,Cl$ C. $CF_2\,Cl_2$ D. CF_4

72. How many significant figures are there in 0.0500 ?
A. One B. Three C. Two D. Four

73. If 16 grams of O_2 react with excess $C_2\,H_6$, how many grams of CO_2 will be formed? The unbalanced equation is :

$$C_2\,H_6\,(g) + O_2\,(g) \rightarrow CO_2\,(g) + H_2O\,(g)$$

A. 22 grams B. 9 grams C. 13 grams D. 12.3 grams

74. Hydrogen atoms are excited to the stationary state designated by the principal quantum number, n = 4 What will be the number of lines (frequencies) emitted by the hydrogen atoms?
A. 1 B. 4 C. 6 D. 16

75. According to the molecular orbital theory, which one of the following is paramagnetic?
A. F_2 B. H_2 C. NO^+ D. NO

76. For the reaction,

Telegram: https://t.me/Extreme Series

$2\,NO\,(g) + O_2\,(g) \rightarrow 2NO_2\,(g)$

The following rate data were obtained What is the rate law for the reaction?

A. Rate = k[NO] $[O_2]$
B. Rate = K[NO] $O_2]^2$
C. rate = K[NO]^2[O_2]
D. rate = K [NO]^2 [O_2]^2

77. The value of Kc for the reaction, $COCl_2\,(g) \leftrightarrows CO\,(g) + Cl_2\,(g)$. Is 2.5×10^{-10} at 300K. Starting with pure COCl2 (g) in a closed vessel, if the concentration of CO (g) is 1×10^{-5} M at equilibrium, what would be the equilibrium concentration of $COCl_2$ (g)

A. 2.5M
B. 2.5×10^{5} M
C. 2.5×10^{-5} M
D. 0.4 M

78. An aqueous solution freezes at -0.186^{0}C (K_f = 1.86^{0}C hg mol^{-1} K_b = 0.512^{0}C hg mol^{-1}) what is the elevation in boiling point (in ^{0}C)?

A. 0.186
B. 0.512
C. 0.512/1.86
D. 0.0512

79. The PKa of acetyl salicylic acid (aspirin) is 3.5. The PH of gastric juice in human stomach is about 2 to3 and the PH in the small intestine is about 8. What one of the following statement is correct? Aspirin is

A. unionized in the small intestine and in the stomach
B. completely ionized in the small intestine and in the stomach
C. Ionized in the stomach and almost unionized in the small intestine
D. Ionized in the small intestine and almost unionized in the stomach

80. A 10 mL sample of tartaric acid (a diprotic acid) is titrated with 20mL of 1.0M Na OH. What is the molarity of the acid?

A. 2.0 B. 1.0 C. 4.0 D. 0.5

Telegram: https://t.me/ExtremeSeries

UNIVERSITY ENTRANCE EXAMINATION (UEE
CHEMISTRY, SENE 2004/ JUNE 2012

BOOKLET CODE :27

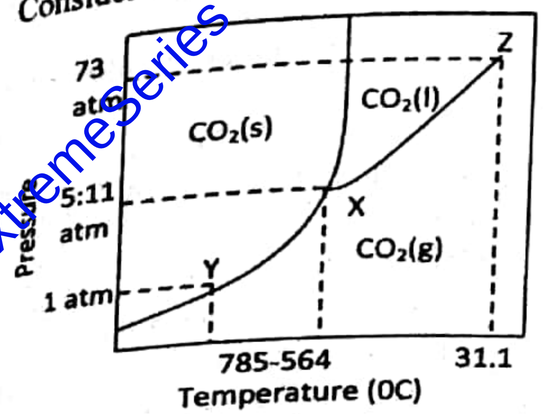
1. The compound CuCl emits blue light having a wavelength of 450nm when heated at about 1200^o C. What is the increment in energy (quantum) that is emitted at 450nm?

A. 2.25×10^{-19} J
B. 4.41×10^{-19} J
C. 8.20×10^{-19} J
D. 16.20×10^{-19} J

2. What is the total number of valence-shell electrons in BrO_3

A. 20 B. 26 C. 32 D. 36

3. Consider the following Phase Diagram for CO_2



What happens when in a CO_2 sample initially at 1 atm and -70 °C,the temperature increases from - 70 °C to -10 °C at a constant pressure of 60 atm?

A. $CO_2\,(g) \rightarrow CO_2(s)$
B. $CO_2\,(s) \rightarrow CO_2(l)$
C. $CO_2\,(g) \rightarrow CO_2(g)$
D. $CO_2\,(g) \rightarrow CO_2(l)$

4. What is the name of a base – promoted ester hydrolysis reaction?

A. Acylation
B. Esterification
C. Condensation
D. Saponification

5. A solution is labeled 0.500 M HCl. What is $[H^+]$?

A. 0.5 B. 1.5 C. 1.6 D. 1.69

6. Consider the following reaction:

$HOCl + H_2O \rightleftharpoons OCl^-$

What willhappen if NaOCl is added to the above reaction at equilibrium?

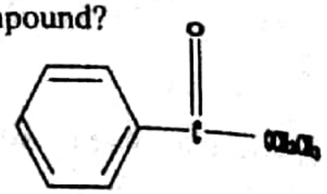
A. The concentrations of both HOCl and H_3O^+ would increase.
B. The concentrations of both HOCl and H_3O^+ would decrease
C. The concentrations of both HOCl would increase and concentration ofH_3O^+ would decrease
D. Theconcentrations of both HOCl would decrease and the concentration of H_3O^+ would increase.

7. Which of the following compounds would be the most basic?

A. 0.1 M acetic acid
B. 0.1 M sodium acetate

C. 0.1 M hydrochloric acid D. 0.1 M ammonium chloride

8. How is a buffer solution prepared?
   A. By mixing a weak acid and a strong acid.
   B. By mixing a weak acid and its conjugate base.
   C. By mixing a strong acid and its conjugate base.
   D. By mixing a strong acid and its conjugate acid.
9. What is the name of the following compound?



   A. Benzoate ester
   B. Ethyl benzoate
   C. Phenyl butyrate
   D. Ethylbenzylketone
10. Consider the following reaction:
    $I_2O_5(s) + 5CO(g) \longrightarrow I_2(s) + 5CO_2(g)$
    What is the magnitude of the change in the oxidation number of the elements?
    A. I, + 5 to 0, C, + 2 to +4 C. I, + 0 to 0, C, + 2 to +4
    B. I, + 5 to 0, C, + 4 to +4 D. I, + 4 to 0, C, + 2 to +4
11. Consider the following oxidation – reduction equation:
    $2H_2O(1) + Al(s) + MnO_4^-(aq) \rightarrow AlOH_4(aq) + MnO_2(s)$
    What are the reducing and the oxidizing agents in this reaction?
    A. Al(s) is the reducing agent and H_2O is the oxidizing agent.
    B. Al(s) is the oxidizing agent and H_2O is the reducing agent.
    C. Al(s) is the oxidizing agent and $MnO_4^-(aq)$ is the reducing agent.
    D. Al(s) is the reducing agent and $MnO_4^-(aq)$ is the oxidizing agent.
12. The dye bromothymol blue (HBb) is a weak acid whose ionization can be represented as follows:
    $HBb(aq) \rightleftharpoons H^+(aq) + Bb^-(aq)$
    Which way will the equilibrium shift when NaOH is added?
    A. To the left
    B. To the right
    C. Initially to the left and after a while to the right
    D. Initially to the left and after a while to the left
13. Which one of the following organic molecules has the highest water solubility?
    A. $HOCH_2CH_2CH_2OH$ C. $HOCH_2CH_2CH_2CH_2OH$
    B. $CH_3CH_2CH_2CH_2OH$ D. $CH_3CH_2CH_2CH_2CH_2OH$
14. Which one of the following substances is a **non-conductor** of electricity?

A. Graphite B. $MgCl_2(s)$ C. silver (s) D. $H_2SO_4(aq)$

15. Which of the following is NOT a solution?
    A. milk B. brass C. whisky D. coca cola drink
16. What is the difference between chlorine – 35 and chlorine – 37?
    A. Chlorine – 37 has two more protons than chlorine -35.
    B. Chlorine – 37 has two more neutrons than chlorine -35.
    C. Chlorine – 37 has two more electrons than chlorine -35.
    D. Chlorine – 37 has one more proton and more neutron than chlorine - 35.
17. Which of the following clement has the highest melting point?
    A. iodine B. tungsten C. mercury D. bromine
18. which of the following compounds is unlikely to contain ionic bonds?
    A. CO B. NaF C. LiC D. $MgBr_2$
19. Which of the following is a chemical formula that represents an amino acid?
    A. CH_4 B. CH_3NH_2 C. CH_3COOH D. NH_2CH_2COOH
20. Which one of the following electromagnetic radiations has the shortest wavelength?
    A. X-rays B. UV rays C. gamma rays D. microwaves
21. Which one of the following statements is **NOT** true about acids?
    A. An acid is a proton donor.
    B. An acid is an electrons pair donor.
    C. An acid is an electron pair acceptor.
    D. An acid is a substance that is ionized in water to produce $H^+(aq)$
22. Which term describes the units that make up compounds with covalent bonds?
    A. ions B. acids C. salts D. molecules
23. What is the number of moles of atoms and the number of atoms in a 10.0-g sample of copper?
    A. 0.08 mol Cu atoms and 2.16×10^{23} atoms
    B. 0.16 mol Cu atoms and 9.63×10^{22} atoms
    C. 0.16 mol Cu atoms and 9.63×10^{23} atoms
    D. 0.31 mol Cu atoms and 4.16×10^{23} atoms
24. Which of the following is the right order of the steps of a scientific method?
    A. Performing experiments – Formulating hypothesis – Making observations
    B. Formulating hypothesis- Making observations –Performing experiments

Telegram: https://t.me/ExtremeSeries

C. Making observations – formulating hypothesis – performing experiments
D. Making observation-Performing experiments- Formulating hypothesis

25. Which group of elements is characterized with ns^2np^2 outer –electron configuration?
A. Group 2A C. Group 4B
B. Group 4A D. Group 3B

26. Consider the following equilibrium:

$$CaCO_3(s) \rightleftharpoons CO_2(g)$$

Which of the following mixture, each placed in a closed container and allowed to stand is not capable of reaching the equilibrium given above?
A. Pure $CaCO_3$
B. $CaCO_3$ and CaO
C. Some CaO and a pressure of CO_2 greater than the value of K_p
D. Some $CaCO_3$ and a pressure of CO_2 greater than the value of K_p

27. Which of the following statements is **true** about esters?
A. Esters can form intermolecular hydrogen bonds.
B. Ester molecules can form intramolecular hydrogen bonds.
C. Ester molecules cannot form intramolecular hydrogen bonds.
D. Esters have higher boiling points than alcohols of comparable molecular weight.

28. Consider the following acids:
I. CH_3COOH, $Ka = 1.8 \times 10^{-5}$
II. HCO_2H, $Ka = 1.8 \times 10^{-4}$
III. HOBr, $Ka = 2.4 \times 10^{-9}$
IV. C_6C_5OH, $Ka = 1.0 \times 10^{-10}$
Which one of the following aqueous solutions will have the highest pH?
A. 0.10 M NaOBr C. 0.10 M C_6H_5OHa
B. 0.10 M HCO_2Na D. 0.10 M CH_3COONa

29. How much water has to be evaporated from 250 ml of 1 M $Ca(OH)_2$ to make it 3 M?
A. 100 ml B. 150 ml C. 167 ml D. 200 ml

30. How many ml of water is required to dilute 50 ml of 3.5 M H_2SO_4 to 2.00 M H_2SO_4?
A. 37.5 B. 45 C. 75 87.5

31. Under what conditions will the enthalpy change of a process equals the amount of heat transferred into or out of the system?

A. Under constant volume.
B. Under constant pressure.
C. Under constant pressure and volume
D. Under constant pressure and temperature.

32. A solution 1t 25°C contains the metal ions Ni^{2+}, Pt^{2+} and Pd^{2+}, all at 1.0 M concentrations. Consider the following standard reduction potentials:

$Ni^{2+} + 2e^- \longrightarrow Ni$ $E^0 = -0.23$ V
$Pt^{2+} + 2e^- \longrightarrow Pt$ $E^0 = 1.20$ V
$Pd^{2+} + 2e^- \longrightarrow Pd$ $E^0 = 0.99$ V

Which metal (s) will plate out first when the solution is electrolyzed?
A. Ni B. Pd C. Pt D. Ni and Pd

33. Which of the following statements is true?
A. A battery is rechargeable.
B. A battery produces electricity spontaneously.
C. A battery has generally no liquid components.
D. A battery produces the same amount of electricity, regardless of composition.

34. Which of the following quantum number/s determine the energy of an electron in a hydrogen atom?
A. n B. n and l C. n, l and m D. n, l, m and s

35. What is the half-life, $t_{1/2}$ for a zero order reaction A → B, (k is rate constant)?
A. ln2k B. $[A]_0/2k$ C. $\ln k[A]_0$ D. ln2 $[A]_0k$

36. There is a strong covalent bond between the N atoms in nitrogen gas, N_2. Why, then, does nitrogen have such a low boiling point of 196°C?
A. The bond between the N-atoms is triple.
B. N is very electronegative, only next to F and O.
C. The strong bond, an intramolecular one, determines the boiling point of the substance.
D. Boiling point is determined by intermolecular force, which in this case is weak as the molecule is non-polar.

37. For elements in the left-most column of the periodic table, properties that have increasing values as the atomic number increases include which of the following?
I. Ionization energy II. Atomic radius III. Atomic mass
A. III only C. I and II only
B. I, II, and III D. II and III only

Telegram: https://t.me/ExtremeSeries

38. Which of the following statement correctly describes a chemical reaction at equilibrium?
A. The concentrations of the products and reactants are equal.
B. The change in the concentrations of the products and reactants is constant.
C. The rate of the forward reaction is less than the rate of the reverse reaction.
D. The rate of the forward reaction is greater than the rate of the reverse reaction.

39. The solubility of sodium selenate, Na_2SeO_4, is 84g/100g of water at 35°C. If a solution is obtained by dissolving 92 g of Na_2SeO_4 in 200g of water at 35°C, what do you call this solution?
A. diluted C. Unsaturated
B. saturated D. supersaturated

40. Which law relates the concentration of a dissolved gas, Cg, to its partial pressure?
A. Henry's law C. Boyle's law
B. Raoult's law D. Ideal gas law

41. What did Rutherford's alpha particle experiment show?
A. Electrons have a negative charge.
B. A proton is a hydrogen atom without electron.
C. Electrons circle the nucleus of an atom in orbits.
D. Most of the mass and all of the positive charge of an atom is found in a tiny nucleus.

42. Which of the statement below best explains why atoms react chemically with each other?
A. When atoms react, they gain protons and are more stable.
B. When atoms react, they lose all their electrons and become more stable.
C. When atoms react, they lose, gain, or share electrons and are then less stable.
D. When atoms react, they lose, gain, or share electrons to attain a full outer energy level and are then more stable.

43. All of the following when added to water will produce an electrolytic solution **EXCEPT.**
A. $N_2(g)$ B. NaI(s) C. HCl(g) D. KOH(s)

44. What is the hydroxide ion concentration for a solution with a pH of 10 at 25°C?
A. 10^{-14} M B. 10^{-10} M C. 10^{-7} M D. 10^{-4}M

45. Which of the following electron transitions requires the smallest energy to be absorbed by the hydrogen atom?
A. from n = 1 to n = 2 C. from n = 3 to n = 4
B. from n = 2 to n = 3 D. from n = 4 to n = 5

46. For an electron that has quantum numbers n = 4 and $m\ell = 0$, which of the following is true?
A. It must have the quantum number n = 0.
B. It must have the quantum number = 0.
C. It must have the quantum number $m_s = +1/2$.
D. It may have the quantum number $\ell = 0,1,2,3$

47. Which of the following species has the smallest H-X-H bond angle where X is the central atom?
A. OH_2 B. NH_3 C. CH_4 D. BH_3

48. What is the hybridization of phosphorus atom in PCl_5?
A. sp^3 B. sp^3d^2 C. sp^3 D. sp^2

49. The organic compound $CH_3C(O)CH_3$ is:
A. aldehyde B. ester C. carbonyl D. ketone

50. which molecule has a Lewis structure that does NOT obey the octet rule?
A. NO B. CS_2 C. PF_3 D. HCN

51. Which of the following explains why, at room temperature, I_2 is a solid, Br_2 is a liquid and Cl_2 is a gas?
A. Ionic bonding C. Hydrogen bonding
B. Hybridization D. London dispersion forces

52. Which molecule listed below has two sigma (δ) and two pi(π) bonds?
A. N_2 B. C_2H_4 C. N_2F_2 D. HCN

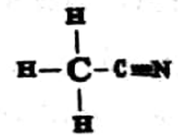
53. Which of the following titrations will have an equivalence point at a pH < 7?
A. strong acid with weak base
B. weak acid with weak base
C. strong acid with strong base
D. weak acid with strong base

54. How many atoms are present in 22 g CO_2?
A. 3.01×10^{23} C. $2 \times 6.02 \times 10^{23}$
B. 6.02×10^{23} D. $1.5 \times 6.02 \times 10^{23}$

55. For which of the following elements is Hund's rule used in writing the electron configuration?
A. C B. B C. Be D. Li

Telegram: https://t.me/ExtremeSeries

56. What is the hybridization of the carbon atom attached to nitrogen in acetonitrile shown below?



A. Sp
B. sp^2
C. sp^3
D. sp^4

57. What mass of aluminum is produced in one hour by the electrolysis of molten $AlCl_3$ with a current of 10A?
A. 1.5 g B. 2.5 g C. 3.4 g D. 4.3 g

58. Consider the following reaction :

$$2S_2O_3^{2-}(aq) + I_2(aq) \longrightarrow S_4O_6^{2-}(aq) + 2I^-(aq)$$

If, in an experiment, 0.05 mol S_2O_32- is consumed in 1.0L of solution each second, at what rates are S_4O_62- and I- produced in this solution?
A. $S_4O_6^{2-} = 0.025$; $I^- = 0.025$
B. $S_4O_6^{2-} = 0.05$; $I^- = 0.05$
C. $S_4O_6^{2-} = 0.025$; $I^- = 0.05$
D. $S_4O_6^{2-} = 0.05$; $I^- = 0.025$

59. All of the following can act as Bronsted – Lowry bases EXCEPT
A. I^-
B. NH_3
C. HCO_3^-
D. H_2CO_3

60. The reaction $2X + Y \rightarrow Z$ was studied and the following data were obtained.

Expt.	[X]	[Y]	Rate (mole. $L^{-1}.S^{-1}$)
1	3.0	3.0	1.8
2	3.0	1.5	0.45
3	1.5	1.5	0.45

What is the proper rate expression?
A. rate = k[X]
B. rate = k[X]
C. rate = $k[Y]^2$
D. 2rate = $k[X]^2[Y]$

61. If the following reaction is at equilibrium, which one of the following changes will shift the equilibrium to the left?

$N_2 + 3H_2 \leftrightarrow 2NH_3$ + heat

A. increasing pressure.
B. decreasing temperature.
C. adding more N_2 and H_2
D. Increasing the volume of the reaction container.

62. The reaction between NO and I_2 is second – order in NO and first-order in I_2. What change occurs in the rate of the reaction if the concentration of NO is doubled and I_2 left unchanged?
A. double B. quadruple C. eight times D. three times

63. Which of the following compounds would give the lowest freezing point depression when 100 g of each are dissolved in 1 kg of water (Kf for water = 1.86°C/m)? Assume complete dissociation.
A. NaCl B. NH_4NO_3 C. $(NH_4)_2SO_4$ D. glucose, $C_6H_{12}O_6$

64. For the reaction A + B → C + D, $\Delta H^\circ = +40$ kJ and $\Delta S^\circ = +50$ J/K. Therefore, the reaction under standard conditions is
A. Spontaneous at all temperatures.
B. non spontaneous at all temperatures.
C. spontaneous at temperatures greater than 800 K.
D. spontaneous only at temperatures between 10 K and 800 K.

65. Which of the following is a conjugate acid/ base pair?
A. HCl/OCl^- B. H_3O^+/OH^- C. NH_4^+/NH_3 D. H_2SO_4/SO_4^{2-}

66. Which of the following is most likely to deviate from ideal gas behavior?
A. He B. Ar C. Cl_2 D. CCl_2F_2

67. Consider the following balanced equation:

$3Ba(NO_3)_2(aq) + Fe_2(SO_4)_3(aq) \rightarrow 3BaSO_4(s) + 2Fe(NO_3)_3(aq)$

The net ionic equation to describe this balanced equation is
A. $3Ba^+(aq) + SO_4^{2-}(aq) \rightarrow 3BaSO_4(s)$.
B. $3Ba^{2+}(aq) + 3SO_4^{2-}(aq) \rightarrow 3BaSO_4(s)$.
C. $6NO_3^-(aq) + 2Fe^{2+}(aq) \rightarrow 2Fe(NO_3)_3(aq)$
D. $3Ba^{2+}(aq) + 2NO_3^-(aq) + 2Fe^{3+}(aq) + 3SO_4^{2-}(aq) \rightarrow 3BaSO_4(s) + 2Fe^{3+}(aq) + 6NO_3^-(aq)$.

68. A reaction is 50% complete in 2 hours and 75% complete in 4 hours. What is the order of this reaction?
A. 0 B. 1 C. 2 D. 3

69. Suppose reactions A → B and B → A are both elementary processes with rate constants of $8 \times 10^2\ s^{-1}$ and $4 \times 10^4 s^{-1}$, respectively. What is the value of the equilibrium constant for the equilibrium?

$$A \rightleftharpoons B$$

A. 2×10^{-2} B. 0.5×10^2 C. 4×10^2 D. 4×10^{-2}

70. Consider the following reaction :

What are the products of this reaction?
A. Sodium acetate and ethanol
B. Sodium acetate and methanol
C. Sodium propionate and methanol

Telegram: https://t.me/ExtremeSeries

D. Methyl propionic acid and methanol

71. What is the molarity of a solution containing 10g of sulfuric acid in 500 ml of solution?
A. 0.02 B. 0.03 C. 0.12 D. 0.2

72. The pH at room temperature of a 0.1 M solution of formic acid ($HCHO_2$) was measured to be 4.50 . What is the hydrogen ion concentration?
A. 3.16×10^{-5} M
B. 3.16×10^{-12} M
C. 6.32×10^{-3} M
D. 6.32×10^{-4} M

73. What is the change in internal energy, ΔE of a system if it absorbs 300 kJ of heat from the surroundings and does 500 kJ of work on the surroundings?
A. 100KJ B. -200 KJ C. 400 KJ D. 500 KJ

74. For the conversion of C (diamond) →C(Graphite) , ΔH = -3 kJ. What does this mean?
A. Both are equally stable
B. Graphite is more stable than diamond
C. Diamond is more stable than graphite
D. Graphite has more stable energy than diamond

75. Why does a pressure cooker reduce cooking time?
A. The heat is more evenly distributed.
B. A stronger flame is used for cooking
C. The high pressure tenderizes the food.
D. The boiling point of water inside the cooker is increased.

76. How can silver can be plated onto nickel?
A. Electricity to a nickel anode in a solution of silver ions.
B. Electricity to a silver anode in a solution of nickel ions.
C. A solution of nickel ions to react with a piece of silver
D. Electricity to a nickel cathode in a solution of silver ions.

77. Which catalyst is used in the hydrogenation of vegetable oils?
A. Iron B.Nickel C. Platinum D. Molybdenum

78. Which of the following is NOT a carbohydrate?
A. Starch B.Glucose C. Glycine D. Cellulose

79. Which element is added to natural rubber to make it harder and reduce its susceptibility to oxidation and chemical attacks?
A. Sulfur B.Silicon C. Carbon D. Nitrogen

80. Which of the following ions is the most abundant in sea water?
A. Na^+ B.Ca^{2+} C. Cl^- D. HCO_3^-

Telegram: https://t.me/ExtremeSeries

ETHIOPIAN UNIVERSITY ENTRANCE EXAMINATION (EUEE) CHEMISTRYGINBOT 2005/JUNE 2013

BOOKLET CODE:27

PHYSICAL CONSTANTS

a) Mass of Electron=$9.11 \times 10^{-31} kg$
b) Gas constant, R=8.314Jmol^{-1}k^{-1}=0.0821L atm mol^{-1}k^{-1}
c) Avogadro's number = $6.023 \times 10^{23} mol^{-1}$
d) Plank's constant , $h = 6.626 \times 10^{-34} J\,s/photon$
e) Faraday's constant, $F = 9.65 \times 10^{4} C mol^{-1}$
f) Rydberg's constant, $R_H = 109{,}678\ cm^{-1}$
g) Velocity of light, $c = 3 \times 10^{8} ms^{-1}$

ATOMIC NUMBERS (Z) AND ATOMIC WEIGHTS (A)

Element	H	He	Li	Be	B	C	N	O	F
Z	1	2	3	4	5	6	7	8	9
A	1.0	4.0	6.9	9.01	10.8	12.0	14.0	16.0	19.0

Element	Ne	Na	Mg	Al	P	S	Cl	Ar	K
Z	10	11	12	13	15	16	17	18	19
A	20.2	23.0	24.3	26.98	31.0	32.1	35.5	39.95	39.09

Element	Ca	Mn	Fe	Cu	Zn	Br	Kr	I
Z	20	25	26	29	30	35	36	80
A	40.08	54.9	55.9	63.55	65.38	79.90	83.80	200.6

1. Which of the following is NOT a basic SI unit?
A) Candela B) Gram C) Mole D) Second

2. Which of the following statements is TRUE?
A) Ultraviolet light has longer wavelength than visible light.
B) The energy of radiation decreases as the wavelength decreases.
C) The frequency of radiation increases as the wavelength decreases.
D) Wave number of an electromagnetic radiation increases as wavelength increases.

3. Which of the following ionic compounds is formed from the reaction between magnesium and nitrogen?
A) MgN_2 B) Mg_2N_2 C) Mg_3N_2 D) Mg_2N_3

4. Which of the following molecules represents a non-polar covalent bond?
A) B-Cl B) C-Cl C) Cl-Cl D) Mg-Cl

5. Which one of the following groups in the periodic table has paramagnetic atoms?

A) Group zero B) Group IIA C) Group IIB D) Group IVA

6. How many types of cubic unit cells are known?
A) 2 B) 3 C) 4 D) 5
7. What is the process that converts liquid vegetable oils to solid fats?
A) Hydration C) Hydrolysis
B) Hydrogenation D) Saponification
8. What would be the solubility of $HOCH_2(CH_2)6CH_2OH$ compared to $CH_3(CH_2)6CH2OH$?
A) Less soluble in water
B) More soluble in water.
C) The same solubility in water.
D) More soluble in a non-polar solvent such as dichloroethane.
9. Which of the following gases is manufactured using the Haber process?
A) Ammonia C) Nitrogen
B) Nitric oxide D) Nitrogen dioxide
10. Which one of the mixture of the following pairs will NOT give a buffer solution?
A) HCN and NaCN C) H_3PO_4 and KH_2PO_4
B) NH_3 and NH_4Cl D) HNO_3 and $NaNO_3$
11. Which one of the following is TRUE for salts formed from strong acids and strong bases?
A) No hydrolysis takes place.
B) Produces ions that are proton donors.
C) Produces ions which are proton acceptors.
D) Depends on the pK_a and pK_b of the parent acids and bases, respectively.
12. Which of the following metals is NOT obtained by commercial electrolytic process?
A) Ag B) Al C) Cu D) Na
13. Which of the following metals has the highest electrical and thermal conductivities?
A) Ag B) Co C) Cu D) Ni
14. Enthalpy is defined as the heat content of the system at constant:
A) heat B) moles C) pressure D) volume
15. Which of the following elements is the second most abundant element in the earth's crust?
A) Aluminum B) Iron C) Oxygen D) Silicon
16. Which of the following plant nutrient will be produced as a result of nitrogen fixation?

A) Carbohydrate C) Mineral
B) Cellulose D) Protein
17. What is the number of significant figures in 0.0030050?
A) 4 B) 5 C) 7 D) 8
18. Which of the following properties of a substance does NOT represent an intensive physical property?
A) Boiling point B) Color C) Density D) Volume
19. What is the mass of one molecule of water?
A) 3.0×10^{-23}g C) 0.0003g
B) 1.8×10^{-22}g D) 18.0g
20. Which of the following quantum number (s) is (are) related to the size and energy of an electron in a hydrogen atom?
A) n B) n,l C) n,l,m D) n,l,m,s
21. An electron has a spin quantum number, $s = +\frac{1}{2}$ and a magnetic quantum number, $m_l = +1$. In which of the following orbital will it NOT be present?
A) s-orbital B) p-orbital C) d-orbital D) f-orbital
22. Which of the following molecules has a trigonal bi pyramidal structure?
A) SF_4 B) IF_5 C) ICl_4^- D) BrF_5
23. Which of the following represents the general configuration of the transition elements?
A) ns^2np^6 B) ns(n-1)d C) ns(n-2)f D)$ns^2np^6(n-1)d^{10}$
24. The total number of electrons participating in the bond formation of carbonate anion, CO_3^{2-} in the molecule of carbonic acid are:
A) 16 B) 10 C) 8 D) 5
25. Which of the following crystals possess high electrical and thermal conductivities?
A) Ionic crystals C) Molecular crystals
B) Metallic crystals D) Covalent network crystals.
26. What is a valid rate expression for the following reaction? $2NO + 2H_2 \rightarrow N_2 + 2H_2O$
A) $\frac{1}{2}\frac{\Delta[H_2]}{\Delta t}$ B) $-\frac{1}{2}\frac{\Delta[H_2O]}{\Delta t}$ C) $-\frac{1}{2}\frac{\Delta[NO]}{\Delta t}$ D) $-\frac{\Delta[N_2]}{\Delta t}$
27. Which of the following statements is TRUE about equilibrium reaction?
A) No more reactants are transformed into products.
B) There are equal amounts of reactants and products.
C) The rate constant for the forward reactions equals that of the reverse reaction.

Telegram: https://t.me/ExtremeSeries

D) The rate for the forward reactions equals that of the reverse reactions.

28. Three gases are in equilibrium in a closed chamber sealed with a piston. The following equilibrium is established:

$$2NH_3(g) \leftrightharpoons N_2(g) + 3H_2(g)$$

What will happen if the piston is pushed into the chamber?

A) The mole fraction of N_2 increases.
B) The mole of N_2 decreases.
C) The mole fraction of N_2 remains the same.
D) The mole fraction of N_2 increases and then decreases.

29. Which of the following reactions will produce an akyl carboxylic acid?

A) Heating a methyl ketone with acid and iodine.
B) Reacting an alkyl halide with hydrogen gas and platinum.
C) Reacting an alcohol with ozone.
D) Oxidation of a primary alcohol with hot permanganate or chromate.

30. Which of the following statements is NOT TRUE?

A) Naturally derived soaps consists of a soluble salt of a long chain fatty acid.
B) Triacylglycerols are esters of glycerol and long chain carboxylic acids.
C) Long chain carboxylic acids are also known as fatty acids.
D) The major acidic component of vinegar is formic acid.

31. What is the the name of triacylglycerol(triglycerides) that is solid at room temperature?

A) Lecithin B) Fat C) Wax D) Oil

32. Which of the following is the most important type of solute-solvent interaction in a solution of n – butanol in water?

A) Dispersion C) Diplole-dipole
B) Ion- dipole D) Hydrogen bonding

33. What is the product of the hydrolysis of easters in the presence of a mineral acid catalyst?

A) Alcohol C) ether
B) Carbon dioxide D) ketones

34. To which organic functional group does the following molecular representation, i.e, R_1R_2CHCOH belong? (R_1 and R_2 represent different alkyl chains)

A) Amides B) Aldehyde C) Ethers D) Organic acid

35. What is the quantity of water, in mL, required to prepare 0.5 M of HCl from a concentrated solution of 3.5Min 50 mL is?

A) 50mL B) 100mL C) 300mL D) 350mL

36. What is the equivalent weight of HNO_3, as an oxidizing agent, in the following balanced reaction?

$$3Fe^{2+} + 4H^+ + +NO_3^- \rightarrow 3Fe^{3+} + NO + 2H_2O$$

A) 10.50 B) 15.75 C) 21.00 C) 31.50

37. Given the following equilibria and equilibrium constants:

(I) $HC_2H_3O_2 + H_2O \rightleftarrows H_3O^+ + C_2H_3O_2^-; K_a = 1.80 \times 10^{-5}$
(II) $H_2CO_3 + H_2O \rightleftarrows H_3O^+ + HCO_3^-; K_a = 4.20 \times 10^{-7}$
(III) $NH_4^+ + H_2O \leftrightharpoons H_3O^+ + NH_3; K_a = 5.6 \times 10^{-10}$
(IV) $HCOOH + H_2O \rightleftarrows H_3O^+ + COOH^-; K_a = 1.80 \times 10^{-4}$

What is the strength of the acids in DECREASING order?

A) I, IV, II and III C) III, II, I and IV
B) II, III, IV and I D) IV, I, II and III

38. Given the reaction:

$$H_2PO_4^- + H_2O \rightarrow H_3O^+ + HPO_4^{2-}$$

Which of the following represents a conjugate acid –base pair?

A) $H_2PO_4^-$ and H_2O C) $H_2PO_4^-$ and H_3O^+
B) $H_2PO_4^-$ and HPO_4^{2-} D) H_2O and $H_2PO_4^{2-}$

39. Which of the following statements is TRUE about colligative properties?

A) Both vapor pressure and freezing point increase when a non-volatile solute is added to a solvent.
B) Both freezing point and boiling point increase when a non-volatile solute is added to a solvent.
C) Both vapor pressure and boiling point decrease when a non-volatile solute is added to a solvent.
D) Colligative properties depend only upon the number of solute particles in a solution and not upon their identity.

40. What is the pH of 0.005 M solution of $Ca(OH)_2$?

A) 2.3 B) 10 C) 12 D) 14

41. The indicator Bromthymol Blue (HBb) is a weak acid with $K_a = 1.0 \times 10^{-7}$ ionizes as follows:

$$HBb(aq, yellow) \rightleftharpoons H^+(aq, colorless) + In^-(aq, blue)$$

Which way will the equilibrium shift when NaOH is added and what will the color of the NaOH solution be containing this indicator?

A) Equilibrium will shift to the right and the color of NaOH solution will turn green.
B) Equilibrium will shift to the right and the color of NaOH solution will turn blue.

Telegram: https://t.me/Extremeseries

C) Equilibrium will shift to the left and the color of NaOH solution will turn yellow.
D) Equilibrium will shift to the left and the color of NaOH solution will turn blue.

42. How many kilo joules of heat are absorbed when 20 g of NaCl(s) is decomposed in to Na(s) and $Cl_2(g)$ at constant pressure according to the following reaction?

$2Na(s) + Cl_2(g) \longrightarrow 2NaCl(s) \quad \Delta H = -821.8kJ$

A) -281.0 B) -140.5 C) +140.5 D) +281.0

43. In the electroplating of nickel from a solution containing Ni^{2+} ion, what will be the weight of the metal deposited on the cathode by a current of 8 A flowing for 500 minutes?

A) 73g B) 103g C) 117.4g D) 145g

44. A 1M solution of $Cu(NO_3)_2$ is placed in a beaker with a strip of Cu metal. A 1M of $SnSO_4$ is placed in a second beaker with a strip of Sn metal. The two beakers are then connected by a salt bridge and the two metal electrodes are connected by wires to a voltammeter. Which of the following electrode serves as the anode and which electrode gains mass?

Given that $E°Cu^{2+}/Cu=0.34V$ and $E^0Sn^{2+}/Sn=-0.14v$

A) Anode, Sn, Sn electrode gains mass.
B) Anode, Sn, Cu electrode gains mass.
C) Anode, Cu, Sn electrode gains mass.
D) Anode, Cu, Cu electrode gains mass.

45. Which of the following metal alloys does NOT contain tin?

A) Brass B. Bronze C) Pewter D) Plumber's solde

46. What structural feature is usually needed to present in order for an addition polymer to be produced?

A) A carbon-carbon sigma bond.
B) A carbon-oxygen pi bond.
C) A carbon-oxygen sigma bond.
D) A carbon –carbon pi bond.

47. What are the raw materials required to synthesize nylon 6,6, a specific kind of nylon?

A) Diacids C) Diacids and Diamines
B) Diamines D) Polyethylene

48. Which of the following substances are added to natural rubber to toughen it?

A) Calcium B) Carbon C) Nitrogen D) Sulfur

49. What is the quantity of heat required to raise the temperature of 80g of ethanol from 25^0C to 75^0C?(specific heat of ethanol=$2.46Jg^{-1}K^{-1}$)

A) 2.46kJ B) 4.0 kJ C) 9.84kJ D) 18.68KJ

50. Consider the following gaseous reaction at 25^0C:

$CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(g) \; \Delta H = -802 \; kJ$

Which energy change would occur if 3.2 moles of CH_4 is completely combusted?

A) $2.57 \times 10^2 kJ$ will be released.
B) $2.57 \times 10^2 kJ$ will be absorbed.
C) $6.43 \times 10^3 kJ$ will be released.
D) $6.43 \times 10^3 kJ$ will be absorbed.

51. What is the number of chloride ions (Cl^-) present in 1.0×10^{-5} mol of $AlCl_3$?

A) 1.80×10^{19} C) 6.02×10^{23}
B) 6.02×10^{18} D)6.02×10^{28}

52. Which of the following is the most important ingredients used for production of DAP fertilizer?

A) Ammonia and phosphoric acid.
B) Nitric acid, urea and phosphoric acid.
C) Phosphoric acid, urea and ammonia.
D) Sulphuric acid, ammonia and urea.

53. Which of the following bio molecules forms a zwitterions at higher or lower pH?

A) Cellulose B) Glucose C) Protein D) Starch

54. Which of the following form of synthetic rubbers can be vulcanized to greatly enhance its mechanical strength?

A) Neoprene C) Styrene-butadiene rubber
B) Isoprene D) Butyl rubber

55. What is the range in the number of carbon atom of the monosaccharides that are found in nature?

A) 3 to 7 B) 4 to 10 C) 4 to 12 D) 5 to 12

56. What are the two principal polysaccharide forms of starch?

A) aldohexose and ketopentose.
B) amylose and amylopectin
C) maltose and cellobiose
D) sucrose and lactose.

57. What is the Kelvin scale (K) corresponding to the temperature readings when the degree Celsius (^0C) is identical to the degree Fahrenheit (^0F)?

A) 0 k B) 37 k C) 233 K D) 273 K

Telegram: https://t.me/ExtremeSeries

58. What is the wavelength associated with an electron of mass, $m = 9.11 \times 10^{-28} g$, travelling at 40% of the velocity of light?
A) $6.06 \times 10^{-15} m$ C) $6.06 \times 10^{-12} m$
B) 2.42×10^{-15} D) $2.42 \times 10^{-11} m$

59. The quantum numbers listed below are meant for four different electrons in an atom:
I. $n = 4,\ 1 = 0,\ m_1 = 0,\ m_s = +1/2$
II. $n = 3,\ 1 = 1,\ m_1 = 1,\ m_s = +1/2$
III. $n = 4,\ 1 = 2,\ m_1 = 0,\ m_s = +1/2$
IV. $n = 4,\ 1 = 1,\ m_1 = 0,\ m_s = -\frac{1}{2}$
When these sets of quantum numbers are arranged in order of increasing energy, one may get:
A) $I < II < III < IV$ C) $II < I < III < IV$
B) $I < III < II < IV$ D) $IV < III < II < I$

60. Which of the following hybrid orbitals is favoring the formation of trigonal bipyramidal?
A) sp^3d B) sp^3 C) sp^3d^2 D) sp^3d^3

61. which one of the following molecules/molecular ions is paramagnetic according to the molecular orbital theory?
A) O_2^{2-} B) O_2 C) F_2 D) O_2^{2+}

62. For the reaction: $2A + B \longrightarrow C$ the following experimental results were obtained:

Experiment	[A]	[B]	Rate (mol $L^{-1}s^{-1}$)
1	0.50	0.50	0.300
2	0.50	0.25	0.075
3	0.25	0.25	0.075

What is the value of the rate constant?
A) 0.6 $molL^{-1}s^{-1}$ C) 1.2 $Lmol^{-1}s^{-1}$
B) 0.6 $L\ mol^{-1}s^{-1}$ D) 2.4 $molL^{-1}s^{-1}$

63. Increase in temperature of rate of a given reaction is due to the increase in the:
A) Extent of molecular dissociation.
B) Activation energy of the reaction.
C) Frequency of collision of the reacting species.
D) Numerical value of the rate constant of the reaction.

64. If a piece of aluminum (Al) foil measuring 24 cm by 31 cm has a mass of 10.35 g, (density of Al=2.70 gcm^{-3}). What is the thickness of the foil in millimeters?

A) 5.15×10^{-3} B) 5.15×10^{-2} C) 3.833 D) 744

65. What sizes of particles and velocities can one consider quantum effect?
A) Particles with very large mass and large velocities.
B) Particles with large mass and small velocities.
C) Particles with very small mass and large velocities.
D) Particles with small mass and small velocities.

66. Which of the following molecules has a dipole moment?
A) XeF_4 B) H_2S C) SO_3 D) CH_4

67. The reaction for the formation of nitrosyl chloride $2NO\,(g) + Cl_2(g) \rightleftharpoons 2NOCl(g)$ was studied at 25°C. The value of K_p for this reaction at 25°C is $1.9 \times 10^3\ atm^{-1}$. What is the value of K at 25°C?
A) $1.9 \times 10^{-3}\ L/mol$ C) $4.6 \times 10^4 L/mol$
B) $3.8 \times 10^{-3} L/mol$ D) $4.6 \times 10^5\ L/mol$

68. Which of the following reactions will convert carboxylic acids to primary amines?
A) Decarboxylation with HBr and peroxide, then reaction of the alkyl bromide with ammonia.
B) Reduction of the acid to the alcohol with NaOH/ formaldehyde then reaction with ammonium chloride and heat.
C) A two-step conversion, first to the amide with ammonia and heat, and then by reduction with lithium aluminum hydride or hydrogen plus catalyst.
D) Lactase fermentation in the presence of ammonia atmosphere

69. A solution was prepared by adding 48 g of methanol (CH_3OH) in to 81 g of water (H_2O). What is the mole fraction of methanol in this solution?
A) 0.25 B) 0.75 C) 1.5 D) 4.5

70. A solution was prepared by dissolving 3.75 g of pure hydrocarbon in 95.0 g of cyclohexane. The boiling point of pure cyclohexane was observed to be 80.70°C and that of the solution was 81.45°C. What is the approximate molecular weight of the hydrocarbon? (K_b for cyclohexane= 2.79°C/m)
A) 71.0 g/mol B) 105g/mol C) 147 g/mol D) 312g/mol

71. A 50 mL solution of H_2SO_4 of 0.205 M is titrated with NaOH solution of unknown concentration. The endpoint against phenolphthalein indicator was signaled when 41.0ml of NaOH was added. What is the concentration of NaOH solution?
A) 0.10M B) 0.25M C) 0.41M D) 0.50 M

Telegram: https://t.me/ExtremeSeries

72. To 0.2 M solution of a weak monoprotic acid , HA, enough quantity of its sodium salt, NaA, was dissolved to give a concentration of 0.2 M of the salt. What will be the acid concentration, $[H_3O^+]$, in the final solutions?
(Ka of HA $= 1.80 \times 10^{-5}$)
A) $3.60 \times 10^{-6} M$
B) $1.00 \times 10^{-5} M$
C) $1.80 \times 10^{-5} M$
D) $1.90 \times 10^{-3} M$

73. Gaseous petrol in a combustion system has done 375 kJ of work during its expansion in the piston. Simultaneously, it absorbed 586 kJ of heat from the engine. What is the internal energy change during the process?
A) +211 kJ B) +961kJ C) -211 kJ D) -961 kJ

74. Which of the following reactions is expected to have negative value of entropy change (ΔS)?
A) $C_6H_6(s) + 6O_2(g) \longrightarrow 6CO_2(g) + 6H_2O(l)$
B) $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$
C) $N_2O_4(g) + Cl(g) \longrightarrow 2NOCl(g) + O_2(g)$
D) $2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$

75. Consider the following un balanced redox reaction in acidic solution: $MnO_4^- + Fe^{2+} \longrightarrow Mn^{2+} + Fe^{3+}$ what is the change in oxidation state for both the substances oxidized and reduced , and the coefficients of Fe^{2+} and Mn^{2+} , respectively, after balancing?
A) 2 and 7, and 2 and 5
B) 3 and 2, and 5 and 1
C) 1 and 5, and 5 and 1
D) 2 and 5, and 5 and 2

76. How many mL conc. HNO_3 & how many mL of water are required to prepare 500mL of 0.1 M HNO_3 from a conc. 13 M HNO_3?
A) 1 mL HNO_3 and 496.15 mL H_20.
B) 3 mL HNO_3 and 500 mL H_20.
C) 3.85 mL HNO_3 and 500 mL H_20.
D) 3.85 mL HNO_3 and 496.15 mL H_20.

77. What is the pH of an aqueous solution prepared to contain $1.3 \times 10^{-3} M$ sodium nitrite ($NaNO_2$) if the acid dissociation equilibrium constant, K_a for nitrous acid (HNO_2) is 5.1×10^{-4} ?
$K_w = 1.0 \times 10^{-14}$
A) 3.1 B) 5.1 C) 6.0 D) 7.3

78. What reactions occur at the anode and cathode when an aqueous solution of Na_2SO_4 is electrolyzed?

		E^0 red
(I)	$S_2O_8^{2-} + 2e^- \longrightarrow 2SO_4^{2-}$	2.01 V
(II)	$O_2 + 4H^+ + 4e^- \longrightarrow 2H_2O$	1.23 V
(III)	$2H_2O + 2e^- \longrightarrow H_2 + 2OH^-$	− 0.83 V
(IV)	$Na^+ + e^- \longrightarrow Na$	− 2.71 V

A) H_2 at cathode and O_2 at anode.
B) Na at cathode and $S_2O_8^{2-}$ at anode.
C) H_2 at cathode and $S_2O_8^{2-}$ at anode.
D) Na at cathode and O_2 at anode

79. Which synthetic polymer is produced from Caprolactam?
A) Nylon -6 B) Nylon 6,10 C) Teflon D) Terylene

80. For the following hypothetical equation, in aqueous solution, what is the correct representation of the cell notation?
$A(s) + B^{2+}(aq) \longrightarrow A^{2+}(aq) + B(s)$
A) $A(s)|A^{2+}(aq)||B^{2+}(aq)|B(s)$
B) $B^{2+}(aq)|B(s)||A^{2+}(aq)|A(s)$
C) $A^{2+}(aq)(s)|A(s)||B(s)|B^{2+}(aq)$
D) $B(s)|B^{2+}(aq)||A^{2+}(aq)|A(s)$

Telegram: https://t.me/ExtremeSeries

ETHIOPIAN UNIVERSITY ENTRANCE EXAMINATION(EUEE)
CHEMISTRY GINBOT 2006/JUNE 2014

BOOKLET CODE:29

Directions: Each of the following questions is followed by four possible alternatives. Read each question and carefully blacken the letter of your best choice on the separate answer sheet provided. You may refer to the information given below when you work on some of the questions.

Physical constants

a. Gas constant, R= 8.314j $mol^{-1}k^{-1}$=0.0821L atm $mol^{-1}k^{-1}$
b. Avogadro's number=$6.023\times 10^{23} mol^{-1}$
c. Plank's constant, h= $6.626\times 10^{-34} Js^{-1}$
d. Speed of light,c=$2.9979\times 10^{8} ms^{-1}$
e. Faraday's constant(F) = 96500C/mol
f. Charge of 1 mole of electrons= 96500C

SI units and Conversion Factors

a. 1 ton=907.185 kg
b. 1metric ton=1000kg
c. $1Å = 10^{-10} m$
d. 1L-atm=101.3J
e. Coulombs=amperes×seconds
f. $T_F = T_c \frac{9^0F}{5^0C} + 32^0F$ (T_F =Temperature in 0_F; T_c = Temperature in 0_C)
g. Vapor pressure of pure water(25°C)=23.8 torr
h. Vapor pressure of pure water(35°C) = 100 torr

ATOMIC NUMBERS (Z) And ATOMIC WEIGHTS(A)

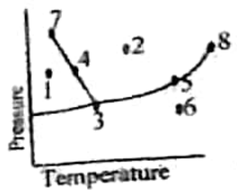
Element	H	He	B	C	N	O	F	Na	Mg	Al	Si	P	S
Z	1	2	5	6	7	8	9	11	12	13	14	15	16
A	1.0	4.0	10.8	12.0	14.0	16.0	19.0	22.98	24.30	26.98	28.08	30.97	32.1

ATOMIC NUMBERS(Z) And ATOMIC WEIGHTS(A)

Element	Cl	Ca	Cr	Mn	Fe	Ni	Cu	Zn	Ag	Cd	Xe	Pb
Z	17	24	24	25	26	28	29	30	47	48	54	82
A	35.5	40.1	52.0	54.9	55.9	58.7	63.5	65.4	107.9	112.4	131.3	207.2

Telegram: https://t.me/ExtremeSeries

1. Which of the following is NOT a conjugate acid- base pair ?
   A. HNO_3/NO_3^-
   B. H_2SO_4/HSO_4^-
   C. NH_3/NH_2^-
   D. H_3O^+/OH^-
2. Which of the following is correct ?
   A. $1L=1dm^3$
   B. $1L=10dm^3$
   C. $10L=1dm^3$
   D. $1L=1m^3$
3. Commercially, liquid vegetable oils are converted to solid fats auch as margarine by:
   A. Hydrogenation
   B. Hydration
   C. Saponification
   D. Oxidation
4. What is the chemical name for Aspirin?
   A. Acetyl salicylic acid
   B. Salicylic acid
   C. Methyl salicylate
   D. Sodium salicylate
5. Which of the following statement(s) is (are) true of an ideal liquid-liquid solution?
   I. It obeys $PV = nRT$
   II. It obeys Raoult's law
   III. Solute-solute, solvent-solvent, and solute-solvent interactions are very similar.
   IV. Solute-solute, solvent-solvent, and solute-solvent interactions are quite different.
   A. I, II and III
   B. I, II and IV
   C. II and III
   D. II and IV
6. The unit cell in a certair lattice consists of a cube formed by an anion at each corner, an anion in the center, and a cation at the center of each face. How many cations and how many anions does the unit cell have ?
   A. 5 anions and 6 cations
   B. 5 anions and 3 cations
   C. 2 anions and 3 cations
   D. 3 anions and 4 cations
7. In a reaction, A+B→Product, the rate is doubled when the concentration of B is doubled, and the rate increases by a factor of 8 when the concentrations of both the reactants (A and B) are doubled, the rate law for the reaction can be written as:
   A. Rate=K[A][B]
   B. Rate=$k[A][B]^2$
   C. Rate=$k[A]^2[B]$
   D. Rate=$k[A]^2[B]^2$
8. Answer the following question using the phase diagram below.



At which point can only the solid and liquid phases coexist?

A. 1 B. 2 C. 3 D. 4

9. Which electron transition in a hydrogen atom releases the largest energy?
   A. n=2→ $n = 1$
   B. n=4→ $n = 2$
   C. n=6→ $n = 3$
   D. n=7→ $n = 6$
10. The simplest formula for a compound containing Mn^{4+} and O^{2-} is
    A. MnO B. MnO_2 C. Mn_2O_4 D. Mn_4O_2
11. What is the ratio of the energy of a photon of 300 nm wavelength radiation to that of 600 nm radiation?
    A. 1:2 B. 1:1 C. 2:1 D. 3:1
12. Which one of the following chemicals is used to disinfect water?
    A. Fluorine B. Nitrogen C. Oxygen D. Chlorine
13. During the titration of a known volume of a strong acid with a strong base, there is
    A. a steady increase in PH
    B. a sharp increase in PH around the end point
    C. a steady decrease in PH
    D. a sharp decrease in PH around the end point
14. A solution with a pH of 7.5 would be described as:
    A. Very basic C. Slightly acidic
    B. Slightly base D. Very acidic
15. The electrolysis of molten NaCl is an industrial process. What does the electrolysis produce?
    A. Na and Cl_2 C. Na^+ and Cl^-
    B. H_2 and O_2 D. NaOH and Cl_2
16. Butane burns in oxygen according to the equation below. $2C_4H_{10}(g)+13O_2(g)\rightarrow 8CO_2(g) + 10H_2O(l)$. If 11.6g of butane is burned in 11.6 g of oxygen, which is the limiting reagent?
    A. Butane C. Neither
    B. Oxygen D. Both oxygen and butane
17. A beaker filled to the 100mL mark with salt (the salt has a mass of 100g) and another beaker to the 100mL mark with water (the water has a mass of 100g) are mixed together in a bigger beaker until the salt is completely dissolved. What will be the mass of the solution?
    A. It will be much more than 200 g

B. It will be much smaller than 200g
C. It will be exactly 200g
D. It will be slightly more than 200g

18. Which one of the following is NOT an intensive property?
    A. Mass B. Temperature C. Colour D. Density
19. Which of the following is natural polymer?
    A. Nylon B. PVC C. Cotton D. Dacron
20. Which of the following metals forms a volatile compound during the extraction process?
    A. Fe B. Co C. Ni D. Cu
21. Which one of the following atoms in its ground state has the greatest number of unpaired electrons?
    A. $_{13}Al$ B. $_{14}Si$ C. $_{15}P$ D. $_{16}S$
22. Which statement is true about chemical reactions at equilibrium?
    A. The forward and backward reactions proceed at equal rates
    B. The forward and backward reactions have stopped
    C. The concentrations of the reactants and products are equal
    D. The forward reaction is exothermic
23. Which compound contains both covalent and ionic bonds ?
    A. Sodium carbonate, Na_2CO_3
    B. Magnesium bromide, $MgBr_2$
    C. Dichloromethane, CH_2Cl_2
    D. Ethanoic acid, CH_3COOH
24. Which set of quantum numbers(n, 1,m_1, m_s) is not possible?
    A. 1,0, 0, $\frac{1}{2}$ C. 1,0,0, $-\frac{1}{2}$
    B. 1, 1, 0, $\frac{1}{2}$ D. 2, 1,-1, $\frac{1}{2}$
25. Which of the following particles contains more electrons than neutrons?
    I. $^{1}_{1}H$ II. $^{35}_{17}Cl^-$ III. $^{39}_{19}K^+$
    A. I only B. II only C. I and II only D. II and III only
26. Which molecule or ion does NOT have a tetrahedral shape?
    A. XeF_4 B. SiCl4 C. BF_4^- D. NH_4^+
27. Why are metals soft and malleable?
    A. Because they are very shiny.
    B. Because they experience electrostatic repulsion
    C. Because of the presence of mobile electrons.
    D. Because the metal cations can slip over each other fairly easily.

Telegram: https://t.me/ExtremeSeries

28. In which region of the Periodic Table would the element with the electronic structure below be located?
$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^6 5s^2$
A. Group 6
B. Noble gases
C. s block
D. d block

29. Which factors will influence the rate of the reaction shown below?
$NO_2(g)+CO(g) \rightleftarrows NO(g)+CO_2(g)$
I. The number of collisions per second
II. The energy of the collisions
III. The geometry with which the molecules collide
A. I only B. II only C. I and II only D. I, II and III

30. How many π bonds are present in CO_2?
A. One B. Two C. Three D. Four

31. Which pair of monomers forms polyesters?
A. Difunctional alcohol, difunctional organic acid
B. Difunctional alcohol, difunctional amino acid
C. Ethylene, ethylene
D. Amino acid, amino acid

32. Which changes will increase the amount of $SO_3(g)$ at equilibrium?
$2SO_2(g)+O_2(g) \rightleftarrows 2SO_3(g) \quad \Delta H^0 = -197kJ$
I. Increasing the temperature
II. Decreasing the volume
III. Adding a catalyst
A. I only B. II only C. I and III only D. I, II and III

33. Which compound is a carboxylic acid ?
A. CH_3COOH
B. $(CH_3CO)_2O$
C. $(CH_3)_2CHCOCH_3$
D. $(CH_3)_2O$

34. The mechanism of a reaction is shown below.
$HOOH+I^- \rightarrow HOI + OH^-$ (slow)
$HOI+I^- \rightarrow I_2 + OH^-$ (fast)
$2OH^- + 2H_3O^+ \rightarrow 4H_2O$ $(fast)$
What is the rate law based on this mechanism?
A. Rate $=k[HOOH]\,[I^-]$
B. Rate $=k[HOOH]^2\,[I^-]$
C. Rate $=k[HOOH]\,[I^-]^2$
D. Rate $=k[HOOH]$

35. What is the equilibrium constant expression for the following reaction?
$2Hg(g)+O_2(g) \rightleftarrows 2HgO(s)$

A. $K=1/([Hg]^2/([O_2])$
B. $K=[Hg]^2[O_2]$
C. $K=[HgO]^2/([Hg]^2[O_2])$
D. $K=[2HgO]/([2Hg][O_2])$

36. A triacylglycerol that is solid at room temperature is called:
A. Lecithin B. Fat C. Wax D. Oil

37. Which compound is an ester?
A. CH_3COOH B. $CH_3OC_2H_5$ C. C_2H_5CHO D. $HCOOCH_3$

38. If a solute dissolves in an endothermic process,
A. H bonds must exist between solvent and solute.
B. Strong ion-dipole forces must exist in the solution.
C. The entropy of the solution must be greater than that of its pure components.
D. The solute must be a gas.

39. A solution is made by dissolving 250.0 g of potassium chromate crystals (K_2CrO_4, molar mass, 194.2g) in 1.00kg of water. What will be the freezing point of the solution? (K_f for water is $1.86^0 cm^{-1}$).
A. -8.87^0C B. -7.18^0C C. -5.73^0C D. -1.86^0C

40. How many moles of sodium hydroxide are present in 2.5L of 0.5 M aqueous solution?
A. 0.2 B. 0.5 C. 1.25 D. 12.5

41. Which species **CANNOT** act as a Lewis acid?
A. NH_3 B. BF_3 C. Fe^{2+} D. $AlCl_3$

42. Which of the following statements is true?
A. A universal indicator is a mixture of indicators that will give a different colour for a different PH.
B. Phenolphthalein is a universal indicator.
C. A universal indicator can only be used in either strongly acidic or basic solution.
D. The colour of a universal indicator is red in a weak acid.

43. Three acids, HA, HB, HC have the following K_a values.
$K_a(HA)= 1\times 10^{-5}$ $K_a(HB)=2\times 10^{-5}$ $Ka(HC)=1\times 10^{-6}$
What is the correct order of increasing acid strength (weakest firs)?
A. HA, HB, HC
B. HC, HB, HA
C. HC, HA, HB
D. HB, HA, HC

44. If the solute- solvent interactions are greater than the solute-solute and solvent-solvent interactions, what will be the total vapor pressure of the solution?
A. Greater than that calculated from Raoult's law
B. Less than that calculated from Raoult's law

Telegram: https://t.me/Extremeseries

C. The same as calculated from Raoult's law
D. Raoult's law cannot be applied for such interactions.

45. Which of the following procedures will produce a buffered solution?
I. Equal volumes of 0.5 M NaOH and 1M HCl solutions are mixed.
II. Equal volumes of 0.5M NaOH and 1M CH_3COOH solutions are mixed.
III. Equal volumes of 1M $NaCH_3CO_2$ and 1M CH_3COOH solutions are mixed.
IV. Equal volumes of 1.0M NaOH and 1M HCl solutions are mixed.
A. I B. III C. I and II D. II and III

46. For which conversion is an oxidizing agent required?
A. $2H^+(aq) \rightarrow H_2(g)$ C. $SO_3(g) \rightarrow SO_4^{2-}(aq)$
B. $2Br^-(aq) \rightarrow Br_2(aq)$ D. $MnO_2(s) \rightarrow Mn^{2+}(aq)$

47. What volume of 0.500M NaOH is required to neutralize 25.0 mL of 1.2 MH_2SO_4? (Assume complete ionization of the acid).
A. 60mL B. 90mL C. 100mL D. 120mL

48. A gas is confined to a cylinder under constant atmospheric pressure. When the gas undergoes a particular chemical reaction, it releases 135 kJ of heat to its surroundings and does 63 kJ of P-V work on its surroundings. What are the values of ΔH and ΔE for the process?
A. $\Delta H = 135kJ, \Delta E = 63kJ$
B. $\Delta H = -135kJ, \Delta E = -63kJ$
C. $\Delta H = 135kJ, \Delta E = 198kJ$
D. $\Delta H = -135kJ, \Delta E = -198kJ$

49. The oxidation numbers of nitrogen in NH_3, HNO_3 and NO_2 are____ respectively.
A. -3, -5, +4 C. -3, +5, -4
B. +3, +5, +4 D. -3, +5, +4

50. Which statement about the following reaction is correct?
$2FeS(s) + 3CO_2(g) \rightarrow Fe_2O_3(s) + 3CO(g)$ $\Delta H^0 = +26.6kJ$
A. 26.6kJ of energy are released for every mole of Fe reacted.
B. 26.6 kJ of energy are absorbed for every mole of Fe reacted.
C. 53.2kJ of energy are released for every mole of Fe reacted.
D. 13.3 kJ of energy are absorbed for every mole of Fe reacted.

51. Which of the following metals is extracted by thermal reduction process?
A. Cu B. Fe C. Al D. Mg

52. The two standard electrode potentials involved in the nickel cadmium rechargeable cell are given below. Calculate the ΔG^0 in kJ of the cells.
$NiO_2(s) + 2H_2S(1) + 2e^- \rightarrow Ni(OH)_2(s) + 2OH^-$ $E^0 = +0.49V$
$Cd(OH)_2(s) + 2e^- \rightarrow Cd(s) + 2OH(aq)$ $E^0 = -0.76V$
A. -184 B. -153 C. -241 D. -206

53. Which of the following is a synthetic rubber produced from Caprolactam (CPL)?
A. Nylon 6,10 B. Terylene C. Teflon D. Nylon6

54. Which one of the following reactions is **NOT** a redox reaction?
A. $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$
B. $2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$
C. $Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$
D. $Cu^{2+}(aq) + Zn(s) \rightarrow Cu(s) + Zn^{2+}(aq)$

55. The half-reaction for formation of magnesium metal upon electrolysis of molten $MgCl_2$is:
$Mg^{2+} + 2e^- \rightarrow Mg$
What is the mass of magnesium formed upon passage of a current of 60.0A for a period of $2.00 \times 10^3 s$?
A. 5.0 B. 10.0 g C. 15.1 g D. 30.2g

56. An aqueous solution is 70% nitric acid (HNO_3)By mass. What is the concentration of HNO_3 expressed in molality?
A. 0.559m B. 8.62m C. 11.1m D. 37.0m

57. Which of the following is a natural polymer?
B. Polythene B. Polysaccharides C. Nylon D. Terylene

58. What is the ionization energy of an iron atom if it requires a radiation of 276 nm to completely remove its outer most electron in the gaseous state?
(Plank's consant, $h = 6.626 \times 10^{-34} Js$, speed of light, $c = 3 \times 10^8 ms^{-1}$)
A. $7.21 \times 10^{-19} J$ C. $7.21 \times 10^{19} J$
B. $7.21 \times 10^{-19} kJ$ D. $7.21 \times 10^{19} kJ$

59. Which of the electron configurations describes the ground state electron configuration of Cl^-?
A. $1s^2 2s^2 2p^6 3s^2 3p^6$ C. $1s^2 2s^2 2p^6 3p^1$
B. 1s22s22p63s1 D. $1s^2 2s^2 2p^6 3s^2 3p_x^2 3p_y^1$

60. When 0.68 is divided by 14.364, the actual answer is 0.0473405. What will be the correct answer?
A. 0.05 B. 0.047 C. 0.0473 D. 0.04734

Telegram: https://t.me/ExtremeSeries

61. What is the correct molecular electronic configuration foe the molecular ion, B_2^+?
A. $\sigma_{1s}^2\,\sigma^*1s^2\,\sigma_{2s}^2\,\sigma^*2s^2\sigma_{2P}^2$
B. $\sigma_{1s}^2\,\sigma^*1s^2\,\sigma_{2s}^2\,\sigma^*2s^2\pi\ 2py_1^2$
C. $\sigma_{1s}^2\,\sigma^*1s^2\,\sigma_{2s}^2\,\sigma^*2s^2\pi\ 2px_1^1$
D. $\sigma_{1s}^2\,\sigma^*1s^2\,\sigma_{2s}^2\,\sigma^*2s^2\pi\ 2Px^1\pi_{2Py}^1$

62. Which of the following molecules or ions will exhibit delocalized bonding?
NO_2^-, NH_4^+, N_3^-
A. NH_4^+ and N_3^-
B. NH_4^+ and N_3^-
C. NO_2^-
D. NO_2^- and NH_4^+

63. Based on molecular orbital theory, the bond orders of H_2, H_2^+, and H_2^- are _____, respectively.
A. 1, 0, and 0
B. 1, $\frac{1}{2}$, and 0
C. 1, 0 and $\frac{1}{2}$
D. 1, $\frac{1}{2}$, *and* $\frac{1}{2}$

64. How many 3d electrons are present in the ground state of chromium atom?
A. 4 B. 5 C. 6 D. 10

65. When the following substances are arranged in order of increasing melting point (lowest melting point first), the correct order is:
A. $CH_3CH_2CH_3, CH_3COCH_3, CH_3CH_2CH_2OH$
B. $CH_3CH_2CH_3, CH_3CH_2\ CH_2OH, CH_3CO\ CH_3$
C. $CH_3COCH_3, CH_3CH_2CH_2OH\ , CH_3CH_2CH_3$
D. $CH_3CH_2CH_2\ OH, CH_3CH_2CH_3, CH_3COCH_3$

66. The half life for the first order decomposition of nitromethane, CH_3NO_2, at 500K is 650 seconds. If the initial concentration of CH_3NO_2 is 0.500M, what will its concentration be (M) after 1300 seconds have elapsed ?
A. 0.125 B. 0.140 C. 0.250 D. 0.425

67. In a zero-order reaction for every 10^0 rise of temperature, the rate is doubled. If the temperature is increased from 10°C to 100°C, the rate of the reaction will become
A. 64 times B. 128 times C. 256 times D. 512 times

68. Which of the following mathematical relationships between K, K_1 and K_2 is correct ?
$CO_2\,(g) + H_2(g) \rightleftarrows CO\,(g) + H_2O(g)$ K
$Fe(s)+CO_2(g) \rightleftarrows FeO(s) +CO(g)$ K_1
$Fe(s)+H_2O(g) \rightleftarrows FeO(s) + H_2(g)$ K_2
A. $K=K_1+K_2$ B. $K= K_1 \times K_2$ C. $K= K_1/K_2$ D. $K=K_2/K_1$

69. Which of the following gives the correct order of decreasing acidity of carboxylic acids?
A. $Cl_3CCOOH, Cl_2CHCOOH, FCH_2COOH, CH_3COOH$
B. $FCH_2COOH, CH_3COOH, Cl_2CHCOOH, Cl_3CCOOH$
C. $CH_3COOH, FCH_2COOH, Cl_2CHCOOH, Cl_3CCOOH$
D. $Cl_2CHCOOH, CH_3COOH, FCH_2COOH, Cl_3CCOOH$

70. The value of K_c for the following equilibrium reaction is 4.0 at a temperature of 373 k.
$CH_3COOH+C_2H_5OH \rightleftharpoons CH_3COOC_2H_5+H_2O$
What mass of ethyl ester ($CH_3COOC_2H_5$) would be present in the equilibrium mixture if 15 g of acetic acid and 11.5 g of ethanol were mixed and equilibrium was established at this temperature?
A. 5.2 B. 10.1 C. 12.6 D. 14.1

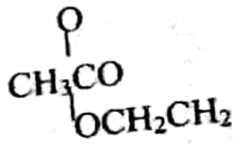
71. Which of the following is true about an open system?
A. A system that exchanges both energy and matter with its surroundings.
B. A system that cannot exchange both matter and energy with its surroundings.
C. A system that only exchanges matter with its surroundings
D. A system that only exchanges energy with its surroundings

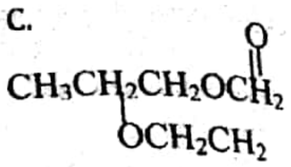
72. Which of the following statements is true about the percent ionization of a weak acid?
A. It decreases with increasing concentration
B. It increases with decreasing concentration
C. It increases with increasing concentration
D. It decreases with decreasing concentration

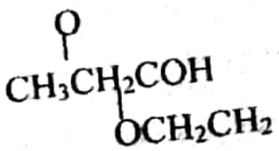
73. What is the value of ΔH for the reaction $S(s) \rightarrow S(g)$?
$S(s)+O_2(g) \rightarrow SO_2(f)$ $\Delta H = -395kJ$
$S(g)+O_2(g) \rightarrow SO_2(g)$ $\Delta H = -6185kJ$
A. -1013kJ C. +223kJ
B. -223kJ D. +1013kJ

74. Which of these compounds is the ester formed from the reaction of acetic acid and 1-Propanol?

Telegram: https://t.me/ExtremeSeries

A. 

C. 

B. 

D.

75. A lab instructor is preparing 5.0 liters of a 0.10 M $Pb(NO_3)_2$ (molecular mass 331) solution. What is the mass required?
A. 165.5 g of $Pb(NO_3)_2$ and add 5.0 kg of H_2O
B. 165.5 g of $Pb(NO_3)_2$ and add H_2O unitl the solution has a volume of 5.0 liters
C. 33.1 g of $Pb(NO_3)_2$ and add H_2O until the solution has a volume of 5.0 liters
D. 33.1 g of $Pb(NO_3)_2$ and add 5.0 liters of H_2O.

76. Which of the following is NOT a characteristic of the electrolytic cell containing aqueous solution of NaCl used in the manufacture of sodium hydroxide?
A. The sodium hydroxide solution is produced in the electrolytic cell.
B. The electrolyte must be a dilute solution of NaCl
C. Hydrogen is produced at the cathode
D. The production of chlorine gas occurs at the anode

77. The kinetic data are for the reaction:

$$A+B \rightarrow C$$

From these data what are the orders of the reaction with respect to A and B?

	[A]	[B]	initial Rate ($moldm^{-3}sec$
1)	0.1	0.1	1×10^{-5}
	0.1	0.2	4×10^{-5}
	0.2	0.1	1×10^{-5}

A. order of A=1 order of B=0
B. order of A=0 order of B=2
C. order of A=0 order of B= 4
D. order of A= 1 order of B=2

78. Which of the following metals has the highest electrical and thermal conductivities of any metal?
A. Ag B. Cu C. Ni D. Co

79. Which of the following is a linear polymer?
A. High density polyethene (HDPE)
B. Low density polyethene (LDPE)
C. Bakelite
D. Vulcanized rubber

80. Which of the following is the most important source for the extraction of iron?
A. Hematite B. Bauxite C. Chalcopyrite D. Sphalerite

Telegram: https://t.me/ExtremeSeries

1. D: The general rule "Link dissolves Like" applies. CCl_4 is non-polar. I_2 is miscible in CCl_4 because both have weak attractive forces and hence strong solute – solvent interaction. Molecular crystals like I_2 are insoluble in polar solvents (Na_2CO_3, CH_3F, $(CH_2)_4O_2$) because polar solvent molecules attract each other too strongly and there is weak solute – solvent interaction.
2. A: A catalyst only helps a chemical reaction reach equilibrium faster. It has no net effect on the state of chemical equilibrium.
3. D: Silk, wool and cotton are examples of a natural fiber. Examples of synthetic fiber include

Acrylic – an addition polymer of a crylonitrile, $CH_2= CH - CN$

Polyester – a condensation polymer of ethylene glycol, $HO - CH_2 - CH_2 - OH$ and dimethylterphthalate, $CH_3O_2CC_6H_4CO_2CH_3$

Nylon – a condensation polymer of hexamethylenediamine, $NH_2-(CH_2)_6-NH_2$ and adipia acid, $HOOC - (CH_2)_4 - COOH$.

4. D: Polyprotic acids contain more than one ionizable hydrogen. Monoprotic acids contain only one ionizable hydrogen. Oxalic acid, $H_2C_2O_4$ contains two ionizable hydrogen

   $H_2C_2O_4 + 2H_2O \rightleftharpoons 2H_3O^+ + C_2O_4^=$

   HClO, HBrO and HIO are monoprotic acids

   Eg: $HClO + H_2O \rightleftharpoons H_3O^+ + ClO^-$
5. C: $\Delta E = q - w$ (W is negative because work is done by the gas (the system) or work is done on the surroundings)

   $= 50 - 25$

   $= 25kJ$
6. D: $C^0(s) + 2H_2O(g) \rightarrow CO_2(g) + 2N_2(g)$

   $C^0 \rightarrow C^{+4}$ (C is oxidized because by losing 4e$^-$s its oxidation n^o is changed from 0 to +4)

   $2N^{+1} \rightarrow N_2^0$ (N is reduced because by gaining 2e$^-$s its oxidation n^o is changed from +1 to 0).
7. D: CaO is an ionic compound. There is a large electronegativity difference between Ca and O. Thus, e$^-$s are transferred from Ca (less electronegative) to O (more electronegative) and the bond is IONIC (electrovalent). If the electron – egativity difference between the atoms is ≥ 2.0, the compound formed is likely ionic.
8. C: At equilibrium,
   i) the rate of the forward reaction equals the rate of the reverse reaction.

   ii) There is no net change in the concentration of the reactants and the products.
9. A: amphiprotic (Amphoteric) species can donate and accept a proton (H^+). They can act as a base and an acid. SO_4^{2-} can accept a proton (it is a base) but cannot donate a proton, hence not amphoteric.

   Anions derived from polyprotic acids are amphoteric.

   Eg: HS^-, $HSO_4^- + H_2O_3^-$

   $HSO_4^- + H_2O \rightleftharpoons SO_4^{2-} + H_3O^+$ (HSO_4^- donates a proton and is an acid)

   $HSO_4^- + NH_3 \rightleftharpoons H_2SO_4 + NH_2^-$ (HSO_4^- accepts a proton and is a base)
10. D: CO_2 is used in the manufacture of washing soda and baking soda. Solid CO_2 (Dry ice) is widely used as a refrigerant. CO_2 dissolved in water is the BASIS of the soft drink and beverage industry to produce carbonated beverages.
11. C: $n=5 \Rightarrow l= 0,1,2,3,4$ (l=0(s), l=1(p), l=2(d), l=3(f), l=4(g))

    $\Rightarrow$ 5s, 5p, 5d, 5f, 5g

    Since there is no atom with g sublevel, we take up to 5g because

    $5f \Rightarrow n=5$, $l=3$, $m_l=+1$ and $l=3 \Rightarrow m_l= -3,-2,-1,0,1,2,3$

    Maximum no of e$^-$s = $5s^25p^65d^{10}5f^{14} = 2+6+10+14=32e^-s$
12. B: $I_3^- = 3 \times 7 + 1 = 22e^-s$ $\left[\ddot{I}-\ddot{I}-\ddot{I}\right]^-$
13. B: K_p is very small (Kp $\ll$ 1) and the equilibrium mixture contains mainly the reactant. Thus, the equilibrium shifts far to the left showing the initial pressure of CO and Cl_2 must decrease. For gases, pressure is directly proportional to concentration.
14. A: $K_c^1 = \frac{1}{k_c} = \frac{1}{3.6\times10^3} = 0.27\times10^{-3} = 2.7\times10^{-4}$

    $\Rightarrow$ The K_c of the reverse reaction ($K_c^{'}$) is the reciprocal of the K_c of the forward reaction.
15. D: $H_2O + (CH_3)_3N \rightleftharpoons (CH_3)_3NH^+ + OH^-$

    acid base acid base

    H_2O, OH^- = conjugate acid – base pairs

    $(CH_3)_3NH^+$, $(CH_3)_3N$ = conjugate acid – base pairs
16. B: Halogens in general are useful. But chlorine is the most abundant of the halogens and hence commercially the most important.

Telegram: https://t.me/ExtremeSeries

17. B: Entropy is the measure of disorder (Randomness). In boiling a liquid is changed to vapor (Gas) where by the randomeness (disorder) increases, i.e. entropy increases. In sublimation a solid is directly changed to a gas which greatly increases the disorder (entropy) In melting a solid is changed to a liquid which increases the entropy. In freezing a liquid is changed to a solid which decreases the disorder and hence the entropy.
N.B: solid < liquid < gas in entropy

18. B: $MnO_4^-(aq) + CN^-(aq) \rightarrow MnO_2(s) + CNO^-(aq)$
Balance the reaction using half – rxn method
$MnO_4^- \rightarrow MnO_2$ (reduction)
$CN^- \rightarrow CNO^-$ (oxidation) } ⇒ splitting the rxn
Add H_2O to the side deficient in O and H^+ to the side deficient in H
$4H^+ + MnO_4^- \rightarrow MnO_2 + 2H_2O$
$H_2O + CN^- \rightarrow CNO^- + 2H^+$ } ⇒ atomic balance
Add e^- to balance the charge and cancel the e^-s by multiplying with a suitable integer.
$3e^- + 4H^+ + MnO_4^- \rightarrow MnO_2 + 2H_2O$
$H_2O + CN^- \rightarrow CNO^- + 2H^+ + 2e^-$ } × 2
$6e^- + 8H^+ + 2MnO_4^- \rightarrow 2MnO_2 + 4H_2O$ } × 3
$3H_2O + 3CH^- \rightarrow 3CNO^- + 6H^+ + 6e^-$
$2H^+ + 2MnO_4^- + 3CN^- \rightarrow 2MnO_2 + 3CNO^- + H_2O$
Since it is in basic solution, add OH^- to remove H^+
$\underbrace{2OH^- + 2H^+}_{2H_2O} + 2MnO_4^- + 3CN^- \rightarrow 2MnO_2 + 3CNO^- + H_2O + 2OH^-$
$2H_2O + 2MnO_4^- + 3CN^- \rightarrow 2MnO_2 + 3CNO^- + 2OH^-$ (Balanced)
⇒ The coefficient of $MnO_2 = 2$

19. B: $1\text{Å} = 10^{-8}$cm
1atom Cr = $2.4\text{Å} = 2.4 \times 10^{-8}$cm
x = 1cm
$$x = \frac{1 \times 1}{2.4 \times 10^{-8}} = 4.16 \times 10^7 \approx 4.2 \times 10^7 Cr \text{ atoms}$$

20. C: 5%(m/m) ⇒ 5g of solute in 100g of solution
5g NaOCl → 100g Berekina

x ← 2500g Berekina
$$x = \frac{5 \times 2500}{100} = 125g$$

21. A: % dissociation $= \frac{2.7 \times 10^{-2} M}{1M} \times 100$
$= 2.7 \times 10^{-2} \times 10^{-2} = 2.7\%$

22. B: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + 890kJ$
16g/mol CH_4 = 16g → -890kJ
10g → x
$$\Rightarrow x = \frac{10 \times -890}{16} = -556.25kJ$$

23. A: 1 mole N at STP = 22.4L
$28gN_2 \rightarrow 22.4L$
x ← 1L
$$\Rightarrow x = \frac{28 \times 1}{22.4} = 1.25\dot{N}_2$$
1L of the gas = 1.25g $\Rightarrow x = \frac{22.4 \times 1.25}{1} = 28g$
22.4L(=1mol) = x
(Emp.Formula mass)$_n$= molecular mass
$$n = \frac{mol.mass}{Emp.formula\ mass} = \frac{mol.mass}{mass\ of\ CH_2} = \frac{28}{14} = 2$$
$(CH_2)_n = (CH_2)_2 = C_2H_4$

24. A: Robert Millikan made an experiment, involving charged oil drops which allowed him to determine the charge on the e^-.

25. C: N = M×Z (Z = n^o of H^+ ions)
N = 2.3 × 2($2H^+$ in H_2SO_4)
= 4.6N

26. C: $2N_2O_5 \rightleftharpoons 4NO_2 + O_2$
$$\text{rate} = \frac{-\frac{1}{2}\Delta[N_2O_5]}{\Delta t} = \frac{\frac{1}{4}\Delta[NO_2]}{\Delta t}$$
Interms of NO_2, rate $= \frac{\frac{1}{4}\Delta[NO_2]}{\Delta t}$

27. B: $CaCO_2(s) \rightleftharpoons CaO(s) + CO_2(g)$

Telegram: https://t.me/ExtremeSeries

$Kp = Kc(RT)^{\Delta n}$ Δn = The sum of the coefficients of gaseous products – the sum of the coefficients of gaseous reactants.

$\Rightarrow$ only CO_2 is gas

$\Delta n = 1 - 0 = 1$

$Kp = Kc(RT)^1 = Kc(RT)$

28. C: $A_2(g) + 2B(g) \rightleftharpoons C(g)$ + heat

$\Rightarrow$ The forward reaction is exothermic. The yield of the product C can be increased by

i) decreasing the temperature because increasing the temperature favors the reverse reaction since it is endothermic.

ii) increasing the pressure because increasing the pressure favors in the direction with smaller n^o of gaseous molecules.

29. B: Glucose is the most important monosaccharide. It is the normal constituent of blood in animals.

30. A: the anions of polyprotic acids are amphoteric (Amphioprotic)

$HCO_3^- + H_2O \rightleftharpoons H_2CO_3 + OH^-$ (HCO_3^- acts as a base)

$HCO_3^- + H_2O \rightleftharpoons CO_3^{2-} + H_3O^+$ (HCO_3^- acts as an acid)

31. B: Acids with relatively higher Ka values are stronger

$Ka_{HCN} < KaCH_3OOH < KaC_6H_5COOH < KaHCOOH$

$\Rightarrow$ HCOOH has the highest Ka value and is the strongest aicd

32. B: $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$

From the balanced chemical equation,

1 milliole of H_2SO_4 reacts with 2 millimoles of NaOH millimoles of

NaOH = 10mL × 1M = 10 millimoles

1millimole $H_2SO_4 \rightarrow$ 2 millimoles NaOH

x $\leftarrow$ 10 millimoles NaOH

$$x = \frac{1 \times 10}{2} = 5 \quad \text{millimoles } H_2SO_4$$

$M_{H_2SO_4} V_{H_2SO_4}$ = 5 millimoles H_2SO_4

$$V_{H_2SO_4} = \frac{5mmol}{M_{H_2SO_4}} = \frac{5mmol}{\frac{1mmol}{mL}} \quad \left(1M = 1\frac{mol}{L} = 1\frac{mmol}{mL}\right)$$

$$= 5mL$$

33. B: since the gas is expanding, it does work on the surroundings, and work done is negative (W < 0)

$W = -P\Delta V = -2atm \times (15 - 5)L$

$= -2atm \times 10L$

$= -20L\ atm$

$1L\ atm = 101.3J$

$20L\ atm = x$ $\quad \Rightarrow x = \frac{20L\ atm \times 101.3J}{1Latm} = -2026J$

$$= \frac{-2026J}{1000\ J/kJ} = -2.03kJ$$

34. B: Elements with diagonal relationship in the periodic Table exhibit similar properties and are usually metalloids. Their compounds (their oxides and hydroxides) exhibit amphoteric properties. Eg Be and Al

N.B: B, Si, As, Ge are Te are elements with diagonal relationships and are typical metalloids.

35. B: Metal ions are involved in complexes with a type of macrocylic ligand called a porphyrin. Porphyrins are derivatives of porphin molecule. The heme group consists of Fe and the chlorophyll group consists of Mg.

36. B: 1mol Cu = 63.5g

1mol Cu $\rightarrow 6.02 \times 10^{23}$ Cu atoms

$x \leftarrow 5.00 \times 10^{20}$ Cu atoms

$$x = \frac{1 \times 5.00 \times 10^{20}}{6.02 \times 10^{23}} = 0.83 \times 10^{-3} = 8.3 \times 10^{-4} moles\ Cu$$

mass or Cu = 8.3×10^{-4}mol × 63.5g/mol

$= 527 \times 10^{-4}g = 5.27 \times 10^{-2}g$ Cu

37. A: $2LiOH(S) + CO_2(g) \rightarrow Li_2CO_3(s) + H_2O(l)$

$48g \quad 44g$

$1kg = 1000g \quad x$ $\quad \Rightarrow x = \frac{1000 \times 44}{48} = \frac{44000}{48} \cong 917g$

38. C: $2NH_3(g) + 3CuO(s) \rightarrow N_2(g) + 3Cu(s) + 3H_2O(g)$

$$n_{NH_3} = \frac{36}{17} = 2.118mol$$

$$n_{CuO} = \frac{180}{79.5} = 2.264mol$$

$2molNH_3 \rightarrow 3mol$ CuO (balanced chemical equation)

$2.118mol\ NH_3 \rightarrow x$

$$\Rightarrow x = \frac{2.118 \times 3}{2} = 3.177mol\ CuO$$

Telegram: https://t.me/ExtremeSeries

$\Rightarrow$ 3.117mol CuO is needed but we have only
2.264mol CuO $\Rightarrow$ CuO is the limiting reactant. It determines the amount of product formed.

3mol CuO $\rightarrow$ 28gN_2

2.264mol CuO $\rightarrow$ x

$$\Rightarrow x = \frac{2.264 \times 28}{3} \cong 21.13 g N_3$$

39. D: Negeterians cannot have zero cholesterol in their bodies because vegetable oils can be converted in to fats /cholesterol by catalytic hydrogenation in their bodies.

40. C: Thee reaction is 2nd order because in 10hours it is reduced from 0.2M to 0.1M. In 20 hours (twice as long) it is reduced from 0.1M to 0.05M.

$[sugar]_0 = 0.2M$. $t_{½} = 10hrs = 36000s$ (1hr = 36000s)

$$t_{½} = \frac{1}{k[sugar]_0}$$

$$k = \frac{1}{t_{\frac{1}{2}} \times [sugar]_0} = \frac{1}{36000s \times 0.2M} = \frac{1}{7200Ms}$$

$$= 0.0001388 M^{-1}s^{-1}$$

$$= 1.38 \times 10^{-4} M^{-1}$$

41. B: $m = \frac{ItE}{F}$ I = current in A

t = time in s

E = Equivalent mass in g

F = 96500c = 96500As

$$m = \frac{15A \times 259200s \times 17.33g}{96500As}$$

$$E = \frac{M}{z}\begin{pmatrix} m = molar\ mass \\ z = char\ ge \end{pmatrix}$$

= 698.23g

$\cong$ 700g

3days = 3×24hrs

=72hrs = 259200s

$$E_{Cr^{3+}} = \frac{M}{z} = \frac{52}{3} = 17.33g$$

42. A: 1mol H_2 = 2gH_2 $\rightarrow$ 12.04 × 10^{23}H atoms

4gH_2 $\rightarrow$ x

$$x = \frac{4 \times 12.04 \times 10^{23}}{2} = 24.08 \times 10^{23} H\ atoms$$

1mol Ca = 40g Ca $\rightarrow$ 6.02 × 10^{23} Ca atoms

1mol I_2 = 254g I_2 $\rightarrow$ 12.04 × 10^{23} I atoms

127g I_2 $\rightarrow$ x

$$\Rightarrow x = \frac{127 g I_2 \times 12.04 \times 10^{23} I\ atoms}{254 g I_2}$$

= 6.02 × 10^{23} I atoms

1mol O_3 = 48gO_3 $\rightarrow$ 18.06 × 10^{23} O atoms

24g O_3 $\rightarrow$ x

$$\Rightarrow x = \frac{24 \times 18.06 \times 10^{23}}{48} = 9.03 \times 10^{23} O\ atoms$$

$\therefore$ 4g of H_2 has the greatest n° of atoms

43. C: In each orbit the angular momentum of the e^- is quantized and has the value given by mvr = $n\frac{h}{2\pi}$. The quantum condition is that angular momentum of the e^- is an integral multiple of $\frac{h}{2\pi}$. It differs in the value of n

For n = 3, mvr = $\frac{3h}{2\pi}$

$\Rightarrow$ The no of spectral lines is directly proportional to the value of n.

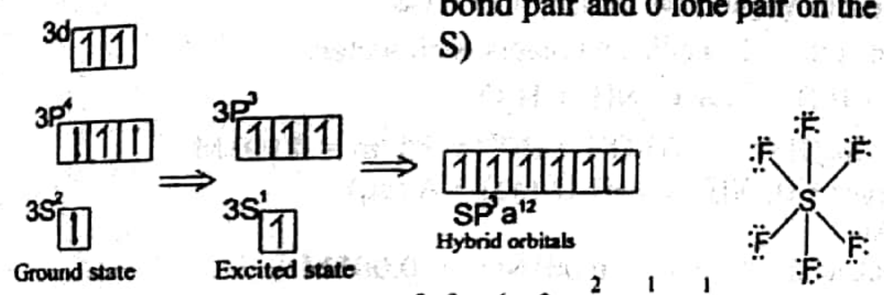
44. D: Z = 20 $\Rightarrow$ Ca: 2, 8, 8, 2 (4 shells)

period = n° of shells = 4 $\Rightarrow$ period 4

group = n° of valence e^-s = 2 $\Rightarrow$ Group IIA

45. D:

* SF_6 has octahedral molecular geometry (6 bond pair and 0 lone pair on the central atom S)



S(z=16): $1s^2 2s^2 2p^6 3s^2 3p_x^2 3p_y^1 3p_z^1$ (Ground state)

Telegram: https://t.me/ExtremeSeries

$1s^2 2s^2 2p^6 3s^1 3p_x^1 3p_y^1 3p_z^1 3d_{xy}^1 3d_{xz}^1$

Sp^3d^2 hybridization

$\Rightarrow$ 6 sp^3d^2 hybrid orbitals are formed that are equivalent

46. A: $O_3 \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} O_2 + O$ ... fast

$O + O_3 \xrightarrow{k_2} 2O_2$ ... slow

Take the fast equilibrium. At equilibrium, the rate of the forward and reverse reaction is equal.

$$K_1[O_3] = K_{-1}[O_2][O]$$

$$[O] = \frac{k_1[O_3]}{k_{-1}[O_2]}$$

The slow step deterimues the rate law for the overall reaction.

Rate = $K_2[O][O_3]$

Substituting [O] by $\frac{k_1[O_3]}{k_{-1}[O_2]}$

$$\text{Rate} = \frac{k_2 k_1[O_3][O_3]}{k_{-1}[O_2]}$$

$$= \frac{k[O_3]^2}{[O_2]}$$

$$= k[O_3]^2[O_2]^{-1}$$

47. D: Salts undergo hydrolysis (Reaction with water).
NH_4Cl is a salt of strong acid, HCl and weak base, NH_4OH

$NH_4Cl(aq) \rightarrow NH_4^+(aq) + Cl^-(aq)$

NH_4^+ is the conjugate acid of the weak base

NH_3 acid is strongly acidic and reacts with water.

$NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$

48. B: $pH = -\log[H_3O^+] \Rightarrow [H_3O^+] = 10^{-pH} = 10^{-3}m = 0.001M$

$HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$

0.1M			
(weak acid)		0.001M	0.001M

$$K_a = \frac{[H_3O^+][A^-]}{[HA]} = \frac{0.001 \times 0.001}{0.1} = \frac{10^{-3} \times 10^{-3}}{10^{-1}}$$

$$= 10^{-6+1} = 10^{-5}$$

49. D: the half – reaction with more positive electrode potential is placed at the cathode.

$\Rightarrow Ni/Ni^{2+}$ - cathode, Fe/Fe^{2+} - anode

$E^0_{cell} = E^0_{cathode} - E^0_{anode}$

$= 0.49v - (-0.88v)$

$= (0.49+0.88)v = +1.37v$

50. D: HF has unusually high bond energy compared with the other hydrogen halides (Hx) due to the strong intermolecular forces (Hydrogen bonding) between HF molecules. Thus, it exhibits the highest enthalpy of vaporization and boiling point among the hydrogen halides. It is least acidic due to the high bond dissociation energy which hinders dissociation

(HF < HCl < HBr < HI in acidity).

Telegram: https://t.me/ExtremeSeries

Extreme Series
t.me/ExtremeSeries

Booklet Code: 28

1. $\Delta s^0 = \left[2s^0_{so_3}\right] - \left[2s^0_{so_2} + s^0_{o_2}\right]$
   $= (2 \times 256.6) - (2 \times 248.1 + 205)$
   $= 513.2 - 701.2$
   $= -188.0 J/k$
2. C. $C_3H_7NO_2 = 3 \times 12 + 7 \times 1 + 1 \times 14 + 32 = 89$
   $1mol\ C_3H_7NO_2 \longrightarrow 89g$
   $x \longleftarrow 159g$
   $$x = \frac{1 \times 159}{89} = 1.78mol$$
3. C. The strength of metallic bonding is variable and depends on the no of free e⁻s available.
4. B. Synthetic fibers are manufactured using chemicals. Eg. Nylon 66 is made from hexamethylene diammine and adipic acid. Acrylics are produced from acrylonitrile.
5. C. Amphioprotic (amphoteric) species can donate and accept a proton (H^+), that is, they can act an acid and a base.
6. A. In general for polyprotic acids (acids containing more than one H^+) much of the H_3^{0+} is the first step (Ka_1)

   $H_3PO_4(aq) + H_2O(l) \quad H_3^{0+}(aq) + H_2PO_4^-(aq)$; $Ka_1 = 7.1 \times 10^{-3}$

   $H_2PO_4^-(aq) + H_2O(l) \quad H_3^{0+}(aq) + H_2PO_4^{2-}(aq)$; $Ka_2 = 6.3 \times 10^{-8}$

   $HPO_4^{2-}(aq) + H_2O(l) \quad H_3^{0+}(aq) + PO_4^{3-}(aq)$; $Ka_3 = 4.2 \times 10^{-13}$

   $\Rightarrow Ka_1 > Ka_2 > Ka_3$
7. A. Alkali metals are the most reactive metals since they have only one valence e- and hence can easily lose this e-.
8. C. Ionic compounds donot conduct electricity in the solid phase since the cations and anions are held by strong electrostatic force of attraction and are not set free to move.
9. C. Decreasing the temperature of the system favors the exothermic r x n (more NOCl is formed).
10. C. In the process of roasting, sulfide minerals are converted to the respective oxide before reducing them to the metal.

11. D. 1st law of thermodynamics states that the total amount of energy in the universe is constant.
    $\Rightarrow$ Energy is neither created nor destroyed but can be transferred to another form.
    Moreover, $\Delta E = q + w$ (ΔE = change in internal energy
    q = quantity of heat gained by the system
    w = work done)
12. D. $1mol\ CaCl_2 = 2molCl^-$
    $= 2 \times 6.02 \times 10^{23}$
    $= 1.20 \times 10^{24} Cl^-$ ions
13. B. Types of solutions
    1. solid in liquid eg. NaCl in water
    2. Gas in Gas eg. Atmospheric air
    3. Gas in Liquid eg. CO_2 in water
    4. Solid in Solid eg. Brass (Cu and Zn)
14. C. $N = Mz$ (M = molarity)
    $= 1 \times 2$ z = total positive valence)
    $= 2N$
15. C. Proteins are condensation polymers of α - aminoacids.
16. D. H_2O, NH_3 and HCO_3^- are amphioprotic. They can donate and accept a proton, i.e., they can act as an acid and a base.
    Eg. $H_2O + HCO_3^- \quad H_3^{0+} + CO_3^{2-}$
    base
    $H_2O + NH_3 \quad OH^- + NH_4^+$
17. C. $CaH_2 + 2H_2O \rightarrow Ca(OH)_2 + 2H_2$
    $CaO + H_2O \rightarrow Ca(OH)_2$
    $N^{3-}(aq) + H_2O(l) \rightarrow NH_3(g) + 3OH^-(aq)$
    Non-metal oxides give acids in water.
    Eg. $SO_3 + H_2O \rightarrow H_2SO_4$
18. C. HB must be a weak acid.
    $HB(aq) + H_2O(l) \quad H_3O^+(aq) + B^-(aq)$
    N.B. HB must be a weak acid and [HB] > [H^+] because weak acids only slightly dissociate.
19. B. $O_2^+ = \sigma_{1s}^2 \sigma_{1s^*}^2 \sigma_{2s}^2 \sigma_{2s^*}^2 \sigma_{2p_x}^2 \pi_{2p_y}^2 \pi_{2p_z}^2 \pi_{2p_y}^{*1}$
    (15e⁻s)
    Bond order = ½($n_b - n_a$), n_b = no of bonding e⁻s
    = ½(10 − 5) n_a = no of antibonding e⁻s

Telegram: https://t.me/ExtremeSeries

$= ½ \times 5$

$= 2.5$

20. C. Antibonding molecular orbital (AMO)
    - MO_s resulting from subtraction of atomic orbitals (AOs)
    - Electron density is concentrated away from the region between the two nucleic
    - The two nucleic repel each other
    - Energy of the antibonding orbitals is greater than the two isolated AOs.
    - Denoted by σ^* or π^*

21. C. $M = \dfrac{no\ of\ moles}{volume\ in\ Liters}$

    500mL = 0.5L

    $M = \dfrac{0.01}{0.5} = 0.02M$

22. A. The rate of a chemical reaction is affected by
    i) concentration of the reactant ⇒ as concentration increases, the rate also increases.
    ii) Surface area of solid reactants ⇒ as surface area increases, the rate also increases.
    iii) Temperature ⇒ as temperature increases, rate also increases.
    iv) Pressure of gaseous reactants ⇒ as pressure increases, the rate also increases.

23. C. $\text{rate} = \dfrac{1}{4}\dfrac{\Delta[NH_3]}{\Delta t} = \dfrac{-1}{7}\dfrac{\Delta[O_2]}{\Delta t} = \dfrac{1}{4}\dfrac{\Delta[NO_2]}{\Delta t} = \dfrac{1}{6}\dfrac{\Delta[H_2O]}{\Delta t}$

    The minus sign in the reactants shows that they are consumed in the course of the rxn.

24. B. The order of the rxn that describes an elementary step is equal to the stoichiometric coefficient for the reactant in the equation representing that step.

    $A + B \longrightarrow P$; rate = $k[A][B]$

    $2A \longrightarrow P$; rate = $k[A]^2$

    $A + 2B \rightarrow P$; rate $k[A][B]^2$

    $A + B + C \rightarrow P$; rate = $k[A][B][C]$

25. C. x = factor, a = order of rxn ⇒ The rxn rate increases by a factor of x^a

    Doubling ⇒ a factor of 2 ⇒ x = 2

$Order = 2 + 1 = 3 \Rightarrow a = 3$

$x^a = 2^3 = 8$

26. A. The value of equilibrium constant varies only with changes in temperature. It is independent of initial concentrations of reactants and products.

27. D. Reverse rxn (1) and multiply by 2 to obtain rxn (2)

$$\left(SO_3(g) = SO_2(g) + \frac{1}{2}O_2(g) = \frac{1}{k}\right) \times 2$$

$$2SO_3(g) = 2SO_2(g) + O_2(g) = \frac{1}{k^2}$$

28. A. Nucleic acids are proteins, and triglycerides are fats and oils. Aldoses, polyhydroxylectones, polyhydroxyaldehydes, disaccharides and aldoses are carbohydrates.

29. D. The Cl atom withdraws one e^- from CH_3 (methylradical) and CH_3 can have a vacant orbital so that it can accommodate the e^- pair coming from Hs^-

⇒ Hs^- is e^- pair donor and is a Lewis base

30. $H_3PO_4 + H_2O \quad H_2PO_4^- + H_3O^+$
    acid base base acid

    $H_2PO_4^- + H_2O \quad HPO_4^{2-} + H_3O^+$
    acid base base acid

    $HPO_4^{2-} + H_2O \quad PO_4^{3-} + H_3O^+$
    acid base base acid

    $\Rightarrow H_3PO_4 / H_2PO_4^-$
    $H_2PO_4^- / HPO_4^{2-}$
    HPO_4^{2-} / PO_4^{3-}
    } are conjugate acid-base pairs.

31. B. The higher the value of pH, the higher the concentration of OH^-. The lower the value of pH, the higher the concentration of H^+.

32. A. More positive emf is cathode.

$$E^0_{cell} = E^0_{cathode} - E^0_{anode}$$
$$= 0.34 - (-0.76)$$
$$= +1.1v$$

33. A. From the combined gas law,

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

Telegram.https://t.me/ExtremeSeries

$V_1 = V_2$ because $P_1 = P_2$ and T_1 and T_2

$\Rightarrow \Delta P = 0, \Delta V = 0$

$\Delta H = \Delta U + P\Delta V + V\Delta P \Rightarrow \Delta H = \Delta U$

34. C. $\Delta H < 0, \Delta S > 0 \Rightarrow$ spontaneous at all temperatures.
$\Delta H < 0, \Delta S < 0 \Rightarrow$ spontaneous at lower temp. and non – spontaneous at higher temperatures.
$\Delta H > 0, \Delta S < 0 \Rightarrow$ non-spontaneous at all temperatures.
$\Delta H > 0, \Delta S > 0 \Rightarrow$ spontaneous at higher temperatures and non-spontaneous at lower temperatures.

35. C: Silicates are not commonly used as sources of metals because of their complex structure and they are very insoluble in water, and hence difficult to concentrate and reduce.

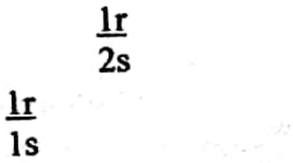
36. B: Ag has the highest thermal and electrical conductivity of all metals. It ranks 2nd in ductility and malleability to gold.

37. D: HNO_3 is used
- as an acid
- in the manufacturing of NH_4NO_3 (ammoniating agent)
- as an oxidizing agent

38. D: An oxidation – reduction (redox) rx_n involves change in oxidation no. Double displacement and acid-base (neutralization) rxns donot involve change in oxidation no and therefore are NOT redox rxns.
In the rxn $2H_2SO_4 + Cu \longrightarrow CuSO_4 + 2H_2O + SO_2$
$CuO \longrightarrow Cu^{2+}$ (Cu is oxidized)
$S^{+6} \longrightarrow S^{+4}$ (S is reduced)

39. C: n = 1, 2, 3, … defines main energy level
$l = 0, 1, 2, 3, \ldots, n-1$
$m_l = -l, 0, +l$
$m_s = \pm \frac{1}{2}$
$n = 3 \Rightarrow l = 0, 1, 2$ (l cannot be 3)

40. A. For any set of orbital that are equal in energy (degenerate orbitals), the configuration with the maximum no of parallel spins results in the lowest e = e⁻ repulsion
The correct figure is



41. A: BCl_3, $SiCl_4$ and PCl_3 all are coralenthy bonded molecules since they are formed by sharing of e⁻s.

42. B: $M = \dfrac{\dfrac{actual\ mass}{mol.mass}}{V(L)}$

$\Rightarrow$ act. Mass = M × V(L) × mol.mass
$= 0.25 \times 1.5 \times 74$
$= 2.78g$

43. C: The equilibrium constant expression equals the concentration of the products divided by the concentration of the reactants each raised to the balancing coefficients.

$$K_C = \frac{[I_2][Cl_2]}{[ICl]^2}$$

44. D: $HSO_4^- + NH_3 \quad NH_4^+ + SO_4^{2-}$
acid base acid base
$\Rightarrow HSO_4^-$ and SO_4^{2-} are conjugate acid – base pairs.

45. B: Cu^{2+} is reduced to Cu
$\Rightarrow E^0_{Cu^{2+}/Cu}$ is the cathode

$E^0_{cell} = E^0_{cathode} - E^0_{anode}$

$E^0_{cell} = E^0_{Cu^{2+}/Cu} - E^0_{M^{2+}/M}$

$E^0_{M^{2+}/M} = E^0_{Cu^{2+}/Cu} - E^0_{cell}$
$= 0.34 - 0.75$
$= -0.41v$

46. A: The Lattice energy of a crystal increases with decreasing ionic radii and increasing ionic charge.
$F^- < Cl^- < Br^- < I^-$ in ionic radii. Crystals with high lattice energies are generally harder and higher melting points.

47. B: CFC's such as $CFCl_3$ and CF_2Cl_2 decompose to form Cl atoms and chlorine oxides like ClO when they absokb sunlight (uV energy)
Have unpaired e⁻s and are unusually reactive. In the atmosphere, they react with ozone (O_3) or with the O atoms that are needed to form O_3 and finally deplete the ozone layer.

48. D: Pt is inert metal (unreactive element) and can exist in the free state.

49. B: Bonds are broken in H_2 (H – H) and N_2 (N ≡ N) and are formed in H – C and C ≡ N.
$\frac{1}{2}H_2(g) + C(graphite) + \frac{1}{2}N_2(g) \longrightarrow HCN(g)$

Telegram: https://t.me/ExtremeSeries

50. C: Sc is a group IIIB element (transition metal) and can form an ionic compound with Cl. CCl_4, NO_2 and ClO_2 are all covalent compounds.

51. B: H_2O^{-2}, $H_2O_2^{-1}$, O_2^{0}, $O^{+2}F_2$

52. D. The ion is $Ca^{2+} \Rightarrow 2, 8, 8$

$Ar \Rightarrow 2, 8, 8$

Therefore, Ca^{2+} is iso electronic with Ar.

53. B. $density = \dfrac{mass(m)}{volume(v)} \Rightarrow m = \rho v$

Mol.mass of $C_{10}H_8 = 128 = 1.6g/mL \times 500mL$

$= 800g = 0.8kg$

$$m = \frac{\dfrac{act.mass}{mol.mass}}{kg\ of\ solvent} = \frac{\dfrac{51.2}{128}}{0.8} = 0.50m$$

54. D. Increasing the temperature of a reaction increases the average kinetic energy of the reactant molecules thereby increasing the collision between them which in turn increase the rate of the rxn.

55. A. In the titration of weak acid with a strong base, the titration starts at a higher pH and the equivalence point occurs on the basic side.

Rapid pH change occurs from about pH = 7 to pH =11

56. B.

		$HX + H_2O$		$H_3O^+ + x^-$
Initial	0.1m	O	O	
Change	-x	+x	+x	
At equilibrium	0.1-x		x	x

0.059% ionized $\Rightarrow x = \dfrac{0.059}{100} \times 0.1 = 59 \times 10^{-5} = [H_3O^+] = [x^-]$

$$k_a = \frac{x^2}{0.1 - x}$$

Assume x is very small

$$k_a = \frac{x^2}{0.1} = \frac{(5.9 \times 10^{-5})^2}{0.1} = \frac{34.81 \times 10^{-10}}{0.1} = 3.5 \times 10^{-8}$$

57. C. More positive $E^0_{red} \Rightarrow Cathode \Rightarrow Au$

$$E^0_{cell} = E^0_{cathode} - E^0_{anode}$$
$$= E^0_{Au^{3+}/Au} - E^0_{Fe^{2+}/Fe}$$
$$= 1.50 - (-0.44)$$
$$= 1.94v$$

58. B. $(C + O_2 \longrightarrow CO_2) \times 2$ (multiply the equation by 2)

$2C + 2O_2 \rightarrow 2CO_2$; $\Delta H^0_f = -787 kJmol^{-1}$

$(H_2 + \frac{1}{2}O_2 \longrightarrow H_2O) \times 3$ (multiply the equation by 3)

$3H_2 + \dfrac{3}{2}O_2 \rightarrow 3H_2O$; $\Delta H^0_f = -857.4 kJmol^{-1}$

$C_2H_6 \longrightarrow 2C + 3H_2$ (Reverse the rxn); $\Delta H^0_f = +84.7 kJ/mol$

Add the equation and the ΔH^0_f's to obtain

$$C_2H_6(g) + \frac{7}{2}O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$$

$\Rightarrow 84.7 + (-787) + (-857.4) = -1560 kJmol^{-1}$

59. C. $64_g SO_2 = 1mol\ SO_2 = 1mol\ s + 2mol\ O$

$= 6.02 \times 10^{23} + 2 \times 6.02 \times 10^{23}$ atoms

$= 3 \times 6.02 \times 10^{23}$ atoms

60. C. Atomic radius decrease across a period because of the increased effective nuclear charge (Z_{eff}).

Mg > Al > Cl in atomic radius.

61. D. P has the smallest atomic size implying the greatest effective nuclear charge.

$\Rightarrow$ k > Ca > Br > p in atomic size

62. HCN has a linear shape

H – C ≡ N:

$\Rightarrow$ There is no lone pair of e$^-$s on the central atom C

$\Rightarrow$ It cannot be bent if there is no lone pair.

63. C. C = Z + 1 = 9 + 1 = 10 $\Rightarrow$ Ne(noble gas)

Noble gases do not react with elements except under special conditions.

(Eg. XeF_4).

B = Z = 9(F)

D = z + 2 = 9 + 2 = 11 (Na)

BD $\Rightarrow$ NaF (IONIC)

Telegram: https://t.me/ExtremeSeries

64. C. Since the energy of product (z) is less than the energy of the product (x), $\Delta H < 0$ which means the forward rxn is exothermic.
Y is the transition state, $\Delta H^0_{rxn} = b - a$,
b = activation energy

65. B. Rate = $k[A]^x[B]^y$
To find x, vary [A] keeping [B] constant.
Take experiments 1 and 2

$$\frac{(rate)_0 \ for \ \exp t-1}{(rate)_0 \ for \ \exp t-2} = \left(\frac{[A]_0 \ \exp t-1}{[A]_0 \ \exp t-2}\right)^x$$

$$\frac{4.0\times10^{-5}}{1.6\times10^{-4}} = \left(\frac{0.1}{0.2}\right)^x$$

$$0.25 = \left(\frac{1}{2}\right)^x$$

$$\frac{1}{4} = \left(\frac{1}{2}\right)^x$$

$$\Rightarrow x = 2$$

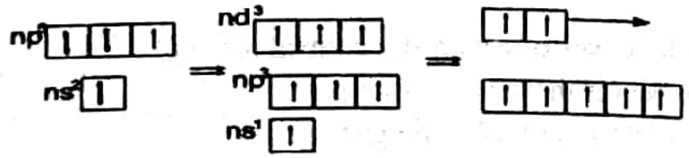
Similarly, to find y, take experiments 1 and 3.

$$\frac{4.0\times10^{-5}}{8.0\times10^{-5}} = \left(\frac{0.1}{0.2}\right)^y$$

$$\frac{1}{2} = \left(\frac{1}{2}\right)^y \Rightarrow y = 1$$

Therefore, Rate = $k[A]^2[B]$

66. B. There is a base-weakening resonance in arylamines. Aryl groups decrease the basicity of nitrogen



Aqueous solutions of amine salts are just on the acid side of neutrality due to hydrolysis.

67. B. $pH = p^{ka} + \log\frac{[base]}{[acid]}$ (Henderson-Hasselbach equation)

$$pH = p^{ka} + \log\frac{[CH_3COO^-]}{[CH_3COOH]}$$

$$pH - p^{ka} = \log\frac{[CH_3COO^-]}{[CH_3COOH]}$$

$$6 - 5 = \log\frac{[CH_3COO^-]}{[CH_3COOH]}$$

$$1 = \log\frac{[CH_3COO^-]}{[CH_3COOH]}$$

$$\Rightarrow \frac{[CH_3COO^-]}{[CH_3COOH]} = 10^1 = \frac{10}{1} \Rightarrow 10:1$$

68. A. Reverse rxns (i), (ii) and (iii)
$2NOCl(g) + Cl_2(g) \longrightarrow 2NOCl(g)\ \Delta H^0 = -75kJ$
$2NO_2(g) \longrightarrow 2NO(g) + O_2(g)\ \Delta H^0 = +113.05kJ$
$N_2O_4(g) \longrightarrow 2NO_2(g)\ \Delta H^0 = +58.03kJ$
Add the equation and ΔH^0's to obtain
$N_2O_4(g) + Cl_2(g) \longrightarrow 2NOCl(g) + O_2(g)$
$\Delta H^0 = -75 + 113.05 + 58.03 = +96.08kJ$

69. A. Using oxidation no change method, 1st balance the equation.
$2KBrO_3 + 10kBr + 2HNO_3 \longrightarrow 12KNO_3 + 5Br_2 + 6H_2O^{-2}$

$$2Br^{+5} \xrightarrow{+10e^-} Br_2 \times 1$$

$$2Br^{-1} \xrightarrow{-2e} Br_2 \times 5$$

$$2Br^{+5} + 10e^- \rightarrow Br_2$$

$$\underline{10Br^{-1} - 10e^- \rightarrow 5Br_2}$$

2mol $KBrO_3 \longrightarrow$ 6mol Br_2 (from the balanced eqn.)
x $\longleftarrow$ 0.07mol Br_2

$$x = \frac{2\times0.07}{6} = 0.0233$$

70. C. The slowest step is the rate determining step
Rate = $k[A_2][A]$
Taking the fast reversible equilibrium

$$A + A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} A_2$$

$k_1[A]2 = k_{-1}[A_2]$

telegram: https://t.me/ExtremeSeries

$$\Rightarrow [A_2] = \frac{k_1[A]^2}{k_{-1}}$$

$$\text{rate} = k\left[\frac{k_1[A]^2}{k_{-1}}\right][A]$$

$$= \frac{kk_1}{k_{-1}}[A]^3$$

$$\text{rate} = K[A]^3$$

71. A. $Ba(OH)_2 \longrightarrow Ba^{2+} + 2OH^-$

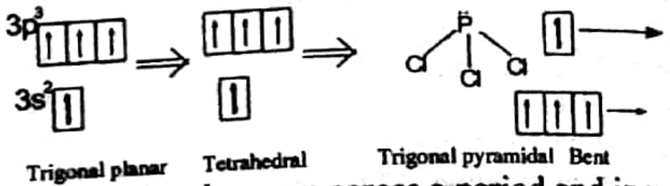
0.05m $\qquad 2 \times 0.05 = 0.1\text{m}$

$pOH = -\log[OH^-] = -\log 0.1 = -\log 10^{-1} = 1$

$pH = 14 - pOH = 14 - 1 = 13$

$[H_3O^+] = {}_{10}^{-}pH = 1.0 \times 10^{-13}\text{M}$

72. A



Trigonal planar Tetrahedral Trigonal pyramidal Bent

73. D. Atomic size deceases across a period and increases down the group.

$\Rightarrow B < Be < Li < Na$

74. B. $\text{rate} = \frac{-1}{4}\frac{\Delta[NH_3]}{\Delta t} = \frac{1}{6}\frac{\Delta[H_2O]}{\Delta t}$

$$\frac{\Delta[NH_3]}{\Delta t} = \frac{4}{6}\frac{\Delta[H_2O]}{\Delta t} \text{ (multiplying both sides by 4)}$$

$$\frac{\Delta[NH_3]}{\Delta t} = 0.67\frac{\Delta[H_2O]}{\Delta t}$$

$$= 0.67 \times 0.025 = 0.017\text{mol min}^{-1}$$

75. A. 75%A converted $\Rightarrow$ 25% A remains.

$$\log\frac{[A]_0}{[A]_t} = \frac{kt}{2.303} \text{ (1st order rxn)}$$

$$\log\frac{[A]_0}{\frac{25}{100}[A]_0} = \frac{kt}{2.303}$$

$$\log\frac{1}{\frac{1}{4}} = \frac{kt}{2.303}$$

$$\log 4 = \frac{kt}{2.303}$$

$$0.602 = \frac{60k}{2.303}$$

$$60k = 0.602 \times 2.303$$

$$k = 0.602 \times 2.303 = 0.023$$

$$t_{\frac{1}{2}} = \frac{0.693}{k} = \frac{0.693}{0.023} \approx 30\,\text{min}$$

76. B. $k_c = \frac{[NO_2]^2}{[NO]^2[O_2]} = \frac{(0.18)^2}{(0.52)^2 \times (0.24)} = \frac{0.0324}{0.0649} = 0.50$

77. C. $M_1V_1 = M_2V_2$

$0.105 \times 25 = M_2 \times 17.23$

$$\Rightarrow M_2 = \frac{0.105 \times 25}{17.23} = 0.1524M$$

78. A. $m = \frac{ItE}{F} = \frac{2I \times \frac{1}{2}tE}{F} = \frac{ItE}{F}$

79. B. Anode rxn: $Pb(s) \longrightarrow Pb^{2+}(aq) + 2e^-(\text{oxd.})$

Cathode rxn: $Cl_2(g) + 2e^- \longrightarrow 2Cl^-(aq)$ (Red).

Overall rxn: $Pb(s) + Cl_2(g) \longrightarrow Pb^{2+}(aq) + 2Cl^-(aq)$

80. B. Nitrogen fixing bacteria are found in the root nodules of leguminous plants like peas, beans and alfalfa. They convert

Telegram: https://t.me/ExtremeSeries

1. D
Pure gold is 24 carat (100% pure)
Then, 24K = 100%

$$18K = x$$
$$X = \frac{18\times100\%}{24} = 75\%$$

2. A and B
Colligative property of a solution is the one that depends on the number of particles dissolved in it, but not on the chemical identify of the particles. The term "colligatves" in latin means "Collection" .There are four properties in which we shall be particularly interested: They are:
- Lowering of Vapor pressure
- Osmotic pressure
- Boiling point elevation and
- Freezing point depression

3. D
When Δs is positive ($\Delta S > 0$) spontaneity is favored. Spontaneity of a chemical reaction is governed by both the energy change and the entropy change.

4. B
$[1s^2, 2s^2, 2p^6] 3s^2, 3p^1$, it is Aluminum (Z = 13)

5. A
Polar bond is a covalent bond in which the bonding electrons are attracted towards the more electronegative atom. Fluorine the most electronegative element and then the order of polarity will be

$$S \rightarrow F > \quad S \rightarrow Cl > S \rightarrow Br > S \rightarrow I$$

6. B

$$Rate = \frac{\Delta\, Concentration}{\Delta time}$$

Unit of rate : $\frac{mol\,L^{-1}}{s} = \frac{M}{s} = Ms^{-1}$ or mol L^{-1}

7. D
Dipole dipole force is the intermolecular force of attraction between polar molecules.
London or Dispersion force is the intermolecular force of attraction between non- polar molecules

8.B

$$\%\ NaOCl = \frac{\text{Mass of NaO Cl (x)} \times 100}{\text{mass of solution}}$$

$$5.2 = \frac{x}{2{,}500\ g} \times 100$$
$$5.2 = \frac{100\,x}{2{,}500\,g}$$
$$X = \frac{5.2\times2{,}500\,g}{100} = 130g$$

9.A
State function is any property of a thermodynamic system, which depends only on the initial and final states and not on the path taken between the states. E.g internal energy, work and heat are both path functions.

10.C

Matal	Oxidation state
Ti	+2, +3, +4
Cr	+2, +3, +6
Mn	+2,+3, +4, +6, +7
Cu	+1,+2

11. D
Naturally occurring polymers include proteins, nucleic acid, starch, cellulose (polysaccharides) and rubber (polyisoprene)

1. B
The balanced chemical equation is
$2\,Al(NO_3)_3 + 3Na_2S \longrightarrow Al_2S_3 + 6NaNO_3$
The coefficients are 2, 3, 1, 6

13. D
1 mol $(NH_4)_2HPO_4$ contains:
2mol . N − atoms
9 mol. H − atoms
1mol. P − atom
4mol O − atoms

14. C
500g sample contains = 0. 2 g $[F^-]$
10^6 g sample contains = X

$$X = \frac{0.2g\times10^6g}{500g} = 400g\ ppm$$

15. D
PH + POH = 14 , [OH-] = 10^{-PoH}
A. POH = 14- 3.21 = 10.79, $[OH^-] = 10^{-10.7g} = 1.6\times10^{-11}$
B. POH = 14-7.93 = 6.07, $[OH^-] = 10^{-6.07} = 8.5\times10^{-7}$

Telegram: https://t.me/ExtremeSeries

C. POH = 14- 9.82 = 4.18, [OH-] = $10^{-4.18}$ = 6.6× 10^{-5}

D. POH = 14 – 12.59 = 1.4, [OH^-] $10^{-1.41}$ = 0.039

16. A

Reduction means decreasing of oxidation number.

($C^{+6}r^{2}O_7^{-2} \longrightarrow Cr^{+3}$) (its oxidation number decrease from +6 to + 3)

17. C

Buffer solution is a solution which contains either a weak acid and its conjugate base or a weak base and its conjugate acid.

18. C

Salts obtained from salts of weak acids and strong bases such as $Na_2 CO_3$ (from NaOH and $H_2 CO_3$) is basic salt

19. A

If we multiply a thermo chemical equation with an enthalpy change of ΔH rxn through out by n, then the value of the enthalpy change for the resulting equation is

$\Delta H_{rxn} = n\Delta H_{rxn}$

$\Delta H_{rxn} = 1.2\times(-434KJ) = 520.8KJ$

20. B

$AlCl_3 \longrightarrow Al^{3+}+3Cl^-$

$Al^{3+} + 3e^- \longrightarrow Al$

$3F \longrightarrow 27g$

$X \longrightarrow 13.5g$

$X = \frac{3F\times13.5g}{27g} = 1.5F$

21. A

When the lead strong battery is in operation (on discharge) the following reactions occur at the electrodes.

Anode reactrons: $Pb(s) + SO_4^{2-}(aq) \longrightarrow PbSO_4(S) + 2e^-$

Cathode reactrons: $PbO_2(s) + 4H^+ + SO4^{2-} + 2e^- \longrightarrow PbSO_4 + 2H_2O$

22. D

The cyanide process is the most widely used system for the production of gold

$4Au + 8CN^- + O_2 + 2H_2O \longrightarrow 4 + 4OH^- \; 4[Au(CN)_2]^-$

The gold complex, $[Au(CN^-)_2]^-$ is recovered by the reaction with zinc

$Zn + 2[Au(CN)_2]^- \longrightarrow 2AU + [Zn(CN)_4]^{2-}$

23. A

A process that is endothermic (Δ H Positive) and that produces a more ordered system (Δs negative) does not proceed spontaneously (both $\Delta H and\ \Delta S$ contribute to a more positive value of ΔG)

24. A

In galvanic cell anode is the negative electrode. Anions move to the anode and oxidized. The electrons move out to the cell at this electrode.

25. C

Hybrid orbitals are orbitals formed by mixing the basic atomic orbitals, Examples sp, sp^2, sp^3, sp^3d, sp^3d^2

$O = \frac{1\ell}{2s}\ \frac{1\ell}{2p_x}\ \frac{\ell}{2p_y}\ \frac{\ell}{2p_z}$ (unhybridized)

$O(\text{in } H_2O) : \frac{1\ell}{2S}\ \frac{1\ell\ 1\ell\ 1\ell}{SP^3}$

Orbital hybridization of oxygen atom in water is SP^3 hybridization

26. B

When we go from left to right across in the same period atomic number increase but the number of shell remains the same. Across a period atomic radius decrease while electro negativity and ionization energy increase.

27. D

n	ℓ	ml (-l to +l)
n = 1	ℓ =0	0
n = 2	ℓ =0,1	-1,0,+1
n =3	ℓ = 0,1,2	-2, -1, 0, +1, +2
n =4	ℓ= 0,1,2,3	-3,-2,-1, 0, +1, +2 +3
n=8	ℓ = 0,1,….7	-7, -6, ……..+7

28. D

Molecular Gas	Bonding pair	Lone pair	Molecular geometry	Example
AF_3	3	0	Trigonal planar	BF_3, BCl_2
	3	1	Trigonal pyramidal	NF_3, NCl_3
	3	2	T – shaped	ClF_3

29. C

Nitrogen trichloride (NCl_3), oxygen hexa fluoride (OF_6) and phosphorus trichloride (PCl_3), nitrylfluoide (NO_2F)

Telegram: https://t.me/ExtremeSeries

30. C
Due to the presence of hydrogen bonding the intermolecular force of HCl is much greater than Br_2 and N_2

31. C.
Molarity $= \frac{0.006 mol}{1L} = 0.006$ M

$$\text{Rate} = \frac{[S_2O_3]}{2\Delta T} = \frac{[I_2]}{\Delta T}$$
$$= \frac{0.006\ M}{2} = \frac{0.006 M}{1}$$
$$= 0.003 \text{ and } 0.006$$

32. A

Trial	[H⁺],m	Rate (M/s)
1	0.005	3.2×10^{-7}
2	0.01	1.7×10^{-7}
3	0.02	0.85×10^{-7}

$$Rate = K[H+]^n$$
$$\frac{Rate-1}{Rate-2} = \frac{K[H^+]^n}{K[H^+]_2^n} \qquad \frac{304\times10^{-7}}{1.7\times10^{-7}} = (\frac{0.005}{0.01})^n$$
$$= 2 = (0.5)^n$$
$$\log_{0.5} 2 = n$$
$$\frac{\log_{10} 2}{\log_{10} 0.5} = \frac{0.30103}{-0.30103} = -1$$

33. A
$O_3(g) \rightarrow O_2(g) + O(g)$
$O_3(g) + O(g) \rightarrow 2O_2(g)$
$O_3(g) + O_3(g) + O(g) \rightarrow O_2(g) + O(g) + 2O_2(g)$
$2O_3(g) + O(g) \rightarrow 3O_2(g) + O(g)$
∴Over all reaction : $2O_3(g) \rightarrow 3O_2(g)$

34. A
For Zero order reaction the rate is constant . Rate does not change with concentration as it does for first order or second order reactions
Rate = $K[A]^0 = K$

35. B
$n = MV = 0.9L \times 0.12 mol L^{-2} = 0.0108 mol\ of\ NaCl$ is needed.
There fore, for choice B nNaCl = 0.638g / 58.58g= 0.0108 mol

36. C
Alcohols such as ethanol, methanol and ethylglycol used as antifreeze (as freezing point depression) and boiling point elevation ethyl glycol(C_2, H_6, O_2) is the main component of automobile

antifreeze. The antifreeze solution that protects the engine coolant from freezing also protects it from boiling in hot weather.

37. A
Order of acidity: $HClO_3 > HBrO_3 > HBrO_2 > HBrO$

38. C
Before any NaOH is added, we have a solution of acetic acid. The governing equilibrium system is :
$CH_3COOH\ (aq) \leftrightarrows H^+ + (aq) + CH_3COO^-(aq)$
$$Ka = 1.8\times10^{-5} = \frac{[H+][CH_3COO-]}{[CH_3COOH^-]} = \frac{x^2}{0.5}$$
$$1.8\times10^{-5} = \frac{x^2}{0.5}$$
$$X = \sqrt{9\times10^{-6}} = 0.003 = [H^+]$$
$$PH = -\log[H+] = -\log(3\times10^{-3}) = 2.5$$

39. B
White $P_4 \rightarrow$ Red P_4 $\Delta H = -17.7\ Kmol$
Because of the negative value of ΔH, we would expect this reaction to occur spontaneously at 25°C and 1atm. Red phosphorus is much less reactive than white phosphorus and need chemical agent.

40. D
A change that takes place with out the input of energy from an external source is said to be spontaneous. One begun, such a change (process) continues on its own.

41. B
If an aqueous solution of $CuCl_2$ is electrolyzed between copper electrodes, Cu^{2+} ions are reduced at the cathode
$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$
The expected products will be copper at the cathode and chlorine gas.

42. B
Oxygen is the most abundance element in the earth's crust. Silicon is the second most abundant element in the earth's crust . Aluminum is the most abundance metal in the earth's crust.

43. B
Apolymer may contain monomers of identical or of different chemical structures if it has identical units then it is known as homo polymer, where as apolymer containing several types of nonnumeric units in its chain is known as copolymer or mixed polymer.

44. C

Telegram: https://t.me/ExtremeSeries

Each amino acid has a different side chain. The side chains vary greatly in their complexity and properties Example the side chain of glycine is simply a hydrogen.

45. C

Solution has three steps:

1. Solute separation (ΔH^0) the process is endothermic because energy is needed to over come the force holding the particles together.
2. Solvent- separation($\Delta H_2{}^0$) is endothermic, the amount of energy needed for the separation of solvent molecule is small.
3. Hydration(salvation)($\Delta H_3{}^0$) – is exothermic process

46. B

The formation of a solution is often accompanied by either an absorption or a release of energy i.e it is accompanied by a temperature change. When H_2SO_4 dissolves in water heat is released to the surrounding (exothermic) solubility increase when temperature decrease.

47. D

$0.2 \text{ decigram} = 0.2\,(10^{-1}) \text{ gram} = 0.2\,(10^{-1})\,(10^{-3}) \text{ mg}$

$= 2\,(10^{-1})\,(10^{-3}) \text{ mg}$

$= 2\times 10^{-5} \text{ mg}$

48. C

$n = MV = 0.1L \times 0.25\ M = 0.025 mol$

0.025 mol X

$Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$

1mole 100g

$$\frac{0.025 mol}{1 mol} = \frac{X}{100 g}$$

$$X = 0.025\, mol \times 100 g = 2.5\, g$$

49. C

Radio wave has low frequency and high wave length. Gamma ray has low wave length and high frequency.

50. A

n	l	Sub shells	ml
4	0	4s	0
	1	4p	-1, 0 +1
	2	4d	-2,-1,0,1,2
	3	4f	-3,-2,-1,0,1,2,3

Telegram: https://t.me/ExtremeSeries

Azimuthal quantum number (ℓ) some times called angular quantum number. It describes the shape of the electron cloud (an orbital).

51. A

The electronic configuration of Ni (Z=28) is $[Ar]\ 4s^2\ 3d^8$

52. A

The formation of acetylene, C_2H_2 there are only two hydrogen atoms to over lap with the two carbon centers. This implies that carbon can not undergo an sp^2 hybridization as in the case of C_2H_4 Instead, one 2s orbital and only one 2P orbital will mix to give two sp hybrid orbitals of equal energy

53. C

$2NO_2(g) \longrightarrow 2NO(g) + O_2(g)$

$$\text{Rate} = \frac{-\Delta[NO_2]}{2\Delta T} = \frac{\Delta[NO]}{2\Delta T} = \frac{\Delta[O_2]}{\Delta T}$$

54. D

The following factors can control the position of a chemical equilibrium. Concentration change pressure change, temperature change, catalyst . A catalyst makes areaction reach equilibrium faster but does not change either the value of the equilibrium constant or the equilibrium concentrations.

55. D $PCl_5(g) \leftrightarrows PCl_3(g) + Cl_2(g)$

The concentration of $PCl_3(g)$ and $Cl_2(g)$ increase by adding more $PCl_5(g)$.

56. B

Molality is the mole of solute in a solution devided by the mass of solvent in Kilogram (Kg)

$$\text{Molality} = \frac{\text{NO of moles of solute (n)}}{\text{Mass of solvent (in kg)}}$$

Solute = 1mole

Solvent (H_2O) = 1000g

$$\text{Mole of } H_2O = \frac{1000 g}{18 g\ mol^{-1}}$$

Mole of solution = 55.56 mol +1mol = 56.56 mol

$X_{solute} = \dfrac{\text{mole of solute}}{\text{Mole of solution}}$

$$= \frac{1}{56.56}$$

$$= 0.018$$

57. A When a solute is dissolved in a solvent the freezing point of the solution is lower than that of the pure solvent. None electrolyte solutes such as urea lowers the vapour pressure of the solvent. Note that both the freezing poing depression and the boiling point elevation are defined as positive

58. D

$MnO_2 + HO_2^- \longrightarrow MnO_4$

Balance H atoms by adding H^+

$MnO_2 + HO_2^- \quad MnO_4 + H^+$

Half reactions

$MnO_2 \longrightarrow MnO_4$

$HO_2^- \longrightarrow H^+$

Balance O- atoms by adding H_2O

$2H_2O + MnO_2 \longrightarrow MnO_4 + 4H^+$

$4H^+ + HO_2 \longrightarrow H^+ + 2H_2O$

Balance the charges on both sides

$2H_2O + MnO_2 \quad MnO_4 + 4H^+ + 4e^-$

$2e^- + HO_2^- + 4H^+ \; H^+ + 2H_2O$

Balance the electron transfer

$2H_2O + MnO_2 \longrightarrow MnO_4 + 4H^+ + 4e^-$

$4e^- + 2HO_2^- + 4H^+ \longrightarrow 2H^+ + 4H_2O$

$2H_2O + MnO_2 + 2HO_2^- + 8H^+ \longrightarrow MnO_4 + 6H^+ + 4H_2O$

$MnO_2 + 2HO_2^- + 2H^+ \longrightarrow MnO_4 + 2H_2O$

Add $2OH^-$ both sides

$MnO_2 + 2HO_2^- + 2H_2O \longrightarrow MnO_4 + 2H_2O + 2OH^-$

$MnO_2 + 2HO_2^- \longrightarrow MnO_4 + 2OH^-$

59. A Cell potential (emf of the cell) is composed of oxidation potential (E_{ox}) and the reduction potential (E_{red})

$$Emf = E_{ccell} = E_{ox} + E_{red}$$

To this point , we have dealt only with standard potentials E_{cell}, E_{red}, E_{ox} . These apply at standard concentrations (1.0M for species, 1.0 atm for gases). How ever, we often carry out redox reactions were the concentrations of one or more species are different from 1.0M

Effect of concentration up on cell potential for

$Zn + Cu^2 \longrightarrow Cu + Zn^{2+}$, $E_{cell} = +1.10v$

$[Cu^{2+}]$ (M)	$[Zn^{2+}]$ (M)	E_{cell} (V)
1.0	0.01	1.16
0.1	0.1	1.10
0.01	1.0	1.042

$$Emf = E_{cell} - \frac{0.0592V}{n} \log Q$$

There are two electrons transferred in the reaction (n =2)

$$Q = \frac{[Zn\,2+]}{[Cu\,2+]} = \frac{0.1M = 1}{0.1M} =$$

$$Emf = 1.1v - \frac{0.0592}{2} \log (1)$$

$$Emf = 1.1v - \frac{0.0592}{2} (0)$$

$$Emf = 1.1V - 0 = 1.10V$$

60. C Proteins belong to a class of organic compound called polyamides. Polyamides are polymers where the monomer units are held together by amide groups (- CO – NH-)

Nylon is a generic name for the polyamides rather than the name of a single polymer

61. A Transition elements (Fe, Cr, Mn, Cu, Zn) can form ionic bonds by losing electrons from both the S and d electrons.

62. B Given

$n = 6.626 \times 10^{-34}$

$c = 3 \times 10^8$

$E = 7.21 \times 10^{-19}$

For electromagnetic radiation of wave length λ, the energy of each photon is given by the formula;

$$E = \frac{h\,C}{\lambda}$$

$$E = \frac{6.626\times10^{-34}\times3\times10^8}{7.21\times10^{-19}} = \frac{19.878\times10^{-26}}{7.21\times10^{-19}}$$

$$= 2.757 \times 10^{-7} = 276nm$$

63. B

When we give the shape of the ozone , O_3, molecule are arranged around the central atom, , not how the electrons are arranged. The central oxygen atom has one lone pair electron and two bonding electrons, Therefore, we do not describe the O_3 molecules as a triangular. Instead, we say that it is non-linear or bent, or angular or V-shaped

64. B $H_2(g) + I_2(g) \leftrightarrows 2\,HI(g)$

$[H] = 4mol/3L = 1.33M$

$[I_2] = 0.6m/3L = 0.2M$

Telegram: https://t.me/Extremeseries

$[HI] = 7mol/3L = 2.33M$

$$KC = \frac{[HI]^2}{[H_2][I_2]} \frac{(2.33)^2}{(1.33)(0.2)} = \frac{5.4289}{0.266} = 20.4$$

65. B $K_c = 0.09 = \frac{[HOCl]^2}{[H_2O][Cl_2O]}$

$$[HOCl] = \frac{0.084}{2L} = 0.042$$

$$[Cl_2O] = \frac{0.080}{2L} = 0.040$$

$$[H_2O] = \frac{0.98L}{2L} = 0.49$$

$$K_c = 0.09 = \frac{(0.042)^2}{(0.04)(0.49)} = \frac{1.764\times10^{-3}}{1.96\times10^{-2}} = 0.09\ K_c$$

66. A Mass of HNO_3 = 1000g of 55/100 = 550g
Mass of H_2O = 1000g – 550 g = 450g H_2O = 450 ml = 0.45L

67. B
During dilution if heat energy is evolved, the process is exothermic ($\Delta H < O$). But for ideal solution $\Delta H_{sol} = 0$

68. C

HF (aq)	H^+ (aq)	+	F^- (aq)
Initial (M) 0.1	0		0
Change (M) 0.1-x	+x		+x
Equilibrium 0.1-x	x		x

$$Ka = \frac{[H^+][F^-]}{[HF]} = \frac{x^2}{0.1-x} = 6.8\times10^{-4}$$

$$\approx \frac{x^2}{0.1} = 6.8\times10^{-4}$$

$x = [H+] = [F-] = \sqrt{6.8\times10^{-5}} = 0.008M$

$[HF] = 0.1M - 0.008M = 0.092M$

$$\% \text{ ionization} = \frac{0.008}{0.092}\times100\% = 8.6\%$$

69. A Equivalent point is the point at which stoichiometrically equal quantities of acid and base have been brought together. The net ionic equation for the reaction between a strong acid and a strong base is

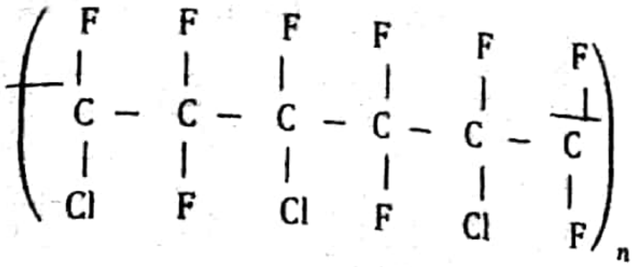
$H^+(aq) + OH^-(aq) \longrightarrow H_2O(l)$

The quantity of base required to reach the equivalence point is the same in both titration.

70. A Neutralization and double displacement don't undergo redox rection.

71. A

$(CFCl = CF_2)_n \rightarrow$ monomer



$\longrightarrow (CFCl = CF_2)_n$

Monomer Polymer

72. C A zero is not a significant figure, if digits other than zero do not precede it. In such cases it is used only to locate the decimal point. E.g 0.010 has only two significant digits.

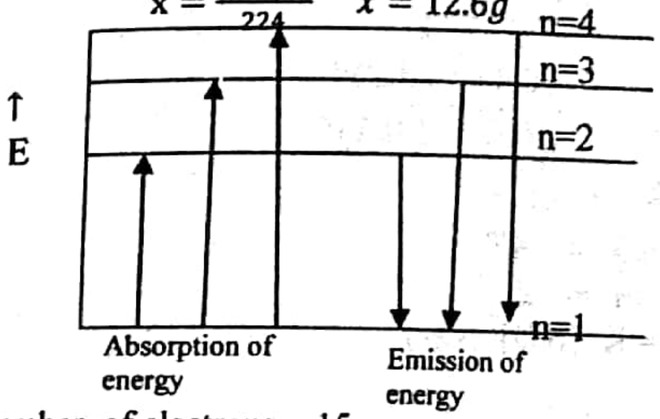
72. C A zero is not a significant figure, if digits other than zero do not precede it . In such cases it is used only to locate the decimal point . E.g 0.010 has only two significant digits.

73. D $2C_2H_6(g) + \underset{224\frac{g}{m}}{7O_2} \rightarrow \underset{176g}{4CO_2} + 6H_2O(g)$ ($16g$ over $7O_2$, x over $4CO_2$)

$$\frac{16g}{224g/mol} = \frac{x}{176g/mol}$$

$$x = \frac{16\times176}{224} \quad x = 12.6g$$

74. C



75. D N O number of electrons = 15

$No = \delta1S^2\ \delta^*1S^2\ \delta S^2\ \delta2S^{*2}\ \delta2P^2X\ \prod2PY^2\ \prod2PZ^2\ \prod2^XPY\ 1.\prod2^XPZ$

NO = molecule is paramagetism, paramagentism material have a week attraction towards magnetic field. It is a property possessed by substances having unpaired electrons. Example NO has un paired electron on $\pi2p^{*\ 1}y$

76. C

Telegram: https://t.me/Extremeseries

$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$			
Trial	[No]	$[O_2]$	Rate
1	0.1	0.1	2.5×10^{-4}
2	0.1	0.2	5×10^{-4}
3	0.4	0.2	8×10^{-3}

Rate = $K[NO]^m[O_2]^n$

$$\frac{Rate2}{Rate\ 1} = \frac{K[NO]_2^M}{K[NO]_1^m}\frac{[O_2]_2^n}{[O_2]_1^n}$$

$$\frac{5\times10^{-4}}{2.5\times10^{-4}} = \frac{(0.1)^m\ (0.2)^n}{(0.1)^m\ (0.1)^n}$$

$$2 = (\frac{0.2}{0.1})^2$$

$$2 = 2^n$$

$$n = 1$$

$$\frac{Rate2}{Rate\ 1} = \frac{K[NO]_3^m}{K[NO]_2^m}$$

$$\frac{8\times10^{-3}}{5\times10^{-4}} \quad \frac{(0.4)^m\ (0.2)^n}{(0.2)^m\ (0.1)^n}$$

$$2 = (\frac{0.4}{0.1})^m$$

$$16 = 4^m$$

$$m = 2$$

Rate = $K\ [NO]^2[O_2]$

77. D Given

$K_c = 2.5\times 10^{-10}$

$[CO] = 1\times 10^{-5}M$

$[COCl_2] = ?$

$COCl_2\ (g) \leftrightarrows CO\ (g) + Cl_2\ (g)$

$$K_c = \frac{[CO][Cl_2]}{[COCl_2]}$$

$$2.5\times10^{-10} = \frac{(1\times10^{-5})^2}{[COCl_2]}$$

$$[COCl_2] = \frac{1\times10^{-10}}{2.5\times10^{-10}}$$

$$[COCl_2] = \frac{1}{2.5} = 0.4M$$

78. D Boiling point elevation (ΔTb) is the difference between the boiling point of the solution (Tb) and that of pure solvent (Tb^0).

$\Delta Tb = Tb - Tb^0 = K_b m$, but $\Delta Tf = K_f M$

$\Delta Tb = K_b m$ substitute, $m = \frac{\Delta Tf}{Kf}$

$= K_b\ (\frac{\Delta T_f}{Kf})$

$= 0.512\ (\frac{0.186}{1.86})$

$= 0.0512$

79. C Salicylic acid (aspirin) is acidic, it ionized in the stomach and increase the acid in the stomach (PH = 2), and almost unionized in the small intestine

80. A Given

V_1 = 10ml

M_1 = ?

V_2 = 20ml

M_2 = 1M

$M_1V_1 = M_2V_2$

$$M_1 = \frac{M_2V_2}{V_1} = \frac{1M\times 20mol}{10mol} = 2M M_1 = M_2V_2$$

Telegram: https://t.me/ExtremeSeries

Booklet Code 27

1. B. $E = h\nu = \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} Js \times 3 \times 10^{8} m/s}{450 \times 10^{-9} m}$
$= 0.0441 \times 10^{-17} J$
$= \underline{4.41 \times 10^{-19} J}$

2. B. In BrO_3^-, Br has 7 valence electrons and oxygen has 6 valence electrons. Hence, $BrO_3^- \Rightarrow 7 + 3 \times 6 + 1 = 26$
Note that one electron is added for each negative charge, and one electron is subtracted for each positive charge

3. B. CO_2 exists as solid at low temperature and high pressure. When the temperature increases at high pressure, it converts to the liquid state. At 5.11 atm and $-56.4°C$, CO_2 (ℓ) and CO_2 (g) are at equilibrium known as the triple point of CO_2.

4. D. Acylation is the addition of an acyl group ($R^1 - \overset{O}{\overset{\|}{C}} -$) to an alcohol (ROH) to form ester.

$$R^1 - \overset{O}{\overset{\|}{C}} - Cl + H - O - R \rightarrow R^1 - \overset{O}{\overset{\|}{C}} - O - R + HCl$$

Acidchloride Alcohol Ester

- Esterification is the reaction between alcohols and carboxylic acids to form ester and water

$$R^1 - \overset{O}{\overset{\|}{C}} - OH + H - \overset{O}{\overset{\|}{O}} - R \rightarrow R^1 - \overset{O}{\overset{\|}{C}} - O - R + H_2O$$

Carboxhylic acid Alcohol Ester

- Saponification is a base – catalyzed ester hydrolysis giving salts of carboxylic acids (soaps) and alcohols

$$R - \overset{O}{\overset{\|}{C}} - O - R^1 + NaOH \rightarrow R - \overset{O}{\overset{\|}{C}} - Ona + R^1 - OH$$

Ester Soap Alcohol

- Condensation is the formation of small molecules like water in the reaction between compounds of different functional groups like carboxylic acids and alcohols.

5. A. $HCl \rightarrow H^+ + Cl^-$
0.5M 0.5M 0.5M
$\Rightarrow [H^+] = 0.5M = 5 \times 10^{-1} M$
$pH = log^{[H^+]} = -log^{0.5} = -log^{5 \times 10^{-1}} = 1 - \log 5 = 1 - 0.5 = \underline{0.5}$

6. C. $HOCl + H_2O \quad H_3O^+ + OCl^-$
$NaOCl \rightarrow Na^+ + OCl^-$

Telegram: https://t.me/ExtremeSeries

NaOCl contains the common ion OCl^-. According to Le chatelier's principle, addition of OCl^- shifts the equilibrium to the left increasing the concentration of HOCl and decreasing the concentration of H_3O^+, known as the common ion effect. The common ion effect decreases the dissociation of HOCl

7. B. Sodium acetate, CH_3COONa is a salt of a weak acid, CH_3COOH (acetic acid) and a strong base NaOH. When dissolved in water, the acetate ion, CH_3COO^- hydrolyzes generating OH^-
$CH_3COO^- + H_2OCH_3COO + OH^-$. Thus, CH_3COONa is basic.

8. B. A buffer solution is a mixture of a weak acid and its conjugate base or a weak base and its conjugate acid.
E.g. CH_3COOH/CH_3COO^- and NH_3/NH_4^+ form a buffer solution
A buffer solution resists P^H changes by reacting with the added acid or base.

9. B. In naming esters, the alkyl name is taken from the alcohol and the acid name ends in – ate. Thus, the ester is formed from benzoic acid and ethanol, and has the name Ethylbenzoate.

10. $\overset{+5}{I_2}\overset{-2}{O_5}(s) + 5\overset{+2}{C}\overset{-2}{O}(g) \rightarrow I_2^o(s) + 5\overset{+4}{C}\overset{-2}{O_2}(g)$
$\Rightarrow$ I from + 5 to O and C from + 2 to +4

11. D. $2H_2O(\ell) + Al(s) + MnO_4^-(aq) \rightarrow Al(OH)_4^-(aq) + MnO_2(s)$
$Al \rightarrow Al(OH)_4^-$ (oxidation)
$MnO_4^- \rightarrow MnO_2$ (reduction)
Al is oxidized and is reducing agent
MnO_4^- is reduced and is oxidizing agent

12. A. $HBb(aq) \rightleftharpoons aq) + Bb^-(aq)$
The conjugate base of a weak acid is strong. Hence Bb^- is a strong base. NaOH is also a strong base. Addition of a strong base shifts the equilibrium to the left according to Le chatelier's principle

13. A. Dihydric alcohols (alcohols containing two OH group) are more soluble in water than monohydric alcohols(alcohols) containing one OH group) due to the increased hydrogen bond. Moreover, the lower the hydrocarbon part (hydrophobic), the greater the solubility. Thus, $HOCH_2CH_2CH_2OH$ has the highest water solubility.

14. B. Silver and graphite are conductors of electricity due to free mobile electrons. Aqueous sulfuric acid is also an electrolyte due to the presence of oppositely charged ions (H^+ and SO_4^{2-}). Solid ionic compounds like $MgCl_2$ are non-conductors because the ions are not set free to conduct electricity.

A. A solution is a homogeneous mixture, but milk is a heterogeneous mixture

16. B. Isotopes are atoms of the same element differing in the number of neutrons, or mass number. Chlorine – 37 has two more neutrons than chlorine- 35

17. B. Iodine is a molecular solid, mercury and bromine are liquids, and hence all have low melting points. Tungsten is a metallic solid and has the highest melting point among them.

18. A. Ionic bonding involves transfer of electrons, and contains metals and non- metals E.G. NaF, LiCl and $MgBr_2CO$ contains a coordinate covalent bond.

19. D. Amino acids are bifunctional containing the amide group ($-NH_2$) and the carboxyl group (_COOH)

$$R-\underset{\displaystyle NH_2}{\overset{\displaystyle H}{\underset{|}{\overset{|}{C}}}}-COOH\ (R = H)$$

20. C. $\underset{\gamma\text{-ray}}{\text{Energy decreases, frequency decreases, wave length increases}}\ \underset{X\text{-ray}}{}\ \underset{UV}{}\ \underset{IR}{}\ \underset{microwave}{}\ \underset{Radio\ waves}{}$

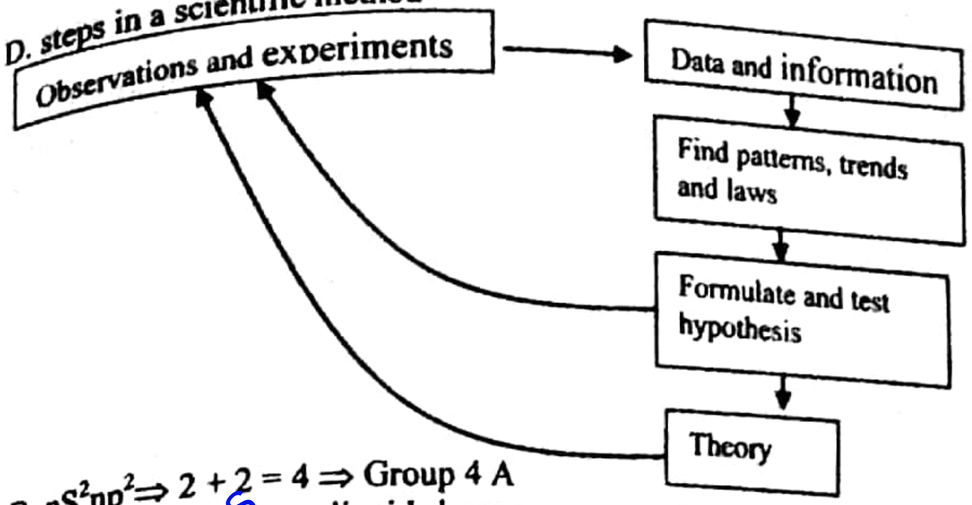
Gamma rays have the highest energy and frequency but the shortest wavelength because $E = h\nu = \frac{hc}{\lambda}$ where E = energy, h= planck's constant, ν= frequency, λ = wave length

21. B. Arrhenius definition : acid – a substance that produces H^+ in water, base – a substance that produces OH^- in water
Bronsted – Lowry definition: acid – a proton donor, base – a proton acceptor
Lewis definition: acid – electron pair acceptor, base – electron pair donor

22. D - Formula units are to ionic compounds as molecules are to covalent compounds.

23. B. 63.5g Cu ⟶ 1 mol Cu
10 g Cu ⟶ x
$\Rightarrow x = \frac{10X1}{63.5} = 0.16$ mol Cu
63.5g Cu ⟶ 6.02×10^{23} Cu atoms
10g Cu ⟶ x
$\Rightarrow x = \frac{10\ X\ 6.02\ X10^{23}\ Cu\ atoms}{635} = 9.5 \times 10^{22}$ Cu atoms

24. D. steps in a scientific method



25. B. $nS^2np^2 \Rightarrow 2 + 2 = 4 \Rightarrow$ Group 4 A

26. B. Pure solids and pure liquids have a concentration of unity (1), and do not appear in the equilibrium constant expression, that is, Kp = P_{CO_2} and $Kc = [CO_2]$ for the reaction $CaCO_3\ (s) \rightleftharpoons CaO\ (s) + CO_2\ (g)$

27. C. Ester molecules cannot form intramolecular and intermolecular hydrogen bonds with each other since they do not contain a hydrogen atom attached to the oxygen atoms.

28. C. Based on the Ka values, $C_6H_5OH < HOBr < CH_3COOH < HCO_2H$ in acidity. The conjugate base of the weakest acid C_6H_5OH is the strongest base and hence has the highest pH . C_6H_5ONa contains the conjugate base $C_6H_5O^-$, which is the strongest base
The basicity increases in the order :
$HCO_2^- < CH_3COO^- < OBr^- < C_6H_5O^-$

29. C. $M_1V_1 = M_2V_2$
$1 \times 250 = 3 \times V_2$
$\Rightarrow V_2 = \frac{1\ X\ 250}{3} = 83.33$ mL
Thus volume of water to be evaporated (x):
$250 - x = 83.33 \Rightarrow x = 250 - 83.33 = 166.7 = \underline{167}$ mL

30. A. $M_1V_1 = M_2V_2$
$3.5 \times 50 = 2 \times V_2$
$\Rightarrow V_2 = \frac{3.5\ X\ 50}{2} = 87.5$ mL
Thus volume of water required = 87.5 mL – 50 mL = $\underline{37.5}$ mL

31. B. Under constant pressure, $\Delta H = q_p$
Under constant volume, $\Delta E = q_v$

32. C. Arranging the metal ions in order of increasing standard reduction potential (E°) gives $Ni^{2+} < Pd^{2+} < Pt^{2+}$

Telegram: https://t.me/Extreme_series

The ion with more positive E° easily discharges first, and hence Pt plates out first.

33. B. A battery produces electricity by itself without the application of any external force, and hence the reaction is spontaneous (occurs by itself). In electrolysis cells, the reaction is non- spontaneous and needs D.C source to occur

34. A. For the H atom, the charge of the nucleus (2) =1, and

$$En = -R_H \times \frac{Z^2}{n^2} = -R_H \times \frac{1}{n^2} = -2.18 \times 10^{-18} J \times \frac{1}{n^2}$$

(n= 1, 2, 3, …)

35. B. For a zero order rxn, $t1/_2 = \frac{[A]_o}{2k}$

For a 1st – order rxn, $t1/_2 = \frac{0.693}{k}$

For a 2nd – order rxn, $t1/_2 = \frac{1}{K[A]_o}$

Where K= rateconstant

36. D. The boiling point of N_2 is low because of the weak intermolecular force (London dispersion force) between the nitrogen molecules

37. D. Ionization energy decreases as atomic number increases in the Column. Atomic radius and atomic mass increase as atomic number increases in the column

38. B. At equilibrium,
- The rate of the forward reaction equals the rate of the reverse reaction
- The concentration of the reactants and products doesnot change with time
- all the reactants and products are present at equilibrium, but their concentration is not equal
- the free energy change (ΔG) is zero.

39. A. $\frac{84^g Na_2SeO_4}{100g\ H_2O} = \frac{92g\ Na_2SeO_4}{x}$

$$\Rightarrow x = \frac{100X\ 92}{84} = 109.5g\ H_2O$$

But we have 200 g of water, and the solution is said to be diluted

40. A. $Cg = K_H P_g$ where K_H = Henry's law constant
P_g = Pressure of the gas
Cg = concentration of dissolved gas

Raoult's law: $P_1 = X_1 P_1^o$ Where P_1 = partial pressure of a solvent over a solution

X_1 = Mole fraction of the solvent in the solution

P_1^o = vapor pressure of the pure solvent

Boyle's law: $P_1V_1 = P_2V_2$ Ideal gas law: $PV = nRT$

41. D. Ernest Rutherford discovered the nucleus by alpha scattering experiment, and found that most of the spae in the atom is empty, and most of the mass and all of the positive charge of an atom is concentrated in the nucleus

42. D. Atoms react chemically with each other by losing, gaining or shaking electrons toattain a stable noble gas electronic configuration. In doing so, they become at lower energy and hence more stable. This is the basis of chemical bonding

43. A. N_2 is non-polar and does not contain positive and negative ion. NaI, HCl and KOH produce positive and negative ions in water and form electrolytic solution, but N_2 does not.

44. D. $[H^+] = 10^{-pH} = 10^{-10}M$

$[H^+][OH^-] = 10 \times 10^{-14}M$

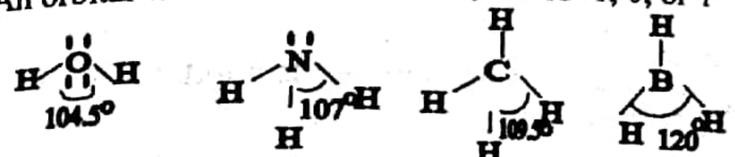
$$\Rightarrow [OH^-] = \frac{1.0 \times 10^{-14}}{[H^+]} = \frac{10^{-14}}{10^{-10}} = 10^{-4}M$$

45. D. Because the difference in energy associated with the jump from initial state (n_i) to the final state (n_f) becomes smaller and smaller as the distance from the nucleus increases (from n = 4 to n = 5 in this case).

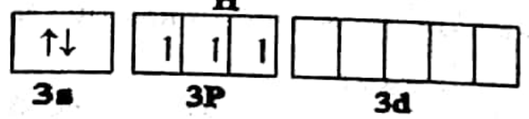
46. B. An orbital with $\ell = 0$ can have only $m_\ell = 0$

An orbital with $\ell = 1$ can have m_ℓ values -1, 0, or 1

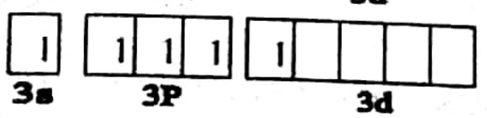
47. A.



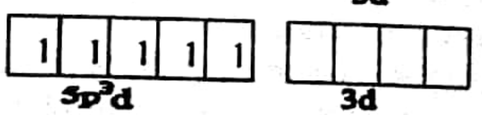
48. A. Ground state P:



Excited state P:



SP^3d hybrid orbitals:



49. D. $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$

Propanone (dimethyl Ketone)

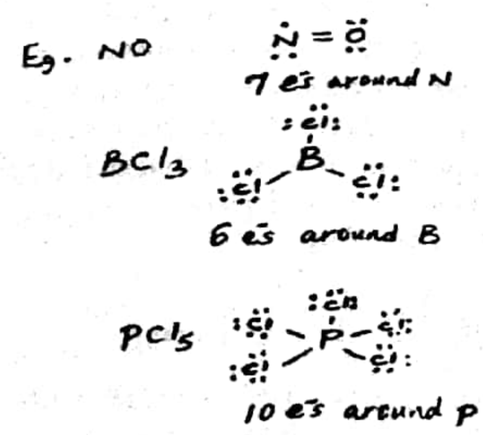
The general formula for Ketones is $R-\overset{O}{\overset{\|}{C}}-R$ or R – COR

$-\overset{O}{\overset{\|}{C}}-$

Telegram: https://t.me/Extremeseries

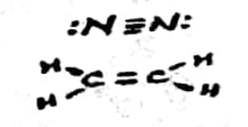
R – CHO – aldehyde R- COOR - ester - Carbonyl

50. A. Chemical species containing odd number of electrons, molecules whose central atoms contain fewer than 8 electrons, and molecules whose central atoms contain fewer than 8 electrons, and molecules whose central atoms have greater than 8 electrons donot obey the octet rule.
E.g NO.



51. D. As molecular size increases, the strength of London dispersion forces increases. Cl2 < Br2 < I2 in London dispersion force, and the states of matter accordingly varies from gas to liquid then to solid

52. D. :N (one σ and two π bonds)



(5σ and one π bonds)

(3σ and one π bonds)

(2σ and two π bonds)

N.B. A single bond contains only sigma (σ) bonds
A double bond contains one sigma (σ) and one Pi (π) bonds
A triple bond contains one sigma (σ) and two Pi (π) bonds.

53. A. The titration of strong acid with weak base has an equivalence point at pH < 7 because the salt formed is acidic.
For strong acid – strong base titration, the equivalence point is at pH > 7 since the salt formed is basic.

For weak acid – weak base titration, the equivalence point varies depending on the degree of hydrolysis of the salt formed.

54. A. 1 mol CO_2 = 44 g CO_2 = 6 .02 x 10^{23} CO_2 molecules

22g CO_2 = x

$$\Rightarrow x = \frac{22g\ CO_2 \times 6.02 \times 10^{23}\ CO_2\ \text{molecules}}{44g CO_2}$$

$= 3.01 \times 10^{23}$ CO_2 molecules (not atoms!)

55. A. $6^C: 1s^2 2s^2 2P_x^1 2P_y^1$

$5^B : 1s^2 2s^2 2P_x^1$

$4^{Be} : 1s^2 2s^2$

$3^{Li} : 1s^2 2s^1$

Hund'srule: Each equal energy orbitals (degenerate) orbitals are each occupied by a single electron before the electrons of opposite spin enters the orbital. Each of the three 2p orbitals ($2P_x$, $2P_y$ and $2P_z$) will hold a single electron before any of them receives a second electron.

56. A. C : $1s^2 2s^2 2P_x^1 2P_y^1 2P_z^o$ (unhybridized)

C : $1s^2 2\underbrace{s^1 2P_x^1}_{SP} 2P_y^1 2P_z^1$ (hybridized)

SP – hybridization

N : $1s^2 2s^2 2P_x^1 2P_y^1 2P_z^1$ (un hybridized)

N : $1s^2 2s^2 2P_x^1 2P_y^1 2P_z^1$ (hybridized)

SP – hybridization

The SP- hybrid orbitals are directed along a straight line, 180° apart (linear shape). Most examples of SP hybridization are found in organic molecules containing triple bonds such as acetonitrile.

57. C. $m = \frac{ItE}{F}$ $I = 10A$

$t = 1\ hr = 3600s$

$E_{Al} = \frac{M_{Al}}{Z_{Al}} = \frac{27}{3} = 9g$

$F = 96500C = 96500AS$

$$m = \frac{10A \times 3600S \times 9g}{96500\ AS} = 3.4g\ Al$$

OR $Al^{3+} + 3e^- \rightarrow Al$

3F 27 g

(3x96500C)

Telegram: https://t.me/ExtremeSeries

$10A \times 3600s = 36000C$

$289500\ C \rightarrow 27g\ Al$

$36000\ C \rightarrow X$

$$X = \frac{36000 \times 27}{289500} = 3.4\ g\ Al$$

58. C. $$r = -\frac{\Delta[S_2O_3^{2-}]}{2\Delta t} = \frac{\Delta[S_4O_6^{2-}]}{\Delta t} = \frac{\Delta[I^-]}{2\Delta t}$$

$$\Rightarrow \frac{\Delta[S_4O_6^{2-}]}{\Delta t} = \frac{1}{2} x \Delta \frac{\Delta[S_4O_3^{2-}]}{\Delta t}$$

$$= \frac{1}{2} \times 0.05\ M/S$$

$$= 0.025 M/S$$

$$\frac{\Delta[I^-]}{\Delta t} = \frac{\Delta[S_2O_3^{2-}]}{\Delta t} = 0.05\ M/S$$

59. D. A Bronsted - Lowry base is a proton acceptor I^-, NH_3 and HCO_3^- can accept a proton and can act as a base H_2CO_3can only donate a proton and acts as an acid

60. C. $r = K[X]^a[Y]^b$

To find a vary the [X] keeping the [Y] constant. Thus, take experiment 2 and 3

$$\frac{\text{rate of experiment } -1}{\text{rate of experiement } -3} = \frac{K[X]^a - \text{expt } 1}{K[X]^a - \text{expt } 3}$$

$$\frac{0.45}{0.45} = \left(\frac{3.0}{1.5}\right)^a$$

$$1 = (2)^a$$

$1 = 2^a \Rightarrow a = 0 \Rightarrow$ The reaction is zero order with respect to x

Similarly, to find b, vary the [Y] keeping the [X] constant. Thus, take experiments 1 and 2

$$\frac{1.8}{0.45} = \left(\frac{3.0}{1.5}\right)^b$$

$$4 = 2^b$$

$2^2 = 2^b \Rightarrow b = 2 \Rightarrow$ The reaction is second order with respect to Y

Therefore, Rate $= K[X]^\circ[Y]^2 = K[Y]^2$

61. D. $N_2(g) + 3H_2(g) \rightleftharpoons I_3 + heat$

The formation of N_2 and H_2 (left equilibrium) is favored by decreasing the pressure, by increasing the temperature, by adding more NH_3, and by increasing the volume of the reaction container, which means according to Boyle's law decreasing the pressure of the reaction container

62. B. if the concentration of a given reactant is increased by a factor, x, the rate increases by a factor X^a where 'a' is reaction order. Concentration of NO doubled $\Rightarrow x = 2$ reaction order in NO $\Rightarrow 2$. Thus rate increases by $X^a = 2^2 = 4$ (Quardruple)

63. D. $\Delta T_f = iK_fm$ where I is the Van't Hoff's factor for electrolytes. NaCl, NH_4NO_3 and $(NH_4)_2SO_4$areelectrolytes and have higher freezing point depression, which is directly proportional to the Van't Hoff's factor. Glucose, $C_6CH_{12}O_6$ is non- electrolyte and has no Van't Hoff's factor, and hence has the lowest freezing point

64. C. ΔG = negative, ΔS = positive, and ΔH = positive means the reaction is spontaneous at high temperatures

$\Delta G = \Delta H - T\Delta S$ where T = temperature

$= 40{,}000\ J - Tx50J/K$

$= 40,000 - 50T\ J/K$

If the temperature is > 800 K , ΔG becomes negative and the reaction will be spontaneous. If the temperature equals 800K, $\Delta G = 40{,}000 - 50x800 = O$, which is equilibrium condition.

65. E. $NH_3 + H_2O \quad NH_4^+ + OH^-$

base acid Conjugate acid Conjugate base

66. D. CCl_2F_2 is a polar molecule and there is a likely deviation from Raoult's law when C Cl_2F_2 is mixed with other substances where the vapor pressure of the solution doesnot exactly fit with the sum of the vapor pressures as predicted by Raoult's law.

67. $3Ba(NO_3)_2(aq) + Fe_2(SO_4)_3(aq) \rightarrow 3BaSO_4(s) + 2Fe(NO_3)_3(aq)$

$3Ba^{2+}(aq) + 6NO_3^-(aq) + 2Fe^{3+}(aqw) + 3SO_4^{2-}(aq) \rightarrow 3BaSO_4(s) + 2Fe3+(aq) + 6NO^{3-}(aq)$

$3Ba^{2+}(aq) + 3SO_4^{2-}(aq) \rightarrow 3BaSO_4(s)$

68. B. The half life of the reaction is 2 hours for a 50% completion. If the remaining 50% takes 2 hours for a 25% completion, the total completion is 75% in 4 hours, which shows the same time interval and consequently the reaction is first order.

69. A. $K_{eq} = \frac{Kf}{Kb}$ Keq = equilibrium constant

K_f = rate constant for forward rxn

K_b = rate constant for reverse rxn

Telegram: https://t.me/Extreme Series

$$A \underset{K_b}{\overset{K_f}{\rightleftharpoons}}$$

$$K_{eg} = \frac{Kf}{Kb} = \frac{8 \times 10^2 S^{-1}}{4 \times 10^4 S^{-1}} = 2 \times 10^{-2}$$

70. C. $R - \overset{O}{\overset{\|}{C}} - OR^1 + NaOH \rightarrow R - \overset{O}{\overset{\|}{C}} - ONa^+ + R^1 - OH$

$CH_3CH_2 - \overset{O}{\overset{\|}{C}} - O\ CH_3 + NaOH \rightarrow CH_3CH_2 - \overset{O}{\overset{\|}{C}} - O\ Na+ + CH_3OH$

Sodiumpropionate methanol

71. D. $M = \frac{\text{actual mass}}{\frac{\text{molar mass}}{\text{volume (L)}}}$

Molar mass of H_2SO_4 = 988/mol, 500 mL = 0.5L

$M = \frac{10q}{\frac{98g/mol}{0.5L}} = 0.2\ M$

72. A. $[H^+]$= antilog (-pH) = antilog (-4.5)

$-4.5 = -5 + 0.5$

$[H^+]$ = antilog (-5 + 0.5)

= antilog (-5) x antilog 0.5

$= 10^{-5} \times 3.16$

$= 3.16 \times 10^{-5}$ M

73. B. $\Delta E = q + W$

q = positive because heat is absorbed by the system

W= negative because work is done by the system

$\Delta E = q - W = 300KJ - 500\ KJ = -200\ KJ$

74. B. $\Delta H = H_{product} - H_{reactant}$

$H_{product} < H_{reactant} \Rightarrow \Delta H < 0$

The heat of the product, that is, C (Graphite) at lower heat (energy) means more stability.

75. D. A pressure cooker reduces cooking time because it increases the boiling point of water inside the cooker. The higher the surrounding pressure (the pressure of the cooker), the higher is the boiling point of the water inside the cooker. The food is easily cooked with the water vapor.

76. D. The Metal to be plated is the cathode

⇒ Nickel is the cathode.

- The plating metal is the anode

⇒silver is the anode

A salt containing the plating metal (silver)

Such as $AgNO_3$ is the electrolyte

77. B. Catalytic hydrogenation of oils at high pressure is called hardening of oils, and it is a process by which oils are converted to fats. The catalyst can be nickel or palladium

78. C. Glycine is a non-polar amino acid with the structure $NH_2 - CH_2 - COOH$. Starch, glucose, and cellulose are carbohydrates

79. A. Vulcanization is heating natural rubber with sulfur to establish cross – links between linear polymer chains. It makes the rubber hard, non – sticky, elastic, insoluble in solvents, not easily oxidized, and have high tensile strength.

80. Sea water largely contains NaCl and $MgCl_2$.
Thus, Na^+ is abundantly found in sea water.

Telegram: https://t.me/ExtremeSeries

ETHIOPIAN UNIVERSITY ENTRANCE EXAMINATION (EUEE) CHEMISTRY ANSWER GINBOT 2005/ JUNE 2013

BOOKLET CODE:27

1. B. The basic physical quantities are length, mass, time, electric current, temperature, amount of substance, and luminous intensity. The basic SI units are length- meter, mass – kilogram, time-second, electric current-ampere, temperature-kelvin amount of substance – mole, and luminous intensity candela. Hence, gram is not a basic SI unit.
2. C Gamma – rays < x-rays < U ltraviolet < visible < Infrared < microwaves < Tv < Radio in wavelength.
   - Radio < TV < Microwaves < Infrared < Visible < Ultraviolet < x-rays < Gamma – rays in frequency and energy.
   - $E = hV = \frac{hc}{\lambda}$ implies energy of radiation is directly proportional to frequency (v) , and inversely proportional to wavelength (λ). That is , as energy decreases, wavelength increases, and frequency decreases.
   - $v = \frac{c}{\lambda}$ implies frequency is inversely proportional to wavelength and it increases as the wavelength decreases.
   - $\overline{v} = \frac{1}{\lambda}$ implies wave number ($\overline{v}$) increases as wavelength decreases.
3. C The oxidation number of magnesium is +2 and that of nitrogen is -3. Using the criss –cross method and dropping the algebraic signs gives Mg_3N_2.

   That is $Mg^{+2} \times N^{-3} \equiv Mg_3N_2$
4. C Non – polar covalent bond is formed between atoms of similar electro negativities. Cl_2 is formed between two Cl atoms which have the same electro negativity value.
   Polar covalent bond is formed between atoms of different electro negativities. E.g B-Cl, C- Cl and Mg – Cl.
5. D Paramagnetic atoms are atoms having unpaired electrons and are attracted by a magnetic field. Group IV A atoms have unpaired electrons and hence are paramagnetic .
   E.g $_{50}Sn = [Kr]5S^2 4d^{10} 5P_x^1 5P_y^1$
   $_{32}Ge = [Ar]4S^2 3d^{10} 4P_x^1 4P_y^1$
   $_{82}Pb = [xe]6S^2 4f^{14} 5d^{10} 6P_x^1 6P_y^1$

   - Group zero, Group IIA and Group IIB all have paired electrons and are diamagnetic, that is , they are weakly repelled by a magnetic field.
6. B A unit cell is the repeating pattern of the structure of a crystal and it is the smallest single unit in a crystal lattice.
   There are three cubic unit cells, that is , simple cubic unit cell, face-centered cubic unit cell, and body-centered cubic unit cell. The coordination number (number of nearest neighbors) in face- centered cubic unit cell is 6. (E.g Nacl), and 8 in body-centered cubic unit cell. (Eg. CsCl)
7. B Hydrogenation is addition of hydrogen to liquid vegetable oils (unsaturated fatty acids) in the presence of Ni or Pd catalyst to convert them to solid fats (saturated fatty acids).
   - Hydration is addition of water to unsaturated organic compounds in the presence of acid catalysts.
   - Hydrolysis (cleavage by water) is the addition of water to a larger molecule of a compound to convert it to smaller molecules of compounds in the presence of acid catalysts.

     E.g Ester + water $\xrightarrow{\text{acid}}$ carboxylic acid + Alcohol.
   - Saponification is alkaline hydrolysis of fats or vegetable oils to produce glycerol and salts of fatty acids (soaps).
8. B Dihydric alcohols (alcohols containing two hydroxyl, OH groups) are more soluble than monohydric alcohols (alcohols containing one hydroxyl group) because more OH group means more hydrogen bonding and hence more solubility in water (a polar solvent)
9. A The reaction of N_2 with H_2 at high pressure and moderate temperature in the presence of iron catalyst to produce ammonia (NH_3) is known as Haber Process.

   $N_2(g) + 3H_2(g) \xrightarrow[400-600^0C, Fe]{200-300atm} 2NH_3(g)$
10. D A buffer solution is a solution that resists changes in PH from the addition of a fixed amount of an acid or a base. It contains a weak acid and its conjugate base or a weak base and its conjugate acid. E.g HCN and NaCN, NH_3 and NH_4Cl, H_3PO_4 and KH_2PO_4. It does not contain a strong acid or a strong base such as HNO_3 or NaOH.
11. A Salts formed from strong acid and strong base do not hydrolyse, that is , neither the cation nor the anion of the salt reacts with water.

Telegram: https://t.me/ExtremeSeries

E.g [illegible] + [illegible] → No rxn

$Cl^- + H_2O \rightarrow$ No rxn

- Salts formed from strong acid and weak base produce ions that are proton donors and form acidic solution.

E.g NH_4Cl, $NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$

$Cl^- + H_2O \rightarrow$ No rxn

- Salts formed from weak acid and strong base produce ions that are proton acceptors and form basic solution.

E.g CH_3COONa, $CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$

$Na^+ + H_2O \rightarrow$ No rxn

- Salts formed from weak acid and weak base depend on the K_a of the cation and K_b of the anion.

That is, $K_a > K_b \Rightarrow$ acidic solution E.g NH_4F

$K_b > K_a \Rightarrow$ basic solution E.g NH_4CN

$K_a = K_b \Rightarrow$ neutral solution E.g CH_3COONH_4

12. A Silver (Ag) is less reactive and less electropositive metal and can be obtained by chemical reduction from its ore. Na and Al are more reactive and electropositive and can be extracted by electrolytic reduction from their compounds. Cu is further purified by electrolytic refining.

13. A Though all metals are good conductors of heat and electricity due to the free mobile electrons, silver exceptionally has the highest electrical and thermal conductivity followed by copper (Cu) and gold (Au).

14. C Enthalpy (H) is the heat absorbed or released at constant pressure. It is the sum total of the internal energy (E) and PV product of a system. That is, $H = E + PV \Rightarrow \Delta H = \Delta E + P\Delta V$

15. D O (45.5%), Si(27.2%), Al (8.3%), Fe (6.2%), Ca (4.7%), all others (5.3%) in abundancy.

16. C Nitrogen fixation is the process of converting atmospheric nitrogen (N_2) to nitrates (NO_3^-), which can be used by plants to make proteins. Nitrates are minerals Industrial processes also fix atmospheric nitrogen to fertilizers, which are also minerals for plant use.

17. B Zero is significant if it is preceded by a non – zero digit (1 through 9). Hence, 0.0030050 has 5 significant figures.

18. D Intensive properties do not depend on the amount of matter present. Eg. Color, density, boiling point, temperature, melting point, etc.

Extensive properties depend on the amount of matter present. Eg. Volume, pressure, enthalpy, mass, internal energy, etc

19. A $18g\ H_2O \rightarrow 6.02 \times 10^{23}\ H_2O$ molecules

$x \leftarrow 1H_2O$ molecule

$$x = \frac{18 \times 1}{6.02 \times 10^{23}} = 2.99 \times 10^{-23} \approx 3.0 \times 10^{-23}\,g$$

20. A The principal quantum number (n) is related to the size and energy of the electron.

⇒ The angular momentum quantum number (ℓ) is related to the number of shells and shape of the orbital of the electron.

⇒ The magnetic quantum number (m_ℓ) is related to the orientation of the orbital of the electron.

⇒ The electron spin quantum number (m_s) is related to the spinning motion of the electron.

21. A If $n = 4$, $\ell = 0, 1, 2, 3$; $m_\ell = -3, -2, -1, 0, +1, +2, +3$; $m_s = \pm\frac{1}{2}$

$\ell = 0\ (s - orbital) \Rightarrow m_\ell = 0$

$\ell = 1\ (p - orbital) \Rightarrow m_\ell = -1, 0, +1$

$\ell = 2\ (d - orbital) \Rightarrow m_\ell = -2, -1, 0, +1, +2$

$\ell = 3\ (f - orbital) \Rightarrow m_\ell = -3, -2, -1, 0, +1, +2, +3$

22. No answer SF_4 (4bp and ℓp) – see saw.

IF_5 (5bp and $1\ell p$) – square pyramidal.

ICl_4^- (4bp and $2\ell p$) – square planar.

BrF_5 (5bp and $1\ell p$) – square pyramidal.

PCl_5 (5bp and no ℓp) – triangonal bi pyramidal

N.B ℓ_p = lone pair . b_p = bond pair

23. B Transition elements are d-block elements with the general electronic configuration of $ns(n-1)d$.

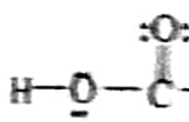
E.g $_{29}Cu = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^9$

$= [Ar] 4s^2 3d^9$

$_{26}Fe = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$

$= [Ar] 4s^2 3d^6$

24. C The Lewis structure of H_2CO_3 (carbonic acid) is



There are four covalent bonds around C to form the CO_3^{2-} anion showing that 8 electrons are involved in the bond formation.

Telegram: https://t.me/ExtremeSeries

25. B Metallic crystals have high electrical & thermal conductivities due to the free mobile electrons.

- Molecular crystals and covalent network crystals (except graphite) do not possess ions or free electrons and are non-conductors.
- Ionic crystals conduct electricity in the molten state or in aqueous solutions, not in the solid state.

26. C $2NO + 2H_2 \longrightarrow N_2 + 2H_2O$

$$\text{rate} = -\frac{1}{2}\frac{\Delta[NO]}{\Delta t} = -\frac{1}{2}\frac{\Delta[H_2]}{\Delta t} = \frac{\Delta[N_2]}{\Delta t} = \frac{1}{2}\frac{\Delta[H_2O]}{\Delta t}$$

27. D A reaction at equilibrium has the following characteristics.

i) The rate of the forward reaction equals the rate of the reverse reaction.

ii) The reaction should take place in a closed vessel to reach equilibrium.

iii) There is no net change in the concentration of the reactants and products of the reaction.

28. B Pushing the piston means increasing the pressure of the gaseous reaction system and decreasing the volume.

$$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$$

$$\Delta n = (1+3) - 2 = 4 - 2 = 2$$

$\Rightarrow$ $\Delta n > 0$ meaning the equilibrium shifts to the left (formation of NH_3 is favored). This means the amount (mole) of N_2 or H_2 decreases.

29. D Carboxylic acids can be prepared by the oxidation of primary alcohols using strong oxidizing agents such as potassium per manganate or potassium di-chromate.

$$R - CH_2 - OH \xrightarrow[K_2Cr_2O_7]{KMnO_4 \text{ or}} R - \overset{O}{\overset{\|}{C}} - OH$$

Primary alcohol (alkyl alcohol) — Alkylcarboxylic acid

30. D Acetic acid is found in vinegar.

- Butyric acid is found in rancid butter.
- Formic acid if found in ants and bees.
- Lactic acid is found in sour milk.
- Citric acid is found in lemon and orange juice.
- Malic acid is found in apples.

31. B Fats and oils are triesters of glycerol (triglycerides) with

Telegram: https://t.me/ExtremeSeries

three long chain fatty acids. Fats are derived from saturated fatty acids and glycerol and are solid esters while oils are derived from unsaturated fatty acids and glycerol and are liquid esters.

32. D Hydrogen bonding is formed when hydrogen is bonded to the three most electronegative atoms F,O and N.
n-butanol and water both have hydrogen bonding and are miscible in each other.

33. A Acid-catlyzed hydrolysis of esters yields carboxylic acid and alcohol.

$$R - \overset{O}{\overset{\|}{C}} - O - R' + H_2O \xrightarrow{H_3O^+} R - \overset{O}{\overset{\|}{C}} - O - H + R' - OH$$

Ester

34. B The general formula for aldehydes is $R - \overset{O}{\overset{\|}{C}} - H$

The structure for $R_1\ R_2CHCOH$ is

$$R_1 - \underset{H}{\overset{R_2}{C}} - \overset{O}{\overset{\|}{C}} - H$$

35. C $M_1V_1 = M_2V_2 \Rightarrow V_2 = \frac{M_1V_1}{M_2} = \frac{3.5 \times 50}{0.5} = 350\text{m}$. The volume of water added is $350 - 50 = 300ml$, because there is already 50ml of solution.

36. C In the reaction $3Fe^{2+} + 4H^+ + NO_3^- \longrightarrow 3Fe^{3+} + NO + 2H_2O$

$$3 \times \left\{ Fe^{2+} \xrightarrow{-1e^-} Fe^{3+} \text{ (oxidation)} \right.$$

$$1 \times \left\{ N^{+5} \text{ (in } NO_3^-) \xrightarrow{+3e^-} N^{+2} \text{ (in NO) (reduction)} \right.$$

$$3Fe^{2+} \xrightarrow{+3e^-} 3Fe^{3+}$$

$$N^{+5} \xrightarrow{+3e^-} N^{+2}$$

$\Rightarrow 3e^-s$ are involved in this balanced redox reaction.

$$E = \frac{M}{Z} \text{ where } \begin{matrix} M = \text{molar mass} \\ Z = \text{no of } e^-s \text{ lost or gained} \end{matrix}$$

$$E_{HNO_3} = \frac{MHNO_3}{3} = \frac{63}{3} = 21$$

37. D The higher the Ka value, the stronger the acid.
That is , $1.8 \times 10^{-4} > 1.8 \times 10^{-5} > 4.2 \times 10^{-7} > 5.6 \times 10^{-10}$
Or $IV > I > II > III$

38. B $H_2PO_4^- + H_2O \rightarrow H_3O^+ + HPO_4^{2-}$
acid base acid base

$H_2PO_4^-$ and HPO_4^{2-} are conjugate acid base pairs.
H_2O and H_3O^+ are conjugate acid-base pairs.
$H_2PO_4^-$ donates a proton (H+) to H_2O and hence $H_2PO_4^-$ is an acid and H_2O is a base.

39. D Properties that depend on the number of particles and not on their nature are referred to as colligative properties.
⇒ Addition of a non volatile solute to a solvent lowers the freezing point and vapor pressure of the solvent and increases its boiling point.

40. C $Ca(OH)_2 \rightarrow Ca^{2+} + 2OH^-$
$0.005M \quad 2 \times 0.005 = 0.01M = 1 \times 10^{-2}M$
$P^{OH} = -\log[OH^-] = -\log 1 \times 10^{-2} = 2 - \log 1 = 2$
$P^H = 14 - P^{OH} = 14 - 2 = 12$

41. C According to the Le Chatelier's principle, addition of a base (NaOH) shifts the equilibrium to the left since In^- is the conjugate base of HIn.

$HIn(aq) + H_2O(\ell) \longrightarrow In^-(aq) + H_3O^+(aq)$
Acid form (yellov *Base form (Blue)*

Hence, the color of NaOH turns yellow.

42. C $2Na(s) + Cl_2(g) \rightarrow 2NaCl(s), \quad \Delta H = -821.8kJ$
$2NaCl(s) \rightarrow 2Na(s) + Cl_2(g), \quad \Delta H = +821.8kJ$
$2\ mol\ NaCl = 117\ g$
$117\ gNaCl \rightarrow 821.8\ kJ$
$20g\ NaCl \rightarrow x$
$\Rightarrow x = \frac{20 \times 821.8}{117} = 140.5kJ$

43. A $Ni^{2+} + 2e^- \rightarrow Ni$
2mol 1mol=58.7g
$1mol\ e^- = 1F = 96{,}500c$
$2mole^- = 2F$
$2F = 2 \times 96{,}500 = 193{,}000c$
$193{,}000c \longrightarrow 58.7g\ Ni$
$240{,}000c \longrightarrow x$
$\Rightarrow x = \frac{240{,}000c \times 58.7g\ Ni}{193{,}000c} = 73g$

44. B The one with more positive electrode potential (E^o) is made

the cathode of the cell, and the one with less positive E^o is made the anode of the cell. Hence the cell diagram is

$Sn(s)/Sn^{2+}(aq)//Cu^{2+}(aq)/Cu(s)$
Anode *Cathode*

Sn is oxidized (Sn loses mass) $\Rightarrow Sn \longrightarrow Sn^{2+} + 2e^-$
Cu^{2+} is reduced (Cu gains mass) $\Rightarrow Cu^{2+} + 2e^- \longrightarrow Cu$

45. A Brass is an alloy of copper and zinc and does not contain tin. Bronze, pewter and plumber's solder contain tin.

46. D For addition polymerization to occur, the monomer should be a compound with multiple bonds, which contain a carbon-carbon pi (π) bond.

47. C Nylon 66 is made of hexamethylene diamine, $H_2N-(CH_2)-NH_2$ &adipic acid, $HOOC-(CH_2)_4-COOH$ and it is a condensation polymer which involves the formation of water.

48. D The process of adding sulfur to rubber so as to make it tough and hard is called vulcanization.

49. C $Q = mc\Delta T$
$= 80g \times 2.46\frac{J}{gk} \times (348 - 298)k$
$= 9840J$
$= 9.84\ KJ$

50. No answer $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$
1mole $CH_4 \longrightarrow$ 802 KJ
3.2mol$CH_4 \longrightarrow$ x
$\Rightarrow x = \frac{3.2 \times 802\ kJ}{1} = 2566.4KJ$
$= 2.566 \times 10^3$KJ
$= 2.57 \times 10^3 KJ$
$\Rightarrow 2.57 \times 10^3 KJ$ will be released

51. A $AlCl_3 \longrightarrow Al^{3+} + 3Cl^-$
$\Rightarrow 1mol\ AlCl_3 \longrightarrow 3\ mol\ Cl^-$
$1.0 \times 10^{-5} mol\ AlCl_3 \longrightarrow x$
$\Rightarrow x = \frac{1.0 \times 10^{-5} \times 3}{1} = 3 \times 10^{-5} mol\ Cl^-$
$1\ molCl^- \longrightarrow 6.02 \times 10^{23}\ Cl^-\ ions$
$3 \times 10^{-5} mol\ Cl^- \longrightarrow x$
$\Rightarrow x = \frac{3 \times 10^{-5} \times 6.02 \times 10^{23}}{1} = 18.06 \times 10^{18} = 1.80 \times 10^{19} Cl^-$

Telegram:https://t.me/ExtremeSeries

52. A $NH_3(g) + H_3PO_4(aq) \longrightarrow NH_4H_2PO_4(s)$

$NH_4H_2PO_4(s) + NH_3(g) \longrightarrow (NH_4)_2HPO_4(s)$

DAP

53. C Proteins are made up of aminoacids. The doubly ionized form of aminoacid is called a zwitterion .

$R-\underset{\underset{NH_3^+}{|}}{CH}-COO^-$ A zwitterion

54. C Styrene – butadiene rubber (SBR) is made by polymerization of butadiene and styrene. Its mechancal strength is enhanced by adding carbon black and sulfur, SBR is widely used in tires more than any other synthetic rubber.

55. A The simplest monosacharides are aldotriose and Ketotriose containing 3 carbon atoms. Pentoses contain 5 carbon atoms and hexoses contain 6 carbon atoms.

56. B Starch is separated into amylose and amylopectin. Amylose is a straight chain glucose polymer with $\alpha-1,4-$ glycosidic linkeagees and is a minor component. It is water soluble. Amylopectin is a highly branched glucose polymer with α-1,4 and α-1,6-glycosidic linkages and is a major component. It is water insoluble.

57. C $O_F = \frac{9}{5}{}^0C + 32 \implies {}^0C = -40$

let $x = {}^0F = {}^0C$ $\quad K = {}^0C + 273$

$x = \frac{9}{5}x + 32$ $\quad = -40 + 273$

$x - \frac{9}{5}x = 32$ $\quad K = 233$

$-\frac{4}{5}x = 32$

$x = 32 \times -\frac{5}{4}$

$= -40$

58. C Using de Broglie's equation,

$$\lambda = \frac{h}{mv} = \frac{6.626 \times 10^{-34} Js}{9.11 \times 10^{-31} kg \times 0.4 \times 3 \times 10^8 \frac{m}{s}}$$

$$= 6.06 \times 10^{-12} m$$

59. No answer I = 4s, II = 3P, III = 4d and IV = 4P

Because $\ell = 0(s), \ell = 1(p), \ell = 2(d)$ the general designation of the electron configuration is $n\ell^x$ where n=principal quantum no , ℓ =magnetic quantum number, & $x = no\ of\ e^-s$

The increasing order of energy is

$3P < 4S < 4P < 4d$ which means

$II < I < IV < III$

60. A $sp^3 - tetrahedral$ $\quad sp^3d^2 - octahedral$

$sp^3d - trigonal\ bipyramidal$ $\quad sp - linear$

$sp^2 - trigonal\ planar$

61. B $\sigma_{1s}^2\sigma_{1s}^{*2}\sigma_{2s}^2\sigma_{2s}^{*2}\sigma_{2px}^2\pi_{2py}^2\pi_{2pz}^2\pi_{2py}^{*1}\pi_{2pz}^{*1}$

$\Rightarrow$ There are $16e^-s$ in O_2 *and* 2 un paired e^-s showing O_2 is paramagnetic according to molecular orbital theory.

62. C $r = k[A]^x[B]^y$ to find x, vary [A] keeping [B] constant. Hence, take experiments 2 and 3.

$$\frac{rate\ of\ expt.\ 2}{rate\ of\ expt.\ 3} = \frac{K[A]^x expt.2}{K[A]^x\ expt.3}$$

$$\frac{0.075}{0.075} = \left(\frac{0.50}{0.25}\right)^x$$

$$1 = 2^x$$

$\Rightarrow x = 0$ *because* $2^0 = 1$

Similarly, to find y take experments 1 and 2.

$$\frac{rate\ of\ expt.\ 1}{rate\ of\ expt.\ 2} = \frac{k[B]^y expt.1}{k[B]^y expt.2}$$

$$\frac{0.3}{0.075} = \left(\frac{0.50}{0.25}\right)^y$$

$$4 = 2^y$$

$$2^2 = 2^y$$

$$\Rightarrow y = 2$$

Hence, the rate law is $r = K[B]^2$

Thus, $K = \frac{r}{[B]^2}$

Take any of the experiments

$$K = \frac{0.075}{(0.25)^2} = \frac{0.075}{0.0625} = 1.2\ L/mols$$

63. C The rate of a chemical reaction increases as temperature

Telegram: https://t.me/ExtremeSeries

increases beacause as temperature increases, the average kinetic energy of the reacting particles increases which inturn increases the collision of the reacting species.

64. B $A = \ell \times h$ $\quad V = \frac{m}{d}$

$V = \ell \times w \times h$

$V = A \times w \quad A \times w = \frac{m}{d}$ where $m = mass \quad A = area$ $d = density, w = width$

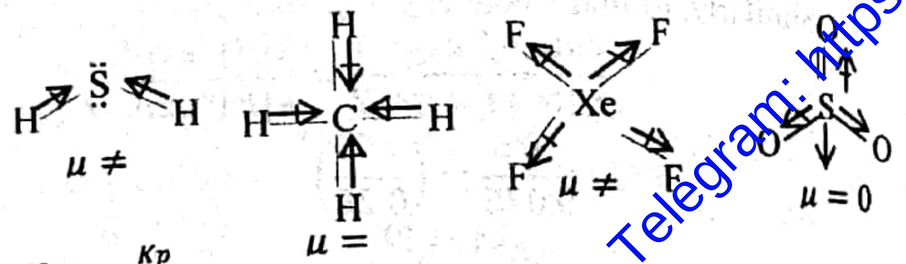
$\Rightarrow w = \frac{m}{Ad} = \frac{10.35g}{24cm \times 31cm \times 2.7g/m^3}$

$= 0.00515cm$

$= 0.0515mm = 5.15 \times 10^{-2}mm$

65. D The central idea of quantum theory is that energy is not continuous but comes in discrete packets called quanta (photons), which correspond to small mass and small velocities. Energy is quantized in a whole number multiple of hv,that is , $\Delta E = nhv$ $(n = 1,2,3, - - -)$

66. B A dipole moment (μ) exists when there is electronegativity difference between atoms. A molecule is said to be non-polar if all the dipole moments cancel. If there is a net dipole moment, it is said to be polar. In xeF_4, SO_3 and CH_4 all the diploe moments cancel in H_2S there is a net dipole moment.



67. D $Kc = \frac{Kp}{(RT)^{\Delta n}}$

$2NO(g) + Cl_2(g) \longrightarrow 2NOCl(g)$

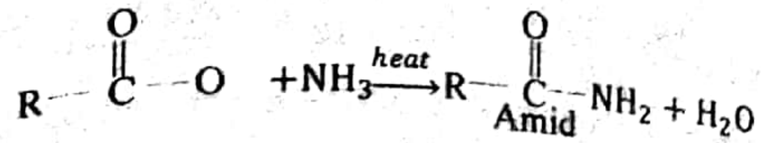
$\Delta n = 2 - (2 + 1) = 2 - 3 = -1$

$K_C = \frac{1.9 \times 10^3 atm^-}{\left(0.082 \frac{L\,atm}{mol\,k} \times 298K\right)^{-1}} = 4.6 \times 10^4$

$K' = \frac{1}{K_c} = \frac{1}{4.6 \times 10^4} = 2.2 \times 10^{-4}$

$K = \frac{1}{k'} = \frac{1}{2.2 \times 10^{-4}} = 0.455 \times 10^4 = 4.6 \times 10^5 L/mol$

68. C



69. A $n_{CH_3OH} = \frac{48}{32} = 1.5$

$n_{H_2O} = \frac{81}{18} = 4.5$

$X_{H_2O} = \frac{1.5}{1.5 + 4.5} = \frac{1.5}{6} = 0.25$

Where n=no of moles

x=mole fraction

70. C $Molality\ (m) = \frac{\frac{actual\ mass}{molar\ mass}}{kg\ of\ solvent}$

$= \frac{3.75g}{\frac{molar\ mass\ (M)}{95}\ \frac{95}{1000}kg}$

$= \frac{3.75}{0.095 \times molar\ mass\ M)}$

$\Delta T_b = K_b m$

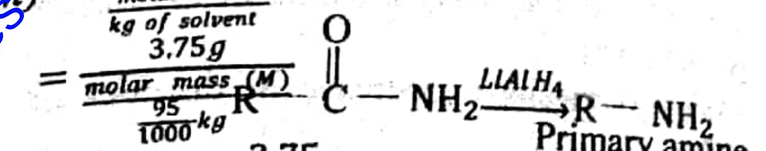
$T_b - T_b^0 = {}_b^k m$, where T_b^0 =boiling point of pure cyclohexane (solvent)

Tb = boiling point of solution

$81.45 - 80.7 = 2.79 \times \frac{3.75}{0.095 \times M}$

$\Rightarrow 0.75 = \frac{2.79 \times 3.75}{0.095 \times M}$

$\Rightarrow M = \frac{2.79 \times 3.75}{0.75 \times 0.095} = 147\ g/mol$



71. B $M_{H_2SO_4} V_{H_2SO_4} = M_{NaOH} V_{NaOH}$

$M_{NaOH} = \frac{M_{H_2SO_4} V_{NaOH}}{V_{NaOH}} = \frac{0.205M \times 50ML}{41mL} = 0.25M$

72. C According to the common ion effect,

$NaA(aq) \longrightarrow Na^+(aq) + A^-(aq)$

$0.2M \qquad 0.2M \qquad 0.2M$

$HA(aq) + H_2O(\ell) \rightleftharpoons H_3O + (aq) + A^-$

Initial,	M	0.2M	0	0.2M
Change,	M	$-x$	$+x$	$+x$
Equilibrium,	M	$0.2 - x$	x	$0.2 + x$

Telegram: https://t.me/ExtremeSeries

$$Ka = \frac{[H_3O^+][A^-]}{[HA]}, \quad 1.80 \times 10^{-5} = \frac{x(0.2+x)}{0.2-x}$$

Assume x is very small, and $0.2 - x \approx 0.2$ and $0.2 + x \approx 0.2$

Hence, $1.80 \times 10^{-5} = \frac{0.2x}{0.2}$

$\Rightarrow x = 1.80 \times 10^{-5} = [H_3O^+]$

73. A Expansion means system loses energy and work (w) is negative $[W < O]$.

- Heat is obsorbed by the system means heat (q) is positive $[q > 0]$.
- Change in internal energy $(\Delta E) = q - w$

$$= (586 - 375)kJ$$

$$= +211kJ$$

74. D Change in entropy (Δs) is a measure of disorder or randomness.

- $(\Delta S) < 0$ means a decrease in disorder or an increase in order.
- $Solid < liquid < gas\ in\ entropy\ (s)$
- In $2SO_2(g) + O_2(g) \longrightarrow 2SO_3(g)$, 3 moles of gas in the left is converted to 2 moles of gas in the right which is decrease in entropy change $(\Delta S < 0)$.

75. C $MnO_4^- + Fe^{2+} \longrightarrow Mn^{2+} + Fe^{3+}$

$Mn^x O_4^{-2-}$

$x - 8 = -1 \Rightarrow x = -1 + 8 = +7$

$Mn^{+7}O_4^- + Fe^{2+} \longrightarrow Mn^{2+} + Fe^{3+}$

$$Mn^{+7} \xrightarrow{+5e^-} Mn^{2+} \quad ----(1)$$

$$Fe^{2+} \xrightarrow{-1e^-} Fe^{3+} \quad ---(2)$$

To cancel the electrons, multiply (2) by 5 and (1) by 1

$$Mn^{+7} + 5e^- \longrightarrow Mn^{2+}$$

$$\underline{5Fe^{2+} - 5e^- \longrightarrow 5Fe^{3+}}$$

$$Mn^{+7} + 5Fe^{2+} \longrightarrow Mn^{+2} + 5Fe^{3+}$$

$\Rightarrow Fe^{2+}$ is oxidizied , and the change in oxidation state is 1.

$\Rightarrow Mn^{+7}$ is reduced, and the change in oxidation state is 5.

$\Rightarrow$ The coefficient of Fe^{2+} is 5 and that of Mn^{+2} is 1.

76. D $M_1V_1 = M_2V_2$

$13M \times V_1 = 0.1M \times 500mL$

$$V_1 = \frac{0.1M \times 500mL}{13M} = 3.85mL\ (volume\ of\ conc.HNO_3)$$

Volume of water required $= 500mL - 3.85mL = 496.15mL$

77. D

	$NaNO_2(aq) \longrightarrow$	$Na^+(aq)$	+	$NO_2^-(aq)$
	$1.3 \times 10^{-3}M$	$1.3 \times 10^{-3}M$		
	$NO_2^-(aq) + H_2O(\ell)$	⟶		$HNO_2(aq) + OH^-(aq)$
Initial, M	$1.3 \times 10^{-3}M$		O	O
Change, M	$-x$		$+x$	$+x$
Equilibrium	$1.3 \times 10^{-3} - x$		x	x

Assume x is small $\Rightarrow 1.3 \times 10^{-3} - x = 1.3 \times 10^{-3}M$

$$K_W = K_aK_b \Rightarrow K_b = \frac{K_W}{K_a} = \frac{1.0 \times 10^{-14}}{5.1 \times 10^{-4}} = 1.96 \times 10^{-11}$$

$$K_b = \frac{[OH^-][HNO_2]}{[NO_2^-]}$$

$$1.96 \times 10^{-11} = \frac{x^2}{1.3 \times 10^{-3}}$$

$$\Rightarrow x^2 = 1.96 \times 10^{-11} \times 1.3 \times 10^{-3}$$

$$= 2.55 \times 10^{-14}$$

$$x = \sqrt{2.55 \times 10^{-14}} = \sqrt{2.55} \times 10^{-7}$$

$\Rightarrow x = [OH^-] = 1.6 \times 10^{-7}M$

$\Rightarrow P^{OH} = -log[OH^-] = -log1.6 \times 10^{-7} = 7 - log1.6 = 6.7$

$P^H = 14 - P^{OH} = 14 - 6.7 = 7.3$

78. A Electrolysis of Na_2SO_4 solution

$Na_2SO_4 \longrightarrow 2Na^+ + SO_4^{2-}$

$2H_2O \longrightarrow H_3O^+ + OH^-$

Cathode rxn

$2H_3O^+ + 2e^- \longrightarrow 2H_2O + H_2$ *OR*

$4H_2O + 4e^- \longrightarrow 2H_2 + 4OH^-$ *(water is reduced)*

Anode rxn

$4OH^- - 4e^- \longrightarrow 2H_2O + O_2$ *OR*

$6H_2O \longrightarrow O_2 + 4H_3O^+ + 4e^-$ *(water is oxidized)*

Hence, the overall reaction is electrolysis of water $(2H_2O \longrightarrow 2H_2 + O_2)$. That is, H_2 is liberated at the cathode and O_2 is liberated at the anode.

79. A Caprolactam polymerizes to Nylon -6 on heating in water.

80. A The oxidized substance is the anode of the cell (left) and the reduced substandce is the cathode of the cell (right).

$A(s) \longrightarrow A^{2+}(aq) - oxidized\ (anode)$

$B^{2+}(aq) \longrightarrow B(s) - reduced\ (cathode)$

The cell notation is

$$A(s)/A^{2+}(aq)\ /B^{2+}(aq)/B(s)$$

Anode *Cathode*

Telegram: https://t.me/Extremeseries

BOOKLET CODE :-29

1. **D** According to Bronsted-Lowry definition of acids and bases, an acid is a proton donor and a base is a proton acceptor. H^+ is a proton. After the acid has donated a proton, the remaining part can be a proton acceptor and is called a conjugate base. When a base accepts a proton, the species formed is called conjugate acid.

$$Acid + Base \rightleftharpoons Conjugate\ Base + Conjugate\ Acid$$

$$HNO_3 + H_2O \rightleftharpoons NO_3^- + H_3O^+$$

$$H_2SO_4 + H_2O \rightleftharpoons HSO_4^- + H_3O^+$$

$$NH_3 + H_2O \rightleftharpoons NH_2^- + H_3O^+$$

Hence $HNO_3/NO_3^-, H_2SO_4/\ H_2SO_4^-$ *and* NH_3/NH_2^- are conjugate acid- base pairs.

H_3O^+/OH^- is not a conjugate acid-base pair but H_2O/OH^- is a conjugate acid-base pair.

Eg:- $\underset{Acid}{H_2O} + \underset{Base}{NH_3} \rightleftharpoons \underset{confugate\ acid}{NH_4^+} + \underset{conjugate\ base}{OH^-}$

2. **A** $1L=1dm^3=0.001m^3=1000mL=1000cm^3$
3. **A** Hydrogenation is the addition of hydrogen at high pressure in the presence of Ni or Pd catalyst. It refers to hardening of oils. It converts oils (unsaturated) to solid fats (saturated).
4. **A** Aspirin is made from salicylic acid and acetic acid. It is a pain killer and its chemical name is Acetylsalicylic acid (ASA).
5. **C** Ideal solutions are characterized by the following properties.
   i) The volume of an ideal solution is the sum of the volumes of its pure components.
   ii) The heat of solution is zero.
   iii) The vapor pressure above the solution equals the sum of the individual partial vapor pressures.
   iv) The intermolecular forces among solute-solute solvent-solvent, and solute-solvent molecules are equal.
   v) The solutions obey Raoult's law.
   vi) The components have similar chemical structures.

Telegram: https://t.me/ExtremeSeries

6. **A** A cubic closest packing (ccp) has faces containing four corners. Since there is an anion at each corner and at the center, we have $4+1=5$ anions. Since there are six faces in a cube and there is one cation at the center of each face, we have $6\times1=6$ cations.
7. **C** When the concentration of B is doubled, the rate is doubled implies the rxn is first order with respect to B. When the concentrations of both reactants (A and B) are doubled, the rate increases by a factor of 8 implies the overall order of the reaction is 3. Because, if the concentration of the reactants is increased by a factor of x, the rate increases by a factor of x^a where a is rxn order.

   ✓ Doubled $\Rightarrow x = 2$
   ✓ A factor of 8 $\Rightarrow x^a = 8$
   $\Rightarrow 2^a = 2^3$
   $\Rightarrow a = 3$
   $Rate = k[A]^2[B]$

8. **D** The curve containing the points 3,4 and 7 represents the boundary between solid and liquid. Along this curve solid and liquid phase coexist and hence known as fusion or melting curve. Point 3 is the triple point in which the solid, liquid and gaseous phase of a substance coexist.
   At point 4, only the solid and liquid phases can coexist.
9. **A** The energy difference associated with jump from initial state (n_i) to the final state (n_f) becomes smaller and smaller as the distance from the nucleus increases.

$$\Delta E = -2.18\times10^{-18}J\left(\frac{1}{n_f^2}-\frac{1}{n_i^2}\right)$$

$$n = 2 \rightarrow n = 1$$
$$= -2.18\times10^{-18}J\left(\frac{1}{1^2}-\frac{1}{2^2}\right)$$
$$= -2.18\times10^{-18}J\left(1-\frac{1}{4}\right)$$
$$= 0.75\times-2.18\times10^{-18}J$$
$$= -1.635\times10^{-18}J\ (largest\ energy)$$
$$n = 4 \rightarrow n = 2$$
$$\Delta E = -2.18\times10^{-18}J\left(\frac{1}{4}-\frac{1}{16}\right)$$
$$= -2.18\times10^{-18}\times0.1875J$$

$$= -4.0875 \times 10^{-19} J$$

$n = 6 \rightarrow n = 3$

$$\Delta E = -2.18 \times 10^{-18} J \left(\frac{1}{9} - \frac{1}{36}\right)$$
$$= -2.18 \times 10^{-18} J \times 0.082$$
$$= -1.7876 \times 10^{-19} J$$

$n = 7 \rightarrow n = 6$

$$\Delta E = -2.18 \times 10^{-18} J \left(\frac{1}{36} - \frac{1}{49}\right)$$
$$= -2.18 \times 10^{-18} J \times 8 \times 10^{-3}$$
$$= -17.44 \times 10^{-21}$$
$$= -1.744 \times 10^{-20} J$$

N.B:- The minus sign indicates that energy is emitted.

10. B Using the criss-cross method,

$Mn^{4+} \not{2} \not{4} O^{2-} \Longrightarrow Mn_2O_4 \Longrightarrow MnO_2$

11. C $E = \frac{hc}{\lambda} = \frac{6.626\times10^{-34} Js \times 3.0\times10^{8}\frac{m}{s}}{300\times10^{-9} m}$

$$= \frac{19.878 \times 10^{-26}}{300 \times 10^{-9}}$$
$$= 0.06626 \times 10^{-17}$$
$$= 6.626 \times 10^{-19} J$$

$$E = \frac{hc}{\lambda} = \frac{6.26 \times 10^{-34} Js \times 3.0 \times 10^{8}\frac{m}{s}}{600 \times 10^{-9} m}$$
$$= 3.313 \times 10^{-19} J$$

$\Rightarrow \frac{6.626\times10^{-19} J}{3.313\times10^{-19} J}$

$\Rightarrow$ 2:1

Or

Let $E_1 = \frac{hc}{\lambda_1}$ *and* $E_2 = \frac{hc}{\lambda_2}$

Where λ_1=300 nm , λ_2=600nm

$$\frac{E_1}{E_2} = \frac{\frac{hc}{\lambda_1}}{\frac{hc}{\lambda_2}} = \frac{\frac{1}{\lambda_1}}{\frac{1}{\lambda_2}} = \frac{1}{\lambda_1} \times \frac{\lambda_2}{1} = \frac{\lambda_2}{\lambda_1}$$
$$= \frac{600nm}{300nm}$$
$$= \frac{2}{1} \Longrightarrow 2:1$$

12. D Chlorine (Cl_2) is disinfectant. That is, it kills germs which are found in water. But chloride ions (Cl^-) are poisonous.

13. B During the titration of a known volume of a strong acid with a strong base

- ✓ The PH is low at the beginning of the titration.
- ✓ The PH changes slowly until just the end point.
- ✓ The PH rises sharply around the end point.
- ✓ The PH rise slowly beyond the end point.
- ✓ Indicators having a PH range from about 4 to 10 can be used to detect the end point.

14. B PH=7 is neutral
PH>7 is basic
PH<7 is acidic
PH=7.5 is slightly basic

15. A Electrolysis of molten (fused) NaCl produces Na at the Cathode and Cl_2 at the anode.

$NaCl \rightarrow Na^+ + Cl^-$

Cathode rxn: $Na^+ + e^- \rightarrow Na$

Anode rxn: $Cl^- \rightarrow \frac{1}{2}Cl_2 + e^-$

Overall rxn: $Na^+ + Cl^- \rightarrow Na + \frac{1}{2}Cl_2$

16. B $2C_4H_{10}(g) + 13O_2(g) \longrightarrow 8CO_2(g) + 10H_2O(l)$

$$n_{C_4H_{10}} = \frac{11.6}{58} = 0.2\ mol$$

$$n_{O_2} = \frac{11.6}{32} = 0.3625\ mol$$

From the balanced chemical equation,

$2mol\ C_4H_{10} \longrightarrow 13\ mol\ O_2$

$0.2\ mol C_4H_{10} \longrightarrow x$

$$x = \frac{0.2 \times 13}{2} = 1.3\ mol O_2$$

But we have only 0.3625 mol of O_2 showing that O_2 is the limiting reagent (limiting reactant).

Hence, O_2 determines the amount of products (CO_2 *and* H_2O) to be formed.

17. C Mass of solution=Mass of solute(salt) +Mass of solvent (water)
=100g+100g
=200g

Telegram: https://t.me/ExtremeSeries

18. **A** Intensive properties – do not depend on the amount of matter present.
Eg:- Color, temperature, density.
Extensive properties-depend on the amount of matter present.
Eg:- Mass, internal energy, volume, pressure, enthalpy

19. **C** Cotton, wool silk, carbohydrates and proteins are well known natural polymers.
✓ Nylon, pvc, Dacron, Perspex, polyethylene, Bakelite, polyurethane, etc are synthetic polymers(artificial polymers)

20. **C** The oxide of nickel is reduced with carbon and refined electrolytically or reduced and purified by the mond process. In the Mond process CO at 80°C is passed over the impure nickel to form volatile nickel carbonyl, $Ni(CO)_4$ a poisonous liquid of boiling point 43°C.

21. **C** $_{13}Al=1S^22S^22P^63S^2\ 3P_x^1 3P_y^0 3P_z^0$ (1 *unpaired* e^-)
$_{14}Si=1S^22S^22P^63S^2\ 3P_x^1 3P_y^1 3P_z^0$ (2 *unpaired* e^-s)
$_{15}P=1S^22S^22P^63S^2\ 3P_x^1 3P_y^1 3P_z^1$ (3 *unpaired* e^-s)
$_{16}S=1S^22S^22P^63S^2\ 3P_x^2 3P_y^1 3P_z^1$ (2 *unpaired* e^-s)
⇒ $_{15}P$ has the greatest number of unpaired e-s.

22. **A** In a chemical rxn at equilibrium,
✓ The forward and backward rxn rates proceed at equal rates.
✓ The rxn takes place in a closed vessel.
✓ All reactants and products exist at equilibrium.
✓ Temperature, pressure, volume and concentration do not change with time.
✓ The reactant and product concentration are not equal.
✓ Equilibrium is dynamic in nature.
✓ Equilibrium state can be found either from the reactant or product side.
✓ Free energy change (ΔG) is zero.

23. **A** In Na_2CO_3, the bond between Na^+ and CO_3^{2-} is ionic, and within CO_3^{2-} the bond is covalent.
✓ $MgBr_2$ contains only ionic bond.
✓ CH_2Cl_2 and CH_3COOH contain only covalent bond.

24. **B** n=1 ⟹l=0 because l has values from o to n-1
✓ $l = 0 \Longrightarrow m_l = 0$ because m_l has values from $-l$ to $+l$.

✓ $m_s = \pm\frac{1}{2}$
✓ $1,1,0,\frac{1}{2}$ is not possible since when n=1, l=0. Thus, 1,0,0,1/2 and 1,0,0,-1/2 are possible.
✓ $n = 2 \Rightarrow l = 0,1\ \&\ l = 1 \Rightarrow m_l = -1,0,1\ \&\ m_s = \pm\frac{1}{2}$
Hence 2,1,-1,1/2 is possible.

25. **A** ^1_1H = 1 proton, 1 electron, 0 neutron (more electron than neutron)
✓ $^{35}_{17}Cl^-$ = 17 protons, 18 electrons, 18 neutrons.
✓ $^{39}_{19}K^+$ = 19 protons, 18 electrons, 20 neutrons.

26. **A** Pentatomic molecules and ions having no lone pairs on the central atom have tetrahedral shape.
Eg:- $SiCl_4, XeF_4^-, NH_4^+, CH_4, CCl_4, etc$ XeF4 has two lone pairs on the central atom Xe and hence has square planer shape.

27. **D** According to the electron – sea model metals have the ability to deform (that is they are malleable and ductile) because metal ions can move without any specific bonds being broken. Meaning they can slip over each other fairly easily. Hardness depends on the no of available mobile electrons. During deformation of metals, new metallic bonds are formed on being distorted.

28. **D** $1S^22S^22P^63S^23P^63d^{10}4S^24P^64d^65S^2$
⇒ 2,8,18,14,2
⇒ The element has incomplete inner shell, that is 14 instead of 32. Hence, the element is transition metal or d-block. It is known as Ruthenium (Ru).

29. **D** The rate of reaction is directly proportional to the number of collisions per second (the frequency of collision). For effective collision, the reacting particles must collide with sufficient energy to break or rearrange bonds. The reacting particles must also be in proper orientation (geometry).
Activation energy for activated complex formed in linear orientation is less than that is formed in bent orientation. Hence linear orientation is favored over bent orientation.

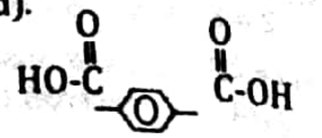
30. **B** In CO_2, $\ddot{O}=C=\ddot{O}$, there are two double bonds. One double bond contains one sigma (δ) and one Pi (π) bonds. Totally, there are 2 sigma bonds and two pi bonds.
N.B:- ***A single bond contains only sigma bond.***

Telegram: https://t.me/Extremeseries

A double bond contains one sigma and one Pi bonds.
A triple bond contains one sigma and two Pi bonds.

31. A Polyesters (Dacron or Terylene) are produced by condensation of ethylene glycol (a di functional alcohol) with terephthalic acid (a difunctional organic acid).

$HO-CH_2-CH_2-OH$
Ethyleneglycol
(1,2 – ethanediol)



Terephthalic acid
(1,4-benzenedicarboxylic acid)

32. B In the reaction $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g);\ \Delta H^0 = -197KJ.$

- ✓ The forward rxn is exothermic. Increasing temperature decomposes SO_3 to SO_2 and O_2. Adding a catalyst has no net effect on the position of the equilibrium.
- ✓ Decreasing the volume means increasing the pressure in line with Boyle's law because volume and pressure are inversely related.
- ✓ increasing the pressure (decreasing the volume) shifts the equilibrium in the direction toward smaller number of molecules (to the right) and increases the amount of SO_3.

33. A Carboxylic acids contain the carboxyl (hydroxyl) groups their general formula is

OR $R-COOH$

E.g:- $R-C(=O)-O-H$ $CH_3-C(=O)-O-H$ Or CH_3COOH

34. A In chemical kinetics regarding reaction mechanisms, the slowest step is the rate determining step.

N.B:- $HCOOH + I^- \rightarrow HOI + OH^-$ (slow)

The rate law depends on the slowest rxn. Thus, Rate=K[HCOOH][I^-]

35. A Solids and Liquids don not appear in the equilibrium constant expression. In the rxn:

$2Hg(g) + O_2(g) \rightleftharpoons 2HgO(s)$

$$K = \frac{[HgO]^2}{[Hg]^2[O_2]} = \frac{1}{[Hg]^2[O_2]}$$

Telegram: https://t.me/ExtremeSeries

36. B Fats and oils are triesters of glycerol (triglycerides/ triacylglycerol).

- ✓ Fats are derived from saturated fatty acids and glycerol and are solid esters.
- ✓ Oils are derived from unsaturated fatty acids and glycerol and are liquid esters.

37. D Esters are compounds having the general formula $R-C(=O)-O-R'$ R=H, alkyl or aryl group R'=alkyl or aryl group

$H-C(=O)-O-OCH_3$ is an ester known as methanoate.

NB:- $CH_3OC_2H_5$- Ethylmethylether methyl.
C_2H_5CHO – Propionaldehyde
CH_3COOH- Ethanoic acid

38. B When a solid dissolves in a liquid, the solute particles must separate, so energy must be added (endothermic process) indicating . $\Delta H^0 Solute > 0$. In most cases, the solubility of a solid substance increases with increase in temperature. Strong ion-dipole force exist in the solution showing that higher heat energy is needed to break them,

39. No answer

$$m = \frac{actual\ mass}{molar\ mass \times kg\ of\ solvent}$$

$$= \frac{250g}{194.2\frac{g}{mol} \times 1kg}$$

$$= 1.29\frac{mol}{kg}$$

$$= 1.29\ m$$

$$\Delta T_f = k_f m$$

$$= 1.86^0 C/m \times 1.29m$$

$$= 2.40^0 C$$

$$\Delta T_f = T_f^0 - T_f \Rightarrow T_f = T_f^0 - \Delta T_f$$

$$= 0 - 2.40$$

$$= -2.40^0 C$$

$M = molarity$

40. C $M = \frac{n}{V(l)}$ where $V = volume\ in\ liters$

$n = no\ of\ moles$

$n = M \times V(L)$

$= 0.5\frac{mol}{L} \times 2.5L$

$= 1.25\ mol$

41. A A Lewis acid is any chemical species that is electron deficient and can accept electron pairs (s). It can be positively charged or a molecule. Eg:- $SiCl_4$, PF_5, Cu^{2+}, Ni.
- ✓ A Lewis base is any chemical species that has an excess of electrons and can donate electron pairs(s). It can be a molecule or negatively charged species.
  Eg:- Cl^-, F^-, NH_3,H_2O,CO
  NB:- Lewis acid-base reactions form chemical species having coordinate covalent bond.
  $NH_3 + BF_3 \rightarrow NH_3BF_3$
  $PCl_3 + O \rightarrow POCl_3$

42. A A universal indicator is a mixture of indicators that gives a different color for a different PH.
- ✓ It gives a yellow orange color in weak acids and red in strong acids. It gives a blue color in weak base and purple in strong bases.

43. C The larger the Ka value, the stronger the acid.
$K_a(HC)[= 1 \times 10^{-6}] < K_a(HA)[= 1 \times 10^{-5}] < K_a(HB)[= 2 \times 10^{-5}]$

44. B If the solute-solvent interactions are greater than the solute-solute or solvent-solvent interactions, the vapor pressure of the solution is less than the sum of the vapor pressures as predicted by Raoult's law which is negative deviation and the mixing is an exothermic process.

45. B A buffer solution is formed between a weak acid and its conjugate base or a weak base and its conjugate acid. It resists PH changes.
Eg:- 1) CH_3COOH (weak acid) and CH_3COONa(CH_3COO-) is the conjugate of CH_3COOH)
Eg:- 2) NH_3 (weak base) and NH_4Cl (NH_4^+ is the conjugate base acid of NH_3)

46. B $2H^+(aq) \rightarrow H_2(g)$(*decrease in oxidation number*)
$2Br^-(aq) \rightarrow Br_2(g)$(*increase in oxidation number*)
$SO_3(g) \rightarrow SO_4^{2-}(aq)$(*decrease in oxidation number*)
$MnO_2(s) \rightarrow Mn^{2+}(aq)$(*decrease in oxidation number*)
$\Rightarrow$ Br- is oxidized and requires oxidizing agent.

47. D $2NaOH + H_2SO_4 \rightarrow Na_2SO_4 + H_2O$

$n_{H_2SO_4} = 25.0 \times 10^{-3}L \times 1.2\frac{mol}{L}\left(\frac{mol}{L} = M\right)$

$= 30 \times 10^{-3}\ mol$

$= 0.03\ mol$

$2mol\ NaOH \rightarrow 1mol H_2SO_4$ (from balanced chemical equation)

$x \leftarrow 0.003mol H_2SO_4$

$x = \frac{2 \times 0.03}{1} = 0.06\ mol\ NaOH$

$V_{NaOH} = \frac{n_{NaOH}}{M_{NaOH}} = \frac{0.06mol}{0.5\frac{mol}{L}} = 0.12L$

$\Rightarrow 0.12L \times \frac{1000mL}{1L} = 120mL$

48. D $\Delta E = q + w$
- ✓ System releases heat means q<0.
- ✓ Work is done by the system means W<0
- ✓ $\Delta E = -q - W = -135kJ - 63kJ = -198kJ$
- ✓ $\Delta H = q_p$ (*constant pressure process*)
  since $q < 0, \Delta H = -q_p = -135kj$

49. D $N^xH^{+1} \Rightarrow x + 4 = 0 \Rightarrow x = +4$
$HN^{+1}{}^x{}^{-2}O_3 \Rightarrow 1 + x - 6 = 0 \Rightarrow x - 5 = 0 \Rightarrow X = +5$
NB:- The oxidation number of H=+1 except in hydrides where it is -1. The oxidation number of O=-2 except in peroxides and superoxides where it is -1 and -1/2 respectively.

50. D $2FeS(s) + 3CO_2(g) \rightarrow Fe_2O_3(s) + 3CO(g); \Delta H^0 = +26.6kJ$ For 2 moles of Fe reacted, 26.6 kJ of energy is absorbed. Hence 13.3kJ of energy is absorbed for every (one) mole of Fe reacted.

51. A Cu is extracted by thermal reduction that is its ore is roasted (heated to a high temperature), in a limited supply of air.
- ✓ Fe is extracted from its ore by chemical reduction using CO in the blast furnace.
- ✓ Al and Mg are extracted by electrolytic reduction.

52. C $E^0_{cell} = E^0_{cathode} - E^0_{anode}$
- ✓ The more positive E^0 is the cathode of the cell.
- ✓ $E^0_{cell} = 0.49 - (-0.76) = 1.25V$
- ✓ $\Delta G^0 = -nFE^0_{cell}$ where n = no of electrons involved

$= -2 \times 96{,}500\frac{j}{vmol} \times 1.25V$

Telegram: https://t.me/Extremeseries

$$= -241250\frac{J}{mol}$$
$$= -241.25\frac{kJ}{mol}$$

53. **D** Nylon 6 is made from caprolactam.
- ✓ Nylon 66 is made from hexamethylene diamine and adipic acid.
- ✓ Terylene is made from ethylene glycol and terephthalic acid.
- ✓ Teflon is made from tetrafluoroethylene.

54. **A** For a reaction to be a redox reaction, there must be a change in oxidation number.
- ✓ In the rxn $Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl(s)$ there is no change in oxidation number.

NB:- Acid-base rxns and double displacement rxns are not redox rxns as there is no change in oxidation number.

55. **C** $m = \frac{ItE}{F} = \frac{60A\times2\times10^3s\times E_{Mg}}{96{,}500\ C}$

$$E_{Mg} = \frac{24}{2} = 12g$$
$$\Rightarrow m = \frac{60A\times2\times10^3s\times12g}{96{,}500\ As} \quad (C = AS)$$
$$\approx 15g$$

56. **D** Assume 100g of solution.

Mass of H_2O=100 − 70 = 30g = 0.03kg

$$m = \frac{actual\ mass}{molar\ mass \times kg\ of\ solvent}$$
$$= \frac{70g}{63\frac{g}{mol}\times 0.03\ kg}$$
$$= \frac{70}{1.89}$$
$$= 37.0\frac{mol}{kg} = 37.0\ m$$

57. **B** Polysaccharides such as starch and cellulose are natural polymers.
- ✓ Polyethene, nylon and terylene are synthetic polymers.

58. **A** $E = \frac{hc}{\lambda} = \frac{6.626\times10^{-34}Js\times3\times10^{8}\frac{m}{s}}{276\times10^{-9}m}$

$$= \frac{6.626\times3\times10^{-17}}{276}$$
$$= 0.072\times10^{-17}$$
$$= 7.2\times10^{-19}J$$

59. **A** Cl^- has 18 electrons. Its electronic configuration is $1s^22s^22p^63s^23p^6$

60. **B** When multiplying or dividing measured quantities the product or the quotient is reported in the measurement with the least number of significant figures.

0.68=Two significant figures

14.364= Five significant figures

The answer should be reported as 0.047 (two significant figures)

61. **C** B(Z=5)

$E^- = 2\times5 - 1 = 10 - 1 = 9$ *electrons*

The molecular electronic configuration is

$\delta_{1s}^2\delta_{1s}^{*2}\alpha_{2s}^2\delta_{2s}^{*2}\pi_{2p_y}^1$

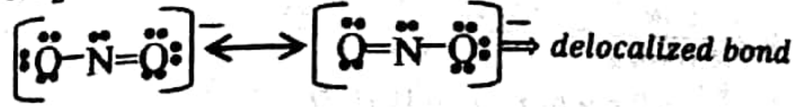
NB:-For simple homonuclear diatomic molecules having total electrons ≤ 14, the order *is*

$$\left.\begin{matrix}\alpha_{1s}\alpha_{1s}^*\alpha_{2s}\alpha_{2s}^*\pi_{2p_y}\\ \pi_{2p_z}\end{matrix}\right\}\delta_{2px}\left.\begin{matrix}\pi_{2p_y}^*\\ \pi_{2p_z}^*\end{matrix}\right\}\delta_{2px}^*$$

For simple homonuclear diatomic molecules having ≥14 electrons, the order is

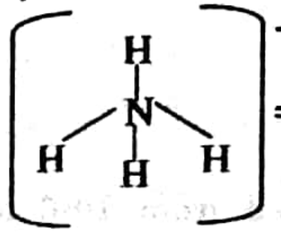
$$\left.\begin{matrix}\delta_{1s}\delta_{1s}^*\delta_{2s}\delta_{2s}^*\delta_{2px}^*\pi_{2py}\\ \pi_{2p_z}\end{matrix}\right\}\left.\begin{matrix}\pi_{2p_y}^*\\ \pi_{2p_z}^*\end{matrix}\right\}\delta_{2px}^*$$

62. **A** $NO_2^- = 5 + 2\times6 + 1 = 18e^-s$



delocalized bond

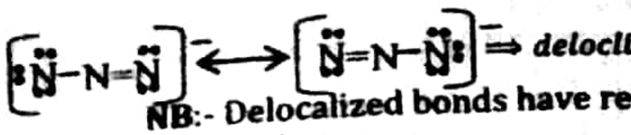
$NH_4^+ = 5 + 4 - 1 = 8e^-s$



$\Longrightarrow$ *No delocalized bond*

$N_3^{-1} = 3\times5 + 1 = 16e^-s$

Telegram: https://t.me/ExtremeSeries


⇒ *deloclized bond*

NB:- Delocalized bonds have resonance structures.

63. **D** $H_2 = 2 \times 1 = 2$ *electrons*

$H_2^+ = 2 \times 1 - 1 = 2 - 1 = 1$ *electron*

$H_2^- = 2 \times 1 + 1 = 3$ *electrons*

$$\text{Bond order} = \frac{1}{2}(n_b - n_a) \text{ where}$$

$n_b = no$ *of bonding electrons*

For H_2, $\delta_{1s}^2\ \delta_{1s}^{*0}$ $n_a = no$ *of antibonding electrons*

$$\text{Bond order} = \frac{1}{2}(2-0) = \frac{1}{2} \times 2 = 1$$

For H_2^+, $\delta_{1s}^1\ \delta_{1s}^{*0}$

$$\text{Bond order} = \frac{1}{2}(2-1) = \frac{1}{2} \times 1 = 0.5 = \frac{1}{2}$$

64. **B** $_{24}Cr = 1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$

⇒ There are five 3d electrons in the ground state.

65. **A** $CH_3CH_2CH_3 < CH_3COCH_3 < CH_3CH_2CH_2OH$ in melting point. $CH_3CH_2CH_3$ is non polar CH_3COCH_3 and $CH_3CH_2CH_2OH$ are polar. Moreover, $CH_3CH_2CH_2OH$ has hydrogen bonding and has the highest melting point.

66. **A** $\log \frac{[CH_3NO_2]_0}{[CH_3NO_2]_t} = \frac{kt}{2.303}$

We can find k from the half-life $t_{\frac{1}{2}} = \frac{0.693}{k}$

$$\Rightarrow k = \frac{0.693}{t_{\frac{1}{2}}} = \frac{0.693}{650s} = 1.06 \times 10^{-3} s^{-1}$$

$$\Rightarrow \log \frac{0.5M}{[CH_3NO_2]_t} = \frac{1.06 \times 10^{-3} s^{-1} \times 1300s}{2.303}$$

$$\Rightarrow \log \frac{0.5}{x} = 0.6 \text{ where } x = [CH_3NO_2]_t$$

$$\frac{0.5}{x} = antilog\ 0.6$$

$$\frac{0.5}{x} = 3.98$$

$$\Rightarrow x = \frac{0.5}{3.98} = 0.125\ M$$

67. **D** When the temperature is increased from 10°C to 20°C it doubles.

$$10^0C \xrightarrow{2^1} 20^0C \xrightarrow{2^2} 30^0C \xrightarrow{2^3} 40^0C \xrightarrow{2^4} 50^0C$$

Telegram: https://t.me/ExtremeSeries

$$\xrightarrow{2^5} 60^0C \xrightarrow{2^6} 70^0C \xrightarrow{2^7} 80^0C \xrightarrow{2^8} 90^0C \xrightarrow{2^9} 100^0C$$

This implies that the rate of the reaction raises by a factor of doubling raised to mine.

$$\Rightarrow 2^9 = 512 \text{ times}$$

68. **C**

$$CO_2(g) + H_2(g) \rightleftharpoons CO(g) + H_2O(g)\ K$$

$$Fe(s) + CO_2(g) \rightleftharpoons FeO(s) + CO(g)\ K_1$$

$$Fe(s) + H_2O(g) \rightleftharpoons FeO(s) + H_2(g)\ K_2$$

$$K = \frac{[CO][H_2O]}{[CO_2][H_2]}$$

$$K_1 = \frac{[CO]}{[CO_2]}$$

$$K_2 = \frac{[H_2]}{[H_2O]}$$

$$\frac{K_1}{K_2} = \frac{[CO]}{[CO_2]} \times \frac{[H_2O]}{[H_2]} = K$$

69. **B** Fluorinated carboxylic acids are strong acids. For example CF_3COOH has $Ka = 5.9 \times 10^{-1}$ where as CH_3COOH has $Ka = 1.8 \times 10^{-5}$ Thus, the order is:

$FCH_2COOH > CH_3COOH > Cl_2CHCOO > Cl_3CCOOH$

70. **No answer**

$$CH_3COOH + C_2H_5OH \rightleftharpoons CH_3COOC_2H_5 + H_2O;\ K_c = 4$$

$$CH_3COOC_2H_5 + H_2O \rightleftharpoons CH_3COOH + C_2H_5OH;\ K_c = \frac{1}{4}$$

$$K_c^1 = \frac{[CH_3COOH][C_2H_5OH]}{[CH_3COOC_2H_5]}$$

$$n_{CH_3COOH} = \frac{15}{60} = 0.25\ mol$$

$$n_{C_2H_5OH} = \frac{11.5}{46} = 0.25\ mol$$

Assume the volume is 1 L.

$$[CH_3COOH] = \frac{0.25mol}{1L} = 0.25\ M$$

$$[C_2H_5OH] = \frac{0.25}{1L} = 0.25M$$

$$[CH_3COOC_2H_5] = \frac{x mol}{1L} = xM$$

$$\frac{1}{4} = \frac{0.25 \times 0.25}{x}$$

$$0.25 = \frac{0.25 \times 0.25}{x}$$

$$x = \frac{0.25 \times 0.25}{0.25}$$

$$= 0.25M$$

Since we assumed 1L of volume,

$$x = \frac{0.25 mol}{1L} = 0.25\ mol$$

$$m_x = m_{CH_3COOC_2H_5} = 0.25\ mol \times 88 \frac{g}{mol}$$

$$= 22g$$

71. A Open system exchanges mass and energy in the form of heat with its surroundings.

- ✓ Closed system transfers energy but not mass with its surroundings.
- ✓ Isolated system does not transfer mass and energy with its surroundings.

72. B The per cent ionization of a weak acid depends on the concentration of the ionized acid per its initial concentration.

- ✓ Moreover, the acid dissociation constant (ka) is a quantitative measure of the strength of the acid and decreases with increasing concentration of the weak acid. That is, for the weak acid HA

$$K_a = \frac{[H_3O+][A^-]}{[HA]}$$

$\Rightarrow K_a \alpha \frac{1}{[HA]}$, meaning the higher the concentration of the weak acid, the lower is its Ka value.

73. C Using Hess's Law reverse the 1st reaction and add the two reactions to get the required reaction.

$$S(s) + O_2(g) \rightarrow SO_2(g);\ \Delta H = -395\ kJ$$

$$SO_2(g) \rightarrow S(g) + O_2(g);\ \Delta H = +618KJ$$

$$S(s) \rightarrow S(g);\ +223KJ$$

74. D $Carboxylic\ acid + Alcohol \rightleftharpoons Ester + water$

$$RCOOH + R - OH \rightleftharpoons RCOOR + H_2O$$

$$CH_3COOH + CH_3CH_2CH_2OH \rightleftharpoons CH_3COOCH_2CH_2CH_3 + H_2O$$

$$\Rightarrow \quad CH_3 - \overset{O}{\overset{\|}{C}} - OCH_2CH_2CH_3$$

73. B $\text{Molarity (M)} = \frac{\text{actual mass (m)}}{\text{molar mass} \times \text{volume of solution (L)}}$

$m = M \times molar\ mass \times volume\ (L)$

molar mass of $Pb(NO_3)_2$=331.2g/mol

$$m = 0.1 \frac{mol}{L} \times 331.2 \frac{g}{mol} \times 5.0L$$

$= 165.6g$

75. B In the electrolysis of concentrated NaCl solution (Brine), H_2 is liberated at the cathode and Cl_2 is liberated at the anode. NaOH is produced in the electrolytic solution. NaOH is manufactured industrially by this process. Dilute NaCl solution cannot produce NaOH as the OH^- oxidizes to O_2 at the anode.

77. B $rate = k[A]^x k[B]^y$ where x and y are orders of the reaction with respect to A and B respectively.

To find x, vary $[A]$ keeping $[B]$ constant. Thus, take the 1st and the 3rd data.

$$\frac{rate - 1}{rate - 3} = \left(\frac{0.1}{0.2}\right)^x$$

$$\frac{1 \times 10^{-5}}{1 \times 10^{-5}} = \left(\frac{0.1}{0.2}\right)^x$$

$$1 = \left(\frac{1}{2}\right)^x \Rightarrow x = 0\ because\ \left(\frac{1}{2}\right)^0 = 1$$

$$\Rightarrow order\ of\ A = 0$$

Similarly, to find, Y, vary $[B]$ keeping $[A]$ constant. Thus take the 1st and 2nd data

$$\frac{1 \times 10^{-5}}{4 \times 10^{-5}} = \left(\frac{0.1}{0.2}\right)^y$$

$$\frac{1}{4} = \left(\frac{0.1}{0.2}\right)^y$$

$$\left(\frac{1}{2}\right)^2 = \left(\frac{1}{2}\right)^y \Rightarrow y = 2$$

$$\Rightarrow order\ of\ B = 2$$

Telegram: https://t.me/ExtremeSeries

78. **A** Pure silver has the highest thermal and electrical conductivity of all metals. The next metal having the lowest electrical resistance is copper.

79. **B** Linear polymers have only one long, un branched chain of their molecules. They are held together by London dispersion forces. They are held together by London dispersion forces. They are much like a molecular solid and are flexible, have low strength and melting point.
Cross-linked polymers have molecules bonded to each other and tend to be hard and non-ductile.
HDPE, Bakelite and vulcanized rubber are hard and are cross-linked polymers.
LDPE is relatively soft and is a linear polymer.

80. **A** The most important ore of iron is hematitle (Fe_2O_3).
The most important ore for
- ✓ Aluminum is bauxite ($Al_2O_3.2H_2O$)
- ✓ Copper is chalcopyrite ($CuFeS_2$)
- ✓ Zinc is sphalerite (Zns)

Telegram: https://t.me/Extremeseries

1. What is the process by which water passes across the cell membrane?
A. Active transport C. Facilitated diffusion
B. Osmosis D. Pinocytosis

2. Which of the following is the most important pigment for photosynthesis?
A. Chlorphyll a. C. Carotenoids
B. Chlorophyll b. D. Xanthophylls

3. How many codons are there in the genetic code?
A. 4 B. 16 C. 64 D. 256

4. Which of the following sources of energy is renewable?
A. Mineral coal C. Petroleum
B. Bio-fuel such as ethanol D. Natural gases such as methan

5. Select the kingdom of life in which the cellular organelles are without membranes around them.
A. Plantae C. Monera
B. Protista D. Fungi

6. To which group of compounds do enzymes belong?
A. Lipoproteins C. Globular proteins
B. Fibrous proteins D. Deoxyribose sugars

7. A goose was observed rolling the egg back to its nest. What kind of behaviour it this?
A. Tropism C. Innate behaviour
B. Learned behaviour D. Social behaviour

8. Which area of biology studies how organisms undergo growth, differentiation and morphological changes?
A. Physiology C. Taxonomy
B. Development D. Cytology

9. Which group of organisms show "double fertilization"?
A. Ferns C. Lower plants
B. Bryophytes D. Angiosperms

10. Which mammalian pest has economic importance in Ethiopia?
A. Quelea C. Rodents
B. Beetles D. desert locust

11. How many carbon atoms are contained in two molecules of glucose?
A. 6 B. 12 C. 16 D. 24

12. Of the constituents of protoplasm, which one is inorganic in nature?
A. Starch B. Protein C. Water D. None

13. In a DNA strand of 3000 bases, 40% of the molecules are known to belong to guanine base. How many adenine base molecules are present in the same DNA strand?
A.120 B. 300 C. 600 D. 1200

14. The strand that is complementary to the DNA sequence given as TAACCG.

A. AUUGGC B. ATTGGC C. TAACCG D. GCCAAT

15. Which group of two words mean the same thing?
    A. Autotroph and producer
    B. Autotroph and consumer
    C. Atutotroph and carnivore
    D. Heterotroph and producer
16. In which biome of the world are epiphytes and lianas commonly found?
    A. Savanna B. Desert C. Tundra D. Tropical rain forest
17. In which of the following is the macromolecule correctly matched with its monomer building blocks?
    A. Carbohydrates – Glycerol
    B. Proteins – Fatty acids
    C. Lipids – Amino acids
    D. Nucleic acids - Nucleotides
18. What are GMOs?
    A. Processed foods
    B. Transgenic organisms
    C. Recombinant DNA
    D. General marine organisms
19. Identify from the following a biological control method of malaria
    A. Use of fishes to feed on mosquito larvae
    B. Use of DDT spraying to kill mosquitoes.
    C. Using crop rotation system in farms
    D. Hand picking and physical removal of weeds and affected plants.
20. In humans, what are the chances that the baby at birth would be a boy or a girl?
    A. 1:1 B. 1:2 C. 1:3 D. 3:1
21. Which group of hominids is called the handy human?
    A. Homo sapiens
    B. Homo erectus
    C. Homo habilis
    D. Australopithecus
22. In alternation of generation, which phase of the life cycle of the plant is the diploid generation?
    A. the gametophyte
    B. The gametes
    C. The sporophyte
    D. The protonema
23. Which of the following is true about the meanings contained in the name HIV and AIDS?
    A. HIV refers to the symptoms that characterize the condition
    B. AIDS refers to the causative agent
    C. AIDS refers to the symptoms
    D. HIV refers to its probable origin from chimpanzees
24. Which of the following is the most abundant form of carbohydrate on earth?
    A. Starch B. Cellulose C. Glycogen D. Chitin
25. If an enzyme has its optimum activity at neutral pH, to which of the following pH values does this correspond?
    A. pH 14 B. pH 12 C. pH 7 D. pH 4
26. We do not usually notice that we are wearing clothes even though the clothes are continuously touching our body. To which type of learned behaviours does this belong?
    A. Innate behaviour
    B. Imprinted behaviour
    C. Habituation
    D. Operant conditioning

27. The half life of carbon – 14 is about 5000 years. What percent of the original amount of carbon -14 is expected to be present in a fossil that is 15,000 years old?
    A. 50 B. 25 C. 12.5 D. 6.25
28. How many sperm cells are formed from 10 primary spermatocytes?
    A. 10 B. 20 C. 40 D. 80
29. Which of the following types of forests in Ethiopia is the poorest in the number of tree species?
    A. Podocarpus forests
    B. Eucalyptus forests
    C. Juniperus forests
    D. Broad – leaved forests of the south west
30. Marsupials contain such animals like marsupial mouse and marsupial cat that correspond to placental mouse and placental cat. What phenomenon of evolution is exemplified?
    A. Adaptive radiation
    B. Parallel evolution
    C. Co – evolution
    D. Phyletic evolution
31. Which of the following physiological properties of life is primarily concerned with the perpetuation of species?
    A. Response to stimuli
    B. Respiration
    C. Excretion
    D. Reproduction
32. Which of the following hereditary phenomena was **not** discovered by Mendel?
    A. Linkage of genes
    B. Dominance of genes
    C. Segregation of genes
    D. Independent assortment of genes
33. Which of the following agricultural practices is expected to impose more genetic changes on crops?
    A. Cultivation B. Domestication C. Sowing D. Harvesting
34. The alkaloid that is consumed when you drink tea is
    A. Nicotine B. Cocaine C. Hashish D. Caffeine
35. Identify the disease that is not correctly paired with the substance whose deficiency causes the disease.
    A. Beriberi – Nicotinic acid
    B. Scurvy – Vitamin C
    C. Night blindness – Vitamin A
    D. Rickets – Vitamin D
36. The process in photosynthesis that can occur both in the light as well as in the dark is
    A. photolysis of water
    B. Glucose formation
    C. ATP production
    D. Excitation of electrons from the pigments
37. Which biotechnological process has practical applications in criminal investigation?
    A. Bio-fuel production

Telegram: https://t.me/ExtremeSeries

B. DNA finger printing
C. Single cell protein technology
D. Use of biosensors to test blood glucose level.

38. What is true about the rate of an enzyme catalysed reaction if the enzyme concentration is kept constant and the substrate is supplied in excess? The rate would
A. rise continuously
B. stay constant
C. decline sharply
D. none of the above

39. What would happen if Rh-positive blood is transfused to Rh- negative individual?
A. The red blood cells of the donor will be agglutinated.
B. The red blood cells of the recipient will be agglutinated
C. The recipient will die of erythroblastosis foetalis
D. The recipient will develop anti – Rh antibody.

40. In order for a population to grow at its biotic potential, which one of the following should be available?
A. Strong environmental resistance
B. Maximum density dependent effect
C. Maximum density independent effect
D. Unlimited supply of resources

41. Based on the five steps listed below, choose the alternative that shows the correct sequence of their application in actual research in biology
1. Performing an experiment
2. Conducting preliminary observation about a particular problem in biology.
3. Statement of a biological theory
4. Collecting of experimental data
5. Hypothesis formulation.

A. 1, 2, 3, 4, 5 B. 5, 4, 3, 2, 1 C. 2, 5, 1, 4, 3 D. None of the above

42. What would be the approximate image size of a cell with the size of 0.03 microns viewed under a compound microscope with magnification powers marked as 10X on the eye piece and 100X on the objective lenses?
A. 10 microns B. 20 microns C. 30 microns D. 40 microns

43. In which of their constituents do the nucleotides of DNA and RNA differ from each other?
A. In the purine bases
B. In the pyrimidine bases
C. In the phosphate group
D. In all of the above

44. When does a hypotonic condition exist in the environment of the cell?
A. When there is equal concentration of solutes outside and inside the cell
B. When the solute concentration outside the cell is higher than inside the cell.
C. When the solute concentration inside the cell is greater than outside the cell.

D. When the net movement of water is from the cell to the outside environment

45. On which of the following evolutionary ideas do Lamarck and Darwin fully agree?
A. Life is a result of natural processes
B. Acquired characteristics are inherited
C. Natural selection leads to evolution
D. Evolution occurs by the use – and – disuse of body parts

46. The name given to the pioneer plants that begin primary succession on very dry substratum like bare rock, sand dune or cooled volcanic lava is
A. Hydrophytes B. Mesophytes C. Halophytes D. Xerophytes

47. Of the following alternatives, which one could be taken as a better explanation for an organism's innate behaviour? The behavior is
A. displayed when the organism is subjected to a stimulus that is neither harmful nor rewarding.
B. Seen during the early life of the organism like the emerging chicks that follow the first object they see.
C. Pre-programmed by the organism's genes
D. Seen when an organism is made to associate an accident with a positively reinforcing reward.

48. Identity the ability of the camel that allows it to survive in desert environments where there is low water supply.
A. Fluctuation of its body temperature to save water by reducing evaporation.
B. Evaporate more water through the skin to keep itself cool
C. Sleep during the day and feed at night
D. Move in flocks to avoid evaporation of water from the body

49. One of the following is true about mutualistic interaction between two species.
A. Both species benefit
B. Both species are harmed
C. One species benefits and the second harmed
D. One species benefits and the second remains unaffected

50. In which biome of the world are ephemerals and succulents more common?
A. Savanna B. Desert C. Deciduous forest D. Temperate grassland

Telegram: https://t.me/ExtremeSeries

ETHIOPIAN HIGHER EDUCATION CERTIFICATE BIOLOGY EXAMINATION SENE 2002/JUNE 2010

SUBJECT CODE: 60

Direction: This exam contains a total of 100 questions. Each of the questions is followed by four possible alternatives. Read each question and carefully **blacken** only the letter of your choice on the separate answer sheet provided.

1. What kind of selection does a bell shaped curve in a variable indicate?
   A. Disruptive selection. C. Divergent selection
   B. Unidirectional selection. D. Stabilizing selection.
2. In which aspect of biodiversity are such measures like richness, abundance and taxonomic diversity in populations commonly applied?
   A. Ecosystem diversity. C. Species diversity.
   B. Cultural diversity. D. Genetic diversity.
3. Which of the following elements is found in proteins and nucleic acids but does NOT occur in carbohydrates?
   A. Carbon. B. Nitrogen. C. Oxygen. D. Hydrogen
4. Which of the following is important to regulate the entry and exit of materials into and out of plant cells?
   A. Cell wall. B. Cell membrane. C. Nucleus. D. Central vacuole.
5. Which of the following terms is synonymous with a producer?
   A. Omnivore. B. Herbivore. C. Carnivore. D. Autotroph.
6. Which of the following steps in cellular respiration can take place in the absence of oxygen?
   A. Electron transport. C. Krebs cycle.
   B. Glycolysis. D. Acetyl-CoA formation.
7. Which of the following processes results in a zygote?
   A. Meiosis. B. Fertilization. C. Cleavage. D. Gastrulation.
8. Which of the following molecules is known as the energy currency of the cell?
   A. Glucose B. Starch C. Fat D. ATP
9. Which of the following processes releases oxygen to the atmosphere?
   A. Respiration. C. Transpiration.
   B. Photosynthesis. D. Burning of fossil fuels.
10. The concentration of which of the following gases is on the increase in the atmosphere in recent years than before?
    A. Carbon dioxide. B. Oxygen. C. Nitrogen. D. Ozone.
11. Which is the source of the oxygen that green plants release during photosynthesis?
    A. Sugar. C. Water.
    B. Carbon Dioxide. D. Chlorophyll.
12. All the living components of the cell are collectively known as what?
    A. Cytoplasm. B. Nucleus. C. Cell membrane. D. protoplasm.
13. With which type of agricultural crops are nitrogen fixing bacteria more associated?
    A. Legumes. B. Cereals. C. Vegetables. D. Fruits.

14. Which of the following is found in both plant and animal cells?
    A. Cell wall. B. Chromoplasts. C. Chromosomes. D. Leucoplasts.
15. What is the stored form of sugar that is found in human tissues?
    A. Cellulose. B. Coenzyme. C. Nucleic acid. D. Glycogen.
16. What happens to a human red blood cell when it is placed in a hypertonic solution?
    A. it becomes turgid. C. It might lose water and shrink.
    B. It might swell and burst. D. It will remain unchanged.
17. Choose the final electron acceptor in the electron transport chain during aerobic respiration of eukaryotic cells.
    A. H_2O B. O_2 C. CO_2 D. NADP
18. Choose the cells of the human body that are results of meiosis.
    A. Bone cells. C. Egg cells.
    B. Brain cells. D. Muscle cells.
19. Where in the plant cell does thee Krebs cycle (citric acid cycle) take place?
    A. Nucleus. C. Mitochondrion.
    B. Cytoplasm. D. Chloroplast.
20. Which organelle of the cell has a function of modifying proteins for secretion?
    A. Golgi body. C. Food vacuole.
    B. Ribosome. D. Lysosome.
21. Choose the alternative that illustrates the medical application of biotechnology.
    A. Bio-fuel production.
    B. Biosensors for testing blood glucose level.
    C. DNA finger printing technology.
    D. Protein production from single cells.
22. Which fossil shows the evolutionary link between reptiles and birds?
    A. Archaeopteryx. C. Cyanognathus.
    B. Australopithecus. D. Semouria.
23. Among the following, identify the molecule that is an enzyme.
    A. Cellulase. B. Starch. C. Cellulose. D. Glycogen.
24. What is the process by which cells like amoeba and white blood cells engulf and internalize particles such as bacteria?
    A. Osmosis. B. Pinocytosis. C. Active transport. D. Phagocytosis.
25. An enzyme known as catalase converts hydrogen peroxide into water and oxygen. In this reaction, which one is the substrate?
    A. Catalase. C. Water and oxygen.
    B. Hydrogen peroxide. D. Water only.
26. In a honeybee colony, which one of the following is the duty of the queen?
    A. Foraging. C. Guarding the hive.
    B. Rearing the young. D. Reproduction.
27. To which of the following molecules is most of the energy released during the Krebs cycle transferred?
    A. ATP. B. FADH2. C. NADH D. ADP.
28. Of the following, which cell type is attacked by HIV?
    A. Red blood cell. C. Egg cell.

Telegram: https://t.me/Extremeseries

B. Sperm cell. D. T-helper cell.

29. Among the following traits of the garden pea studied by Gregor Mendel, identify the one that did NOT express itself in the F1 generation?
A. Yellow pod colour. C. Round seed shape.
B. Violet flower colour. D. Axial flower position.

30. In the Miller and Urey's simulation experiment to study organic molecules formation in primeval atmosphere of the earth, all the following gases were used except which?
A. Ammonia. B. Hydrogen. C. oxygen. D. Methane.

31. The bones in human hands, flippers of the whale and wings of the bird are known as what?
A. Linker fossils. C. Analogous structures.
B. Homologous structures. D. Vestigial structures.

32. Which of the following classes of crops is a good source of proteins?
A. Leafy vegetables. C. Pulses.
B. Cereals. D. Roots and tubers.

33. Which of the following is a polysaccharide?
A. Sucrose. B. Glucose. C. Cellulose .D. Fructose.

34. What makes phospholipids molecules arrange themselves into a bilayer in a water medium?
A. The double bonds of the fatty acids.
B. The length of the fatty acid chains.
C. The presence of hydrophilic and hydrophobic groups.
D. Their high solubility in water.

35. In the cell cycle, which of the following phases is divided into G1, S and G2 stages?
A. Interphase. B. prophase. C. Metaphase. D. Telophase.

36. Which of the following plant groups is a good pioneer colonizer in primary successions that start with bare rocks?
A. Mosses. B. Lichens. C. Liverworts. D. Weeds.

37. Where in the mitochondria does the Krebs cycle take place?
A. On the cristae.
B. In the matrix.
C. Between the outer and inner membranes.
D. On the inner surface of the outer membrane.

38. Regarding seed characteristics, suppose a heterozygous round yellow pea plant of RrYy genotype is selfed and produced 128 seeds, how many of the seeds are expected to be round yellow?
A. 72 B. 64 C. 32 D. 16

39. Which of the following does NOT happen during the cleavage of a zygote?
A. The cells double their contents before division.
B. The nucleus divides mitotically.
C. The number of cells increases.
D. The ratio of the size of the nucleus to the cytoplasm increases.

40. If one regularly includes an orange in his daily diet, which deficiency disease would be avoided?
A. Night blindness. B. Pellagra. C. Beriberi. D. Scurvy.

41. Which of the following diseases of the livestock is associated with tsetse flies?
A. Rinderpest. C. Blackleg disease.
B. Trypanosomiasis. B. Anthrax.

42. Which of the following methods of avoiding predation by animals is mimicry?
A. Resembling another organism.
B. Producing warning colouration.
C. Fighting away the enemy.
D. Running-away from the enemy.

43. In Pavlov's classical conditioning experiments with dogs, which one of the following is the conditioned stimulus?
A. The sight of food. C. The sound of the bell.
B. the smell of food. D. The salivation of the dog.

44. Identify the enzyme that is used to cut DNA molecules into small pieces.
A. Urease. C. DNA polymerase.
B. Reverse transcriptase D. Restriction endonuclease.

45. Suppose a lady has given birth to four boys in a row, what is the chance that her next child will be a girl?
A. 100% B. 75% C. 50% D. 25%

46. Which of the following mechanisms can separate two genes located on the chromosome?
A. Backcrossing. C. Segregation.
B. Crossing over. D. Independent assortment.

47. According to Lamarck's theory of evolution, what is the origin of new structures that are necessary for evolution?
A. Mutation C. Natural selection.
B. Gene recombination. D. Use-and-disuse of body parts.

48. Which stage of the life cycle of a lepidopterous insect is the pest?
A. Egg B. Larva. C. Pupa. D. Adult.

49. Suppose in a certain population, natural selection removes individuals with average height, leaving tall and short individuals behind, which of the following modes of selection is in operation?
A. Stabilizing selection. C. Directional selection.
B. Normalizing selection. D. Disruptive selection

50. If the DNA of the following animals is hybridized to human DNA, with which one of them would human DNA hybridize more?
A. Horse. B. Chimpanzee. C. Fish D. Mice.

51. In an ecosystem, the presence of which of the following groups is essential in order for the others to be present?
A. Autotrophs. B. Herbivores. C. Decomposers. D. Omnivores.

52. What are the three components of a DNA molecule?
A. Glucose, Nitrogen and Phosphorus.
B. Sugar, phosphate group, nitrogenous base.

Telegram: https://t.me/ExtremeSeries

C. Protein, carbohydrate and fat with base.
D. Monosaccharide, disaccharide, polysaccharide.

53. What is a biome?
A. A group of interacting individuals of the same species.
B. The total life zone of the earth's surface.
C. A group of populations that interact with the physical environment.
D. A region of the earth characterized by distinctive life forms.

54. Which of the following defines ATP?
A. Adenine + Adenosine + a phosphate.
B. Adenosine + ribose sugar + double phosphate.
C. Adenine + ribose sugar + triple phosphate.
D. Nitrogen + sugar + monophosphate.

55. Among the following, which one can be an example of prokaryotic organisms?
A. Paramecium. B. Amoeba. C. Liverwort. D. Bacteria.

56. Which alternative best explains speciation?
A. Members of a species diverge genetically and produce new species.
B. It is characterized by minor extinction events.
C. A process that only results from interspecific hybridization.
D. Two different species are able to interbreed in captivity.

57. Choose the alternative that shows the monomer and polymer of carbohydrates.
A. Nucleotide, amino acid.
B. Polypeptide, polynucleotide.
C. Monosaccharide, polysaccharide.
D. Nucleic acid, amino acid.

58. Which of the following is expected to produce new combinations of genes?
A. Hybridization of a tall variety of "Teff" with a short variety.
B. Keeping a male goat and a female sheep in the same room.
C. Vegetative propagation of fruit trees.
D. Asexual reproduction of unicellular animals.

59. What is implied by biological magnification?
A. Increase in the size of organisms with increasing trophic levels.
B. Increase in the number of organisms at higher trophic levels.
C. More concentration of chemicals in organisms of higher trophic levels.
D. Capture of small organisms by larger organisms.

60. What is the cause of infectious diseases?
A. the parental genes. C. Carcinogenic substances.
B. Microorganisms. D. Bad lifestyles.

61. Which of thee following compounds is quickly absorbed through the walls of the stomach?
A. Fat B. Protein. C. Sugar. D. Alcohol.

62. Coffee and tea are normally grouped into which type of substances?
A. Pulses. B. Perfumes. C. Stimulants D. Grains.

63. What does the feeding relationship in an ecosystem consists of?
A. Many stages of community succession.
B. Always a network of interconnected food chains

C. One organism feeds on another but itself eaten by another one.
D. Recycling of solar energy between the source and the trophic levels.

64. Which process of respiration helps to release most of the energy stored in glucose?
A. Oxidative phosphorylation. C. Fermentation reaction.
B. Glycolysis D. Anaerobic reaction.

65. Which of the following stages happens after all the others during the development of the human embryo?
A. Cleavage. C. Formation of gastrula.
B. Formation of blastula. D. Organogenesis.

66. Indicate the oldest relative of the modern horses.
A. Equus. B. Mesohippus. C. Eohippus. D. Piiohippus.

67. To which biological species do Neanderthal humans belong?
A. Homo erectus. C. Homo sapiens.
B. Homo habilis. D. Ardi ramidus.

68. When trying to view an object under the microscope and after switching to the high power objective, which adjustment knob do you use in order to see a clear picture?
A. The coarse adjustment. C. The fine adjustment.
B. The middle adjustment. D. The arm of the microscope.

69. Which one of the following belongs to the category of heterotrophic organisms?
A. Algae. B. Fungi. C. Higher plants. D. Mosses.

70. Which of the following is the most abundant polysaccharide found in plants?
A. Glycogen. B. Sucrose. C. Cellulose. D. Starch.

71. To which category of enzymes do the digestive enzymes that breakdown food substances in the human alimentary canal belog?
A. Intracellular enzymes. C. Extracellular enzymes.
B. Globular enzymes. D. Fibrous enzymes.

72. Of the following information, which one was most important to the development of the techniques of modern genetic engineering?
A. The knowledge about recessive characters.
B. The discovery of lethal genes.
C. The formulation of Punnett squares.
D. The knowledge about the structure of the DNA molecule.

73. Among the following human diseases, identify the one that is transmitted through a vector?
A. Leprosy. B. Chancroid. C. Gonorrhea D. Leishmaniasis.

74. Which of the following molecules is capable of mutation?
A. Nucleic acids. B. Proteins. C. Carbohydrates. D. Lipids.

75. If you observe a piece of paper on which several small squares are drawn, under which of the following magnifications of the microscope do you count more number of the squares?
A. 40X. B. 100X C. 400X. D. 1000X

Telegram: https://t.me/ExtremeSeries

76. A number of viruses, including HIV, have more than one strain. This shows that viruses are:
A. capable of living outside their hosts.
B. capable of mutation.
C. capable of reproduction.
D. obligate parasites.

77. Identify the statement that more correctly explains why each cell contains so many different types of enzymes? Enzymes are:
A. consumed in the process of catalysis.
B. under over and over again.
C. specific to their substrates.
D. easily synthesized in the cell.

78. When is a behaviour referred to as stereotyped? When:
A. an animal responds similarly to all stimuli.
B. all members of a species respond in the same way to a stimulus.
C. a behaviour is easily earned.
D. an animal ignores a stimulus because it is familiar with it.

79. Which of the following is a mismatch between a biome and what is contains?
A. Savanna – large herbivores.
B. Desert – cacti.
C. Deciduous forest – evergreen trees.
D. Tropical rainforest – high biodiversity.

80. In an ecosystem that consists of grasses, birds, grasshoppers and frogs, which of the following food chain is possible?
A. Grass -> frog -> Grasshopper -> bird.
B. Grass -> Grasshopper -> frog -> bird.
C. Grass -> bird ->Grasshopper -> frog.
D. Bird -> frog -> Grasshopper -> Grass.

81. If an animal inhales a radioactive form of oxygen, in which of the following products of the cellular respiration would the radioactivity be detected?
A. Water. B. Carbon dioxide. C. ATP. D. NADH.

82. If pieces of fresh potato are kept for some time in sugar solutions of 20%, 10%, 5% and distilled water, which piece will gain the highest percentage of weight?
A. The one in 20% solution. C. The one in 5% solution.
B. The one in 10% solution. D. The one in distilled water

83. Suppose a lost and found baby is claimed by the four families whose blood genotypes are shown below. If the baby has blood type O, which of the families is the possible parent?
A. AO and BB. B. AA and OO. C. AO and BO. D. AB and OO.

84. For what objective do geneticists engineer the Bt gene of Bacillus thuringiensis into crop plants? To make the plant:
A. resistant to insect pests. C. resistant to herbicides.
B. resistant to pesticides. D. produce food of improved quality.

85. Maize plant has 2n = 20 chromosomes. What is the number of chromosomes in the endosperm tissue of maize?

Telegram: https://t.me/Extreme_Series

A. 10 B. 20 C. 30 D. 40

86. Among the following fossil forms of the human evolutionary ancestors, which one was the first to develop the ability to walk upright?
A. Homo erectus. C. Ardipithecus ramidus.
B. Homo habilis. D. Australopithecus afarencis.

87. If a forest is protected in order to conserve wild coffee in the forest, what sort of conservation do we call this?
A. In situ conservation. C. On farm conservation.
B. Ex situ conservation. D. Cryopreservation.

88. Suppose a girl, after watching her mother making 'enjera' on several occasions, successfully did it herself, what type of learning is this?
A. Sensitization. C. Habituation.
B. Imprinting. D. Latent learning.

89. In which agricultural system of Ethiopia does the production of grain crops dominate over other crop categories?
A. Shifting cultivation. C. Enset-planting complex.
B. Seed farming complex. D. Pastoral complex.

90. Assuming that each of the following is separately analyzed, which one is expected to give more variety of organic molecules?
A. A virus. C. A mitochondrion.
B. A molecule of DNA. D. A polysaccharide.

91. Which evolutionary evidence was very unlikely to have been used by Darwin in the formulation of the theory of evolution?
A. Comparative anatomy. C. Fossils.
B. Comparative biochemistry. D. Geographical distribution.

92. Which of the following alternatives best defines gene flow?
A. The movement of genetic material from the nucleus to the cytoplasm.
B. The transmission of genetic information from the cytoplasm to the nucleus.
C. Thee transmission of genetic information from one population to another.
D. The movement of genes from one chromosomes to another.

93. Which one of the following corresponds to the first step of the scientific method?
A. Analyze the data collected. C. Undertake the research.
B. Identify a topic to be studied. D. State the results.

94. Which of the following makes a food web of an ecosystem different from a food chain?
A. Presence of many tropic levels.
B. Clear sign of the cycling of materials.
C. Presence of many interconnected food chains.
D. Presence of many organisms through which energy passes.

95. What term is used to describe the chemical process of living organisms that involves both the joining up of monomers to produce polymers and the splitting of polymers to monomers?
A. Anabolism. B. Catabolism. C. Metabolism. D. Polymorphism.

96. Choose the environmental factor that decreases both with increasing altitude and latitude?
A. Temperature. B. Sunlight. C. Moisture. D. Fertility of soil.

97. The carbon dioxide produced during the respiratory cycle and other processes is largely absorbed by which of the following?
A. The ozone layer. B. Animals. C. Plants. D. The sky

98. Which process of life is grouped into autotrophic in some organisms and heterotrophic in others?
A. The transport system C. Anaerobic respiration.
B. The aspect of Nutrition. D. the coordination system.

99. How do the two strands of DNA combine to form a molecule?
A. A base of one strand pairs with a complementary base of the other strand.
B. The phosphate of one strand combines with the phosphate of the other strand.
C. The sugar of one strand combines with the complementary sugar of the other strand.
D. A ribose sugar combines with a deoxyribose sugar on the same DNA strand.

100. Which of the following is true about the first stage of photosynthesis?
A. Light-dependent. C. ATP-driven.
B. Temperature-dependent. D. Glucose-driven.

Telegram: https://t.me/Extremeseries

INSTITUETE OF EDUCATIONAL RESEARCH (IER) ADDIS ABABA UNIVERSITY (AAU) UNIVERSITY ENTRANCE ECAMINATION (UEE)BIOLOGY, SENE 2003/JUNE 2011

DIRECTIONS: Each of the following questions is followed by four possible alternatives. Read each question carefully and **blacken** the letter of your choice on the separate answer sheet provided.

1. Which one of the following refers to a change in a gene?
A) Mutation C) Protein synthesis
B) Replication D) Transcription

2. Which one of the following disciplines is studied in ethology?
A) Heredity C) Reproduction
B) Physiology D) Behavior

3. Which one of the following is true about an innate behavior?
A) It is non- adaptive behavior
B) It is learned behavior
C) Members of a species behave similarly.
D) It is environmentally determined.

4. After dividing plant materials into two groups, an experimenter has applied a fertilizer to group A plants and not to group B plants, what are group B plants called?
A) Control group C) Test group
B) Experimental group D) Variable group

5. Which one of the following has a unidirectional flow in an ecosystem ?
A) Nitrogen C) Phosphorus
B) Experimental group D) Variable group

6. The cells of which group of micro – organisms can be described as prokaryotic?
A) Viruses B) Protozoa C) Algae D) Bacteria

7. In the scientific method, which one of the following is used to test a hypothesis?
A) Observation B) Theory C) Experiment D) Prediction

8. Which one of the following is largely made up of phospholipids?
A) Cell wall C) Nucleus
B) Cell membrane D) Chromosomes

9. How many daughter cells are produced from a single mitotic division?
A) Two B) Four C) Six D) Eight

10. Which of the following are rod – shaped bacteria?
A) Cocci B) Spirochetes C) Spirilla D) Bacilli

11. Choose the name of the researcher/ scientist who introduced the term cell for the first time?
A) Aristotle C) Robert Hooke
B) Anton Van Leeuwenhoek D) Robert Brown

12. What is the total magnification power of a microscope with a 10X ocular lens and a 25X objective lens?

A) 2.5X B) 35X C) 250X D) 2,500X

13. What does an ornithologist study?
A) The biology of mammals. C) The biology of fishes
B) The biology of frogs & reptiles D) The biology of birds

14. What are bacteriophages?
A) Bacteria B) Viruses C) Protozoa D) Fungi

15. What is the plant growth substance which causes them to bend towards the light?
A) Starch B) Photosynthesis C) Auxin D) Chlorophyll

16. The unicellular organism, Euglena, swims using its flagellum towards areas of increased light intensity. What is this kind of response to stimulus called?
A) Negative kinesis C) Negative phototaxis
B) Positive kinesis D) Positive photaxis

17. Which one of the following best defines modern biology?
A) The study of animals
B) The study of plants and animals
C) The study of all living things
D) The study of transfer of energy

18. In aerobic respiration of cells, in which cellular part does the Krebs cycle (citric acid cycle) take place?
A) Chloroplasts B) Mitochondria C) Nuclei D) Lysosomes

19. Which alternative correctly matches the polymer with its monomer?
A) Carbohydrate – Amino acid C) Nucleic acid – Nucleotide
B) Protein – Monosaccharide D) Lipid – Fat

20. Which term better captures the concept of variety of organisms present in a whole continent as well as all the genes found in them?
A) Niche B) Biodiversity C) Ecosystem D) Biome

21. Which one of the following is the major killer of AIDS patients?
A) Pneumonia C) Anemia
B) Opportunistic infections D) Malnutrition

22. Among the following discoveries in biology which one is the latest of all?
A) The law of heredity
B) The cell theory
C) The double helix nature of the DNA
D) The binomial system of classification

23. Which of the following molecules requires a template molecule for its synthesis?
A) Cellulose B) Starch C) Protein D) Glycogen

24. All the following carbohydrates are polysaccharides **EXCEPT** which?
A) Glucose B) Cellulose C) Starch D) Glycogen

25. Which of the following is a process by which cells take in fluid by means of vesicles?
A) Pinocytosis B) Endocytosis C) Osmosis D) Phagocytosis

26. Which of the following structures divides the cytoplasm of a plant cell into two halves ?
A. Nuclear membrane C. Spindle
B) Cell plate D) Cleavage furrow

27. The hormone insulin balances the level of blood glucose in the body. This is an example of which of the following properties of life?
A. Metabolism B. Respiration C. Homeostasis D. Response

28. In the life cycle of sexually reproducing organisms, which of the following processes restores the chromosome number from haploid to diploid state?
A. Fertilization C. Chromosome replication
B. Meiosis D. Mitosis

29. If an enzyme is provided with its normal substrate plus another molecule having the same shape and size as the substrate, which one of the following would happen?
A. Allosteric inhibition C. Irreversible inhibition
B. Competitive inhibition D. End product inhibition

30. Which one of the following sugars do babies get from the milk of their mothers?
A. Maltose B. Sucrose C. Lactose D. Fructose

31. It is said that a goat mother accepts and nurses as her own kid any young that she smells at a certain critical period, and rejects all others. This is an example of
A. habituation B. imprinting C. insight learning D. sensitization

32. According to the theory of evolution as proposed by Charles Darwin, which one of the following is the most important driving force of evolution?
A. Mutation C. Genetic recombination
B. Over-reproduction D. Natural selection

33. In Pavlov's classical conditioning experiment on dogs, which one of the following is the unconditioned response?
A. The sound of the bell before presenting food
B. The sight of the food from far away.
C. The salivation of the dog at the sight of food.
D. The salivation of the dog at the sound of the ball.

34. What percentage of the amount of carbon 14 originally present in a fossil would be left after its second half life?
A. 75% B. 50% C. 25% D. 12.5%

35. All other factors remaining the same, which of the following decreases the size of a population?
A. Birth rate that is greater than death rate.
B. Emigration rate that is less than immigration rate.
C. Immigration rate that is greater than death rate.
D. Emigration rate that is greater than death rate.

Telegram: https://t.me/ExtremeSeries

36. Which one of the following modes of natural selection was responsible for the fast increase in the number of the black form of moths in Europe during Industrial Revolution?
A. Stabilising selection C. Normalizing selection
B. Disruptive selection D.Directional selection

37. Among the following infectious human diseases, identify the one that is caused by a virus.
A. Yellow fever B. Pneumonia C. Cholera D. Typhoid fever

38. In plant hybridization experiment, which one of the following is the best way to prevent self-pollination
A. To remove the stamens C. To cross- pollinate flowers
B. To cover flowers with bags D. To make reciprocal crosses

39. Among the following, which one is a learned behavior?
A. Reflex action B. Habituation C. Imprinting D. Fixed action pattern

40. Among the following choices, identify the ones that have identical DNA fingerprints?
A. Parents and children C. Fraternal twins
B. Non-twin sibs D. Monozygotic twins

41. What is biomass?
A. The number of species in a given biome.
B. The total weight of living matter in a given area.
C. The density of a species in a given area.
D. The total energy found in plants and animals in an ecosystem.

42. Which of the following is expected to have a positive effect on the biodiversity of Ethiopia?
A. Diverse ecology C. Overgrazing by livestock
B. Extensive deforestation D. Logging and fuelwood removal

43. Which of the following contains phosphate group in its molecule?
A. Nucleotide B. Protein C. Cellulose D. Simple sugar

44. What is the final electron acceptor in the electron transport chain of cellular respiration?
A. O_2 B. H_2 C. $NADP^+$ D. NAD^+

45. In spite of their ability to form hybrids when mated, which one of the following factors is more important to keep the horse and the donkey as separate species?
A. Their hybrid is not viable C. They cannot form hybrid zygotes
B. Their hybrid is sterile D. They are ecologically isolated

46. Which one of the following terms is most appropriate to collectively refer to all the plants, animals and micro- organisms found in Munesa Forest in Ethiopia?
A. Ecology B. Ecosystem C. Population D. Community

47. Which of the following is the right vector to transfer alien genes to bacteria through genetic engineering?
A. Plasmids B. Snails C. Mosquitoes D. Bacteria

48. Why are T-lymphocytes more vulnerable to HIV infection? Because they possess:
A. thin cell membrane C. HIV receptor protein
B. gp120 on their surface D. large pores in their cell membrane

49. In a given ecosystem, members of which one of the following feeding groups release nutrients that are locked up in organic molecules?
A. Herbivores B. Producers C. Carnivores D. Decomposers

50. Suppose in a monohybrid cross 80 F_2 plants were produced, what is the number of the plants that are expected to have the dominant and the recessive phenotypes?
A. 70 dominant + 10 recessive C. 50 dominant + 30 recessive
B. 60 dominant + 20 recessive D. 40 dominant + 40 recessive

51. Which of the following pairs are both organelles concerned with energy transformation?
A. Nucleus and nucleolus C. Chloroplasts and vacuole
B. Mitochondria and nucleus D. Chloroplasts and mitochondria

52. In which of the following biomes are epiphytes typically present as characteristic elements?
A. Tundra B. Boreal forest C. Cold desert D. Tropical rainforest

53. What is the collective name for plants that are well adapted to live in very dry environments?
A. Epiphytes B. Xerophytes C. Mesophytes D. Hydrophytes

54. The mature fruits of a flowering plant burst open when touched, forcefully ejecting the seeds contained in the pods. Why is this advantageous to the plant?
A. It attracts insects that aid in pollination
B. It aids in the dispersal of the species
C. It prevents seed germination
D. It can cause genetic changes to occur

55. Choose the breeding method used by humans to produce new and improved varieties of plants and animals?
A. Crossing-over C. Artificial selection
B. Natural selection D. Independent assortment

56. Which of the following is more likely to ensure the conservation of all the rest ?
A. Soil conservation C. Wildlife conservation
B. Plant conservation D. Watershed conservation

57. Which of the following statements is true about bacteria?
A. Bacteria found in the human body are all parasitic
B. All bacteria are harmful because they spoil foods
C. All bacteria cause diseases to plants, animals and humans in some ways
D. Some bacteria living in the human body make vitamins needed by

Telegram:https://t.me/ExtremeSeries

the body

58. Which group of organisms is found at a lower level of evolution according to the modern principles of biology?
A. Protists B. Fungi C. Animals D. Plants

59. How do animals obtain nitrogen?
A. By absorbing nitrates and ammonia from the soil
B. From the proteins in the organisms they consume
C. By absorbing nitrogen gas from the atmosphere
D. Through a mutualistic relationship with nitrogen- fixing bacteria

60. Which of the following organelles are likely to be more abundant in active cells such us the muscle cells of human heart?
A. Lysosomes B. Chromosomes C. Mitochondria D. Golgi bodies

61. Which of the following is true about both DNA and RNA?
A. Both are single stranded
B. Both have a five carbon sugar
C. Both are polymers of amino acids
D. Both contain the same four nitrogenous bases

62. Which one is the correct direction of transfer of genetic information in most living things ?
A. Protein => DNA => mRNA C. Protein=> RNA=> DNA
B. DNA=> protein=> tRNA D. DNA=> RNA=> protein

63. If it is known that the total amount of DNA in a cell contains 30% nucleotides and adenine contributes to 80 of these, how many nucleotides are to guanine?
A. 35 B. 70 C. 80 D. 140

64. What proportion of the number of chromosomes found in a normal cell of an angiosperm plant is expected in its endosperm tissue ?
A. Half B. Same C. Two times D. Three times

65. A field biologist who studies the behavior of primates in the forest is most likely expected to frequently use which method of data collection?
A. Experiment C. Observation
B. Animal dissection D. Comparative biochemistry

66. How is it possible for different species to coexist in an ecosystem?
A. Each species has a specific biome of its own
B. Each species occupies a different niche within the ecosystem
C. There is no competition for resources among species in an ecosystem
D. All resources are equally shared among species in an ecosystem

67. Which of the following are NOT included in lipids?
A. Nitrogenous bases B. Waxes C. Steroids D. phospholipids

68. Sodium ions pass from a region where they are at lower concentrations to a region where they are found at higher concentrations in human cells. This is an example of which of the following processes?
A. Osmosis C. Passive transport
B. Simple diffusion D. Active transport

Telegram: https://t.me/ExtremeSeries

69. Choose the structure that is usually present only in the cells of animals
A. Vacuole B. Cell wall C. Nucleus D. Centriole

70. What is the best term that refers to the maximum number of individuals that a given environment can support over a long period of time without degradation?
A. Density C. Carrying capacity
B. Climax community D. Limiting factor

71. If AIDS patients are given antiretroviral drugs that are targeted against the protease enzyme, which one of the following stages of the HIV life cycle would be inhibited?
A. Entry into the host cell
B. Reverse transcription
C. Integrating of viral DNA into host DNA
D. Assembly of viral parts into a whole virus

72. In Ethiopia, which of the following groups of the vertebrates is the most diverse in terms of species number?
A. Mammals B. Birds C . Reptiles D. Amphibians

73. If one keeps on adding a substrate to a fixed amount of an enzyme, when does the reaction rate reach a plateau?
A. When all the substrate is converted to products
B. When all the active sties of the enzyme are occupied
C. When most of the enzyme molecules are consumed
D. When much of the of the products are accumulated

74. Which one of the following types of mutations is responsible for sickle cell anaemia?
A. Addition of a base pair
B. Substitution of a base pair
C. Deletion of a base pair
D. A shift in the reading frame of the genetic code

75. Which of the following is true about those bacteria which dwell in hot springs?
A. They are not metabolically active
B. They use non-protein enzymes.
C. They get energy of activation from the spring heat
D. Their enzymes do not denature at that temperature

76. Which of the following land biomes occupies a large part of Brazil, parts of equatorial West Africa and Southeast Asia?
A. Acacia forest
B. Deciduous temperate forest
C. Coniferous forest
D. Tropical rainforest

77. Which one of the following objective and ocular combinations of the light microscope, respectively, gives the largest field of vision?

A. 10 X and 10 X
B. 4 X and 10 X
C. 40 X and 12 X
D. 100 X and 10 X

78. Which property of water keeps the bottoms of lakes and the life therein from being frozen during cold winters in the temperate regions?
A. Its highest density at 4^0C
B. Its high heat of vaporization
C. Its high boiling point
D. Its high latent heat of fusion

79. Which of the following phases of mitosis if blocked would produce a cell with twice as many chromosomes as the mother cell?
A. Interphase B. prophase C. Antaphase D. Telophase

80. Which one of the following is the mode of reproduction in bacteria?
A. Mitosis B. Binary fission C. Conjugation D. Lysogenization

81. Which of the following statements is consistent with the concept of the gene pool?
A. Species with large gene pool have low potential of adaptation
B. Endangered species usually have small gene pool.
C. Large gene pool is a hindrance for livestock improvement
D. Species with large gene pool are prone to diseases

82. Why is it that the typical diploid chromosome number of many organisms including human beings is an even number?
A. It is only a coincidence
B. Chromosomes duplicate before cell division
C. Both parents contribute equal number of chromosomes
D. Meiosis reduces chromosome number

83. With which of the following hibernation is most associated in animals?
A. Migration
B. Daily biological cycle
C. Annual biological cycle
D. Availability of food in the store

84. Which of the following helps to keep the biological membranes in a fluid state?
A. Waxes B. Phospholipids C. Cholesterol D. Water

85. Which of the following is a threat to the survival of small populations?
A. Habitat destruction
B. Breeding in captivity
C. Disease resistance
D. Absence of competition

86. Choose the disease that is caused by what is known as the "droplet infection"
A. Flu B. AIDS C. Cholera D. Malaria

87. If, due to incomplete dominance, the F_1 plants from a cross of red flowered x white flowered parents are pink, which of the following ratios are expected in the F_2 generation?
A. 1 red : 2 pink : 1 white
B. 2 red : 1 pink : 1 white
C. 1 red : 1 pink : 1 white
D. 3 red : 1 pink : 1 white

88. From which of the following sources did Charles Darwin get the idea that selection can change living organisms? From
A. the work of plant and animal breeders
B. his observation of the Galapagos birds
C. the evolution theory of Lamarck
D. the book by Malthus

89. Wood lice increase their movement in different directions in response to increased light intensity. This is an example of which of the following?
A. positive taxis B. Negative taxis C. Kinesis D. Phototropism

90. Which one of the following is most important for speciation to occur?
A. Hybridization among populations
B. Isolation of the gene pool.
C. Migration between populations
D. Free flow of genes between populations

91. Why is it that the leaves and the soft young stems of plants that have started wilting become stiff again when they are provided with water? This is because of
A. fast intake of minerals
B. the cooling effect of water
C. increased turgor pressure
D. increased rate of photosynthesis

92. Animals without close evolutionary relations are sometimes seen to have similar structures adapted for the same function. Which evolutionary principle is illustrated by this observation?
A. Convergent evolution
B. Divergent evolution
C. Sympatric evolution
D. Allopatric evolution

93. Which of the following descriptions is true about the action of selective pressures in populations? Selective pressure that acts:
A. around the mean value is stabilizing
B. around the mean value is disruptive
C. at both ends of the distribution is directional
D. at one ends of the distribution is stabilizing

94. What of the following is NOT an important unifying theme in biology?
A. The evolution of life
B. The diversity and unity of life
C. The interdependence of living organisms
D. The variation in climatic factors

95. In what ways are the modes of nutrition of animals and saprophytes similar?
A. No enzymes are produced
B. Digestion is external to their cells
C. They build up proteins from light
D. They are autotrophic organisms

96. Which one is biotic factor that can influence a plant?
A. The PH of the soil
B. A pollinating animal
C. CO_2 concentration
D. The amount of radiant energy

Telegram: https://t.me/ExtremeSeries

97. Which of the following compounds is an important component of the exoskeleton of arthropods such as insects?
A. Oil B. Sugar C. Protein D. Chitin

98. If it is known that the total number of the purine bases account for 50% of a DNA molecule and if each of the remaining bases are known to have the same proportion, what proportion is accounted for by Thymine alone in the same molecule?
A. 25% C. 75%
B. 50% D. 100%

99. Which of the following is based on sufficient supporting data?
A. Experimentation C. Theory
B. prediction D. Hypothesis

100. What is the term used to express the ability of a microscope to show details clearly?
A. Enlargement C. Magnification
B. Reduction D. Resolution

Telegram: https://t.me/ExtremeSeries

1. During seasons of reproduction, the males of some species of birds produce colourful feathers to attract females. What do ethologists call this method of communication in animals?
A. Visual B. Chemical C. Auditory D. Touch

2. What do we call the substance upon which an enzyme acts?
A. Product C. Activation energy
B. Substrate D. Enzyme-substrate complex

3. Which one of the following is **NOT** a mutation?
A. DNA replication to form two daughter DNAs.
B. Gain of an extra chromosome by a cell.
C. Deletion of a base pair from DNA.
D. Loss of a chromosome by a cell.

4. Which factor has a more negative effect on the functions of enzymes than the others?
A. Neutral pH.
B. Very high temperature.
C. Optimal amount of salt concentration.
D. Optimal amount of substrate concentration

5. Which group of organisms has a system of protein synthesis in which transcription and translation take place at separate times?
A. In all eukaryotic organisms.
B. In multicellular animals only.
C. Only in prokaryotic organisms.
D. In both prokaryotic and eukaryotic organisms.

6. To which genus of human – like organisms does Lucy belong?
A. The genus *Homo*. C. The genus *Sahelanthropus*.
B. The genus *Ardipithecus*. D. The genus *Australopithecus*.

7. How many different kinds of amino acids are there for protein synthesis?
A. Twenty B. Twenty-four C. Twenty – six D. Thirty-two

8. Which of the following compounds in an important component of the bacterial cell wall?
A. Chitin B. Peptidaglycan C. Cellulose D. Pectin

9. Which of the following is **NOT** recycled between organisms and the environment in an ecosystem?
A. Energy B. Carbon C. Nitrogen D. Phosphorus

10. As was shown by Gregore Mendel in garden pea, what percentage of the F_2 generation of a monohybrid cross has the recessive phenotype?
A. 75% B. 50% C. 25% D. 12.5%

11. Which of the following statements is in agreement with the modern cell theory?
A. Cells come from nothing.
B. Cells comeform existing cells.

C. Cells come from non-living material.
D. Cells arise by means of spontaneous generation.

12. What is the term that refers to all parts of the earth where living things are found?
A. Population B. Ecosystem C. Biosphere D. Environment

13. In which kingdom of life are the unicellular eukaryotes grouped?
A. Monera B. Protista C. Plantae D. Animalia

14. Select the function that living things are NOT capable of performing.
A. Maintain their internal body environment.
B. Pass genetic information to their offspring.
C. Respond to other organisms found in their surroundings.
D. Determine the amount of radiation reaching the environment.

15. What is the role of the worker honey bee just after it emerges?
A. Forage for nectar, pollen and water.
B. Clean out dirty honeycomb.
C. Guard the hive.
D. Build honeycomb.

16. How many carbon atoms are contained in a single molecule of sucrose?
A. 6 B. 12 C. 24 D. 48

17. Which of the following is the best collective name for all bacteria with spherical shapes?
A. Cocci B. Bacilli C. Spitochaetes D. Streptococci

18. Which tool of the biologist is more suitable for culturing bacteria in the laboratory?
A. Test tubes B. Microscopes C. Perti dishes D. Beakers

19. Which of the following corresponds to the beginning step of a scientific work?
A. Testing hypotheses C. Conducting experiments
B. Making observations D. Drawing conclusions

20. What do geneticists call the genotype in which the two alleles of a pair are identical?
A. Dominant B. Recessive C. Homozygous D. Heterozygous

21. In the process of amino acid condensation, which one of the following happens?
A. Oxygen is used up.
B. Carbon dioxide is released.
C. Water is released as a byproduct.
D. Protein is broken down into amino acis.

22. In which one of the following points does the induced-fit model of enzyme action differ from the lock-and-key model?
A. Enzymes lower the energy of activation.
B. Substrates bind at the active site of the enzyme.
C. During the reaction, an enzyme- substrate complex is formed.
D. The shapes of the substrate and active site are complementary.

23. In an experiment that is testing the effect of temperature on the germination rate of certain seeds, in which one of the following should the experimental and the control groups differ?
A. The amount of water they receive.
B. The age of the seeds assigned to them.
C. The number of seeds assigned to them.
D. The temperature at which they are kept.

24. Which of the following processes of photosynthesis does NOT require the presence of light to take place?
A. The splitting of water. C. Reduction of NADP.
B. ATP formation. D. Carbon fixation.

25. Which one of the following is referred to as the first law of Mendel?
A. The occurrence of alleles in pairs.
B. The dominance of one allele over the other.
C. The equal contribution of alleles by both parents.
D. The separation of alleles during gamete formation.

26. Which one of the following types of microscopes is the best to show the details of the surface of an object?
A. Optical microscope. C. Scanning electron microscope.
B. Compound microscope. D. Transmission electron microscope.

27. Which of the following mechanisms do cells use to regulate enzyme catalyzed reactions in metabolic pathways?
A. Enzyme denaturation. C. End product inhibition.
B. Irreversible inhibition. D. Competitive inhibition.

28. Which of the following modes of material transport across the cell membrane is NOT governed by the concentration gradient of the transported material?
A. Simple diffusion C. Osmosis
B. Facilitated diffusion D. Active transport

29. Which one of the following is NOT one of the roles that microorganisms play in ecosystems?
A. Nutrient recycling C. Nitrogen fixation
B. Carbon fixation D. Energy recycling

30. Which one of the following demographic factors affects the number of human population globally?
A. Natality B. Migration C. Emigration D. Immigration

31. Which of the following laboratory equipment is used to separate the organelles of the cell according to their density?
A. Incubator C. Measuring cylinder.
B. Centrifuge. D. Filter paper with fine pores.

32. Which of the following idea is NOT a part of Darwin's Theory of Evolution?
A. Over reproduction C. Existence of heritable variation.
B. Use – and disuse of body parts. D. Competition for scarce resources.

33. Which of the following types of movements in response to a stimulus has no specific direction?

Telegram: https://t.me/ExtremeSeries

A. Taxis B. Kinesis C. Gravitropism D. Phototropism

34. Which of the following is the best way to check whether an individual having a dominant phenotype is homozygous or heterozygous for the trait?
A. To self the individual.
B. To cross it to a heterozygous individual.
C. To cross it to homozygous recessive individual.
D. To cross it to a homozygous dominant individual.

35. Which of the following pairs of elements are found in all carbohydrates in addition to the element carbon?
A. Nitrogen and oxygen. C. Hydrogen and nitrogen.
B. Hydrogen and oxygen. D. Phosphorus and nitrogen.

36. When an enzyme is denatured by heat or extreme pH, which one of the following does it lose?
A. The peptide bond. C. Secondary structure.
B. Primary structure. D. Tertiary structure.

37. Which of the following ideas in the cell theory was contributed by Rudolf Virchow?
A. All plants are made up of cells.
B. All animals are made up of cells.
C. Cells are the structural unit of life.
D. Cells come from pre-existing cells.

38. When the muscle cells are in short supply of oxygen, which of the following compounds would be accumulated in them?
A. Ethanol B. Acetic acid C. Lactic acid D. Carbon dioxide

39. Which one of the following is the main source of the greenhouse gases that are concentrating in the atmosphere of the earth?
A. Burning of fossil fuels.
B. Plants growing in greenhouses.
C. Photosynthesis by aquatic plants.
D. Respiration by animals and plants.

40. Which of the following fossils is the nearest to the common ancestor of the hominids and the apes?
A. Homo habilis. C. Ardipithecusramidus
B. Homo erectus D. Australopithecus afarensis.

41. Viruses are better characterized as:
A. decomposers. B. producers. C. carnivores. D. parasites.

42. Which means of particle transport requires input of energy by the cell?
A. Simple diffusion C. Osmosis
B. Facilitated diffusion D. Active transport

43. Which of the following is NOT one of the stages in cellular respiration?

A. Calvin cycle C. Electron transport
B. Glycolysis D. Krebs cycle

44. Among the molecules found in cells, which of the following contains less energy?
A. A glucose molecule C. A triglyceride
B. An amino acid D. A water molecule

45. What is the correct equation for cellular respiration?
A. $6CO_2 + 6H_2O + Energy = 6O_2 + C_6H_{12}O_6$.
B. $6O_2 + C_6H_{12}O_6 = 6CO_2 + 6H_2O + Energy$.
C. $6O_2 + C_6H_{12}O_6 + Energy = 6CO_2 + 6H_2O$.
D. $6CO_2 + 6H_2O = 6O_2 + C_6H_{12}O_6 + Energy$

46. What amount of net gain in ATP does glycolysis provide to a cell?
A. 2 ATP molecules. C. 18 ATP molecules
B. 4 ATP molecules. D. 36 ATP molecules.

47. What is the term for the process by which organisms keep their internal conditions at a fairly constant state?
A. Catabolism. B. Evolution. C. Homeostasis. D. Photosynthesis.

48. Which of the following requires expenditure of ATP?
A. Osmosis C. Simple diffusion
B. Facilitated diffusion D. Endocytosis.

49. What is the most probable selection pressure responsible for the evolution of green skin colour in frogs inhabiting tropical rainforests?
A. Climate C. Infection by pathogens
B. Reproduction D. Predation

50. In which part of the cell do Gram-positive and Gram-negative bacteria differ regarding their staining property with Gram's stain?
A. Cytoplasm C. Cell wall
B. Cell membrane D. Nucleus

51. When do populations of living organisms show exponential growth?
A. When the resources are plentiful.
B. Whenever they enter a new environment.
C. When they face strong competition from other species.
D. When the carrying capacity of the environment is reached.

52. Which of the following terrestrial biomes experiences hot days and cold nights?
A. Tundra C. Desert
B. Tropical rainforest D. Grasslands

53. In DNA cloning technology, which of the following molecules serves as a vector of gene of interest to be transferred to bacterial host?
A. Bacterial DNA. C. Nuclear DNA.
B. Plasmid DNA. D. Mitochondrial DNA

Telegram: https://t.me/ExtremeSeries

54. To which of the following groups does HIV belong?
A. Plasmids
B. Retroviruses
C. DNA viruses
D. Bacteriophages

55. Which group of organisms found in Ethiopia is represented by the highest number of endemic taxa?
A. Mammals B. Amphibians C. Birds D. Plants

56. What is the main mode of transmission of diseases such as cholera and typhoid fever?
A. Sexual intercourse.
B. Bites of animal vectors.
C. Blood – to – blood contact.
D. Drinking contaminated water.

57. What are the individual strands of a fungal mycelium called?
A. Fungus B. Spores C. Hyphae D. Hydra

58. To which major area of relevance and applications of biology is the production of transgenic organisms related?
A. Agriculture
B. Environment
C. Biotechnology
D. Medicine

59. Which of the following is NOT in agreement with the scientific method?
A. Putting forward testable hypotheses.
B. Carrying out experiments in duplicates.
C. Putting forward personal value judgments.
D. Analyzing results and drawing conclusions

60. Which of the following properties of water makes sweat an effective body cooler?
A. Its high specific heat.
B. Its high surface tension.
C. Its low density when frozen.
D. Its high heat of vaporization.

61. A cell was examined under a compound light microscope with an eyepiece lens marked 13x and an objective lens marked 40x. How many times larger would the cell appear to the viewer?
A. 40 times. B. 53 times. C. 400 times. D. 520 times.

62. How many moles of ATP will be generated as a result of the oxidation of one mole of $FADH_2$ in an actively respiring mitochondrion?
A. 0 B. 3 C. 2 D. 6

63. Which alternative contains only crops known to have been domesticated within Ethiopia?
A. *Guizotiaabyssinica, Zea mays, Pisumsativum.*
B. *Viciafaba, Carica papaya, Musa paradisiaca.*
C. *Oriza sativa, Triticumaestivum, Solanumtuberosum.*
D. *Coffea Arabica, Eragrostistef, Enseteventricosulum.*

64. Which of the following is true about the evolutionary origin of groups of organisms?

A. The dinosaurs appeared before the origin of the land plants.
B. The earliest *Homo sapiens* appeared before the flowering plants.
C. The first photosynthetic organisms appeared before the oldest eukaryotes.
D. The first animals appeared before the formation of free O_2 in the atmosphere.

65. Why are fossils of soft-bodied organisms usually relatively rare in the environment?
A. They are generally small in size.
B. Their bodies decompose readily.
C. They all lived in environments where sedimentation did not occur.
D. They were never common in environments in which they lived.

66. Which of the following is true for cellular respiration?
A. Restricted to plant cells.
B. Restricted to animal cells.
C. Occurs in all eukaryotic cells.
D. Occurs in prokaryotic cells only

67. What could be the main reason behind the currently observed slow or stable rate of population growth in the industrialized countries?
A. Good family planning.
B. Increasing death rate.
C. Poor health conditions
D. High rate of child death.

68. What is the average projected rate of loss of biodiversity every 50 years?
A. 5% B. 10% C. 20% D. 50%

69. A genetic cross between two F1-hybrid pea plants having yellow seeds (dominant) will yield what percent green-seeded (recessive) plants in the F2 generation?
A. 0% B. 25% C. 50% D. 75%

70. Which of the following statements is true about the nitrogen cycle?
A. Plants fix nitrates from atmospheric nitrogen.
B. The nitrogen used by animals largely comes from plants.
C. Nitrogen is consumed by bacteria and removed from the soil.
D. Nitrogen-fixing bacteria reduce the total amount of available nitrogen.

71. Which one of the following steps in the life cycle of HIV is blocked if an antiretroviral drug that inhibits the reverse transcription enzyme is given to an AIDS patient?
A. Formation of DNA from RNA.
B. The entry of HIV into CD4 cells.
C. The assembly of parts into HIV particle.
D. The integration of HIV DNA into host chromosome.

Telegram: https://t.me/Extremeseries

72. Carbon 14 has a half life of about 5730 years. Suppose a fossil contains only 12% of the amount of carbon 14 normally present in living organisms, how old is the fossil?
A. 5730 years. B. 11460 years. C. 17190 years. D. 22920 years.

73. In competitive inhibition, which one of the following factors determines the rate of the inhibition?
A. The reaction temperature.
B. The enzyme concentration.
C. The substrate concentration.
D. The ratio of inhibitor to enzyme concentration.

74. In which one of their structural parts do different molecules of amino acids differ from one another?
A. In their R group. C. In their carboxyl group.
B. In their amino group. D. In their alpha-carbon group.

75. In an experiment designed to test the effect of different concentrations of a fertilizer on the growth rate of a plant, which one of the following is the dependent variable?
A. The growth rate of the plant.
B. Plants assigned to the control group.
C. Concentrations of the fertilizer applied.
D. Plants assigned to the experimental group.

76. What is the reason that Louis Pasteur used a swan-necked flask instead of straight necked one in his experiment designed to disprove the theory of spontaneous generation?
A. To allow free passage of air to the broth inside the flask.
B. To prevent the escape of any microorganism form the flask.
C. To keep the broth in the flask hot to kill microorganisms.
D. To trap particles from the air that might enter the flask before reaching the broth.

77. If one mixes a sample of a fruit juice and some drops of Benedict's solution and obtains a brick – red precipitate upon warming the mixture, what does the juice contain?
A. Starch B. Reducing sugar C. Sucrose D. Protein

78. In cyclic photophosphorilation, what is the source of the recycled electron?
A. Reduced NADP. C. Adenosine triphosphate.
B. Chlorophyll molecule. D. Photolysis of water molecules.

79. Suppose we consider four hypothetical cells (designated A, B, C, and D) having cubic shape with their sides measuring 2, 4, 6, and 8 arbitrary units, respectively which of these cells has the largest surface area to volume ratio ?
A. Cell A B. Cell B C. Cell C D. Cell D

Telegram: https://t.me/ExtremeSeries

80. Suppose 25% of the molecules of an enzyme are inhibited by a non-competitive inhibitor, which one of the following would happen if the amount of the substrate is increased by 50%?
A. The reaction rate would double.
B. More enzyme molecules would get inhibited.
C. The rate of the reaction would decrease by 50%
D. The rate of the reaction would remain unchanged.

81. What would most likely result if mitosis fails to be accompanied by cytoplasmic division?
A. Two cells without nuclei.
B. One cell without a nucleus.
C. Two cells each with one nucleus.
D. One cell with two identical nuclei.

82. A biologist applied the scientific method repeatedly, gathered a large amount of supporting experimental data and finally described a pattern or relationship between different factors. What is the best term that refers to facts established in this way?
A. Theory B. Hypothesis C. Prediction D. Law

83. In the classical conditioning experiment performed by Pavlov on dogs, which of the following alternatives is the unconditioned stimulus?
A. The sound of the bell.
B. The smell of the food.
C. The salivation at the sound of the bell.
D. The salivation at the smell of the food.

84. Which stage in a primary ecological succession contains more biodiversity?
A. The third seral stage. C. The climax community.
B. The second seral stage. D. The pioneer community

85. If there were no free oxygen to breathe, which one of the following steps of the respiration process can operate in our body?
A. Glycolysis. D. Reaction that links glycolysis and Krebs cycle.
B. Krebs cycle.
C. Electron transport chain.

86. Suppose three potato cylinders are kept for some time in 15%, 8% and 4% sucrose solutions, respectively, and the fourth cylinder is kept in distilled water, which of the cylinders will be more flaccid?
A. The cylinder in 4% solution. C. The cylinder in 15% solution.
B. The cylinder in 8% solution. D. The cylinder in distilled water.

87. Which of the following pairs of molecules can give information about how much two species are evolutionary related to one another?

A. DNA and proteins.
B. Starch and cellulose.
C. Lipids and carbohydrates.
D. Carbohydrates and proteins.

88. Among the following couples whose ABO blood genotypes are shown, which one can produce children of A, B, AB and O blood types?
A. OO and AB. BO and AA. C. BO and AO D. BB and AO.

89. Which of the following is true about the antiretroviral drugs currently used to treat AIDS patients?
A. They cure AIDS.
B. They stop HIV transmission.
C. They serve as anti – HIV vaccines.
D. They slow down HIV multiplication .

90. What type of molecules CANNOT pass across the cell membrane by simple diffusion?
A. Charged molecules.
B. Non-polar molecules.
C. Lipid soluble molecules.
D. Molecules of very small size.

91. Why is it that the woodlice are typically found under logs, stones, bark and amongst leaf litter?
A. To be sheltered in a dry windy environment.
B. To run away from the area where the air is humid.
C. To make sure that they are in the hottest place all the time.
D. To reduce the rate at which water is lost from their bodies.

92. What are the environmental advantages of using enzymes in industry?
A. It makes high production possible with less input of heat.
B. It makes high production possible with high input of heat.
C. It makes high production possible with emission of more CO_2.
D. It helps high production with supply of more heat and emission of more CO_2

93. How do human beings increase biodiversity?
A. By reducing species richness.
B. By increasing genetic variability.
C. By promoting habitat uniformity.
D. By narrowing ecological variability.

94. Which of the following can be given as a good reason for finding large numbers of plant and mammal species in Ethiopia today?
A. Lack of ecological disturbance.
B. Environment free from predators.
C. Presence of many biomes and habitats.
D. Good ecological and biodiversity management.

95. Which parts of the angiosperm flower are both essential for the success of hybridization experiments?
A. Sepal and petal.
B. Stamen and petal.
C. Pollen and filament.
D. Gynoecium and andoecium

96. Which factors are involved in the determination of climax vegetation?
A. Temperature and precipitation.
B. Grazing and browsing animals.
C. Radiation and reflection.
D. Predators and preys.

97. What does the structural similarity between the flippers of whales and arms of humans show?
A. Whales evolved from the human species.
B. Whales are older than the human species.
C. The human species began life in the oceans.
D. Whales and humans had a common ancestry.

98. What if the source of the oxygen that is produced during the process of photosynthesis by higher plants?
A. CO_2 B. H_2O C. ATP D. Chlorophyll

99. Which of the following terms mean stages in an ecological succession?
A. Pioneers B. Climaxes C. SeresD. Niches

100. The tropical rainforest largely found in South America and Africa can be best characterized by a combination of which environmental features?
A. Low rainfall and low temperature.
B. High rainfall and high temperature.
C. Low rainfall and high temperature.
D. High rainfall and low temperature.

Telegram: https://t.me/ExtremeSeries

INSTITUTE OF EDUCATIONAL RESEARCH (IER) ADDIS ABABA UNIVERSITY (AAU) ETHIOPIAN UNIVERSITY ENTRANCE EXAMINATION (EUEE) BIOLOGY, GINBOT 2005/JUNE 2013

1. Which one of the following is the main constituent of biological membranes?
   A. Phospholipids B. Glycoproteins C. Glycolipids D. Cholesterol
2. One of the following biomes in Africa is supporting large wild mammals such as elephants, giraffes and lions.
   A. The Congo Rainforest
   B. The Rain Forest of Western Ethiopia
   C. The Savanna Grassland
   D. The Sahara Desert
3. Which of the following steps in the scientific method comes following observation?
   A. prediction B. Hypothesis C. Experiment D. Theory
4. The main components of the plant cell wall are _____?
   A. Starch B. Cellulose C. Protein D. Chitin
5. Which of the following classes of enzymes digests carbohydrates?
   A. Amylases B. Lipases C. Proteases D. Nucleases
6. Which of the following kingdoms of life is consisting of prokaryotic organisms?
   A. Fungi B. Monera C. Protista D. Plantae
7. Which component of soil fertility is improved when farmers grow legumes in crop rotation?
   A. Phosphorus B. Nitrogen C. Sulfur D. Carbon
8. Before making crosses, which part of the flower did Mendel remove to avoid self pollination?
   A. Stigma B. Ovule C. Ovary D. Stamens
9. Which one of the following do bees use to inform other bees about location and distance of a new source of nectar they discover?
   A. pheromones B. Waggle dance C. Buzzing noise D. Vibration of wings
10. One of the following molecules is the building units of an enzyme molecule.
    A. Amino acids B. Glucose C. Nucleotides D. Fatty acids
11. What are the specialists in biology called if he/she studies fossils to generate new knowledge on the origin and evolution of living things of past geologic periods?
    A. Geneticist C. Paleontogist
    B. Ecologist D. Comparative Biochemist
12. All proteins contain carbon, hydrogen, oxygen and what other element?
    A. Chlorine B. Fluorine C. Nitrogen D. Sulphur
13. Which of the following is the correct F2 phenotypic ratio of a monohybrid cross?
    A. 1:2 B. 1:1 C. 3:1 D. 2:2
14. How many carbon atoms are there in one disaccharide molecule?
    A. 6 B. 12 C. 18 D. 24
15. Which unit is best to use for measuring the smallest cells and organelles?
    A. Micrometer B. Milliliter C. Millimeter D. Nanometer
16. Which energy rich organic compound contains adenine in its molecule?
    A. Lipid B. Carbohydrate C. Glucose D. ATP
17. What are the most frequent causative agents of food poisoning?
    A. Bacteria B. Protozoa C. Viruses D. Worms
18. What are the most likely causes of variations within species?
    A. Mitosis and a sexual reproduction.
    B. Overpopulation and overproduction.
    C. Vegetative propagation and cloning.
    D. Mutations and sexual reproduction.
19. In which hominid species did scientists find the smallest brain size (cranial capacity)?
    A. Homo Sapiens C. Homo erectus
    B. Homo habilis D. Homo neanderthalensis
20. Which of the following is **NOT** classified as a learned behavior?
    A. Insight B. Innate C. Latent D. Conditioned
21. In which one of the following aspects is the tropical rainforest biome poor?
    A. Species diversity C. Annual precipitation
    B. Amount of sunlight D. Soil fertility
22. If the size of a cell increases, which one of the following gets smaller?
    A. The volume of the cell C. surface area to volume ratio of the cell
    B. The surface area of the cell D. volume to surface area ratio of the cell
23. What makes unsaturated fatty acids different from saturated fatty acids?
    A. The presence of long chain of carbon
    B. The presence of one or more double bonds.
    C. The presence of large number of hydrogen atoms.
    D. Their occurrence as solid at room temperature
24. Which of the following pairs of molecules are known to have catalytic activity?
    A. lipids and proteins C. Protein and RNAs
    B. Carbohydrates and proteins D. Protein and DNAs
25. If a suspension of a mixture of cellular organelles is spun in a centrifuge, which organelle settles to the bottom first?
    A. Mitochondria B. Nuclei C. Chloroplasts D. Ribosomes
26. During chemiosmosis, what substance diffuses from one side to the other side of the membrane?
    A. water molecules B. Protons C. Electrons D. ATP molecules
27. Which of the following human diseases can be prevented by taking proper diets?

https://t.me/ExtremeSeries Telegram

A. Degenerative diseases C. Social diseases
B. Genetic diseases D. Deficiency diseases

28. When two species are compared, which of the following sources of evidence is least informative about the degree of relationships between the species?
A. Nucleotide sequence of DNAs
B. Amino acid sequences of proteins
C. Glucose sequence of polysaccharides
D. DNA-DNA hybridization

29. Which of the following practices does **NOT** normally transmit HIV?
A. Sexual intercourse through anus C. Blood transfusion
B. Sharing injection needles D. Shaking hands

30. What is the important role played by microorganisms such as bacteria and fungi in the ecosystem ?
A. Antibiotic production C. Forming organic substances
B. Recycling of nutrients D. Supplying energy to the ecosystem

31. Which of the following theories explains evolutionary changes of living things in terms of changes in their allele frequencies?
A. Darwin's natural selection
B. Spontaneous generation
C. Inheritance of acquired characteristics
D. Neo-Darwinism

32. Which of the following is an important function of the Golgi apparatus?
A. protein synthesis
B. Packaging of proteins
C. Removing of debris from cell
D. Storage of waste materials not needed by the cell

33. Which of the following is **NOT** usually true as an ecological succession progresses to advanced seral stages?
A. More ecological niches are formed. C. The depth of the soil increases.
B. Species become more diverse. D. Less population is supported.

34. One of the following is an important cause of gene mutation.
A. Old age B. Alcoholic drinks C. Smoking D. Radiation

35. What is the reproductive isolating mechanism called if two species of frogs do not interbreed because they cannot understand the mating calls of one another?
A. Seasonal isolation C. Temporal isolation
B. Behavioral isolation D. isolation by distance

36. How many chromosomes do humans inherit from each of their parents?
A. 23 chromosomes C. 46 chromosomes
B. 23 pairs of chromosomes D. 46 pairs of chromosomes

37. In which one of the four phases of population growth is the number of the population the highest?

A. Lag phase B. log phase C. Constant phase D. Decline phase

38. Which of the following is **NOT** true about the gene called SRY?
A. It is found on the Y-Chromosome.
B. It determines maleness.
C. Testes develop in its presence.
D. Females have two copies of this gene.

39. To which of the following groups do those bacterio phages that integrate their DNA into the chromosome of their bacterial host belong?
A. Virulent viruses C. Lysogenic viruses
B. Lytic viruses D. Non-parasitic viruses

40. What do we call the learned behavior if a mouse that had just escaped from the mouth of a cat jumped violently at a slight touch by a trivial object?
A. Latent learning C. Conditioning
B. Sensitization D. Imprinting

41. In which of the following features are eukaryotic cells distinguished from prokaryotic cells?
A. They have mitochondria C. They have no DNA
B. Their nuclei lack membrane D. They have smaller ribosomes.

42. What will happen if human red blood cells are kept in a hypotonic solution?
A. Lose water by osmosis and burst
B. Lose water by osmosis and shrink.
C. Take in water by osmosis, swell and burst
D. Take in water by osmosis, swell and remain turgid.

43. Which of the following field equipment is used to determine an exact location of a place?
A. Theodolite C. Field microscope
B. Global postioning system D. field PH kit

44. The sheep 'dolly' is an example of which biotechnological manipulation of animals by human?
A. Transgenic animal C. Cloned animal
B. Genetically engineered animal D. Hybrid animal

45. Among the following, which one is the best criterion to show that two population belong to same species?
A. Morphological similarity
B. Physiologically similarity
C. Inhabiting the same geographic area
D. Production of fertile offspring

46. In the process of the evolution of life on earth, which of the following four Processes evolved last?
A. Photosynthesis C. Chemosynthesis
B. Aerobic respiration D. Photo-autotrophism

47. What is a theory in biology?

Telegram: https://t.me/ExtremeSeries

A. The outcome of an experiment
B. A proof that shows the hypothesis is true
C. A hypothesis supported by experiments and/or observations.
D. An opinion or educated guess resulting from observations.

48. For which of the following is a theodolite used in biology education?
A. Measuring the height of trees.
B. Recording positions where a species is found.
C. Measuring the rate of flow of water in a cell.
D. Measuring the PH of water or soil.

49. What is the name of the sugar found in milk?
A. Glucose B. Lactose C. Maltose D. Sucrose

50. To which class of enzymes do the digestive enzymes belong?
A. Esterases B. Transferases C. Hydrolyses D. Isomerases

51. Which of the following crops is considered to be the best choice for a better balance of essential amino acids as a human diet?
A. Maize B. Quinoa C. Rice D. Wheat

52. What is the main reason for the high species richness of pants and mammals observed in Ethiopia ?
A. lack of predators
B. lack of disturbance
C. Presence of several biomes within the country
D. Efficient management of the ecological resources.

53. In which type of solution is the water potential more negative than in the cells?
A. Hypotonic B. Hypertonic C. Isotonic
D. Equal solute and solvent concentration

54. Which of the following paired organelles are membrane-bound?
A. Ribosomes and peroxisomes C. Mitochondria and ribosome
B. Chloroplasts and ribosomes D. Chloroplasts and mitochondria

55. Most cell membranes are primarily composed of which compounds?
A. proteins and lipids C. Chitin and Starch
B. DNA and ATP D. Nucleotides and amino acids

56. Which group of micro-organisms causes the disease known as athlete's foot?
A. Bacteria B. Fungi C. protozoa D. Viruses

57. Choose the one that is different from all the others.
A. Genetically modified organisms C. Pathogenic organisms
B. Genetically engineered organisms D. Transgenic organisms

58. Under which of the following groups can the fungi be more conveniently placed?
A. Autotrophs B. Heterotrophs C. Prokaryotes D. plants

59. What is the type of community called when it has reached the final and most complex stage of a succession?
A. Pioneer community C. Climax Community
B. Seral Community D. Secondary Community

60. Which of the following is an ecosystem?
A. A Tropical Rainforest
B. The African continent
C. All the organisms in a given area
D. The non- living components of an environment

61. Which of the following characteristics can show the evolutionary relationships among organism?
A. Structures having similar functions C. Structure having common origin
B. Structures having same size D. Structure having different origins

62. The group of the Ethiopian wolf does NOT contribute to territory marking with their urine containing pheromones is _____.
A. Adult males C. sub-adult males
B. Adult females D. sub-adult females

63. In which of the following are flowers and fruits found?
A. Ferns and relatives C. Mosses and conifers
B. Gymnosperms and ferns D. monocots and dicots

64. In a cross between heterozygote what proportion is expected to be homozygous recessive?
A. 25% B. 50% C. 75% D. 100%

65. Gene silencing is the function of one of the following molecules.
A. dsRNA B. mRNA C. siRNA D. tRNA

66. Which process is held responsible for chronic myelogenous leukemia?
A. Translocation B. Translation C. Transcription D. Duplication

67. Two parents of genotype Aa are cross-bred. The alleles show complete dominance. What proportion of the offspring will phenotypically look like their parents?
A. 0 B. ¼ C. ½ D. ¾

68. Which one of the following concepts contains all the others?
A. Species B. Genus C. population D. community

69. Choose the one that is NOT a requirement of all living things.
A. Ability to think C. Response to the stimuli
B. organization of parts D. Maintenance of internal constancy

70. What is the advantage of using HAART (highly active antiretroviral therapy) for the treatment of HIV?
A. It gives a lasting immunity to HIV.
B. It prevents mutation of HIV.
C. It prevents re-infection by HIV.
D. It helps to break the life cycle of HIV.

71. In which of its contents RNA differs from DNA?
A. Deoxyribose and guanine C. Ribose and thymine
B. Ribose and uracil D. phosphate and adenine

Telegram: https://t.me/ExtremeSeries

72. Which of the following cell types can be rich in lysosomes?
A. Red blood cells B. Nerve cells C. Phagocytic cells D. Muscle cells

73. What can one conclude about the contents of the foodstuff if a mixture of Benedict solution and foodstuff remain blue after heating?
A. It contains proteins C. It contains fats
B. It lacks starch D. It lacks reducing sugar

74. Which of the following types of enzyme inhibitions can be removed when the end product of the metabolic pathway is depleted?
A. Allosteric inhibition C. competitive inhibition
B. Non-reversible inhibition D. Reversible competitive inhibition

75. If red blood cells shrink when placed in a certain solution, what is the strength of the solution relative to the strength of the protoplasm of the cells?
A. Hypotonic B. Isotonic C. Hypertonic D. Isoosmotic

76. Which of the following is NOT true about photo system –II?
A. Its reaction center molecule is P680.
B. It passes its excited electrons to photo system-I
C. The energy lost from its excited electrons reduces NADP
D. It replenishes its lost electrons from photolysis of water.

77. On which one of the following principle are most of the anti-HIV drugs currently in use working?
A. Inhibition of enzyme action C. Digesting of viral particles
B. Degradation of viral RNA D. Phagocytosis of the virus.

78. Among the following mating, where the ABO blood genotypes of the partners are shown, identify the mating in which all the children will have the same blood type.
A. AO x BO B. AA x OO C. AB X BO D. BB X AO

79. If a new mutant allele arises in a certain population, which of the following factors determines if the allele is going to be adaptive or non-adaptive
A. The environment in which the population lives.
B. The rate at which the gene mutates.
C. The population in which the gene is found
D. The use and disuse of the gene by the population.

80. Which of the following is the correct route that connects a stimulus and a response?
A. Receptor → coordinator → effector
B. Coordinator → receptor → effector
C. Receptor → effector → coordinator
D. Effector → receptor → coordinator

81. In which of the following compounds are both members of the pair are polymers of carbohydrates that naturally occur in plants only?
A. Starch and Chitin C. Glycogen and cellulose
B. Starch and Cellulose D. Glycogen and chitin

82. Which of the following is generally expected to give better health benefits when present in human diet?
A. Animal fat C. Saturated fatty acid
B. Monounsaturated fatty acid D. Polyunsaturated fatty acid

83. What is the importance of chemiosmosis in photosynthesis and cellular respiration?
A. Splitting of water molecules C. Combining hydrogen and carbon
B. Operating the proton pump D. Synthesizing ATP

84. If the age pyramid of a certain country is narrowing at the base, what does this tell about the trend of the population size of the country? The population size is:
A. declining B. growing fast C. increasing slowly D. stabilizing

85. Suppose the amino acid coding region in a mRNA is 1200 nucleotides long, how long is the protein in terms of amino acid number?
A. 1200 amino acids C. 400 amino acids
B. 600 amino acids D. 300 amino acids

86. If a new anti-HIV drug is to be developed to prevent the virus from entering the host cell, which one of the following processes should the drug target?
A. Reverse transcription
B. Binding of Gp 120 and CD4
C. Integration of viral DNA into host DNA
D. Assembly of viral parts into a whole virus

87. Which one of the following factors is NOT important for evolutionary change of a population?
A. Over reproduction C. Existence of heritable variation
B. Insufficiency of natural resource D. Survival of all that are born

88. One of the following would be harder to see under the ordinary light microscope that is more likely to be available in school laboratories.
A. Nucleus B. A bacterium C. A mitochondrion D. A ribosome

89. In some human liver cells there are 92 chromosomes per cell. What is the ploidy level of such cells?
A. Haploid B. Diploid C. Tetraploid D. Hexaploid

90. Which of the following groups of substances are all inorganic?
A. Water, sugar, calcium carbonate
B. Sugar, fatty acid, amino acid
C. Water, calcium carbonate, carbon dioxide
D. Carbon dioxide, amino acid, fatty acid.

91. To what kind of animal behavior can the spinning of a web by a spider be classified?
A. Learned behavior C. Instinctive behavior
B. Experiential behavior D. Accidental behavior

92. Which of the following is a recently developed active area of research in biology today?

Telegram: https://t.me/Extremeseries

A. Taxonomic study
B. Study about the cell theory
C. Ecological research
D. Stem cell research

93. Where does the light dependent reaction of photosynthesis occur in the chloroplast?
A. In the thylakoid membrane
B. In the fluid of the stroma
C. In all parts of the chloroplast
D. In the stomatal opening

94. For which of the following is the sugar produced by photosynthesis NOT used?
A. To produce biomass.
B. To make new DNA
C. To produce ATP in respiration
D. To produce enzymes

95. In which process is ATP generated during short distance high speed running?
A. Aerobic respiration
B. Mitochondrial energy transformation
C. Anaerobic respiration
D. The Krebs cycle

96. Which one of the following choices shows the end products of a mitotic cell division?
A. Gametes and ordinary cells
B. Four genetically different cells
C. Two genetically different cells
D. Two genetically identical cells

97. What exactly happens during a substitution mutation?
A. One base is replaced by a different base.
B. An extra base is added during DNA replication.
C. A base is missed out during DNA replication.
D. Changes occur in the arrangement or structure of a chromosome.

98. During the course of evolution, which of the following events came before all the others?
A. Origin of dinosaurs and other reptiles.
B. Availability of free oxygen in the atmosphere.
C. Origin of the oldest eukaryotic organisms
D. Origin of the first multicellualr animals and plants

99. What kind of natural selection is at work in a situation where individuals at both extreme ends of the range get better survival advantages than those around the middle?
A. Stabilizing selection
B. Directional selection extremes
C. Disruptive selection
D. Selection that removes the

100. In which geologic period does the fossil record show more diverse and relatively higher forms of organisms?
A. Devonian B. Cretaceous C. Jurassic D. Permian

Telegram: https://t.me/Extremeseries

ETHIOPIAN UNIVERSITY ENTRANCE EXAMINATION (EUEE)
BIOLOGY GINBOT 2006/JUNE 2014

1. Which of the following do viruses share with cells?
A. Cell wall
B. Cytoplasm
C. Nucleic acid
D. Protoplasm

2. Which one of the following cellular forms did Robert Hooke observe under his crude microscope?
A. Bacteria
B. Protozoa
C. Yeast
D. Empty cell wall

3. In the case of enzymatic chemical reactions, what do you call the substance that is acted up on by an enzyme?
A. Polypeptide
B. Coenzyme
C. Vitamin
D. Substrate

4. Choose the organisms that belong to the eukaryotes?
A. Bacteria
B. Blue-green algae
C. Amoeba
D. Virus

5. While conducting research, which of the following should a biologist do first?
A. Generate a hypothesis
B. Conduct experiments
C. Communicate the results
D. Identify a research question/problem

6. Which of the following structures is **NOT** present in animal cells?
A. Cell wall
B. Nucleus
C. Protoplasm
D. Nucleic acid

7. An experimental animal stopped responding to a stimulus that has been repeated so many times. What type of learned behavior is this?
A. Latent learning
B. Habituation
C. Imprinting
D. Operant conditioning

8. Behavior that is repeated on a daily basis is referred to as:
A. Lunar
B. Circadian
C. Circannual
D. Seasonal

9. Which molecule in the cell is the constituent of the gene?
A. Nucleic acid
B. Protein
C. Lipid
D. Carbohydrate

10. What is the branch of biology that studies about the origin and gradual changes of living things?

A. Microbiology
B. Evolution
C. Mutation
D. Reproduction

11. What is the general term for the part of the protoplasm that lies outside the nucleus?
A. Cytosol
B. Cytoplasm
C. Central vacuole
D. Plasma membrane

12. How many net ATP molecules are generated through anaerobic respiration, when a single glucose molecule is changed to pyruvate in the human body?
A. Two
B. Three
C. Four
D. Six

13. Which of the following groups of bacteria consists of rod shaped cells?
A. Gram-positive
B. Bacilli
C. Cocci
D. Spirilla

14. How many carbon and oxygen atoms are there in a molecule of maltose?
A. 6 carbon and 6 oxygen
B. 12 carbon and 11 oxygen
C. 11 carbon and 12 oxygen
D. 24 carbon and 12 oxygen

15. On which of the following organelles of the eukaryotic cell does protein synthesis take place?
A. The nucleus
B. The ribosome
C. The chloroplast
D. The mitochondrion

16. Which of the following parts of plant roots is harboring nitrogen fixing bacteria?
A. Root nodule
B. Root hair
C. Root tip
D. Root cap

17. Which of the following is true about protozoa?
A. They are a group of bacteria
B. They are one-celled organisms
C. They represent both plant and animals
D. They are unicellular animals

18. Which of the following is a distinguishing feature of amino acids that is **not** found in carbohydrates?
A. Carbon
B. Oxygen
C. Hydrogen
D. Nitrogen

19. Which of the following statements is true about matter and energy in the ecosystem?
A. Both matter and energy are recycled
B. Matter is not recycled; energy is recycled
C. Matter is recycled; energy is not recycled
D. Energy is recycled more than matter does

20. Which one of the following molecules in living things is **not** an organic molecule?
A. Sugar
B. Nucleotide
C. Amino acid
D. Water

21. At which stage is most of the ATP generated in aerobic respiration?
A. Glycolysis
B. Link reaction
C. Krebs cycle
D. Electron transport

22. Which of the following is the common characteristic of all enzymes?
A. They act inside cells only
B. They are proteins
C. They require cofactors
D. They operate at any PH

23. In a cross between round green pea of RRyy genotype wrinkled yellow pea of rrYY genotype, what is the expected genotype of the F1?
A. RRyy B. rrYY C. RRYY D. RrYy

24. Which of the following cellular structures is possessed by all cells?
A. Cell membrane
B. Nucleus
C. Cell wall
D. Golgi apparatus

25. During the Krebs cycle, which of the following molecules temporarily stores most of the energy released from food molecule?
A. ADP B. ATP C. NADH D. FADH

26. Which of the following carbohydrate has structural function?
A. Cellulose
B. Glycogen
C. Starch
D. Sucrose

27. Which of the following forms of nitrogen are readily utilizable by green plants?
A. Atmospheric nitrogen
B. Nitrite
C. Ammonium ion
D. Nitrate

28. In some crosses of maize, the progeny produces better yield than the parents. What is this phenomenon known as?
A. Dominance
B. Heterozygosity

Telegram: https://t.me/ExtremeSeries

C. Hybrid vigor D. Inbreeding depression

29. Which of the following techniques is used to separate DNA fragments according to their size on a gel?
    A. Southern blotting
    B. Electrophoresis
    C. Radioactive labeling
    D. Digestion by restriction enzyme

30. Which of the following groups of animals have males with ZZ and females with ZW sex chromosome constitution?
    A. Grasshoppers C. Mammals
    B. Birds D. Honey bee

31. The molecules of which pigment is located at the reaction center of a photosystem?
    A. Chlorophyll a C. Carotenoid
    B. Chlorophyll b D. Accessory pigments

32. According to Lamarck's theory of evolution, what is the mechanism by which evolving organisms acquire new structures?
    A. Mutation
    B. Hereditary variation
    C. Use-and –disuse of parts of the body
    D. Recombination of ancestral genes

33. In which of the activities of the honey bee colony are worker bees involved?
    A. Laying eggs C. Reproductive functions
    B. Serving as queens D. Taking care of the larvae

34. In which one of the following ways the Tundra biome differs from desert biome?
    A. It is a treeless biome
    B. It has little biodiversity
    C. It receives little precipitation
    D. It has permanently frozen subsoil

35. Which one of the following is the mechanism by which two genes located on the same chromosome are separated?
    A. Independent assortment C. Linkage
    B. Segregation D. crossing over

36. If present in the fermentation system, which one of the following would negatively affect alcohol production by yeasts?

A. Water C. Glucose
B. Oxygen D. Fermentation enzyme

37. Which of the following can be taken as case for evidence of evolution from the field of plant & animal breeding?
    A. Artificial breeding always creates new species
    B. Artificial breeding only produces new organisms with higher yield.
    C. Artificial breeding shows that natural selection can produce variations
    D. Artificial breeding can produce new variations in organisms

38. Which of the following mechanisms moves digested amino acids and glucose across the plasma membrane of the cells lining the wall of the small intestines?
    A. Osmosis C. Facilitated diffusion
    B. Simple diffusion D. Active transport

39. Where exactly in the cell does the Krebs cycle take place?
    A. Mitochondrial matrix
    B. Inner mitochondrial membrane
    C. Cytoplasmic fluid
    D. Outer mitochondrial membrane

40. What is the ultimate source of energy for all organisms living on the earth?
    A. Water B. Oxygen C. Nitrogen D. The sun

41. Which of the following substances is **NOT** necessary for photosynthesis to take place?
    A. Chlorophyll C. oxygen
    B. Carbon dioxide D. water

42. Which group of organic compounds includes the enzymes?
    A. Proteins C. starches
    B. Lipids D. carbohydrates

43. Intowhich of the following organic compounds can lipase, maltase and sucrase be grouped?
    A. Hormones C. Nucleic acids
    B. Carbohydrates D. Enzymes

44. Choose the characteristics that viruses share with living things.
    A. They are made up of many specialized cells
    B. They contain genetic material

Telegram: https://t.me/ExtremeSeries

C. They reproduce by mitosis
D. They contain chlorophyll

45. Which one of the following eukaryotic cell organelles was a free living cell before eukaryotic cells evolved?
A. Nucleus
B. Ribosome
C. Chloroplast
D. Nucleolus

46. If a radioactive substance that weighs one kilogram has a half-life of 100 years, what would be the percentage of the substance left after 300 years?
A. 50 B. 30 C. 25 D. 12.5

47. Which of the following is more true about the male of the honey bee?
A. It has no father
B. It is sterile
C. It is diploid
D. It regulates the colony

48. Which of the following PH values represents the strongest base?
A. 2 B. 7 C. 13 D. 15

49. In Ethiopia, which one of the following vertebrate groups has the largest number of genera?
A. Mammals
B. Birds
C. Reptiles
D. Amphibians

50. Which of the following is an accurate representation of the relationships between the terms metabolism, anabolism and catabolism?
A. Anabolism=Catabolism
B. Metabolism=Catabolism
C. Anabolism=Catabolism + Metabolism
D. Metabolism=Catabolism +Anabolism

51. What do you call structures that have the same evolutionary origin but now are different in structure or have different functions?
A. Endemic
B. Analogous
C.Homologous
D. Indigenous

52. Which term describes the process by which water is lost by evaporating through the stomata of leaves?
A. Transport
B. Transpiration
C. Anchorage
D. Adhesion

53. Which of the following is true about saturated fatty acids?
A. They have single bonds between carbon atoms
B. They are the same as polyunsaturated fatty acids

C. They are the same as monounsaturated fatty acids
D. They have double bonds between the carbon atoms

54. For a biological detergent to effectively remove oily and greasy dirt, which of the following enzymes should it contain?
A. Cellulase
B. Protease
C. amylase
D. lipase

55. In which kingdom of life is it most likely to find many examples of organisms that can undergo metamorphosis?
A. Monera
B. Fungi
C. Plantae
D. Animals

56. During protein synthesis, where in the cell does transcription take place?
A. Ribosome
B. Nucleus
C.Endoplasmic reticulum
D. Golgi apparatus

57. What is a perfect flower?
A. A flower with petals and pistil
B. A flower with stamens and sepals
C. A flower with stamens and pistil
D. A flower with sepals and petals

58. Which molecules carry the instructions for protein synthesis?
A. Carbohydrates & lipids
B. Amino acids
C. DNA & RNA
D.Enzymes

59. Which region of the human digestive tract contains enzymes that perform well at low PH?
A. Mouth
B. Stomach
C.Small intestine
D. Large intestinal

60. Which of the following resources is needed by an ecosystem from outside in order to sustain itself?
A. Producers
B. Macronutrients
C. Micronutrients
D. sunlight

61. Which of the following is true about the substance represented by the chemical formula "$C_2H_5NO_2$"?
A. Simple sugar
B. Inorganic substance
C. Fatty acid
D. Amino acid

62. Which one of the following characteristics is UNNECESSARY for an object to be considered a living thing?
A. Ability to respond to stimuli
C.Ability to grow

Telegram: https://t.me/ExtremeSeries

B. Ability to reproduce D.Ability to move

63. Which field studies the way in which individual traits of organisms are transmitted from one generation to the next?
    A. Genetics C. Evolution
    B. Ecology D. Morphology

64. In which of the following ways is the carbon dioxide of the atmosphere fixed into the carbon found in organic molecules?
    A. In the breathing processes of all animals
    B. In the decomposition of organic molecules
    C. In the process of photosynthesis by green plants
    D. In all the cellular respiration processes of organisms

65. Which of the following is NOT a term used to describe organisms that have had foreign genes added to them?
    A. Genetically modified organisms
    B. Transgenic organisms
    C. Genetically engineered organisms
    D. Pathogenic organisms

66. Which of the following is prevented from taking place if populations are separated by a geographic barrier?
    A. Mutation C. Gene flow
    B. Evolution D. Natural selection

67. In protein synthesis, what is produced during transcription?
    A. Protein C. DNA
    B. mRNA D. polypeptide

68. If a cell fails to clear its cellular debris, which one of its organelles is most likely NOT functioning?
    A. Nucleus C. Endoplasmic reticulum
    B. Mitochondria D. Ribosomes

69. Which of the following applies to the tropical rainforest?
    A. Low rainfall and high temperature
    B. Low rainfall and low temperature
    C. High rainfall and high temperature
    D. High rainfall and low temperature

70. Which stage in the life cycle of HIV is inhibited by the antiretroviral drug known as integrase-inhibitor?
    A. Conversion of viral RNA to DNA
    B. Replication of the viral genome

C. Insertion of the viral genome into host genome
D. Release of the viral progeny from the cell

71. Which of the following groups of plants carry out light dependent and light independent reactions of photosynthesis in separate cells?
    A. C-3 plants
    B. C-4 plants
    C. CAM plants
    D. Plants without chlorophyll

72. Among the following identify the organelle in which nucleic acid is NOT found?
    A. Mitochondria C.Ribosome
    B. Chloroplast D. Golgi apparatus

73. Which of the following can create strong inter-specific competition if shared by two or more species?
    A. Habitat C. Niche
    B. Ecosystem D. predator

74. The half-life of carbon -14 is about 5730 years. If a fossil is 17200 years old, about what percent of its original carbon-14 is still present in the fossil?
    A. 75% B.50% C.25% D. 12%

75. Which of the following pairs of individuals have identical DNA fingerprints?
    A. Father and son C. Brother and sister
    B. Mother and daughter D. Monozygotic twins

76. Among the following which one has little or no contribution to the loss of biodiversity?
    A. Overgrazing by domestic animals
    B. Clearing forests to grow crops
    C. Growing crops of improved verities
    D. Growing crops of land

77. Suppose sites A,B,C and D have Simpson's Index of Diversity (d) values of 20,8,16 and 3, respectively, which site is dominated by one or just a few species?
    A. Site A B. Site B C. Site D.Site D

78. Which of the following types of chromosome mutations gives rise to an extra chromosome?
    A. Inversion C.Deletion

Telegram: https://t.me/ExtremeSeries

B. Translocation D. Non-disjunction

79. Suppose a hypothetical cube-shaped cell has sides of 10 micrometers, what is the surface area to volume ratio of this cell?
A. 6:10 B. 10:10 C. 3:6 D. 4:8

80. Which of the following pairs are analogous structures?
A. Wing of a bird & wing of a butterfly
B. Front leg of a horse & a human arm
C. Wing of a bird & wing of a bat
D. Front leg of a frog & wing of a bat

81. How are unsaturated fats distinguished?
A. They are made up of glucose & fructose
B. They are made up of amino acids & glycerol
C. They have double bonds in their carbon chains
D. They have only single bonds at the points of attachment

82. Which of the following mating produces children all having the same ABO blood phenotypes?
A. AO x AB C. AO x BO
B. BB x OO D. AA x BO

83. Which of the following is the largest ecological unit?
A. A community C. A population
B. A biome D. An ecosystem

84. Among the following fossil hominid species, which one is the oldest of all?
A. Homo erectus
B. Ardipithecus ramidus
C. Australopithecus afarensis
D. Australopithecus africanus

85. During aerobic respiration, what is the route through which protons return from the mitochondrial inter-membrane space back to its matrix?
A. Proton pump C. Ion channel
B. ATP synthase D. Membrane lipid

86. From which of the processes of cellular respiration is the majority of the ATP generated?
A. Anaerobic fermentation
B. Glycolysis & link reaction
C. Electron transport & chemiosmosis

D. Krebs cycle & glycolysis

87. Choose the one that represents a chemical formula of an organic compound?
A. $C_{12}H_{22}O_{11}$ B. $Mg(OH)_2$ C. NH_3 D. NaCl

88. In which of the following solutions does an animal cell undergo hemolysis?
A. In hypotonic solution
B. In hypertonic solution
C. In isotonic solution
D. In both hypotonic & hypertonic solutions

89. On which of the following does the rate of algal photosynthesis in a lake depend?
A. The oxygen content of the water
B. The nitrogen content of the water
C. The elevation where the lake is found
D. The amount of light that penetrates the lake water

90. What does the absence of chloroplasts in a unicellular organism indicate?
A. Reproduction by sexual means
B. Autotrophic nutrition
C. Heterotrophic nutrition
D. Complete lack of metabolic processes

91. Which of the following did the early heterotrophs do that probably assisted the origin of autotrophs?
A. They added oxygen and chlorophyll to the environment
B. They added carbon dioxide to the environment
C. They stored energy in the bonds of inorganic compounds
D. They manufactured food from carbon dioxide and oxygen

92. If DNA contains 10% thymine, what is the percentage of cytosine in the DNA?
A. 10% B. 30% C. 40% D. 90%

93. Which of the following conclusions can be drawn from the structural similarities observed between the flippers of whales and the arms of humans?
A. The human species began life in the ocean
B. The human species and whales share a common ancestry
C. Whales are older than the human species

Telegram: https://t.me/ExtremeSeries

D. Whales have evolved from early humans that went back to the ocean

94. Which of the following will happen if a plant cell is kept in a solution that is stronger than its protoplasm?
   A. The cell will become turgid
   B. The central vacuole will expand
   C. The protoplasm will get plasmolysed
   D. The cell will swell and burst

95. Which of the following is produced when a lipid molecule is hydrolyzed?
   A. Amino acids & water
   B. Amino acids & glucose
   C. Fatty acids & glycerol
   D. Glucose & glycerol

96. What is the ecological succession in which plant communities establish after an area has been completely demolished by fire?
   A. Tree succession
   B. Secondary succession
   C. primary succession
   D. Bare rock succession

97. What is the difference between HIV and AIDS?
   A. AIDS is the virus and HIV is the disease
   B. Both AIDS and HIV refer to the disease
   C. AIDS is the disease and HIV is the virus
   D. Both AIDS and HIV refer to the virus

98. From which organism is the first antibiotic used in medicine produced?
   A. Gram negative bacteria
   B. Members of the fungi
   C. Genetically modified protozoa
   D. Higher plants

99. Biotechnology can be applied in environmental remediation. To which of the major application areas of biotechnology is this most related?
   A. Medicine and nutrition
   B. Medicine and industry
   C. Biodiversity and industry
   D. Agriculture and environment

100. Suppose in an experiment conducted to test a new drug, 20 people were given the drug and 20 people a placebo. What is the control of the experiment?
   A. People who received the placebo
   B. People who received the new drug
   C. The new drug itself
   D. The new drug and the placebo.

Telegram: https://t.me/Extremeseries

1. B: **Osmosis** is a special kind of diffusion in which water is the substance diffusing from a dilute solution to a concentrated solution across a semi – permeable membrane.

2. A: Plants obtain their energy from the sun. Plants have pigments, which enable them to absorb sunlight. The most important pigment in plants is **chlorophyll** a. Other pigments in photosynthetic cells are **chlorophyll** b, **Carotenes** and **Xanthophylls.**

3. C: Protein molecules are synthesized from 20 different amino acids. **Triplets of bases** are the smallest units that can code for all the amino acids. If the DNA code words consist of triplets with each arrangement of 3 consecutive bases specifying an amino acid, example AGT, there can be 64(4^3) possible code words.

4. C: **Renewable natural resources** are resources that can be replaced. Example, all living things, water, soil and air. Therefore, **biofuels** or fuels from organisms of biological origin are renewable natural resources. Fossil fuels such as **coal, petroleum** and natural gas are non-renewable resources.

5. C: The kingdom **Monera** contains prokaryotic unicellular organisms, **bacteria** and **blue green algae.** In prokaryotic cells such as bacteria and blue green algae the nuclei and other organelles lack membranes around them.

6. C: Enzymes are proteins. Proteins can be either **fibrous** or **globular.** In fibrous proteins the polypeptide chains are arranged in flat sheets. In globular proteins, the polypeptide chains are tightly folded into compact, complicated spherical shapes. **Enzymes** are globular proteins.

7. C: The female goose would retrieve an egg that has rolled just outside her nest by reaching beyond it with her bill and rolling it towards her with the underside of the bill. This kind of behaviour is called an **innate behaviour.**

8. B: **Developmental biology** is the study of the process by which organisms grow and develop. It studies the genetic control of **cell growth, differentiation** and **morphogenesis** which is the process that gives rise to tissues, organs and anatomy.

9. D: **Angiosperms** are also known as flowering plants. They are plants that produce seeds inside a flower. Each pollen grain has two nuclei. These two nuclei are called the **tube nucleus** and the **generative nucleus.** The generative nucleus divides to produce two **sperm nuclei.** After a pollen grain lands on a stigma, a tube called a **pollen tube,** grows out of the pollen gain. The pollen tube

then passes through a tiny opening called a **micropyle** in to an **embryosac** with two **polar nuclei** and an **egg.** One sperm unites with the two polar nuclei and forms endosperm nucleus. The other sperm unites with the egg to form a zygote. Since there are two unions within an embryo sac, it is called **double fertilization.**

10. C: **Rodents** are mammalian pests that include rats, mice and moles.
11. B: One molecule of glucose ($C_6H_{12}O_6$) contains six carbon atoms. Therefore, two molecules of glucose contain twelve carbon atoms.
12. C: The elements that compose the living material of the cell **(protoplasm)** occur in living things in the form of both **inorganic** and **organic compounds**. The most important inorganic compounds in living cells include water, salts, inorganic acids and bases. The organic compounds include carbohydrates, fats, proteins, nucleic acids and vitamins.
13. B: 40% the base molecules in a DNA strand of 3000 bases are guanine base molecules. Therefore, the number of guanine base molecules

$$= \frac{40}{100} \times 3000 = 1200$$

In the DNA of any organism, the amount of adenine equals the amount of thymine and the amount of guanine equals the amount of cytosine. Hence, the amount of cytosine will be 1200.

The sum of guanine and cytosine base molecules (1200+1200) is equal to 2400. The total number of bases (3000) minus 2400 is equal to 600, will be the total number of adenine and thymine. Therefore, the amount of adenine base molecules in the DNA strand will be 300.

14. B: A DNA molecule is made up of two chains of nucleotides held together through their bases. The bases holding the two chains together are paired in a specific manner. Adenine can only pair with thymine and guanine can only pair with cytosine.
Therefore, a DNA strand with the sequence ATTGGC would be complementary to the DNA sequence given as TAACCG.
15. A: **Autotrophs (producers)** are organisms that can make their own food from simple inorganic substances. They include **green plants, algae** and **some bacteria.**
Heterotrophs (consumers) are organisms that cannot prepare their own food. They obtain their food directly or indirectly from autotrophs. They include **animals, fungi** and **most bacteria.**
16. D: The flora in the **tropical rain forest** includes **epiphytes,** plants that grow on other large plants and **lianas,** climbing plants which use the trees to support growth into the canopy.
17. D: The basic structural units of carbohydrates are called **simple sugars.** Proteins are compounds that are made up of small units called **amino acids.** Lipids are made up of units called **fatty acids** and **glycerol.**

18. B: **Transgenic organisms** that have been genetically altered using the technique of genetic engineering are called **genetically modified organisms** or **GMOs.**
19. A: Use of fishes to feed on mosquito larvae is a biological control method of larva.
20. A: The sex chromosomes in males are XY and the sex chromosomes in females are XX. In the production of eggs the two sex chromosomes segregate. All the egg cells will thus carry an X chromosome. Similarly, in the production of sperm cells the two sex chromosomes segregate. Therefore, 50% of all the sperm carry an X chromosome and the other 50% carry a Y chromosome. There is a 50% chance of conceiving a girl and a 50% chance of conceiving a boy:

	Female	Male
Phenotype of parents		
Genotype of parents	XX	XY
Gametes	X and X	X and Y

Eggs

Sperms	X	X
X	XX	XX
Y	XY	XY

XX (Female) = 50%
XY (male) = 50%

21. C: It is believed that the **Australopithecus** gave rise to a new species called **Homo habilis,** so called because they were able to use tools. The Homo habilis is called the handy human. It is thought to be the ancestor of **Homo erectus.**
22. C: Plants have an **alternation of generations.** There is a **diploid sporophyte generation,** which alternates with a **haploid gametophyte.**
23. C: The abbreviation **HIV** stands for the **Human Immunodeficiency Virus,** which is the name of the organism that causes AIDS and the abbreviation **AIDS** stands for the **Acquired Immune Deficiency Syndrome,** which is the name of the symptoms of this disease.
24. B: **Cellulose** is the most abundant polysaccharide. It accounts for fifty percent or more of all the carbon in plants.
25. C: PH is the measure of the **acidity** and **alkalinity (basicity)** of a solution. The PH value of an enzyme that has its optimum activity at neutral PH is 7.

Telegram: https://t.me/Extremeseries

26. C: **Habituation** dis a kind of learning in which we no longer notice or react to the various stimuli, example, traffic passing the school and noise of other students talking near us.

27. C: In two half – life time (10,000 years) only one quarter of the original amount of C_{14} would remain. Therefore, if the age of the fossil is estimated at 15,000 years old, only one eighth (12.5%) of the original amount of ^{14}C would remain.

28. C: The **testes** have **seminiferous tubules,** which produce sperm. The sperm mother cells in the seminiferous tubule are called **spermatogonia**. The sperm cells are produced by the spermatogonia lining the seminiferous tubules of the testes. The spermatogonia divide by mitosis to form **primary spermatocytes.** Each spermatocyte undergoes meiosis I to produce two **secondary spermatocytes**, that undergo meiosis II and produce four sperm cells.
Therefore, if four sperm cells are produced from 1 spermatocyte, 40 sperm cells will be formed from 10 primary spermatocytes.

29. C: The **Juniperus forests** are the poorest in the number of tree species. The species of trees include Juniperus procera, Euphorbia species, Olea europaea subspecies Cuspidate, Hagenia abyssinica and Arundinaria alpina.

30. D: In **phyletic speciation** a species undergoes changes over a long period of time involving many generations. Eventually a new form of species arises which is different from the ancestor species.

31. D: **Reproduction** is the process by which living organisms give rise to new generations and ensure perpetuation of the species.

32. A: **Sex – linked traits** in Drosophila were discovered by **Thomas Hunt Morgan.** Genes carried on the sex chromosomes are said to be sex-liked. Females have two genes for each trait. That is because females have two X chromosomes. Males have only one gene for each trait. When the X and Y chromosomes are together, the genes on the X chromosomes control the trait. Some sex – linked characteristics in humans are colour blindness, haemophilia, and certain types of baldness.

33. B: Man has experimentally proved that it is possible by **selective breeding** to produce desirable characteristics. He has produced new strains of plants for example, which are resistant to diseases. New strains of animals, for example, animals that yield more meat or that yield large quantities of milk have been raised by man.

34. D: **Caffeine** is a chemical substance found in coffee, tea and cola drinks. **Nicotine** is found in tobacco and cocaine.

35. A: **Beriberi** is a deficiency caused by a shortage of **Vitamin B_1 (thiamine)**

Scurvy is a disease due to deficiency of vitamin C.
Night blindness is a deficiency disease caused by a lack of vitamin A.

36. B: **Rickets** is caused by deficiency in the level of vitamin D.

37. B: **Glucose** is the main product of photosynthesis. Glucose formation can occur both in the light as well as in the dark.

38. D: Each person's fingerprints are unique. Not even identical twins are identical. When a criminal commits his crime and leaves his prints behind it is almost like a document with his name and signature guilt.

39. A: The **substrate** is the substance upon which the enzymes act. If the amount of substrate is increased while the concentration of enzyme remains the same, the rate of the reaction will increase. An increase in the concentration of the substrate will increase the rate of the reaction until the available active sites are full, any further increase in substrate concentration would not speed up the rate of reaction.

40. D: The A, B, AB and O blood types are one factor in blood. The Rh factor is another factor. Those individuals with the Rh factor are known as Rh positive. Those lacking the factor are known as Rh negative. An Rh positive person can receive blood from an Rh negative individual. However, if a person who is **Rh negative** receives blood from an **Rh positive** person, the recipient's blood would produce **antibodies** that react with the **Rh antigen** on the red blood cells and cause the red blood cells of the Rh positive blood to agglutinate or clump.

41. C: A population can grow at its maximum rate when there are unlimited resources. The maximum rate at which a population could increase under ideal conditions, when there are unlimited resources is called the **biotic potential.** This would refer to the theoretical maximum birth rate under an ideal environmental condition or when given unlimited environmental resources.

The scientific method follows the following basic steps

1. **observation.** A scientific study begins with observations of objects
2. **Identifying the problem.** Scientists ask questions from what they observe.
3. **Forming a hypothesis.** Observations lead to the formulation of hypothesis. Thee hypothesis is a tentative theory or an intelligent guess. It can be supported or disproved after the experiment.
4. **Making predictions.** A prediction is a logical deduction from a hypothesis, without any kind of experimentation.

Telegram: https://t.me/ExtremeSeries

5. **Conducting an experiment.** Hypotheses are often tested by experimentation. After the experiment the hypothesis can be rejected or accepted.
6. **Drawing a conclusion from the experiment.** If the hypothesis is supported by an experiment it becomes a **theory.**

42. C: In compound microscopes the object that is magnified by the objective is again magnified by the eyepiece. Therefore, the total magnification of a microscope is the product of the magnifying power of the eye piece and the particular objective used. Hence, the image size of a cell with the size of 0.03 microns when viewed under a compound microscope would be 0.03 microns × 10x × 100x = 30 microns

43. B: There are two main types of nitrogen bases, **pyrimidines** and **purines.** The members of the pyrimidine family are cytosine, thymine and uracil. The members of the purine family are adenine and guainine. Thymine is found only in DNA while uracil is found only in RNA.

44. C: When a cell is surrounded by pure water or a solution whose solute concentration is lower than that of the cell's contents, water flows into the cell by osmosis, the cell swells up and the solution is said to be **hypotonic** to the cell.

45. A: **Lamarck's theory of evolution** was based on the theory of use and disuse and the theory of acquired characteristics. **Darwin** proposed that **natural selection** is the mechanism by which new species arise from pre – existing ones. Both Lamarck and Darwin share the view that life is purely a result of natural processes.

46. D: A succession developing on newly emerged land or water is termed a **primary succession.** The **pioneer plants** that are found on a very dry substratum, like bare rock are called **xerophytes.** These plants are adapted and specialized to **with stand** water shortage.

47. C: **Innate behavior** is the pattern of inherited pre-set behavioral responses that develop along the organism. It is the behavior that we inherit from our parents. It is an inbred behavior that is pre-set and cannot be learnt.

48. A: A **camel** mostly lives in **desert** where there is hardly any water and the temperatures rise to **extremes** because of the rays of the sun. Camels have evolved a number of adaptations.
- They can survive even if they lose up to thirty percent of their ideal water content.

- When a camel runs short of water, its body temperature changes **(fluctuates)** by up to ten degrees Celsius. This drop in temperature enables the camel to save the water that would have been evaporating to keep it cool.
- The hump of the camel is used for storing fat, but it also releases water for the camel to use during periods of drought.
- Prevents the loss of water by concentrating its urine.
- It also passes only small quantities of urine at a time.

49. A: **Mutualism** is a type of symbiotic relationship where two organisms live together and both benefit from the relationship.

50. B: In deserts, scattered drought resistant **herbs, euphorbias** and **ephemerals** are found. Euphorbias are **succulent plants** which can store water example, **cacti.** Ephemerals are plants which can grow and produce seeds within a few days.

https://t.me/ExtremeSeries
Telegram

1. D
There are three types of natural selection operating in organisms
a. **Stabilizing (normalizing) selection** selects against phenotype extremes. Individuals with an average of intermediate phenotype are preferred. Thus, this kind of selection can be indicated by a **bell shaped curve.**
b. **Directional selection** favors one extreme phenotype from the population. Thus reducing variations and leads to a shift in the population average for the selected character.
c. **Disruptive selection** favors two or more phenotypes within the population. It pushes the phenotypes away from the average phenotype two wards the extremes of the population. There will be no intermediate ones and thus result in the formation of two distinct species.

2. C
Ecosystem diversity includes the bread differences between ecosystem types and the diversity of habitats and ecological processes occurring within each ecosystem.
Cultural diversity is manifested by diversity in language, religious beliefs, art, music, social structure, diet and many other aspects of human society.
Species diversity refers to the variety of species within a region. It can be measured in **species richness, species abundance** and **taxonomic diversity Genetic diversity** refers to the variation of genes within species.

3. B
Carbohydrates are made up of carbon hydrogen and oxygen
Proteins are compounds containing carbon, hydrogen, oxygen, nitrogen and other elements such as sulphur or phosphorus.
Nucleic acids are organic compounds containing carbon, hydrogen, oxygen, nitrogen and phosphorus.

4. B
The **cell membrane** is a thin covering found in all plant and animal cells. It controls what moves in to and out of the cells.
All plant cells are surrounded by a **cell wall.** It provides protection, shape and support to plant cells. Water and other substances pass freely through the cell wall. It is completely permeable

5. D
Based on their mode of nutrition organisms can be classified as **autotrophs** and **heterotrophs**

a. **Autotrophs** (**producers**) are organisms that can make their own food from inorganic substances. This takes place in **green plants, algae** and **some bacteria.**
b. **Heterotrophs (consumers)** are organisms which are incapable of manufacturing their own food from inorganic materials, hence depend upon other organisms as their source of food, as **animals, fungi** and **most bacteria.** The heterotrophs are further classified based on their methods of obtaining food.
Omnivores are animals that feed both plant and animal foods as man.
Herbivores animals that eat plants only as cow, horse, sheep, etc
Carnivores are animals such as lions that live on flesh.

6. B
The break down of glucose during respiration occurs in three stages.
a. **Glycolysis** is the first stage of cellular respiration. It is anacrobic, because it does not require oxygen. During glycolysis, glucose is broken down into two three carbon compounds called **pyruvic acid.**
b. **The krebs cycle.** In the mitochondria, the molecules of pyruvic acid are broken down to **carbon dioxide, water Coenzyme A.** Then acetyl coenzyme A combines with oxaloacetic acid, a four carbon compound to form citric acid, a six carbon compound.
c. **The electron transport system.** The second, the third stages and the formation of acetyl coenzyme A take place in the presence of oxygen.

7. B
Meiosis is a type of cell division which produces daughter cells containing only half the number of chromosomes of the parent cell
Fertilization is the fusion or union of an egg cell will a sperm cell to form a zygote
Cleavage means dividing a zygote (fertilized egg) into many cells by means of mitosis, i.e. the zygote first is one cell, then two, then four, eight, sixteen, and soon.
Gastrulation is the process by which a blastula develops into a gastrula

8. D
The energy released when food is burned inside the cells is stored in a molecule called **adenosine triphosphate (ATP).** This compound occurs in all living cells and is referred to as the energy **currency of the cell.**

9. B
Photosynthesis is the process by which green plants use light energy from the sun to make food. Carbohydrates, oxygen and water are produced during photosynthesis.

Telegram: https://t.me/ExtremeSeries

10. A
The concentration of carbon dioxide in the atmosphere is increasing in recent years than before. As a result, the temperature of the earth is slowly increasing. This is called **global warming.**

11. C
The source of the oxygen that green plants release during photosynthesis is water. Photosynthesis is divided into two steps. The first step is called the light reaction. In the light reaction the light energy splits water molecule into hydrogen and oxygen

$$12H_2O \xrightarrow[chlorophyl]{light} 24H + 6O_2 \uparrow$$

12. D
Protoplasm is the living matter of every cell and is composed of the cytoplasm and the nucleus

13. A
There are nitrogen – fixing bacteria called **rhizobium** which are found in the **root nodules** of **leguminous plants** like beans, peas, clover and alfalfa. The bacteria in these plants convert the nitrogen gas into ammonia. The bacteria get sugar from the legumes and supply them with nitrates in return

14. C
The **cell wall** is a non living layer that surrounds the cell membranes of plants, fungi and bacteria. Animal cells have no cell wall.
Chromoplasts are plastids that contain yellow, orange, blue and red pigments. They are organelles found only in plants.
Chromosomes are thread like structures that determine the characteristics of living things. They are found both in plant and animal cells.
Leucoplasts are colourless plastids that serve as store houses of nutrients in plant cells. They are found only in plant cells.

15. D
Glycogen or **animal starch** is a polymer of glucose molecules. It is stored in the liver and muscle of animals and is used by the animals as food reserves.

16. C
When the red blood cell is placed in a hypertonic solution, water flows out of the cell and the cell shrinks. A **hypertonic solution** is a type of solution which contains a higher concentration of solutes and therefore a relatively low concentration of water.

17. B
Thee final electron acceptor in the electron transport chain during aerobic respiration is **oxygen.**

18. C
Meiosis is concerned with gamete formation and it occurs only in gamete producing organs such as ovaries, testes and anthers.

19. C
The **Krebs cycle** or **Citric acid cycle** takes place in **mitochondria.**

20. A
The **Golgi apparatus** are flattened sacs found within the cytoplasm. They are used in collecting and packaging cell products such as proteins and lipids. They also transport materials within the cell and out of the cell.

21. B
Biosensors have been produced using **biotechnology** to detect minute amounts of chemicals. A good example of this is a biosensor that can detect glucose levels in the blood. This is very useful for a diabetic to check his or her blood sugar level. In this way, diabetics can control their disease by administering the right treatment.

22. A
Archaeopteryx has features of both reptiles and birds. Therefore it is regarded as one of the intermediate forms and could have been the link between **reptiles** and **birds.**
Australopithecus africanus is the missing link between apes and humans
Cyanognathus is a fossil link between reptiles and mammals
Semouria is a creature that is assumed to be a link between amphibians and reptiles

23. A
Some herbivorous animals like rabbits diget the cellulose with the help of enzyme called **cellulose** which is produced by the symbiotic bacteria that live in the caecum of these animals.

24. D
1. **Osmosis** is the diffusion of water molecules from a dilute (weak) solution to a concentrated (strong) solution across a semi-permeable membrane.
2. **Pinocytosis** means cells drinking and the material is taken up in liquid form.
3. **Active transport** is an energy consuming transport of molecules or ions across a membrane against a concentration gradient.
4. **Phagocytosis** means cell eating. The material is taken up in solid form. For example, when the amoeba feeds on small organisms or when a white blood cell engulfs bacteria.

25. B
Hydrogen peroxide is a substance or a **substrate** which is extremely toxic to cells. An enzyme known as **catalase** which is found in most

Telegram: https://t.me/ExtremeSeries

living tissues such as potato, liver, kidney and muscles catalyses the break down of H_2O_2 to **oxygen** and **water.**

$$2H_2O_2 \xrightarrow{Catalase} 2H_2O + O_2$$

26. D

In **honeybees,** only one female is reproductive, the **queen.** A queen bee is developed from fertilized eggs and the larva fed on the royal jelly. Its job is to lay eggs and to secrete pheromone.

27. C

Most of the energy released during the **Krebs cycle** is transferred to NADH. One glucose molecule causes two turns of the Krebs cycle. The two turns produce **six NADH,** two $FADH_2$ two ATP and four CO_2 molecules.

28. D

When HIV enters the blood stream, it attacks **T-helper cells,** the cells which protect our body against infection.

29. A

The trait **yellow pod colour** is not expressed or masked in the F_1 generation.

30. C

Oxygen molecule was thought to be totally absent from the primaeval atmosphere. The gases of the primitive atmosphere were ammonia, methane, carbon dioxide, water vapor, nitrogen and hydrogen sulphide.

31. B

Homologous structures are organs with the same origin and development but may or may not have the same function. For example, the human fore arm, fore leg of horse, flippers of the whale and wings of the bird are examples of homologous structures

32. C

Pulses are also called **legumes.** Legumes are rich sources of proteins. The dominant pulse crops are horse bean, chickpea, haricot bean, field pea and lentils

33. C

The different classes of carbohydrates are monosaccharides, disaccharides and polysaccharides

Glucose, fructose and galactose are monosaccharides, sucrose, lactose and maltose are disaccharides and starch, glycogen, chitin, cellulose and murein are polysaccharides.

34. C

The membrane has a **phospholipids bilayer.** A phospholipids molecule is composed of a phosphate group and two lipid fat layers. Each fat layer is composed of molecules that have hydrophilic or polar heads and hydrophobic tails. The two layers of phospholipids arrange

Telegram: https://t.me/Extreme_series

themselves in such a way that the heads face water on both sides and the hydrophobic tails avoid contact with water.

35. A

The **cell cycle** is conveniently divided into four phases. These are referred to as G_1, S, G_2 and M. **Interphase** includes G_1, S and G_2. **Mitosis** is referred to as the **M-stage.**

36. B

Primary succession is a succession developing on newly emerged land or water. On such bare surfaces **lichens** and **mosses** are the first to appear. They are the first **colonizers** and are called the **pioneer community.**

37. B

The **Krebs cycle** is a cyclic series of chemical reactions which take place in the **mitochondria,** in the presence of oxygen. The Krebs cycle takes place in the **matrix** of the mitochondrion. The acetyl coenzyme A enters into the **matrix** of the mitochondrion and combines with oxaloacetic acid to form citric acid.

38. A

F_1 Rr Yy

$F_1 \times F_1$ RrYy × RrYy

F_1 gametes RY, Ry, rY, ry × Ry, rY, ry

When crossed the F_2 generation produced 9 round and yellow seeded plants, 3 round and green – seeded plants, 3 wrinkled and yellow-seeded plants and 1 wrinkled and green seeded plant that is a phenotypic ratio of 9: 3: 3: 1

If there are 9 plants with round and yellow seeds out of 16 plants, then there will be $\left(\frac{9}{16} \times 128\right)$. Which is equal to 72 plants with round and yellow seeds out of 128 plants.

39. A

The stages in the development of an embryo are **cleavage, gastrulation** and **organogenesis.** Fertilization is followed by cleavage, which is a succession of rapid cell divisions that produces a ball of cells from the zygote. The cell division that occurs during cleavage is mitosis. Cleavage can be characterized as the period of development in which

a. the number of cells increases
b. no growth occurs
c. the ratio of the size of the nucleus to the cytoplasm increase
d. the nucleus divides mitotically and therefore more nuclei are produced

40. D

Night blindness is a deficiency disease caused by a shortage of vitamin A in the diet. Foods rich in vitamin A are peppers, carrots and Vegetables

Pellagra is caused by the deficiency of niacin or Vitamin B_3. Diets that include vegetables and foods rich in protein like meat, eggs, milk and cheese can prevent pellagra

Beriberi is a deficiency disease caused by a shortage of thiamine or vitamin B_1. The sources of thiamine are cereals, legumes, lean meat, nuts and yeast.

Scurvy is a disease due to deficiency of vitamin C. A diet rich in fruits like orange, tomatoes and vegetables are rich in vitamin C.

41. B

Rinderpest is a Viral disease of the livestock

Trypanosomiasis is caused by protozoan called **trypanosome.** It is a disease transmitted by the bite of **tsetse flies. Blackleg disease** is a disease caused by **Clostridium chauvoei.** It affects all domestic animals including camels.

Anthrax is an animal disease caused by a bacterium called **Bacillus anthracites.**

42. A

Mimicry is a type of protective resemblance. In mimicry one animal looks like another animal. This keeps many predators away.

43. C

Pavlov found that dogs salivate at the sight of food (**unconditioned stimulus).** If Pavlov always range a bell **Conditioned stimulus,** when he offered food, the dogs associate the conditioned stimulus with food. Eventually the sound of the bell alone could cause salivation

44. C

If a part of a DNA strand is damaged the damaged part is cut out and new complementary nucleotides are added in the enzyme that is used to cut DNA molecules in to small pieces is called **DNA polymerase**

45. C

There is a 50% chance of conceiving a girl and a 50% chance of conceiving a boy.

Phenotype of parents		**Female**		**Male**
Genotype of parents		**XX**		**XY**
Gametes		**X and X**		**X and Y**
		Eggs		
		X	**X**	**XX = 50%**
Sperms	**X**	**XX**	**XX**	**XY = 50%**
	Y	**XY**	**XY**	

46. C

Back cross is a method used to determine wheather unknown individual is a hybrid or pure dominant for a given trait by crossing with recessive parent.

Crossing over is an exchange of parts between two homologous chromosomes during meiosis

Segregation is the separation of two members of each pair of genes during meiosis

Independent assortment states that the inheritance of one pair of gene is independent of the inheritance of any other pair of genes.

47. D

Lamarck's theory of evolution was based on the use and disuse of body parts. He suggested that if certain organs were used more than others, they would become more developed and more effective. If organs were not used, they weakened and become vestigial organs

48. B

The larval stages of **Lepidopterous** are called **caterpillars.** It these larval forms or caterpillars that cause much damage to agricultural crops.

49. D

Disruptive selection favors two or more phenotypes within the population. It pushes the phenotypes away from the average phenotype towards the extremes of the population.

50. B

The human DNA would hybridize more with the DNA of **Chimpanzee.** Comparison between DNA of humans and other primates

51. A

The presence of autotrophs or plants is essential for the other because autotrophs synthesize their own food that could be used by heterotrophs. In fact, all animals depend on green plants for food.

52. B

The components of the **DNA molecule** are sugar, phosphate and an organic base

53. D

Biomes are large geographical regions in different parts of the world which have their own climate and particular organisms. A certain biome has its own typical plants called **flora** and its own animal species called **fauna.**

54. C

ATP is a nucleotide made up of the base **adenine** and the sugar **ribose** (together called **adenosine)** and three **phosphate groups.**

55. D

Bacteria and **blue green algae** are prokaryotic, unicellular organisms that belong to the kingdom **Monera.**

Telegram: https://t.me/ExtremeSeries

56. A
Speciation is the formation of new species. A **species** is group of individual organisms that can interbreed and produce fertile offspring. New species are formed as the result of adaptive radiation, interspecific hybridization and isolation.

57. C
Living organisms are made up of large organic molecules known as **Polymers.** The biological molecules are made of smaller sub-units known as **monomers. Polysaccharides** are polymers made up of monomers called **monosaccharides.**

58. B
Hybridization is the mixing of genes of different organisms of related species. Although members of different species are usually not interfertile, occasionally members of two different but closely related species may interbreed to produce a third species by hybridization. The outcome is that the hybrid offspring is more vigorous and bigger in size than either of the homozygous parents.

59. A
A pyramid of numbers is an ecological pyramid based on the number of organisms at each trophic level. There are fewer organisms at the second trophic level than at first, fewer at the third than at the second and so on. Thus, as the number of individuals decreases, the actual size of the individuals increase. This is called **biological magnification.**

60. B
Infectious diseases are diseases that easily spread or passed from one person to another. Infectious diseases are caused by **microorganisms** like bacteria, viruses, protozoa, fungi and worms.

61. D
Alcohol is quickly absorbed through the walls of the stomach in to the body without changing its form. It starts entering the blood within two minutes after it is swallowed. Then it is rapidly carried to the tissues.

62. C
Stimulants are drugs that increase the activity of the nervous system. They make you more alert and keep you awake. Examples are caffeine in coffee, tea and cola drinks the nicotine in tobacco and cocaine.

63. B
The feeding relationships which exist between the various food chains found within an ecosystem can be shown by the **food web.** Food chains are interconnected to make a food web. The food web is the sum total of all the food chains in an ecosystem.

64. A

Oxidative phosphorylation helps to release most of the energy stored in glucose. During oxidative phosphorylation, 34 molecules of ATP are produced from one glucose molecule.

65. D
The zygote undergoes a number of sequential changes to form a multicellular individual. The basic pattern of sequential changes.

$$Zygote \xrightarrow{cleavage} Morula \xrightarrow{Blastulation} Blastula \xrightarrow{Gastrulation} Gastrula \xrightarrow{Organogenesis} Individual$$

66. C
The earliest or oldest relative of the modern horse, genus **Equus** and **Eohippus.** There are many intermediate species between the primitive Eohippus and the modern Equus.

67. D
The **Cro-Magnon** human and the **Neandertal** man are now considered members of the **Homo sapiens.** The Neandertals and Cro-Magnons were Homo sapiens, as are modern humans.

68. C
The **Fine –adjustment** is essential for perfect focusing especially when using high power objective lens

69. B
Heterotrophs are organisms that cannot make their own food from inorganic substances as **animals fungi** and **most bacteria.** Green **plants, algae** and **some bacteria** are autotrophic organisms which can make their own food.

70. D
Starch is a polysaccharide molecule that is used for energy storage in plants.

71. C
Enzymes which leave the cell to perform their functions outside the cell are known as **extracellular enzymes.** Digestive enzymes which break down food substances in the alimentary canal are extracellular enzymes.
Enzymes which are made inside the cell and remain inside it to catalyze chemical reactions are called **intracellular enzymes.** Respiratory enzymes inside mitochondria and in thee cytoplasm and enzymes of photosynthesis in chloroplasts are intracellular enzyme.

72. D
Genetic engineering is a technique of altering an organism's genotype by inserting genes from other organisms into its DNA. Genetic engineering can produce cells that are transformed.

73. D

Telegram: https://t.me/ExtremeSeries

Leishmaniasis is a Vectore borne disease and is passed on by **sandflies** when they bite humans.

74. A
Nucleic acids are found in the nucleus of the cell. There are two types of nucleic acids found in living organisms, DNA and RNA. A **gene** is a portion of a DNA molecule that is capable of replication and mutation and passes on a trait from parent to offspring

75. A
As the magnification power of a microscope increases the diameter of the microscope decreases and hence the field of vision (area seen) decreases. Higher magnifications have less field of vision and therefore the number of objects seen under the microscope decreases. One can see more number of the squares under the microscope with the magnification power of 40X

76. B
HIV Virus is a highly variable virus which mutates very readily. This means that there are many different strains of HIV. Therefore, viruses are capable of mutation and have more than one strain.

77. C
Each cell of the body contains so many different types of enzymes. This is because enzymes are specific, that is, for one particular reaction there is one particular enzyme.

78. B
An **innate behaviour** is **stereotyped** and **complete.** Stereotyped means that it is always performed in exactly the same way by members of a secies.

79. C
The **temperate deciduous forests** are dominated by broad leaved trees that shede periodically. Therefore, the deciduous forests are not evergreen trees.

80. B
The grass is the producer. The grasshopper eats the grass, so it is the primary consumer. The frog is the secondary consumer because it eats the grasshopper and the birds are the tertiary consumers because they eat the frog

Grass ⟶ Grasshopper ⟶ Frog ⟶ Birds

81. B
The radioactivity would be detected in carbon dioxide produced during cellular respiration

$C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O$

82. D

Telegram: https://t.me/Extremeseries

Pieces of fresh potato kept in distilled water will gain the highest percentage of weight. The potato will gain water through the cell member and gain weight.

83. C

A. AO × BB

	Man	
	X	A
Woman	B	AB
	B	AB

50% AB and 50A

B. AA × OO

	Man	
	X	A
Woman	O	AO
	O	AO

100% A

C. AO × BO

	Man	
	X	A
Woman	B	AO
	O	AO

25% AB, 25% B
25% A and 25O

D. AB × OO

	Man	
	X	A
Woman	O	AO
	O	AO

50% A and 50% B

Therefore, the possible blood geneotype of the parents of the baby will be AO and BO

84. A
Bacillus thuringiensis is known to kill a wide range of pests. Biotechnological treatment involving the insertion of the bacterial gene in to the plants produces plants that are resistant to **insect pests.**

85. C
The pollen grain produces two nuclei, the tube nucleus and the generative nucleus. The generative nucleus divides to form two sperm cells. One of the sperms fertilizes the egg cell to form a zygote. The second unites with the two polarnuclei to form a 3n endosperm tissue. Thus, the number of chromosomes in the endosperm tissue of maize is 3×10 which is equal to 30.

86. C
Homo erectus was lived between 1.9 and 0.3 million years ago. It evolved to modern man
Homo habilis occurred between 2 and 1.9 million years ago. It is thought to be the ancestor of Homo erectus.
Ardipithecus ramidus was lived 4.4 million years ago.
Australopithecus afarencis known as **Lucy** is believed to be 3.5 to 2.8 million years old.

87. A
Conservation can either happen **insitu** or **exsitu.** In situ literally means in place. When we refer to something as being in situ, we mean that it is still in its original place and has not been removed elsewhere

or lost. It refers to the maintenance of biodiversity in its wild state and within natural habitats.
Ex situ means out of place. When we refer to something as being ex situ, we mean that it is not in its original place. Ex situ conservation is conservation outside of the natural habitat and ecosystem of the biodiversity.

88. C
Sensitization is learning to be hypersensitive to a stimulus.
Imprinting is a type of learned behaviour in which birds like chickens will follow the first moving object they see for two or three days after hatching and will continue to show this following response as they mature. **Habituation** is a progressive loss of responsiveness to repeated stimulation.
Latent learning is the association of stimuli or situations without reward.

89. B
Ethiopia is classified into four major groups of agricultural systems.
Shifting cultivation is deforestation technique where large areas of forest are cleared for agriculture purposes
Seed-farming complex is characterized by crops grown by seeds. These seeds are cereals, pulses, oil crops and potatoes.
Enset-planting complex is characterized by the cultivation of enset.
Pastoral complex is characterized by livestock husbandry.

90. C
A **Virus** is made up of a single type of nucleic acid, either DNA or RNA surrounded by a protein coat. A molecule of DNA is made of sugar, a phosphate and an organic base.
A **mitochondrion** contains enzymes, proteins, lipids, carbohydrates, ribosomes, coenzymes like NAD, NADP, ATP, FAD, coenzyme A and Various dissolved substances.
A **Polysaccharide** is a carbohydrate made up of many simple sugars, such as starch, cellulose and glycogen.

91. B
The evidence from **comparative biochemistry** was very unlikely to have been used by Darwin in the formulation of the theory of evolution.

92. C
Gene flow is the migration of individuals or their gametes from one population to another and interbreed with the existing population.

93. B
The **scientific method** is a way of solving a problem in a scientific way. Identifying a topic to be studied is the first step of the scientific method.

94. C

A set of linkages in which [illegible] is eaten by a carnivore and so on is called a **food chain**. Food chains are usually interconnected to make a **food web**.

95. C
Metabolism refers to all the processes occurring in a cell or an organisms. The metabolic reactions in a living body are of two kinds.
a) **Anabolism** is the process by which cells synthesize materials they need for the proper functioning of the organism
b) **Catabolism** is a process that results in breaking down or changing of more complex substances into simpler ones.

96. A
The amount of light reaching different parts of the earth from the sun varies at different latitudes. The equator receives the greatest intensity of light. As one goes from the equator to higher latitudes that is north and south poles, the amount of light reaching the earth decreases.
Temperature not only changes with latitude but also changes with changes in altitude. **Altitude** refers to the height of an area above sea level. As altitude increases the average temperature decreases.

97. C
During respiration, plants and animals release carbon dioxide to the atmosphere. The CO_2 in the atmosphere is absorbed by plants and is incorporated into organic molecules during photosynthesis.

98. B
Based on their **mode of nutrition** organisms can be classified as autotrophs and heterotrophs.

99. A
A DNA molecule is made up of two chains of nucleotides held together through their bases.

100. A
Photosynthesis is divided into two steps or phases. The first phase is called thee light reaction because it requires light. In the light reaction the light energy splits water molecule into hydrogen and oxygen.

Telegram: https://t.me/Extremeseries

BIOLOGY ANSWER WITH EXPLANATION 2003/ 2011

1. **A** Gene mutation is a change in a gene of a DNA which may alter or destroy the trait associated with it.

2. **D** Heredity is the study of the transmission of hereditary characteristics from parents to offsprings and from generation to generation.Physiology is the study of the normal functions of living things. Reproduction is the process by which organisms produce new individuals and make sure the continuous existence of each species. Ethology is the study of behaviour

3. **C** Innate behavior is the behavior that we inherit from our parents. It is not learned but inherited. It is always performed in exactly the same way by members of a species. It is species characteristic. It is adaptive that is helps the animal to survive.

4. **A** A carefully planned experiment consists of two components, the experimental group and the control group.In the control experiment, all factors except the one to be tested are the same as in the first experiment. In this way, the control shows the importance of the missing experimental factor

5. **D** In the ecosystem, the movement of energy is unidirectional or linear. The flow of energy is not cyclic. Energy is not recycled. This linear passage of energy is referred to as an energy ,flow. Energy flows from the sun to producers. Unlike energy nutrients like carbon, hydrogen, oxygen, nitrogen, phosphorus and sulphur are recycled in a natural ecosystem

6. **D** Bacteria and blue green algae are described as prokaryotic. All other organisms are called eukaryotic except viruses. Viruses do not have a cellular structure.

7. **C** Hypotheses are often tested by experimentation. After the experiment the hypothesis can be rejected or accepted

8. **B** The cell membrane is made up almost entirely of proteins and phospholipids

9. **A** Mitosis is a type of cell division which produces two daughter cells from a single cell. Thus, two daughter cells are produced from a single mitotic division

10. **D** Cocci are round or spherical shaped bacteria Spirochetes are spiral shaped bacteria .Spirilla are bacteria which have aspiral shape

11. **C** Robert Hooke, an English scientist observed a thin section of dead cork and saw that it was made up of many box-like structures. Hooke introduced the term cell to describe the tiny box-like structures he observed

Telegram: https://t.me/Extremeseries

C The total magnification is the product of the magnifying power of the ocular lens (eye piece) and the particular objective used. The total magnification power of a microscope with a 10X ocular lens and a 25X objective lens will be 10X25 = 250X

B Ornithology is the study of birds. Thus, ornithologist is a person who studies birds

B Bacteriophages are viruses that parasitize bacterial cells

C Auxin is the plant growth substance that causes the parts of plants to grow towards or away from the stimulus

B Kinesis is a non-directional movement response in which the rate of movement is related to the intensity of the stimulus and not the direction of the stimulus.Taxis is a movement towards or away from a stimulus. It is a movement of the organism in response to an external directional stimulus

C Biology is the study of all living things, because if one can say biology studies only plants and animals, organisms like fungi which are not either plants or animals cannot be included in biology.

B The Krebs cycle takes place in the mitochondria of the cell

C Carbohydrates are polymers made up of monomers called monosaccharides (simple sugars) Proteins are polymers made up of monomers called amino acids. Nucleic acids are polymers made up of monomers called nucleotides

Lipids are polymers made up of monomers called fatty acids and glycerol

B **Biodiversity** refers to the **number, variety** and **variability** of living organisms and it includes all the **genes** found in living organisms.

A Since the immune system of AIDS patients is so weak opportunistic infections, such as **pneumonia, diarrhea** and **tuberculosis** are the major killers of AIDS patients

C **The law of heredity** was discovered by Gregor mendel in the small monastery garden in 1856

The cell theory was proposed by M.J.schleiden and Theodore Schwann in 1839.The doublc helix nature of the DNA was discovered byWatson and Crick in the 1950s. The binomial system of classification was developed by a Swedish biologist, Carl Von Linnaeous (1707-1778)

C DNA is used as a template (bench) for the synthesis of mRNA in the nucleus. Then, m RNA leaves and enters the cytoplasm. In the cytoplasm, mRNA is used as a template for the building of amino acids into proteins

A Starch, cellulose and glycogen are examples of polysaccharides while glucose is an example of monosaccharides.

25. **A** **Pinocytosis** is a process by which cells take in fluid by means of vesicles

Endocytosis is the transport of materials (solid or liquid) through membranes into the cells

Osmosis is the diffusion of water molecules across a selectively permeable membrane.

Phagocytosis is the process by which solid materials like bacteria are taken in by a cell.

26. **B** Animal cells divide the cytoplasm into two cells by forming a **cleavage furrow** while plant cells divide the cytoplasm into two halves by forming a **cell plate**

27. **C** **Homeostasis** is the maintenance of a constant internal environment. Some examples are blood sugar levels water regulation and temperature regulation

28. **A** The normal number of 46 chromosomes (2n) is restored at fertilization when a sperm fuses with an ovum(n).

Osmosis is the diffusion of water molecules across a selectively permeable membrane.

phagocytosis is the process by which solid materials like bacteria are a retaken in by a cell.

28. **B** In **Allosteric inhibition,** substances that bind to the allosteric site of an enzyme inhibit enzyme action. In **competitive inhibition** the substrate and another molecule, the competitive inhibitor, resemble each other.

In **Irreversible inhibition,** irreversible inhibitors bind tightly and permanently to enzymes and destroy their catalytic properties.

In **end product inhibition,** the end product inhibits the engyme.

30. **C**

Lactose is a disaccharide or double sugar present in milk composed of glucose and galactose. Also called milk sugar.

31. **B**

Habituation is the fall or elimination of response to frequently occurring stimuli that have no effect on the animal'swell being.

Imprinting is a type of learned behavior acquired during a limited critical period.

Insight learning is the highest form of learning based on thought and reasoning.

Sensitization is learning to be hypersensitive to a stimulus.

32. **D**

According to the theory of evolution proposed by Charles Darwin **natural selection** is the most important driving force of evolution.

33. **C**

Telegram: https://t.me/Extreme_series

Pavlov found that dogs automatically salivate at the sight of food. The food is the **unconditioned stimulus** to which the animal did not have to learn the response to it. If Pavlov always rang a bell, when he offered food, the dogs began slowly to associate the sound of the bell with food. Eventually the sound of the bell alone could cause salivation. The sound of the bell is the **conditioned stimulus.**

C

Half life is the time it takes for half of the radioactive isotope to change into daughter elements. When the plants and animals die the ^{14}C decays to the non-radioactive isotope ^{14}N. The rate of decay of ^{14}C to ^{14}N is such that half of an amount of ^{14}C will be converted into ^{14}N in about 5,730 years. This is its half-life. In two half- life time (11,460 years) only one quarter of the original amount of ^{14}C would remain.

D

Immigration is the movement of animals into an area. This causes an increase in population size.

Emigration is the movement of animals out of an area. It causes a decrease in population.

D

Directional selection favors one extreme phenotype from the population. A good example of directional selection was noted in the peppered moth that is common in Britain. This moth is found in two color variations. There are light moths and dark moths. The light moths were much more common in Britain until the 19th century. Its color blended with the trunks and it was protected from predatory birds. At this time, black moths were rare.

However, during the period of the industrial revolution, the industries polluted the air with soot that made the tree trunks black. The light moths now became easy prey for birds, but the black moths blended with the dark trunks and escaped the sharp eyes of the birds. In these new surroundings, the dark moths were better adapted and were selected for survival.

A

Yellow fever is a tropical disease caused by a virus. Pneumonia, cholera and typhoid are all caused by bacteria.

A

The best way to prevent self-pollination in plant hybridization experiment is to remove the stamens.

B

Habituation is a learned behavior. It is the fall or elimination of response to frequently occurring stimuli that have no effect on the animal's well being.

D

DNA fingerprinting is a process whereby bits of DNA can be collected, for example at a crime scene, and matched with the DNA of suspects. Because DNA has been shown to be unique to an individual as are fingerprints, this is where the term DNA fingerprinting gets its name.

41. **B**

Biomass is the total weight of living things which an area of land can support.

42. **A**

Diverse ecology has a positive effect on the biodiversity of a given country. All other factors have a negative effect on the biodiversity.

43. **A**

A **nucleotide** consists of a five carbon sugar, a phosphate and an organic base.

44. **A**

The final electron acceptor in the electron transport chain of cellular respiration is the **oxygen molecule**.

45. **B**

The mule is an example of a sterile hybrid. If the mule was fertile, then the horse and donkey species would gradually become less and less distinct. This keeps the horse and the donkey as separate species.

46. **D**

Community is a group of different organisms living together in a given habitat like Munesa Forest in Ethiopia.

47. **A**

Plasmids are small circular pieces of DNA found in some bacteria that are used as vectors to transfer genes to bacteria through genetic engineering.

48. **C**

HIV infects cells in the immune system and the central nervous system. The main cell that HIV infects is called a **T-helper lymphocyte**. HIV infects the T helper cell because it has the **protein CD_4** on its surface, which HIV uses to attach itself to the cell before gaining entry. This is why the T helper cell is also referred to as a CD_4 lymphocyte.

49. **D**

Decomposers include **bacteria** and **fungi** that obtain their energy by breaking down dead bodies of plants and animals into nutrients that plants can use again.

50. **B**

In complete dominance, the dominant and recessive phenotype ratio in the monohybrid cross is 3:1 . Thus, the number of plants that have

the dominant phenotype is $\left(\frac{3}{4} \times 80\right)$ which is equal to the number of plants that have the recessive phenotype is $\frac{1}{4} \times 80$ equal to 20.

B

Chloroplasts contain the green pigment chlorophyll that enables plants to absorb sunlight. They are the site of photosynthesis. **Mitochondria** are the centers of respiration in the cell. These two organelles are concerned with energy transformation.

D

Plants in the tropical rainforest include **epiphytes**, plants that grow on other large plants and **lianas**, climbing plants which use other plants to support growth into the canopy.

B

Epiphytes are plants that grow on the branches of large trees.
Xerophytes are plants growing in dry or desert regions. Their leaves are reduced and specialized to withstand water shortage.
Halophytes are plants that occupy neither extremely wet or extremely dry surroundings.
Hydrophytes are plants that live in water rich areas.

B

Some fruits like legumes split open suddenly and fling out the seeds. This helps in the dispersal of fruits and seeds. Fruits and seeds are dispersed to avoid the competition among the seedlings for food and space.

C

Artificial selection is a deliberate act of breeding carried out by man on animals and plants. The main objective of artificial selection is to produce new and improved varieties of plants and animals.

B

Plants bind with their roots and reduce water runoff, thus allowing water to be absorbed into the soil as well as preventing soil erosion.Plants also recycle water through their leaves. Plants are habitat for many types of animal and plant species.

D

Most bacteria are harmless, while some are useful. Others are harmful causing diseases in humans, other animals and plants. Some bacteria are agents of food spoilage and decay. Not all bacteria found in the human body are parasitic. There are bacteria that live in the human intestine which make vitamins like vitamin K and vitamin B_{12}.

A

Protists include unicellular eukaryotes like plasmodium, paramecium and amoeba. They are found at a lower level of evolution.

Telegram: https://t.me/ExtremeSeries

The nitrates in the soil absorbed by the roots of plants are used by plants to make their proteins. Animals obtain their proteins by eating plants. Animals that eat the plants digest and form their own proteins.

60. **C**

The number of mitochondria varies from cell to cell. Cells that expend more energy contain more mitochondria. For example, muscle cells, liver cells and secreting cells.

61. **B**

RNA like DNA consists of nucleotides made up of nitrogenous base, pentose sugar and a phosphate group. The sugar in RNA is **ribose**. The sugar in DNA is **deoxyribose**. With regard to the nitrogenous bases. **Thymine** is found only in DNA while **uracil** is found only in RNA. RNA is usually single stranded whereas DNA is double stranded.

62. **D**

The genetic information in a cell flows from the genes or DNA to RNA, copy of the genes. A copy of the genes (RNA) is made in the nucleus along a strand of DNA and then taken into the cytoplasm of the cell to make proteins. Therefore, DNA controls protein synthesis through RNA. This is known as **Central Dogma** of genetic information flow.

DNA → RNA → Protein

63. **B**

In the DNA of any organism, the amount of adenine equals the amount of thymine and the amount of guanine equals the amount of cytosine. Therefore, if the number of adenine bases in a cell is 80, then , the number of thymine bases will be 80. The sum of thymine and adenine base molecules is equal to 160. The total number of base molecules (300) minus the sum of thymine and adenine base molecules (160) which is equal to 140, will be the total number of guanine and cytosine. Therefore, the amount of guanine base molecules in the DNA strand will be 70.

64. **D.** One sperm nucleus(n) unites with the two polar nuclei (2n) and forms endosperm nucleus (3n). The endosperm nucleus grows into the endosperm tissue of the seed, an area in which food is stored

65. **C.** Describing animal behavior requires careful observation over a long period of time and simple language to convey what has been observed.

66. **B.** Niche includes all the resources that an organism will draw on for living. Each population will occupy a very specific niche in an ecosystem. When populations occupy similar niches, then there is usually competition for one or other resource.

67. **A.** Lipids include neutral fats, phospholipids, steroids, carotenoids and waxes. Nitrogenous bases are found in the nucleotides of DNA and RNA

68. **D.** Active transport is an energy consuming transport of molecules or ions across a membrane against a concentration gradient .With the use of energy ,particles can be forced to move from their place of low concentration to their place of high concentration (against concentration gradient)

69. **D.** Centrioles are found only in animals cells these are two rod-like structures situated at right angles to each other just outside the nuclear membrane

70. **C.** The carrying capacity is the ability of a given environment to support the maximum number of population

71. **D.** HIV can only replicate inside human cells. The Viral core contains there enzymes required for HIV replication called reverse transcriptase, integrase and protease.
Once inside the cell, the reverse transcriptase converts the viral RNA into DNA . Then the viral DNA is joined into the human DNA by HIV enzyme integrase. The viral DNA is transcribed to Viral RNA, which produces Viral proteins including the enzyme reverse transcriptase. Then, the RNA, proteins and reverse transcriptase are assembled by the cell into new HIV particles which are then released from the cell. The enzyme protease plays a role in the assembly of viral parts into a whole virus. Thus , if the antiretroviral drugs affect the enzyme protease ,the assembly of viral parts into new HIV particles will be inhibited

72. **B.** Birds are the most diverse groups of vertebrates in terms of species diversity

Groups of organisms	No. of species	No. of endemic species
Terresterial animals	277	30
Birds	861	16
Reptiles	201	10
Amphibians	63	30
Fresh water fish	150	4
Butterflies	324	7
Higher plants	6500	650

73. **B.** The substrate is the substance upon which the enzymes act. If the amount of substrate is increased while the concentration of enzyme remains the same, the rate of the reaction will increase. An increase in the concentration of the substrate will increase the rate of the reaction until the available active sites are full, any further increase in substrate concentration would not speed up the rate of reaction

Telegram: https://t.me/ExtremeSeries

74. **B**. Sickle cell anaemia is an example of gene mutation caused by substitution of a base pair in one of the genes coding for haemoglobin.

75. **D**. If the bacteria are able to dwell in hot springs, it means that their enzymes do not denature as that temperature

76. **D**. Temperate rainforests are found in the equatorial regions in south America, Brazil, around the Amazon River equatorial West Africa and South East Asia

77. **B**:Diameter is directly proportional to the field of vision. Low power objective lenses have a wider diameter. Thus, microscopes with low magnification power of lenses such as 4x and 10x will have the largest field of vision.

Chromosome from the mother, there is one which is similar in shape, size and containing the same genetic material from the father. Such pairs of identical chromosomes in body cells are called homologous chromosomes.

78. **A** The density of water decreases below 4°C and ice therefore tends to float. The ice at the top of ponds' and lakes protects the water under it from freezing. This helps the organisms in the water to survive the cold winter in temperate and arctic regions.

79. **C** In anaphase, the sister chromatids of each chromosome separates from one another and move to the opposite poles of the spindles. Thus, if the anaphase phase of mitosis is blocked, the cell with twice as many chromosomes as the mother would be produced.

80. **B** Bacteria reproduce asexually by binary fission.

81. **B** Endagered species usually have small gene pool.

82. **C** The diploid chromosome number of many organisms including human beings is always an even number because for every chromosome from the mother, there is one which is similar in shape, size and containing the same genetic material from the father. Such pairs of identical chromosomes in body cells are called homologous chromosomes.

83. **D** Hibernation is a response of certain animals to cold seasons like winters. Animals go into a very deep sleep called hibernation. Hibernation allows animals to survive at a time when food is scare.

84. **B** The cell membrane is made up almost entirely of proteins and phospholipids. The phospholipid bilayer is flexible and can move fairly freely. The phospholipids molecules are capable of much movement. Constant movement of these molecules gives fluidity to the membrane.

85. **A** One of the factors threatening populations is habitat destruction for cultivation, Settlement and plantations by man.

86. **A** Flu is an airborne disease transmitted through droplets of moisture produced by coughing, sneezing and breathing.

A Incomplete dominance is a type of inheritance, where the heterozygote shows an intermediate phenotype. In this case the dominant gene is not able to fully suppress the effects of the recessive gene. In incomplete dominance, two genes are partially dominant, thereby interacting to produce a new phenotype.

Incomplete dominance is observed in a cross between pure red flowered and pure white flowered plants. All the F_1 offspring produced pink flowers. When the F_1 plants were self-pollinated, the resulting F_2 generation produced a phenotypic ratio of 1 red: 2 pink: 1 white flowered plants.

F_1 pink colored flower (Rr)x F_1 pink colored flower (Rr) Gametes R and r

	R	r
R	RR	Rr
r	Rr	rr

The breeding experiments conducted by man on domesticated animals and plants show that characteristics of populations of organisms are not rigidly fixed. The characteristics of a population can be altered by breeding experiments. If human beings are able to alter the characteristics of organisms in order to produce new varieties, then, it can be argued that nature can do even much more through the process of natural selection.

Kinesis is a non- directional movement in response to the intensity of the stimulus. Taxis is a directional movement towards or away from a stimulus in response to an external directional response.

Two populations are geographically isolated by physical barriers. Geographical isolation inaibits gene flow or exchange. The prevention of gene flow would then bring about genetic differences which prevent interbreeding if members of the two populations come together. This phenomenon would result in the formation of new species.

If the plant cell is put into water, water enters by osmosis and the cell swells up. As more water enters the cell, the vacuole swells exerting pressure on the cytoplasm ,which in turn exerts pressure on the cell wall. This is referred to as turgor pressure.

Convergnt evolution is the evolutionary change that occurs when two or more distantly related species become more alike as they develop similar survival techniques. It resulted in completely different features coming together to produce similar features.

Telegram: https://t.me/ExtremeSeries

93. **A** Selective pressure that acts around the mean value is stabilizing. Stablizing selection leads to a reduction in variance by getting rid of the extremes from the population. Individuals with an average or intermediate phenotype are preferred.

94. **D** The variation in climatic factors is not an important unifying theme in biology. Instead, it is a diversifying factor. All others are unifying factors.

95. **B** In higher animals such as man digestion takes place in the alimentary canal outside the cells of the body. In saprophytic fungi and bacteria also digestion occurs outside their bodies when they secrete enzymes into their environments. This is called extacellular digestion.

96. **B** A pollinating animal, like an insect is a biotic factor that can influence a plant, for example, helping in cross pollination. All others are abiotic factors.

97. **D** Chitin is an important constituent of the exoskeletons of insects.

98. **A** There are two types of nitrogen bases, pyrimidines such as cytosine and thymine and purines like adenine and guanine. If the number of the purine and pyrimidines bases is the same, it means that the number of each base accounts for 25% of a DNA molecule.

99. **D** Hypothesis is an explanation or trail answer for an observation which is based on supporting data.

100. **D** Resolution refers to the scattering of parts of the object so that the details are seen clearly.

Telegram: https://t.me/Extremeseries

1. A. This method of communication in animals is called visual
2. B The substrate is the substance upon which the enzymes act
3. A. A mutation is any spontaneous change in a gene or a chromosome of an organism. Gene mutation is a change in a gene of a DNA. In chromosomal mutation a piece of a chromosome may be broken off, or the order of some genes on the chromosome may be changed, or the number of chromosomes per cell may increase or decrease compared to the normal. Mutation is an error during DNA replication that results in a change in the sequence of bases in the genes that make up DNA. The addition of an extra base, or the substitution of one base for another. A change in the total number of bases, which occurs if a base is added or deleted, is an error that alters the genetic material.
4. B. Enzymes are denatured (destroyed) at high temperature. At low temperatures, enzymes work slowly
5. A. In prokaryotes, transcription and translation are coupled. Prokaryotic m RNA does not need post-transcriptional processing. In eukaryotes, transcription and translation are separated. Transcription occurs in the nucleus and translation occurs in the cytoplasm
6. D. Lucy belongs to the genus Australopithecus
7. A. There are about twenty different kinds of amino acids. All the twenty amino acids are used to make all the different proteins
8. B. All bacteria do have a cell wall but it is not made from cellulose like plant cell walls and instead is made from a substance called peptidoglycan, which makes it rigid.
9. A. Materials can be recycled over and over again but energy can only be used once. There is no energy cycle. Energy flows from the sun to producers and from producers to the different level feeders
10. C.
Gregor Mendel found that the percentage of the F_2 generation of a monohybrid cross with the recessive phenotype is 25% and the dominant phenotype 75%
11. B. The major ideas of the cell theory are
    - All organisms are composed of one or more cells
    - Cells are the smallest structural and functional units of life
    - All cells arise from pre-existing cells
    - The cell is the basic unit of heredity
12. C. The biosphere is the part of the earth which includes the atmosphere, land and water within which life can exist
13. B. - Kingdom Monera contains prokaryotic, unicellular organisms, bacteria and blue green algae

- Kingdom protista contains microscopic unicellular eukaryotes
- Kingdom plantae includes all multicelular plants that are eukaryotes
- Kingdom Animalia includes all multicellular animals that are eukaryotes

14. D. Living things cannot determine the amount of radiation reaching the environment
15. B. TheWorker honey beeperforms various functions namely
    - Clean out dirty comb just after it emerges
    - Feed larvae (young)
    - Build honeycomb
    - Guard the hive
    - Forage for nectar, pollen, water
16. B. Sucrose has the chemical formula of $C_{12}H_{22}O_{11}$
17. A. Bacteria have three shapes, spherical (cocci), rods (bacilli) and spirilla
18. C. Microorganisms can be cultured on a nutrient contained in a petridish or in a test tube which contains liquid nutrient
19. B. Procedures of a scientific method
    1. Making observations
    2. Identifying the problem
    3. Forming a hypothesis
    4. Making predictions
    5. Conducting an experiment
    6. Drawing a conclusion from the experiment
20. C. Homozygous refers to a pair of alleles responsible for a particular trait that are alike or similar as TT or tt (homozygous tall or short respectively)
21. C. Cells build their proteins from amino acids. Amino acids joined together in any order to make a protein molecule. The amino acids are joined together by peptide bonds. These bonds are formed by a chemical process called condensation reaction. Condensation is a reaction in which two or more molecules unite to form a large molecule, with the separation of water
22. D. There are two models of enzyme action, the lock- and- key model and theinduced- fit model. Both of these models suggest that the enzyme catalyses the reaction by lowering the activation energy. However, they differ in explaining how the substrate binds to the active site of the enzyme
    Lock – and – key model proposes that the shapes of the substrate molecules are complementary to that of the active site
    Induced-fit model suggests that the active site and the substrate are not complementary in shape but the binding of substrate molecules

produces a change in shape in the active site, allowing the substrate to fit the active site

23. D.A hypothesis is tested by an experiment. An experiment consists of two components, the experimental group and control group. In the control experiment, all factors except the one to be tested are the same as in the first experiment
24. D. The splitting of water, ATP formation and reduction of NADP require the presence of light while carbon fixation does no require the presence of light
25. D. Mendel's first law is called "The law of segregation. This law states that a pair of parental factors or genes separate or segregate during the formation of gametes or sex cells so that one factor is present in each gamete
26. C. The type of enzyme regulation in which the formation of a product inhibits an earlier reaction in the sequence is called end product inhibition or feedback control.
27. The type of enzyme regulation, in which the formation of a product inhabits an earlier reaction in the sequence, is called end product inhibition or feedback control.
28. D. Simple diffusion is the movement of molecules from a region of higher concentration to a region of lower concentration down a concentration gradient
    Facilitated diffusion is a form of diffusion in which the molecules diffuse across the membrane from a region of high concentration to a region of low concentration with the help of transport proteins that lie across the membrane
    Osmosis is the diffusion of water molecules across a selectively permeable membrane.
    Active transport is the energy consuming transport of molecules across a membrane from a region of low concentration to a region of high concentration, against concentration gradient
29. D. Microorganisms are involved in the recycling of nutrients, carbon fixation & nitrogen fixation but do not play a role in energy recycling
30. A. Natality (birth rate) is the number of offspring of a species born in a certain period of time. It affects the number of human population globally
31. B. The centrifuge is a laboratory tool used to separate the organelles of the cell based on their density.
32. B. Charles Darwin's theory of evolution has three main parts
    1. Over reproduction (fecundity). Organisms have the potential to produce a greater number of offsprings than the number that survive, reproduce and live to maturity

Telegram: https://t.me/ExtremeSeries

2. Variation. All the individuals of a species differ from each other in their characteristics
3. Competition (struggle for existence). Individuals compete with one another for the limited resources that enable them to survive such as food, space and light. The use and disuse of body parts is a part of Lamarck's theory of evolution

33. B. Kinesis is a non -directional movement response in which the rate of movement is related to the intensity of the stimulus and not the direction of the stimulus
34. C. We can determine whether an individual having a dominant phenotype is homozygous or heterozygous for the trait by doing a test cross or back cross.
35. B. Carbohydrates are made up of carbon, hydrogen and oxygen.
36. D. When a protein like an enzyme is denatured by heat or extreme PH, it loses its tertiary structure
37. D. Rudolf Virchow stated that cells come from pre-existing cells
38. C. When muscle cells are in short supply of oxygen, glucose is broken down by anaerobic respiration in muscles into lactic acid, with the release of very little energy.
39. A. Burning of fossil fuels is the main source of the greenhouse gases like carbon dioxide that results in global warming
40. C. Ardipithecusramidus is the nearst fossil to the common ancestor of the hominids and the apes
41. D. Virsuses are the smallest and simplest of living organisms. Viruses cannot reproduce by themselves. They can survive and reproduce only inside a living cell all viruses are parasitic
42. D
Acitive transport is the energy consuming transport of mulecules across a membrane from a region of low concentration to a region of high concentration using energy
43. A. Glycolysis, Krebs cycle and electron transport are the stages of cellular respiration where ascalvin cycle is a photosynthetic pathway by which carbon dioxide is reduced to form glucose
44. B. An amino acid molecule contains less energy
45. B. The process by which energy is released from food by the oxidation of food inside the cells of living organisms is known as cellular respiration. The equation of cellular respiration is $C_6H_{12}O_6+6_2O$
$6CO_2 + 6H_2O$ + Energy
46. A. Glycolysis is the first stage of cellular respiration. During glycolysis, glucose is broken down into two three carbon compounds called Pyruvic acid. Glycolysis produces a net result of 2ATP molecules

Telegram: https://t.me/ExtremeSeries

Glucose $\xrightarrow{Enzymes}$ 2 pyruvic acid + 4ATP
2ATP

47. C. Homeostasis is the ability of living organisms to maintain a constant internal environment
48. D. Endocytosis is an active process involving the bulk transport of materials which are too large to be transported by protein carriers into cells across the plasms membrane
49. A The most probable selection pressure responsible for the evolution of green skin colour in frogs inhabiting tropical rainforests in climate
50. C Bacteria are classified based on their shape and whether or not they are coloured by Gram's stain. The bacteria that are stained purple by Gram's stain are called Gram positive and the bacteria that are stained pink by Gram's stain are called Gram negative. The difference is due to the structure of the cell wall of the different bacteria
51. A Exponential population growth is the constant growth rate that occurs when the organisms are in ideal conditions or when the resources are plentiful
52. The days are hot and the nights are cold in a desert
53. B. Plasmids are small circular pieces of DNA found in some bacteria. They are used as vectors of agene to transfer into a host cell
54. B. A retrovirus is an RNA virus that converts its genetic information from RNA into a DNA after it has infected a host. HIV virus belongs to retroviruses
55. D.

Groups of organisms	Number of species	Number of endemic species
Terrestrial mammals	277	30
Birds	861	16
Reptiles	201	10
Amphibians	63	30
Fish	150	4
Butterflies	324	7
Higher plants	6500	650

56. D. Cholera is a food or water borne disease. It is transmitted through foods and water contaminated with faeces.
Typhoid fever is a disease spread through contaminated water, milk, food and vegetables
57. C. The hypha is the individual filament or strand of a fungus
58. C Transgenic organisms are organisms that have been genetically altered using the technique of genetic engineering. Genetic engineering is one of the most important areas of biotechnology

59. C. The series of step scientists use to solve problems in a scientific way is known as the scientific method.Procedures of a scientific method
    1. Observation
    2. Identifying the problem
    3. Forming a hypothesis
    4. Making predictions
    5. Conducting an experiment
    6. Drawing a conclusion
60. D. Due to its high heat of vaporization, water cools the bodies of organisms when they overheat through sweating
61. D. The total magnification of a microscope is the product of the magnifying power of the eyepiece and the particular objective used. Thus, the cell would appear 13x x40x (520) times larger to the viewer
62. C.Each $FADH_2$ molecule results in the production of two ATP molecules
63. D. Crops known to have been domesticated only within Ethiopia are Coffeaarabica, Eragrostistef and EnseteVentricosum
64. C. The first organisms appeared were prokaryotes that gave rise to oldest eukaryotes. The cyanobacteria (blue green algae) which are prokaryotes are the first photosynthetic organisms appeared before the oldest eukaryotes
65. B. Fossils of soft bodied organisms are usually relatively rare in the environment because their bodies decompose readily
66. C. Cellular respiration occurs in all eukaryotic cells
67. A. In developed countries like Europe, North America and few others there is a low birth rate and a low death rate due to the changed cultural and social attitudes like good family planning. In these countries the population is stable
68. C. We are losing 10% of our biodiversity evey 50 years
69. B.The percent of green seeded (recessive) plants in the F_2 generation when two F_1 hybrid plants are self-pollinated is 25%
70. B. The nitrogen used by animals largely comes from plants
71. A. Once inside the cell, the HIV enzyme, reverse transcriptase converts the Viral RNA into DNA, which is compatible with human genetic material
72. C. Half life refers to the time taken for the radioactive isotope to fall by half. Thus if a half life of carbon 14 of a certain fossil is 5730 years
    - 50% of the original amount of carbon 14 remains after 5730 years
    - 25% of the original amount of carbon 14 remains after 11460 years
    - 12.5% of the original amount of carbon 14 remains after 17190 years

    Therefore, the fossil is 17190 years old
73. D. In competitive inhibition, the substrate and the inhibitor resemble each other. Because the inhibitor has a structural resemblance to the substrate, the inhibitor binds with the active site of the enzyme and prevents the substrate from binding. Thus, the rate of the inhibition is determined by the ratio of inhibitor to enzyme concentration

74. A. The R groups of different amino acids are different. However, the amino and carboxyl groups are common or identical to all amino acids
75. A. The dependent Variable is a variable that changes due to an independent variable. In this case, the growth rate is the dependent variable because it depends on the concentrations of a fertilizer & the concentration of a fertilizer is an independent variable
76. D. Louis Pasteur used a Swan- necked flask instead of straight necked one in his experiment to trap particles from the air that might enter the flask before reaching the broth
77. B. The production of a brick – red when a fruit juice is tested with Benedict's solution precipitate indicates the presence of the reducing sugar in the food
78. B. The source of the recycled electron in cyclic photophosphorilation is the chlorophyll molecule
79. A. As the length of the sides of the cubic cells increases the surface area to volume ratio of the cells decreases. Thus, cell A with less length of the sides has the largest surface area to volume ratio.
80. D. Non competitive inhibitors do not compete with the substrate molecules for the active site. Because the inhibitor is not competing with the substrate for the active site, increasing the substrate concentration does not affect the rate of reaction
81. D. At the end of telophase the cell begins to constrict at the center dividing the cytoplasm into two halves, each with its own nucleus. If the cytoplasm does not divide during mitosis, one cell with two identical nuclei results.
82. D.The term that refers to facts established by applying the scientific method repeatedly & gathering a large amount of supporting experimental data is called law or principle.
83. B. Unconditioned stimulus is a stimulus to which the animal did not have to learn the response to it. In the classical conditioning experiment carried out by Pavlov on dogs, the smell of the food is the unconditioned stimulus and the sound of the bell is the conditioned stimulus.
84. C. The final, most complex and stable state of a succession is the climax community. The species diversity will rise through the succession until the climax is reached.
85. A. Glycolysis is a first stage of cellular respiration that does not require oxygen to break down glucose to pyruvic acid. The krebs cycle, electron transport chain and reaction that links glycolysis and krebs cycle take place in the presence of oxygen
86. D. A 15% sucrose solution is more concentrated than the rest of the sucrose solutions. Thus, a potato cylinder kept in a 15% sucrose

Telegram: https://t.me/ExtremeSeries

solution loses more water by osmosis and becomes more flaccid than potato cylinders kept in the other solutions

87. A. The degree of closeness of organisms can be quantified by biological compounds in the body like DNA and proteins

88. C.

A. 00 x AB

x	A	B
O	AO	BO
O	AO	BO

B. BO x AA

X	A	A
B	AB	AB
O	AO	AO

C BO X AO

x	A	O
B	AB	BO
O	AO	OO

D. BB X AO

x	A	O
B	AB	BO
B	AB	BO

Therefore, the couple with blood groups BO and AO can produce children of A, B, AB and O

89. D. Antiretroviral drugs currently used to treat AIDS patients slow down HIV multiplication

90. B. Charged molecules, lipid molecules and small non-pollar molecules can pass across the cell membrane but larger polar molecules like amino acids, sugars, fatty acids and glycerol cannot pass across the cell membrane by simple diffusion

91. D. Woodlice are small land dwelling crustaceans. Because of their flattened shape and small size, they have a relatively large surface area to volume ratio. This means that they tend to lose water quickly through their body surface. They are typically found under logs, stones, bark and amongst leaf litter to reduce the rate at which water is lost from their bodies.

92. A. One of the advantages of using enzymes in industry is that they allow the reactions involved in the processes to be carried out at much lower temperatures

93. B. Biodiversity refers to the number, variety and variability of living organisms. Human beings can increase biodiversity by increasing genetic variability.

94. C. The wide range in altitude and climate, the isolation of the highlands of Ethiopia, and the fact that there are so many different biomes present in the country are some of the reasons for its high biodiversity

95. D. The parts of the angiosperm flower that are essential for hybridization experiments are gynoecium and androecium

96. A. The factors that are involved in the determination of climax vegetation are temperature and precipitation

97. D. Ancestrally related organisms have similar structures known as homologus structures. Homologous structures have the same origin and development but may or may not have the same function. For example, the limbs of humans and the flippers of whales are homologous structures that have five digits. Homologous structures tell us that these organisms with homologous structures provably had a common descent and most likely evolved from acommon ancestor.

98. B.The source of oxygen produced during photosynthesis in plants is water

99. The community passes through a sequence transitional stages called seres, ultimately achieving a relatively stable community called the climax community.

100. B. The tropical rain forest is characterized by a high rainfall and high temperature. A constantly hot humid climate produces this forest. The amount of rain fall in tropical rain forests is generally between 2000 to 4500mm per year with high temperatures.

Telegram: https://t.me/Extremeseries

1. **A:** Though other structures are components, **Phospholipids bilayer** is the basis for biological membranes and is found in large amount.
2. **C:** Important **Savanna Grassland mammals** include lions, leopards, giraffes, zebras, and wolves, as well as elephants, hippopotamuses, and chimpanzees.
3. **B: Steps in the scientific method** begin by observation and continue through gathering information, hypothesizing, testing through the experiment, evaluating data, drawing conclusion and formulating a theory.
4. **B: Cellulose** is a complex carbohydrate, the chief constituent of the cell wall of all plant cells. In plants, cellulose is normally combined with woody, fatty, or gummy substances.
5. **A: Amylase** is a digestive enzyme that starts to break down carbohydrates in food. It could be secreted in mouth by salivary gland or in pancrease.
6. **B: Prokaryotes** are relatively simple unicellular organism, such as a bacterium, characterized by the absence of a true nucleus and other specialized cell structures. They are categorized under **kingdom Monera**.
7. **B:** A common feature in the legumes is the presence of root nodules containing bacteria of the genus *Rhizobium*. These bacteria are capable of converting atmospheric nitrogen, which cannot be used by the plants, into nitrate (NO_3^-), a form that can be used. Legumes are often planted specifically to renew nitrogen supplies in soils.
8. **A: Enzymes** are proteins which are made up of amino acid building blocks.
9. **B: Worker bees** communicate with each other in a very special way to convey information about a source of nectar. Foragers perform a **'wag-dance'** on the honeycomb to inform other workers of the direction of the nectar source and its distance. The dance takes the form of a **'figure of eight'** on the vertical face of the honeycomb.
10. **D:** Mendel removed **stamens** from the flowers of the plant (so that these flowers could not pollinate themselves) and the process is known as **emasculation**.
11. **C: Paleontogist** studies fossils to generate new knowledge on the origin and evolution of living things of past geologic periods.
12. **C: Proteins** are extremely important substances that are needed to form all living cells. **Their molecules contain the elements carbon, hydrogen and oxygen** (like carbohydrates and lipids), and **nitrogen.**
13. **C: A ratio of 3:1 in the F1 generation** (always 3 of the one that had **appeared in the F1 and 1 of the one that hadn't)** is found in monohybrid **cross.**

Telegram: https://t.me/Extremeseries

14. **B: A disaccharide molecule** like maltose, lactose and sucrose are made by condensation reaction between two **hexose sugars**. A molecule of water (H_2O) is formed from a hydroxyl group from one monosaccharide and a hydrogen atom from the other. This allows a bond to be formed between the two monosaccharide units to make a disaccharide. The formula is $C_{12}H_{22}O_{11}$.
15. **D: Nanometer** is the best unit to measure the smallest cells and organelles because most of the size of bacteria is few nanometers long.
16. **D:** The full name for ATP is Adenosine Tri-Phosphate. **ATP is essentially the adenine nucleotide with two extra phosphate groups added on — making three in all.** Adding the extra phosphates requires energy, particularly when the third phosphate is added. As a result, energy is stored in the ATP molecule and when the bonds that hold this third phosphate in place are broken, the energy is released again.
17. **A: Food poisoning** is commonly caused by certain **bacteria or their toxins**, which are poisonous proteins produced by these bacteria. One toxin-producing bacterium is *Staphylococcus,* which occurs almost everywhere and grows readily in foods stored at room temperature, especially processed meat and fish, milk, and cream-filled foods.
18. **D: Crossing over, segregation and random assortment in meiosis** together with **random fusion in fertilization** reshuffle existing genetic material, and **mutation** produces new genetic material.
19. **B:** ***H. habilis*** was thought to have a larger brain than australopithecines and smaller than hominid species. The braincase of *H. habilis* measured from 590 to 690 cubic centimeters.
20. **B: Innate behaviour is** behaviour that is present (potentially) at birth or hatching. It does not have to be learned.
21. **D: The soils are nutrient poor** because many of the minerals from the soil remain 'locked up' in the huge trees. The only recycling of nutrients that occurs on a regular basis takes place when leaves fall. There is no accumulation of detritus as decomposers rapidly break down the leaves and release the mineral ions they contain. The roots take these up, leaving few mineral ions in the soil.
22. **C:** As cells increase in size, the volume increases faster than the surface area. **So, the surface-area-to-volume ratio decreases.**
23. **B:** Hydrocarbon chains with the same number of carbon atoms can have different numbers of hydrogen atoms. This is because of the nature of the bonding between the carbon atoms in the chain. If all the carbon—carbon bonds in the hydrocarbon chain are **single bonds**, the fatty acid is a **saturated fatty acid.** If one or more carbon—carbon **double bond** exists in the chain, then it is an **unsaturated fatty acid.**

24. **C:** Recently, it has been shown that some RNA molecules can catalyse some biological reactions in addition to proteins.
25. **B:** The cells are homogenized in a blender and filtered to remove debris. The homogenized sample is placed in an ultracentrifuge and spun at low speed and **the nuclei settle out**, forming a pellet. The supernatant (the suspension containing the remaining organelles) is spun at a higher speed — **chloroplasts settle out** (if plant tissue is used). The supernatant is spun at a higher speed still — **mitochondria settle out.** The process works based on the mass and density of the organelles.
26. **B: The diffusion of hydrogen ions (protons) through the ATP synthase is chemiosmosis.**The oxidation of one molecule of reduced NAD results in six protons passing through ATP synthase and so leads to the synthesis of three molecules of ATP. The oxidation of one molecule of reduced FAD results in four protons passing through ATP synthase and so leads to the synthesis of just two molecules of ATP.
27. **D: Deficiency Diseases** are disorders caused by lack of specific essential substances such as vitamins, minerals, or amino acids.
28. **C: Glucose sequence of polysaccharides** is nothing to do with the genetic makeup of an organism. Species that are closely related have the most similar **DNA and proteins**; those that are distantly related share fewer similarities.
29. **D:** It is identified that HIV infections spread through **sexual intercourse with an infected person, contact with contaminated blood,** and **transmission from an infected mother to her child before or during birth or through breast-feeding.**
30. **B: Decomposers (bacteria and fungi)** are key in **returning nutrients** to the ecosystem. Important mineral elements such as nitrogen and phosphorus are returned to plants as a result of the action of decomposers.
31. **D: Neo-Darwinism theory** explains evolutionary changes of living things in terms of changes in their allele frequencies because it considers the roles of genetics in evolution.
32. **B: Golgi apparatus**, also Golgi body or Golgi complex, network of stacked sacs found within nucleated cells that **store, package,** and **distribute the proteins and lipids** made in the endoplasmic reticulum.
33. **D:** As an ecological succession progresses to advanced seral stages, **different types of vegetation enter the area,** they affect the amount and **depth of soil.** This, in turn, allows other types of plant with more complex root systems to enter. The increasing complexity of the plant community will create more and **more ecological niches** and so more

animals will also enter the area.**The species diversity will rise through the succession**, until the climax is reached.

34. **D:** Genes themselves are constantly being modified through a process called mutation—a change in the structure of the DNA in an individual's cells. While some mutations occur spontaneously during replication, others are caused by factors in the environment, known as mutagens. Examples of mutagens that affect human DNA include **ultraviolet light, X rays, and various chemicals.**
35. **B: Behavioral Isolation** is if related species of organisms such as birds often differ in their courtship rituals, which tends to keep these species distinct in nature even if they inhabit the same places.
36. **A: Humans inherits 23 chromosomes** from each parents so that they would have 46 chromosomes altogether.
37. **C:** The number of population reaches the carrying capacity which is the highest in **stationary phase (constant phase).**
38. **D:** It is the **Y chromosome** that appears to determine a person's sex because it carries **the SRY gene** — that determines the formation of testes.
39. **C: In Lysogenic cycles,** the virus's DNA becomes incorporated into the DNA of the host cell. Each time the cell divides, the DNA is replicated, and each daughter cell gets a copy of the cell's DNA, which now includes the virus DNA.
40. **B: Sensitisation** is an increase in the response to a harmless stimulus when that stimulus occurs after a harmful stimulus.
41. **A:**

Feature	Prokaryotic cells	Eukaryotic cells
Size	1—10 pm	10—100 pm
Nucleus	No membrane-bound nucleus	Nucleus surrounded by nuclear envelope
DNA	• In a continuous Loop • Not associated with protein to form chromosomes	• Linear DNA • Associated with histone proteins in chromosomes
Mitochondria	Absent	Present
Chloroplasts	Absent (but some prokaryotic cells contain a kind of chlorophyll and can photosynthesise)	Present in some cells (some plant cells and some a (gal cells)
Ribosome	Present, but smaller than in eukaryotic cells (70_s)	Present, but larger than in prokaryotic cells (80_s)

Telegram: https://t.me/ExtremeSeries

Cell wall	•Always present Not made from cellulose (often made from peptidoglycan)	• Present in plant cells, algal cells and fungal cells • Cellulose in plant cells, various materials in other cells

42. **C:** When **red blood cells** are placed in **hypotonic solution, the cells gain water by osmosis and swell.** The pressure will eventually **burst** the weak plasma membrane of **RBCs** this is called **haemolysis.**
43. **B: GPS (Global Positioning System)** is equipment that makes it possible to record positions quickly and extremely accurately.
44. **C: Dolly the sheep** was the first mammal to be cloned. **Clone** of an organism is a group of organisms that are genetically identical to each other and to the organism from which they were derived.
45. **D: A population** belong to the same species if they able to reproduce and give rise to fertile individuals.
46. **B:** In the process of the evolution of life on earth, chemosynthetic bacteria that produce organic compounds thought to evolve first. Later photosynthetic autotrophs evolved that released atmospheric oxygen which in turn enable aerobically respiring organism to evolve.
47. **C: Theory** is a well tested hypothesis which is supported by experiments and/or observations.
48. **A: Theodolite** is an item of field equipment that measures the height of a tree.
49. **B: Lactose** is the sugar with formula $C_{12}H_{22}O_{12}$ present in **milk.** Lactose is obtained from milk in the form of hard, gritty crystals of composition $C_{12}H_{22}O_{11}\cdot H_2O$, by evaporation of the whey remaining after removal of the fat and precipitation of the casein.
50. **C: Digestive enzymes** are all belong to hydrolases because they break down large polymers with the addition of water (hydrolysis)to form small monomer units.
51. **B:** Meat, fish, poultry, eggs and milk are **animal sources** of protein that provide **a good balance of all eight essential amino acids. The best non-animal sources are quinoa** (high protein grain), buckwheat, hempseed and amaranth, although these contain lower overall amounts of protein.
52. **C:** Ethiopia is an important regional centre of biological diversity. The wide range in altitude and climate, the isolation of the highlands of Ethiopia, and the fact that **there are so many different biomes present in the country** are some of the reasons for its high biodiversity.

53. **B:** Pure, liquid water has a higher water potential than any other system. All other systems (cells, solutions and suspensions) have a water potential that is lower than that of water. Therefore, their water potential values must be negative.
54. **D: Chloroplasts and mitochondria** are both double membrane bounded organelles.
55. **A:** The plasma membrane is composed primarily of two types of molecules—**lipids**, which are fatty or oily molecules, and **proteins**.
56. **B: Athlete's Foot**, also called tinea pedis or ringworm of the foot, a **contagious fungal infection** occurring most often between the toes and on the soles of the feet. A condition that tends to recur, athlete's foot is caused by several kinds of fungi that thrive in warm, damp places, such as the floors of showers, swimming pools, and gymnasiums.
57. **C: Genetic engineering** is a process in which the genome of an organism is altered, usually by having an extra gene from a different organism added. The organism is then a **genetically modified** or a **transgenic organism** where as pathogenic organisms are diseases causing organisms.
58. **B: Fungi lack chlorophyll**, the green pigment that enables plants to make their own food. Consequently, fungi cannot synthesize their own food the way plants do. In order to feed, fungi release digestive enzymes that break down food outside their bodies (**hetrotrophic mode of nutrition**). The fungus then absorbs the dissolved food through its cell walls.
59. **C: Climax community** is the final, most complex, state of a succession characterized by more stable community of plants and animals.
60. **A: Ecosystems** exhibit a constant exchange of matter and energy between **the biotic and abiotic community**. Example: - A Tropical Rainforest, desert, pond and others.
61. **C: Homologous structures** are structures with the same basic anatomy and a common evolutionary origin but having a different function and it best provides evolutionary relationship of organisms
62. **D:** The Ethiopian wolf is a social animal; the wolves live and hunt in packs. As a result, they maintain a group territory by marking with urine (containing pheromones). All adult animals (male and female) contribute to this marking behavior, particularly during patrols of the territory. Some of the sub adult (younger) males occasionally mark but **sub adult females never mark.**
63. **D: Angiosperms** are flowering plants with seeds in fruits. They can be monocots or dcots.

Telegram: https://t.me/Extremeseries

64. **A:** In a cross between heterozygote; 25% of the offspring are homozygous dominant, 50% are hetrozygote and 25% are **homozygous recessive**.

65. **C: Short interfering RNA** (siRNA) molecules are very short — only about 21 to 23 nucleotides long — and are double stranded. They don't act on the gene itself, but they 'interfere with' or 'silence' the mRNA once it has been transcribed from the DNA.

66. **A: Chromosomes** sometimes undergo changes called **translocations**, in which part of one chromosome breaks off and attaches to another chromosome. A translocation involving chromosomes **9** and **22** is linked to a type of leukemia called **chronic myelogenic leukemia.**

67. **D:** If two heterozygote are crossed the offspring resulted are:

X	A	a
A	AA	Aa
a	Aa	aa

Three fourth of the offspring contains the allele **A** which determine the phenotype of the offspring.

68. **D: Community** is the living, or biotic, parts of an ecosystem, such as the plants, animals, and bacteria which includes all the population and species found in them.

69. **A:** Life is considered a characteristic of organisms that exhibit all or most of the following characteristics or traits; Homeostasis, Organization, Metabolism, Growth, Adaptation, Response to stimuli and Reproduction.

70. **D:** There are several different Anti-retroviral drugs that act in different ways at **different stages of the cycle of reproduction.** Because the drugs act on different stages of the HIV life cycle, the most effective treatment is obtained when they are used together. This is called High **Activity Anti-Retroviral Treatment (HAART).**

71. **B:**

Telegram: https://t.me/ExtremeSeries

Feature	DNA	RNA
Size	**Huge** — allows the molecule to carry the code for many different proteins in the genes.	**Much smaller** — only code for one protein, small size allows RNA to move out of the nucleus
Stability	**Very stable** — ensures that the genes remain the same over generations.	**Less stable** — degraded quickly so does not carry on coding for a protein.
Number of strands	**Two strands** — allow coding of genes and replication during cell division.	**Single stranded** — does not replicate
Sugar subunit	**Deoxyribose**	**Ribose**
Types of bases	Guanine, Cytosine, Adenine, and **Thiamine**	Guanine, Cytosine, Adenine, and Uracil

72. **C:** Lysosomes contain hydrolytic enzymes that digest worn-out or damaged organelles as well as engulfed bacteria. **Lysosomes are particularly abundant in phagocytic white blood cells.** Here, enzymes from the lysosomes digest foreign cells that have been engulfed.

73. **D:** Reducing sugars are called reducing sugars because they will act as **reducing agents in an alkaline solution**. This is the basis of the Benedict's test. In Benedict's test, reducing sugars reduce copper (II) ions (blue) in Benedict's reagent to copper (I) ions (brick red). If the foodstuff has no effect, one can confidentially state that it lacks reducing sugar

74. **A: Allosteric inhibition** is if a certain substance occupies the allosteric site of the enzyme so that it would bring conformational change of active site to halt catalysis. In the metabolic pathway the end product is an allosteric inhibitor.

75. **C:** If red blood cells shrink, it means that it has lost its water content. Since water move from hypotonic solution to hypertonic solution, the

. C

Viruses are considered entities that lie between living and non-living matter. They are made up of genetic core made of nucleic acid covered by protein coat the feature which they share with cell.

:. D

Robert Hooke, with one of the earliest compound microscopes, makes drawings of cork and sees tiny structures that he calls "cells". It was the cell wall of dead plant cells.

:. D

Substrate is any substance that can be acted up on by enzyme to be either made or degrade.

. C

Living organisms broadly classified as Prokaryotes and Eukaryotes. Prokaryotes include Bacteria and blue-green algae under kingdom monera. The rest of four kingdoms namely; Protista, Fungi, Plantae and Animalia are eukaryotes possessing true nucleus.

:. D

The scientific method follows the following basic steps

1. **Observation.** A scientific study begins with observations of objects
2. **Identifying the problem.** Scientists ask questions from what they observe.
3. **Forming a hypothesis.** Observations lead to the formulation of hypothesis. The hypothesis is a tentative theory or an intelligent guess. It can be supported or disproved after the experiment.
4. **Making predictions.** A prediction is a logical deduction from a hypothesis, without any kind of experimentation.
5. **Conducting an experiment.** Hypotheses are often tested by experimentation. After the experiment the hypothesis can be rejected or accepted.
6. **Drawing a conclusion from the experiment.** If the hypothesis is supported by an experiment it becomes a **theory.**

:. A

Telegram: https://t.me/Extremeseries

Animal cells can change shape due to the lack of a **cell wall** and are usually rounded whereas plant cells have a fixed shape kept by the presence of the cell wall.

7. B

Habituation is an extremely simple form of learning, in which an animal, after a period of exposure to a repeated presentation (stimulus), stops responding.

8. B

Circadian rhythms are patterns that are repeated daily. Example: Sleep/wake cycles in animals and plants. Circadian rhythms appear to be timed by an internal biological clock.

9. A

Gene is a small piece of the genome. It is the fundamental unit of heredity. Genes are found on chromosomes and are made of DNA which is a nucleic acid. Different genes determine the different characteristics, or traits, of an organism.

10. B

Evolution is the study of genetic change in a population from one generation to another or changes in the gene pool of a population from generation togenerationby such processes as mutation, natural selection, and geneticdrift.

11. B

The part of protoplasm that lies inside the nucleus is known as **nucleoplasm** while the part outside of the nucleus is known as **cytoplasm.**

12. A

Glycolysis is the first stage of splitting glucose. The net result of glycolysis is a gain of 2 ATP, an "investment" into 2 NADHs, and the breaking apart of glucose into two pyruvate molecules.

13. B

Bacilli are the group of bacteria categorized based on their rod shape.

14. B

Disaccharide molecule like maltose, lactose and sucrose are made by condensation reaction between two **hexose sugars**. A molecule of water (H_2O) is formed from a hydroxyl group from one monosaccharide and a hydrogen atom from the other. This allows a bond to be formed between the two monosaccharide units to make a disaccharide. The formula is $C_{12}H_{22}O_{11}$.

15. B

They are special organelles that are directly involved in protein synthesis. They are non-membranous organelles. They are made of **RNA (ribonucleic acid)** and protein and are manufactured in

the nucleus (from a DNA template), then go out into the cytoplasm to function.

16. A

An important defining characteristic of legumes is the fact that they form a symbiotic relationship with bacteria of the genus *Rhizobium*. In this relationship, the *Rhizobium* bacteria colonize plants roots causing the formation of **nodules**. From inside these nodules, the bacteria absorb nitrogen gas (N2) from the atmosphere and then transform it into forms of nitrogen (**nitrate:** NO3- or **ammonia:** NH3) that are usable by itself and the host plant.

17. B

Protozoais a collective name for animal-like, single-celled organisms, some of which may form colonies. In the classification, the protozoa are placed in the kingdom Protista with other **single-celled organisms** that have membrane-enclosed nuclei.

18. D

Amino acids are made of the four elements C H O N and some may have S. Carbohydrates are made up of elements called C, H, and 0 only.

19. C

The energy is passed on from trophic level to trophic level while nutrients are continuously recycled between biotic and abiotic components of the ecosystem.

20. D

Water is a very important biological molecule – it is found in large percentages in all living forms. Water is the most abundant component of protoplasm (60 - 80), followed by protein (9%), carbohydrates (2-3%) and lipids (1.5 - 2). Water is an inorganic molecule made up of hydrogen and oxygen.

21. D

About 90% of cellular ATP is generated from the transport of electron across inner membrane of mitochondria.

22. B

Proteins that function as biological catalysts are called enzymes. They are required for almost every reaction in the body. Enzymes are the key to life. ***All known enzymes are globular proteins.***

23. D

Parental gametes would be Ry and rY

X	Ry	Ry
rY	RrYy	RrYy
rY	RrYy	RrYy

From the cross shown All F1 progeny are RrYy.

24. A

All cells are composed of nuclear material, cytoplasm and cell membrane.

25. C

Most of the energy released from glucose is stored temporarily in the form of NADH a coenzyme that further release energy during electron transport system.

26. A

Cellulose is homopolysaccharide of β-glucose joined by β-1, 4-glycosidic bonds, formed by condensation reaction. Have chains of cellulose molecules linked by hydrogen bonds both intermolecular and intra molecular. This results in the formation of cellulose **micro fibrils.** It forms a structural component of cell wall in plants.

27. D

Most plants obtain the nitrogen they need as **inorganic nitrate** from the soil solution. Ammonium is used less by plants for uptake because in large concentrations it is extremely toxic.

28. C

Hybrid vigor-occurs when unrelated breeds or lines are cross-breed. Because it is the result of increased heterozygosity, the term hybrid vigor is sometimes replaced by **heterosis.** In many cases the offspring from these crosses are **more productive** (higher milk yields, more beef per carcass) than the average of their parent breeds.

29. B

Electrophoresis is techniques used to separate DNA fragments according to their size on a gel. This technique is the biotechnology equivalent of screening sand through progressively finer mesh screens to determine particle sizes.

30. B

Sex is determined inbirds by female. Itisfemaleswhoaretheheterogametic sex (ZW) andmalesthe homogameticsex (ZZ).

31. A

A photosystem consists of a number of pigment molecules all clustered around one particular chlorophyll molecule called ***reaction center molecule*** where light-dependent reactions begin. ***The reaction center molecule is always chlorophyll 'a'.***

32. C

According to Lamarck, changes in the environment may lead to changed patterns of behavior which can necessitate new or increased use or disuse of certain organs or structures. The

Telegram: https://t.me/ExtremeSeries

constant use of an organ increases its efficiency and size and leads to its better development. On the contrary, if an organ is not used for a long time it leads to the reduction in efficiency and size and ultimately leads to its degeneration.

33. D

Worker bees do all the different tasks needed to maintain and operate the hive. They work in and outside the hive doing comb ***construction, brood rearing, tending the queen and drones, cleaning, temperature regulation and defending the hive.***

34. D

Tundra is largely frozen treeless area with a short growing season. It receives very little rainfall 200mm / year. It is referred to as **cold desert** due to very low vegetation (sparse vegetation) and plants possess **xerophytic** characteristics.

35. D

Crossing over occurs between the paired DNA strands, creating the chromosomal configurations known as chiasmata. This enables genes on the same chromosome to separate during meiosis.

36. B

Yeast undergoes alcoholic fermentation in the absence of oxygen. If oxygen availability is there they shift their mode to aerobic respiration.

37. C

Selective breeding experiments have shown that genetic and physical modification of species is possible and so should be possible as a result of natural selection (rather than human selection).

38. D

Soluble digested food molecules are absorbed by the lining of the plasma membrane through **endocytosis** (Active transport).

39. A

Krebs cycle takes place inside the matrix of mitochondrion whereas ETC occurs on the inner membrane of mitochondrion.

40. D

The sun is an ultimate source of energy for living organism in that it provides solar energy for green plants which in turn prepares their own food and transfer chemical energy for the rest of organisms.

41. C

The necessary materials for photosynthesis to take place in green plants include **Water, Carbon dioxide sun light** and **Chlorophyll.**

42. A

All enzymes are globular proteins.

43. D

Enzymes are globular proteins which can be named based on the substrate they act on. For example: Lipase act on lipids, Maltase act on maltose and sucurase (invertase) act on sucrose.

44. B

Virus is made up of genetic material surrounded by a protein coat. Having genetic material they show certain resemblance with living things.

45. C

Endosymbiosis theory suggests that Chloroplast and mitochondria which are a eukaryotic organelle by now were free living prokaryotes.

46. D

After 300 years or three half-life's there could be only 12.5 Kg radioactive substance is left.

47. B

Drones (males of the honey bee) are sterile because they are developed parthenogenetically. So they are haploid organism.

48. E

PH value ranges from 0-14 and inversely proportional to POH which measures the strength of bases. So the higher the PH value the stronger the bases are.

49. B

Intermsofthebiodiversityofitsavifauna(birds),Ethiopiaisoneof themostsignificantcountriesinmainlandAfrica. It is also characterized by having the largest number of genera (306) among vertebrates.

50. D

The sum total of all chemical reactions taking place in living organism is called **metabolism.** The reactions which **build up** products are known as **anabolic reactions** whereasthe reactions that **break down** substances are known as **catabolic reactions.**

51. C

Homologous structures are structures that share a common developmental (evolutionary) origin but may serve different functions in modern species.

52. B

Transpiration is an ultimate loss of water from the surface of leaves of a plant.

53. A

Telegram: https://t.me/ExtremeSeries

Saturated fatty acids have only **single bonds** between carbon atoms. Hence they contain the maximum number of hydrogen atoms possible.

54. **D**

Oily and greasy dirt is most effectively be removed by detergent containing lipase which act on lipids.

55. **D**

Metamorphosis is a gradual change of the body and periodic shading of their skin in some animals like insect and snake.

56. **B**

During protein synthesis, two important and related processes are taken place. **Transcription** which is copying genetic information from DNA to mRNA takes place inside nucleus. Following this **translation** or transferring genetic messages to protein synthesis takes place in the cytoplasm.

57. **C**

Perfect flower is the one that contains both male (stamen) and female (pistil) reproductive structures.

58. **C**

Refer Question No 56.

59. **C**

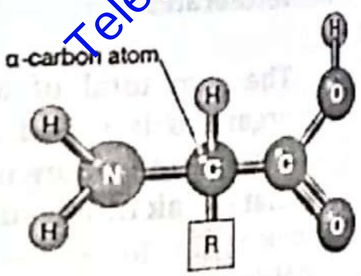
Enzymes in our mouth works best at around neutral pH while those in our stomach works best at relatively high PH. Those in our small intestine work best at low PH.

60. **D**

Ecosystem is self - sufficient environment except the need for sun light from outside.

61. **D**

The substance shown by the chemical formula above would be represented by the molecular formula shown below. **R** group is replaced by **H** if the amino acid is the simplest one.



62. **D**

An object may move from place to place or back and forth with an external force which is unnecessary to define life.

63. **A**

Genetics is the branch of biology that deals with heredity, especially the mechanisms of hereditary transmission and the variation of inherited characteristics among similar or related organisms.

64. **C**

Carbon is the basic building of the carbohydrates, fats, proteins, nucleic acids such as DNA and RNA, and other organic compounds necessary for life. Producers absorb CO2 from the atmosphere or water and use photosynthesis to convert it into complex carbohydrates.

65. **D**

Pathogenic organisms are organisms that cause disease. The other three refers to the same thing to mean organisms whose genetic material is altered.

66. **C**

Physical barrier s like Continental split, rise in sea level, formation of mountain range, advance of glacier and change in habitat prevents gene flow between populations of a species.

67. **B**

Transcription is the process that converts genetic information from a *DNA code into an mRNA code.*

68. **C**

Endoplasmic reticulum is responsible for removing cellular debris by forming vesicles.

69. **C**

Tropical rainforest is characterized by high rain fall and high temperature.

70. **C**

Non-nucleotide reversetranscriptase inhibitors (NATI's) attach toHIV'senzyme preventingitfrom convertingRNAtoDNA.

71. **B**

In C_4 plants, CO_2 is taken up in mesophyll cells and malate, a reduced form of oxaloacetate, is pumped into the bundle-sheath cells; here CO_2 enters Calvin cycle. C_4 plants have advantage over C_3 plants in hot and dry weather because photorespiration does not occur.

72. **D**

Mitochondria and chloroplast are semi-autonomous organelles capable of duplicating themselves. They have their own smaller DNAs. Ribosomes also are made of protein and RNA components.

73. **A**

Whentwodifferentspeciescompete forthesame resourcein thesamehabitat , interspecific competition becomes strong.

74. **D**

Half-life refers to the time taken for the radioactive isotope to fall by half. Thus if a half-life of carbon 14 of a certain fossil is 5730 years

Telegram: https://t.me/ExtremeSeries

- ➲ 50% of the original amount of carbon 14 remains after 5730 years
- ➲ 25% of the original amount of carbon 14 remains after 11460 years
- ➲ 12.5% of the original amount of carbon 14 remains after 17200 years

75. **D**

Withtheexceptionofidenticaltwins (monozygotic twins) ;an individual's fingerprintsareunique. They donotchange throughoutlife. **Genetic fingerprinting** is a technique efor comparing the DNA of different people.

76. **D**

Growing crops of land contribute less for biodiversity loss.

77. **D**

Alowvaluefortheindexofdiversity,suggestingonlyafew successfulspecies,couldbetheresultofahostileenvironmentwith onlyafeworganismsbeingreallywelladaptedtothatenvironment.

78. **D**

Nondisjunction ("not coming apart") is the failure of chromosome pairs to separate properly during meiosis stage I or the failure of sister chromatids to separate duringmeiosis II. The result of this error is a cell with an imbalance of chromosomes. Such a cell is said to be **aneuploid**-Cell that has extra chromosomes or chromosomes missing.

79. **A**

The total surface area of this cell is calculated by the equation: ***height × width × number of sides*** which is 10*10*6=600 whereas the total volume of this cell is calculated by the equation: ***height × width × Length*** which is 10*10*10=1000. So the surface area to volume ratio of this cell is 6:10.

80. **A**

Analogous structures are a contrast to homologous structures. They serve the same function between organisms but are different in internal anatomy.

↳ Such as **the wings of birds and butterflies** or the eyes of lobsters and fish.These structures are of no use in classifying organisms or in working out their evolutionary relationships with each other.

81. **C**

The difference between hydrocarbon chains in fatty acids is because of the nature of the bonding between the carbon atoms.

➲If all the carbon-carbon bonds in the hydrocarbon chain are single bonds, the fatty acid is a ***saturated fatty acid.***

➲If one of the carbon – carbon bonds is a double bond, then it is a ***monounsaturated fatty acid.***

➲If more than one carbon – carbon bond is a double bond, then the fatty acid is a ***polyunsaturated.***

12. **B**

Mating between BB and OO produces offspring with only BO (B blood groups).

13. **D**

Ecosystem is a wider concept consisting of the living components plus the non-living factors of the environment.

14. **C**

The oldest hominid fossil is Australopithecus (Australopithecus afarensis (Lucy))

She had a structure – ape like above the waist and human like below waist.She was bipedalism and she lived in Africa – Ethiopia 3.2 MYA.

15. **B**

ATP synthase is an enzymatic protein that traverses the inner membrane of mitochondria so that protons (H^+) are moved through back to matrix to generate ATP.

16. **C**

About 90% of the ATP generated during aerobic respiration is generated by electron transport & chemiosmosis.

17. **A**

$C_{12}H_{22}O_{11}$is a formula for disaccharide molecules like sucrose and maltose which are groups of organic molecule called carbohydrates.

18. **A**

Animal cells in the hypertonic solution, the cells lose water and ***shrink***. In the hypotonic solution, the cells gain water and ***swell***. The pressure will eventually burst the weak plasma membrane. This is called ***hemolysis***. In the isotonic solution, there is no change.

19. **A**

The rate of photosynthesis in the water body depends on the **amount of light** entering or passing through the water body.

20. **C**

Chloroplast is an organelle used in a cell to manufacture food. If this critical organelle is lacking in a cell, it (the cell) is having heterotrophic nutrition.

21. **B**

Early heterotrophs probably added carbon dioxide to the atmosphere which causes the depletion of oxygen. This in turn causes the origin of autotrophs.

Telegram: https://t.me/ExtremeSeries

92. C

The percentage of Thymine and Adenosine is equal so that they account about 20% and the rest of the bases account for 80% of the total from which Cytosine contains 40%.

93. B

Homologous structures are structures that share a common developmental origin but may serve different functions in modern species. These structures are evidence that organisms with similar structure evolved from a common ancestor.

Examples:- the forelimbs of a variety of mammals such as human, cat, whale and bat.

94. C

In the hypertonic solution, the cytoplasm of plant cells loses water and ***shrinks***. The cell is said to be ***flaccid***. If cytoplasm shrinks too much, it loses contact with the cell wall and we say cell has been ***plasmolysed***.

95. C

The commonest lipid called triglycerides is hydrolyzed in to fatty acids and glycerol.

96. B

Successions that occur on soils that have already been formed, but have suddenly lost their community are referred to as **secondary successions**.

97. C

The abbreviation HIV stands for the Human Immunodeficiency Virus, which is the name of the organism that causes this disease and the abbreviation AIDS stands for the Acquired Immune Deficiency Syndrome, which is the name of the set of symptoms that characterize this condition.

98. B

The first antibiotic made were **penicillin** from the **fungi** called penicillum by Alexander Fleming.

99. D

The use of organism to remove pollutants from water, air, and soil is called **bioremediation.** Bacteria are also effective in cleaning up harmful wastes through bioremediation. In this process bacteria and other microorganisms convert toxic or otherwise objectionable wastes, such as pesticides and oil spills, to harmless or even useful products.

100. A

Acontrolgroupactsasa 'standard 'for comparison. Itis used to'isolate' the factor we are investigating and show that changes are due to this factor.

Telegram: https://t.me/ExtremeSeries

The principle of democracy which asserts that the power of the government derives from the "freely given consent" of the people is known as _____.

A. Democratic election
B. Universal suffrage
C. Popular sovereignty
D. Transparency

The present Ethiopian foreign policy is based on the following principles **except**:

A. Promoting development
B. Building democratic system.
C. National pride and globalization
D. Territorial expansion

Which of the following belong to the factors accounted for the creation of the Regional states in Ethiopia?

A. Religious diversity
B. Ethnic diversity
C. Gender diversity
D. Racial diversity

The mechanism of conflict resolution which involves the coming together of those who are in conflict to make discussion is known as _____.

A. Negotiation
B. Arbitration
C. Mediation
D. Litigation

In Ethiopia the Affirmative Action Policy is important for bridging the gender gap in _____.

A. higher education institutions
B. unfair division of labor at the household level
C. physical abilities between men and women
D. intellectual capacities between men and women

Citizen's devotion and commitment for the protection of democratic values and ideals is known as _____.

A. Patriotism
B. Nationalism
C. Constitutionalism
D. Voluntarism

How may Constitutions were promulgated in the Ethiopian constitutional history?

A. One B. Two C. Three D. Four

The state of being answerable for an obligation is known as _____.

A. Tolerance
B. Transparency
C. Accountability
D. Responsibility

Active involvement of citizens in the socio-economic and political spheres is known as _____.

A. Political participation
B. Social participation
C. Economic participation
D. Community participation

The state of possessing truths, facts, principles or other objects of perceptions through instruction, study and research is known as _____.

A. Information
B. Knowledge
C. Communication
D. Data

All citizens of a democratic country are bound to be abided by the

constitution of their country. They should do so even though
A. they believe it to be totally undemocratic.
B. some of them disagree with some of its provisions.
C. they are convinced it serves the interests of a certain group.
D. it totally fails to reflect their national aspirations and goals.

12. In democratic systems citizens give state officials authority so that the officials could
A. Use the state apparatus for their own ends.
B. Rule the country on behalf of the electorate.
C. Help them become popular politicians.
D. Represent them on various international forums.

13. Who empowers the executive organ of the F.D.R.E. to issue rules and regulations?
A. The House of Peoples' Representatives.
B. The House of Federation.
C. The Prime Minister of the country
D. The President of the country.

14. The principle of constitutional law, which protects the life, liberty and property of a person, is known as _____.
A. Criminal law
B. Civil law
C. Due process of law
D. Habeas Corpus

15. In Ethiopia true patriotism had for long meant a person's readiness to:
A. fight for ones democratic rights.
B. defend the defenseless people.
C. sacrifice oneself for the country.
D. give respect to the national signs.

16. Among the four type of peaceful conflict resolution methods which one is the most formal mechanism?
A. Arbitration. B. Litigation C. Mediation D. Negotiation.

17. To whom does the F.D.R.E. constitution give the power to ratify international treaties?
A. The Executive Branch of the Government.
B. The House of Federation.
C. The House of Peoples' Representatives.
D. The President of the Federation.

18. Civil disobedience is different from other types of protests because it essentially
A. aims to overthrow a government by force.
B. excludes all types of military action.
C. utilizes mainly illegal methods of protest.
D. follows legally prescribed procedures.

19. Which of the following is **not** one of the functions of courts?
A. Imprisoning convicted criminals.
B. Keeping national domestic peace.

C. Deciding controversies between litigants.
D. Enforcing judicial decisions.

20. Which of the following phenomenon could be example of an environmental hazard that is caused by a country's unwise utilization of its resources and which will also damage the environment of other states?
A. Land degradation.
B. Flood and erosion.
C. Air pollution.
D. Land sliding.

21. Which of the following explains the system in which elected public officials and made responsible to the people that elected them?
A. Accountability of state.
B. Transparency of state.
C. Responsibility of state.
D. Sovereign power of state.

22. Which of the following indicates the similarity between parliamentary and presidential democracy?
A. Absence of separation of legislative and executive branches of government.
B. Fusion of power between legislative and executive branches of government.
C. The President is the head of government or the chief executive.
D. The judiciary is independent from the legislative and executive branches.

23. Which of the following is the sources of authority of a democratic government?
A. The consent of the governed.
B. The lineage of the rulers' family.
C. The will of God/Allah.
D. The custom of the society.

24. Which of the following explains the necessity of tolerance of difference for Ethiopian unity?
A. It helps the people appreciate their own cultures, traditions and practices.
B. It may lead to conflict of interests among different linguistic, ethnic and cultural groups.
C. If fosters a variety of viewpoints, new ideas and ways of looking at and solving problems.
D. It enables the people to know their own arts, music, literature, and the like.

25. Which one of the following is an exclusive power and function of the Federal government Ethiopia?
A. Administering land.
B. Collecting taxes.
C. Regulating foreign exchange.
D. Administering police forces.

26. Which of the following indicates the importance of Accountability and Transparency for development of a county?
A. Developing wasteful and corrupt administration.
B. Developing ineffective development strategies.
C. Developing inefficient allocation of resources.
D. Developing clear objectives and monitor performance.

Telegram: https://t.me/ExtremeSeries

→ $V \times m = 2mV_f$

→ $V_f = \frac{V}{2} = 0.5\,V$

12. **A.** The block decelerates up the inclined plane . The net force acting on the block down the block inclined plane is equal to the sum of mg $\sin\theta$ and F_f=15N by newton's 2nd law:-

$Fn = ma$

→ $Mg\sin\theta + 15N = ma$

→ $(4kg)(10m/s2)\sin 30) + 15N = (4kg)\,a$

→ $\vec{a} = 8.75\ m/s^2$

Finally to get the distance the block moves up before stopping can be obtained by:-

$$S = \frac{V_f^2 - V_i^2}{2(a)} \quad V_f = 0 \;\&\; a = \text{negative}$$

$$S = \frac{64m^2/s^2}{(2\times 8.75m/s^2)} = 3.66m$$

13. **D.** The sum of the forces 1N and 7 N can not be equal to 5 N in any way.

14. **A.** In a quasi – static process of gas expansion the work done is found using the following relation which applies an integral calculation

$$W = \int_{v_1}^{v_2} pdv = \int_{V_1}^{V_2} (dv)dv$$

$$= \alpha\frac{1}{2}V^2]_{1m^3}^{2m^3}$$

$$W = 5\times 10^{+5}\text{N/m}^5 \times \frac{3}{2}\,m^6 = 0.75\,mJ$$

15. **B.** After the ball is released from the hand of the thrower it will be under the control of the earth's gravity hence its acceleration will be equal to g

16. **C** When the block is sliding down the inclined plane the net force on it is equal to mg $\sin\theta$ - F_f. By newton's 2nd law its acceleration is

$F_{net} = m\,a$

→ $Mg\sin\theta - F_f = ma$

→ $Mg\sin\theta - mg\cos\theta\mu_k = ma$

→ $a = g\sin\theta\,(1-\mu_K)$ .....(1) $\sin\theta = \cos\theta$, for θ 45^0

When the block is projected up, the net force on the block should be mgsin $\theta + F_f$ By newton's 2nd law the upward acceleration is

$F_{net} = ma$

→ $Mgsin\,\theta + mg\cos\theta\mu_k = ma$

→ $a = g\sin\theta\,(1+\mu_k)$ .... (2)

Dividing (1) by (2) gives us the following

$$\frac{mgsin\,\theta(1-\mu_k)}{mg\,sin\theta(1+\mu_k)} = \frac{1-\mu_k}{1-\mu_k}$$

17. **B.** When a small mass m is suspended from one end of a vertical string the string becomes inclined to the vertical. The other three choices A, C, & D are not correct

18. **D.** If A+ B –C = 0

→ (3i -2j) + (-i -4j) –C =0

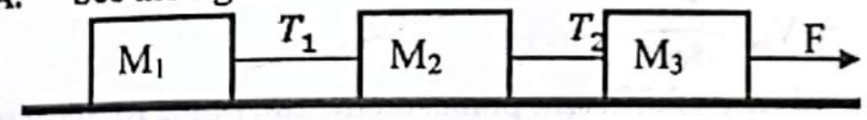
→ (2i – 6j) – C = 0

→ $\vec{C} = 2i - 6j$

Then the unit vector inthe direction of C is

$$\hat{C} = \frac{C}{|C|} = \frac{2i-6j}{2\sqrt{10}} = \frac{1\,i}{\sqrt{10}} - \frac{3}{\sqrt{10}}\,j$$

19. **A.** See the figure



The system of the three blocks acceleration in the direction of the force F.

There fore by newton's 2nd law you can write

→ $F - T_2 = m_3 a$. .... (1)

→ $T_2 - T_1 = m_2 a$.... (2)

→ $T_1 = m_1 a$ ..... (3)

→ $F - (m_2 a + T_1) = m_3 a$, substitute for T_2

→ $F - (\frac{m_2}{m_1}T_1 + T_1) = \frac{m_3}{m_1}T_1$

→ $F = T_1(\frac{m_3}{m_1} + \frac{m_2}{m_1} + 1)$

→ $F = T_1(\frac{m_1+m_2+m_3}{m_1})$

→ $T_1 = \frac{m_1 F}{m_1+m_2+m_3}$

20. **D.** If C is a vector along x axis Then $C = C_x i$

Telegram: https://t.me/ExtremeSeries

27. Rules and laws can be evaluated by using the following criteria **except:**
   A. fairness or free from prejudice, favoritism and self-interest.
   B. Easy to understand and well-designed to achieve its purpose.
   C. Clear as to what is expected to be achieved.
   D. Violating other laws and impossible to follow.
28. Which of the following explains the reciprocal relationship between rule of law and democratic system in a country?
   A. Prevalence of good governance.
   B. Prevalence of political instability
   C. Prevalence of social instability
   D. Prevalence of economic instability
29. Which of the following distinguishes limited governments from unlimited government?
   A. The government has an absolute power.
   B. The constitution is the highest law.
   C. Political power is usually accumulated in the hands of the few.
   D. Absence of division and sharing of governmental power.
30. Which of the following does **not** belong to the consequences of the absence of the rule of law?
   A. Prevalence of peace and stability.
   B. Anarchy and arbitrary rule.
   C. Absence of predictability.
   D. Absence of independent Judiciary.
31. Which one of the following is the consequence of corruption?
   A. Promoting the rule of law.
   B. Enhancing social development.
   C. Promoting good governance.
   D. Discouraging the prevalence of the rule of law.
32. The Nations, Nationalities and Peoples of Ethiopia have equal right to:
   A. share the results of other regional states' development
   B. share the results of the country's development
   C. disrespect others' cultures and traditions.
   D. impose their value systems on others.
33. Affirmative action is important to build the ability of the nation and nationalities of Ethiopia to:
   A. self-administration.
   B. Enhance the development of previously advantaged regions.
   C. support the federal government in terms of human resource.
   D. fight against factors contributing to previously advantaged regions.
34. Which of the following indicates the negative impact of terrorism on the effort to build a democratic society?
   A. It disrupts peace & security.
   B. It promotes rule of law.
   C. It promotes political stability.
   D. It promotes social stability.
35. Which of the following is the reason why the judiciary is set up as an independent organ of the Ethiopian state?
   A. Administer justice in consultation with the executive and legislative organs.
   B. Administer justice without the interference of the executive and legislative organs.
   C. Administer justice independent from the courts and judges.
   D. Exercise its absolute power independent of any organ of government.
36. Which of the following is **not** a patriotic action?
   A. Maintaining domestic peace.
   B. Defending national interests.
   C. Disseminating the state's secrets.
   D. Standing for the truth.
37. The present Ethiopian Agricultural Development Led Industrialization (ADLI) economic strategy primarily aims at developing:
   A. nomadic farming system
   B. peasant farming system
   C. large-scale production
   D. small-scale industries
38. Which of the following actions should be taken to prevent our historical heritages from destruction?
   A. Allowing the tourists to visit our historical heritages.
   B. Proper protection, registration and documentation of our historical heritages.
   C. Irresponsible utilization of our historical heritages.
   D. Improper registration and documentation of our historical heritages.
39. Which of the following belongs to the national responsibilities of a democratic government?
   A. Protecting international cultural and historical heritages.
   B. Protecting the nation's cultural and historical heritages.
   C. Maintaining international peace and security.
   D. Protecting international natural and human resources.
40. Which of the following explains women's disadvantaged position in work places?
   A. Female derogative stereotypes.
   B. Male derogative stereotypes.
   C. Gender neutral socialization.
   D. Gender sensitive work places.
41. Which of the following indicates the relationship between work and time for improving productivity?
   A. Efficient use of time in work places improves productivity
   B. The productivity of work is the result of an inefficient use of time in work places.
   C. Time use has nothing to do with the productivity of work.
   D. Time budgeting hinders the productivity of work
42. Self-respect **does not** prevent a person from appreciating the good work of others because he/she knows that:

Telegram: https://t.me/ExtremeSeries

A. Individuals can acquire different skills and specializations.
B. Individuals can have similar skills and specializations.
C. Every one can exclusively rely on his/her own skills.
D. No one can rely on his/her knowledge and skills.

43. Ethiopia's continued dependency on foreign aids can be accounted for:
A. well-conceived and well-designed donations and interventions.
B. ill – conceived and ill-designed donations and interventions.
C. lack or shortage of natural resources or material goods.
D. absence for shortage of human resources or skilled human power.

44. Which of the following indicates the impact of extravagant practices on the national economy?
A. Reducing the capacity for investment.
B. Building the capacity for investment.
C. Dependency on local products.
D. Reducing foreign aids and loans.

45. A person is **not** expected to pay tax only because he/she,
A. has income from certain legally specified source except pension.
B. owns a certain amount of property that brings profit.
C. earns an income by engaging in a certain economic activity.
D. is a citizen of a certain national state.

46. The major obstacle of the implementation of the various protocols to reduce global warming is:
A. China's increased level of carbon emission in violation of the protocols.
B. lack of willingness to sign these protocols among European Union.
C. refusal of the United States of America to agree to the protocols.
D. India's refusal to sign the Protocols and fail to respect their articles.

47. The cultural and historical heritages of Ethiopia have invaluable importance for the peoples of the country in a sense that they:
A. make them proud of their socio-cultural place in the history of humankind.
B. enable them to earn money from tourists that come to visit them.
C. provide them with recreation places to pass their leisure time.
D. enable them earn their living by being employed as their protectors.

48. The HIV/AIDS pandemic is different from other types of pandemics like malaria because it,
A. Mostly affects peoples of Sub-Saharan Africa.
B. Mainly attacks peoples of countries found in the tropics.
C. Poses threat to the development of the entire human race.
D. Is the deadliest among Vector – born diseases so far known.

49. Disciplinary measure may **not** be taken on a worker who fails to:
A. Properly handle instruments of labor.
B. Come to work on time regularly.
C. Establish good relationships with other workers.
D. Meet work deadlines repeatedly.

50. "Leisure refreshes the mental and physical capacity of workers. As a result it increases their efficiency and productivity." If you agree to this statement, then it is possible to conclude that:
A. It is in the best interest of employers to give sufficient leisure time to their workers.
B. Sufficient leisure time is required only for people who are employed in productive industries.
C. Workers who are proved to be efficient alone should be given sufficient leisure time.
D. Employers should arrange sufficient leisure time only for workers who are productive.

51. One good thing that could be said about the League of Nations is that it was:
A. The first attempt of the international community to cooperate for peace.
B. Capable of solving political disputes between member countries.
C. able to serve as voice of the oppressed and voiceless peoples of Africa.
D. used as a common front of European struggle against the United States.

52. Nationalism is different from patriotism in that by its very nature it discourages,
A. honoring one's own national symbols.
B. co-existence with other nations or groups.
C. cultural superiority of a group over others.
D. resentment and rivalry between groups.

53. One worker may get satisfaction from a work that pays well irrespective of the position he/she attained, while another worker may get satisfaction from a job that offers opportunities for growth in position irrespective of the pay. This shows that:
A. workers' do not like a job that does not offer opportunity for both growth and good pay.
B. it is impossible to create a job that pays well and offers opportunity for growth to satisfy workers.
C. workers' job satisfaction is based on their attitude and the value they give to the job.
D. it is necessary to create jobs that satisfy both workers' need for higher payment and growth in position.

54. How do ethical rules of a certain profession enable workers to minimize misunderstanding between them? By enabling them to:
A. differentiate the right and wrong way of doing things by themselves
B. realize the similarity and difference between various professions
C. benefit from the guidance of the more experienced professions.
D. evaluate their work in the light of the performance of other workers.

Telegram: https://t.me/ExtremeSeries

55. Any type of assistance should be executed properly to produce the desired result. Otherwise, it may back fire and may end up in:
   A. enriching the beneficiary communities.
   B. reinforcing the dependency of the beneficiary.
   C. making the beneficiaries loathe the donors.
   D. closing the gap between beneficiary and donor.
56. Citizens are advised to buy domestically produced goods as much as possible. This is because these goods:
   A. enable the country to save foreign currency and accumulate capital.
   B. are produced by qualitatively superior materials to imported ones.
   C. enable the country to stop importing goods from other countries.
   D. should be consumed locally as they cannot be exported anywhere.
57. Which of the following international organization is one of the Britton Woods institutions?
   A. The United Nations Organization
   B. The World Trade Organization
   C. The International Monetary Fund.
   D. The International Court of Justice.
58. The Agriculture Development Led Industrialization policy of Ethiopia emphasizes the need to:
   A. make industry the backbone of agricultural development of the country.
   B. disregard other sectors of the economy until agriculture fully developed.
   C. encourage private investors to participate in the agricultural development.
   D. urgently replace the small farmer holdings by large mechanized farms.
59. Which of the following action is a proper understanding of saving:
   A. Accumulating wealth for the sake of having it.
   B. Refraining from spending money on basic necessities.
   C. Using natural resources in a planned manner.
   D. Limiting consumption to the level one earns.
60. Microfinance institutions are different from the conventional banks in that they:
   A. safeguard money for their clients and use it for loans and investment.
   B. extend credit, in cash or kind, to the economically active poor people.
   C. are engaged in giving financial service to rural people alone.
   D. are insured against the loss of money in a robbery or other accidents.
61. Banks get the money they lend to various investors from:
   A. personal savings of individuals and proceeds from their own investments.
   B. other profitable banks that are willing and capable to lend them.
   C. the National Bank of the country that lends money for such purposes.
   D. the International Monetary Fund and the World Bank.

62. The most common mode of political participation in which all legal citizens can engage in and that requires nothing more than willingness on the part of the participant is:
   A. running for political office.
   B. engage in political campaigning.
   C. organizing interest groups.
   D. voting in elections.
63. Money is considered as one of the important factors that could increase the chance of a party to win the election. This is because it would enable the party to:
   A. promise financial rewards to people who vote for it.
   B. disseminate its policy and agenda among the electorate easily.
   C. convince the people it will not bother them with huge taxation.
   D. win the confidence of the electorate in the incorruptness of its candidates.
64. It is believed that people would make effective political participation if they work in a group. In this context group can best be defined as two or more people who,
   A. interact to achieve a defined objective.
   B. live in a defined geographical area.
   C. have relatively equal economic status.
   D. speak one common language.
65. Assume that you do not like the students nominated for the position of president of a club to which you are a member and refused to vote for any one of them. Anyways, a president is elected by the members of the club. What should be the correct thing for you to do in such a situation?
   A. Boycott activities that are organized by the elected president.
   B. Patiently work to discredit the president in the eyes of others.
   C. Respect the decision of the majority and accept her/his leadership
   D. Stop being a member of the club until another president is elected.
66. An effective leader gets others to perform a job willingly if, above all other traits, he/she has
   A. aggressiveness.
   B. persuasiveness
   C. decisiveness.
   D. inventiveness.
67. Modern psychologists termed human needs for food, shelter, security, etc. as:
   A. actualization needs.
   B. transformational needs.
   C. societal needs.
   D. maintenance needs.
68. Stereotyping a group of people as having a certain good or bad trait in common is usually wrong because it:
   A. disregards differences in the individuals that make up the people.
   B. is related only to bad traits that people have in common.
   C. is difficult to define what people mean at a certain point in time.
   D. disregards the socio-economic differences between people.

Telegram: https://t.me/Extremeseries

69. Identify an **invalid** deductive conclusion that is based on false premises.
   A. All living things need air to live. Rose is a living thing. Therefore, rose needs air to live.
   B. All Ethiopians are good runners. Mohammed is an Ethiopia. So, Mohammed is a good runner.
   C. All children have the right to education. Alemitu is a child. Therefore, Alemitu has the right to education.
   D. All languages have speakers. Geez is a language. Therefore, Geez has speakers.
70. Individual and group rights overlap in the area of:
   A. protection of the right to equality.
   B. realization of the right to collective development.
   C. protection of the right to privacy.
   D. realization of the right to personal development.
71. Which of the following is the advantage of a federal state structure for Ethiopia?
   A. Providing absolute power to regional governments.
   B. Providing absolute power to the central government.
   C. Limiting the authority of regional government.
   D. Securing peoples' right to self-governance.
72. Which of the following is **not** true about the importance of providing equal opportunities for all peoples of Ethiopia?
   A. It leads to conflict of interests leading to violence.
   B. It fosters a variety of viewpoints, new ideas and ways of looking at and solving problems.
   C. It helps people appreciate cultures, traditions and practices other than their own.
   D. It develops respect for diversities which is necessary for peaceful coexistence and peoples' progress.
73. One of the ways by which Ethiopia protects her national interest is:
   A. enlarging her territory or sphere of influence.
   B. reducing a foreign threat or danger.
   C. striving to dominate weaker states.
   D. utilizing her neighbors' natural resources.
74. Which of the following is a valid reason about the necessity of fighting corruption?
   A. Creates equality among citizens.
   B. Retards economic and social development.
   C. Promotes good governance.
   D. Discourages the prevalence of the rule of men.
75. Which of the following does **not** justify the need for limiting the power of the Federal Government of Ethiopia?
   A. It secures people's right to self-governance.
   B. It provides people with the opportunity to work together for their common good.
   C. It limits the authority of the central government.
   D. It provides absolute power to regional governments.

76. Which of the following is a valid statement about the personal benefits of standing for the common good of society?
   A. The common good of society is the basis for personal well-being.
   B. The common good of society has nothing to do with personal benefits.
   C. Individual members of the community can maintain their personal benefits apart from the common good.
   D. The common good is the sum total of personal benefits.
77. Which of the following practices contribute to environmental degradation?
   A. Irresponsible use of natural resources.
   B. Proper utilization of natural resources.
   C. Protection of natural resources.
   D. Planting trees.
78. The international community gives a very great concern to HIV/AIDS pandemic because it is a:
   A. regional issue affecting African people.
   B. third World problem affecting people in developed countries.
   C. first World problem affecting people in developed countries.
   D. global issue affecting the human race as a whole.
79. Which of the following is an indicator of people with lack of self-confidence?
   A. Absence of fair assessment of one's potentials and limitations.
   B. Ability to accept both compliments and criticisms.
   C. Having personal visions and ambitions.
   D. Value their qualities and give honor to themselves.
80. Which of the following distinguishes political organizations from civic societies?
   A. working to promote the common good.
   B. Organizing people to discuss on community problems.
   C. Monitoring and influencing public policy.
   D. Organizing election campaigns.
81. Which of the following belongs to the roles played by civic societies in the current political system of Ethiopia?
   A. Formulating public policy.
   B. Enacting public policy.
   C. Forming public agenda.
   D. Changing public policy.
82. Participating in a civic society enables you to:
   A. protect your rights and interests.
   B. improve your academic performance.
   C. win political campaigns.
   D. avoid your responsibilities.
83. Which of the following belongs to the process of gathering relevant information for acquiring knowledge?

Telegram: https://t.me/Extreme_series

A. Communicating, interpreting and evaluating information.
B. Questioning, paraphrasing, allowing comments and suggestions.
C. Observing, interviewing and recording information.
D. Recording, Transcribing and Analyzing Information.

84. The Constitution of the F.D.R.E. gives the Federal Government the right to intervene in the internal affairs of any member state of the federation when:
A. the member state fails to promulgate its own constitution.
B. Violation of the human rights of citizens is reported in that state.
C. a law that is suited to the peculiar condition of the state is enacted.
D. that particular state fails to collect enough tax to cover its expenses.

85. On what ground do International Human Rights Organizations persistently campaign for the abolition of capital punishment?
A. The fact that it is mainly used by totalitarian governments to punish political opponents.
B. Capital punishment targets minority groups who cannot defend themselves against the authorities successfully.
C. The right to life is respected by international conventions as an inalienable and inviolable right of the individual.
D. It is most barbaric form of punishment still being used in the most uncivilized nations.

86. A country is accepted as new member of the United Nations Organization only if,
A. the country is recognized as an independent state by some members.
B. the General Assembly accepts its application by two-third majority.
C. the Security Council members fully accept the application.
D. the country has won its independence with the help of the United Nations.

87. "Gender roles and expectations are learned." If you hold this statement to be true,
A. it is possible to see variations in them between and within cultures.
B. they are similar in a given culture but different between cultures.
C. they are always similar in societies that have a good education system.
D. there is no gender role and expectation in societies that are uneducated.

88. Corruption could have direct adverse effect on a country's economic development mainly because it,.
A. reduces domestic investment by increasing the cost of doing business.
B. erodes public trust in the various branches of government.
C. weakens a government's ability to invest in vital social services.
D. jeopardizes citizens' opportunity to get fair and just treatment.

89. Article 20/2 of the Ethiopia Constitution states: "Accused persons have the right to be informed with sufficient particulars of the charge brought against them and to be given the charge against them". This provision is

important because it,
A. helps the accused to remember where and when the incident happened.
B. proves that the accused is guilty of doing the crime stated in the charge.
C. shows that the prosecutor has sufficient legal ground to charge the accused.
D. enables the accused to gather evidence and prepare his/her defense.

90. In many countries high level government officials are required to register their income and property before they assume office. This practice is believed to contribute towards reducing corruption because:
A. it would be relatively easy to identify officials who are living beyond their means and take appropriate measures in time..
B. it would deter them from becoming illegally rich as they realize that their excess wealth would be easily known.
C. people who are willing to make their wealth public are usually honest and would not engage in corrupt practices.
D. such people know that the government would confiscate the registered property if they are found to be corrupt.

91. An act is considered a crime only if it is:
A. committed by a person who has knowledge of the law.
B. causes great humanitarian suffering or material damage.
C. considered to be illegal by the leaders of government.
D. declared as prohibited by an international or national law.

92. Some organizations that claim to be fighting for the freedom of their people are categorized by the United States of America and its allies as terrorist organization mainly because they:
A. were involved in the September 11 attack against that country.
B. are willing to use anything to make their target society live in fear.
C. usually target countries that are strategic allies of the United States.
D. Are avowed enemies of the United States and its allies.

93. It is seen that the financial crisis that began in the United States of America towards the second half of 2008 has triggered economic crush in many countries. This situation **could not** have happened if the:
A. ability of the United States to provide aid to other countries has not been hampered by the effects of the economic crisis.
B. economies of the countries of the world had been independent of each other and been self sufficient by themselves.
C. other economies had been able to extricate themselves from the domination of the United States economy.
D. world Bank and the International Monetary Fund had acted swiftly to help the United States financial sector.

94. Some people see the growth of the influence of Western culture in developing countries as harmful because they believe that

Telegram: https://t.me/Extremeseries

9. To serve the public fairly and equally government officials should necessarily be
A. non-partisan. C. party members.
B. respectable. D. elderly people.
10. The necessary precondition that people who want to form an association should have is common
A. objective or goal. C. ethnic background.
B. educational standard. D. religious persuasion.
11. Among the various conflict resolution methods which is the most formal and binding one?
A. Arbitration B. Litigation C. Mediation D. Negotiation
12. Ethiopia is/was home to the Head Quarters of all of the following organizations EXCEPT?
A. The United Nations Economic Commission for Africa (UN-ECA).
B. EGAD.
C. AU Parliament.
D. African Union (AU)
13. One of the following is among the widely known extravagant practices in Ethiopia
A. Costly marriage ceremonies. C. Travelling to visit places.
B. Extending alms to beggars. D. Partying.
14. The habit of spending more money than one can afford is referred to as _____.
A. Generosity B. Extravagance C. Charity D. Planning
15. Identify the correct assumption regarding the ability of HIV positive persons to engage in production activities
A. HIV positive persons are lazy to work.
B. HIV positive persons are physically weak and powerless; thus, unproductive.
C. HIV positive persons should not be allowed to go to the work place as hey will add nothing to productivity.
D. HIV positive persons can still innovate despite their health problems.
16. The current constitution of FDRE was promulgated in
A. 1985. B. 1955 C. 1995 D. 1991.
17. Which specialized organ of the government is concerned with interpreting the laws of the state?
A. The Legislative organ. C. The executive organ.
B. The judiciary. D. The parliament.
18. Among the following, which one is a minimum standard to make a trial fair? The accused:
A. has no right of appearing in the court room.
B. is encouraged to admit that he/she is guilty before proved.
C. has no right to be presumed innocent before proven guilty.

D. has the right to use his/her language and even to examine witnesses against him/her.
19. An idea is labeled as 'scientific' when it is:
A. divinely B. legendary C. experimented D. imaginary
20. Which of the following is NOT a traditional method of saving in Ethiopia?
A. Iqub. C. Savings in tangible assets.
B. Idir. D. Banking
21. Which of the following powers is one that is expressly given to the regional states by article 52 of the F.D.R.E. Constitution? Power to:
A. enact laws regulating the possession and bearing of arms.
B. patent inventions and protect copy rights.
C. enact and execute state constitution.
D. determine matters relating to nationality.
22. In a democratic system protection for the rights of minorities are guaranteed by
A. the goodwill of the majority group.
B. the laws and institutions of the country.
C. their own active political participation.
D. the party that forms the government.
23. The positive measure that governments take to guarantee equal chance of opportunities for citizens that have been deprived of opportunities is called.
A. law of equality. C. democratic action.
B. affirmative action. D. social equality.
24. One mark of the 1955 revised constitution of Ethiopia is that is
A. established a bicameral parliament for the first time.
B. was drafted, debated and approved by representatives of the nobility.
C. was presented as a grant given by the emperor to "his subjects"
D. reflected the true desire of the people of the country.
25. Article 89/2 provides that "Government has the duty to ensure that all Ethiopians get equal opportunity to improve their economic conditions and to promote equitable distribution of wealth among them". This provision obliges the government to
A. make all citizen have equal means to participate in development
B. enable all citizen have equal right to make use of their resources
C. fully control the economy so that citizens benefit from it equally.
D. distribute the wealth of the nation among citizens on equal basis.
26. Many parents of disabled children used to hide their children rather than sending them to school primarily because they
A. fear that the children would be exposed to various road accident.
B. fear the societal stigma caused by ignorance about disability.
C. believe that the children would be gainfully employed at home
D. think that their children could not do as well as able bodied children.

Telegram: https://t.me/ExtremeSeries

27. Government policies that promote right to equality among citizens may infringe on the liberty of some individuals. Nevertheless, such policies are introduced because governments are first and foremost desirous of
A. doing away with individual liberties until equality is achieved.
B. delaying the enjoyment of the right to liberty until citizens are equal.
C. dealing with the less complicated issue first than the issue of liberty
D. realizing the common good of society rather than individuals.

28. The Ethiopian penal law, civil law and property law are primarily designed to
A. make the people obey the desires of a government.
B. keep societal peace and do away with anarchy.
C. create awareness among the public about rule of law.
D. establishing a law abiding and honest citizens.

29. During the initial years of the Dergue regime binding decisions on the life, liberty and property of Ethiopian was mostly taken by
A. various level of courts.
B. Workers' Party of Ethiopia
C. committees at various levels.
D. Chairman of the Dergue

30. An agreement of treaty that is signed by more than two countries is called
A. bilateral treaty.
B. unilateral treaty.
C. multilateral treaty.
D. regional treaty.

31. Globalization has resulted in greater
A. prosperity of all countries of the world.
B. interdependence among nations of the world.
C. flow of aid and loans to the poor countries.
D. unrest and civil wars in many parts of the world.

32. According to article 20/1 of F.D.R.E. Constitution, a court may hear a case in closed session. Which of the following is NOT sufficient ground for hearing a case in closed session?
A. Protecting the privacy of the parties concerned.
B. When the case is considered to be harmful to public moral.
C. When the case is believed to jeopardize national security.
D. If and when the presiding judge is afraid of public scrutiny.

33. One way by which the Ethiopian Federal Ethics and Anti-Corruption Commission is trying to combat corruption is by
A. raising the awareness of the citizens about the evilness of corruption.
B. imprisoning those officials caught red-handed in corrupt activities.
C. organizing and training anti-corruption police force at federal level.
D. raising the salary of officials who are susceptible to corruption.

34. According to article 521 of the Ethiopian Penal Code, a person who deliberately transmits deadly disease to a healthy person could be
A. accused of committing homicide.
B. asked to cover the medical expense of the victim.
C. handed lenient imprisonment of not more than a year.
D. reprimanded for being negligent.

35. At present countries of the world collaborate to fight terrorism because they consider it as an act that is
A. directed against oppressive governments.
B. a thereat to global peace and security.
C. carried out by some religious fundamentalists.
D. a source of social unrest in the Middle East.

36. When the House of Peoples' Representatives of the F.D.R.E. ratifies an international agreement signed by the executive, that agreement becomes
A. organized and written into the Constitution.
B. available for all to read and comment.
C. an integral part of the law of the land.
D. ready for domestication and use in the land.

37. Our natural environment would be protected effectively only if:
A. there are numerous environment protection clubs in the country
B. the government introduces proper environment protection laws
C. everybody strives for its protection actively and willingly.
D. there is a goodwill among American leaders to protect it.

38. Which one of the following factors was considered for demarcating the boundary of the regional states of FDRE?
A. Religious similarity
B. Similarity of their economic activities.
C. Cultural similarity.
D. Agro-ecology similarity.

39. Which one of the following statements is true regarding the historical and cultural heritages of Ethiopia? They are
A. found in Tigray and Amhara regions only.
B. recognized as the oldest in the world.
C. recorded as world heritage by UNESCO.
D. found scattered throughout the country.

40. According to the Pragmatists' theory truth
A. is what works at the time.
B. cannot be properly defined.
C. is one and the same always.
D. is accepted by all everywhere

41. Women's Associations are established with the main aim of struggling for:
A. achieving all rounded equality of women with men.
B. attaining social, political and economic equality of the sexes.
C. satisfying the economic and political ambitions of their leaders
D. recreating the matriarchal age where women would be dominant

42. Which of the following actions was one of the negative legacies of the French Revolution?
A. Establishment of a federal system of government.
B. Separation of the power of state and religion.
C. use of terror as means of political repression and social control.
D. State sponsorship for brilliant students who are from poor families.

Telegram: https://t.me/Extreneseries

A. it could cause the target people to repudiate their local identities.
B. peoples of the developing nations are opposed to this cultural invasion.
C. western culture is mostly offending to the sensibilities of other peoples.
D. western culture has no good quality to be learned from by other people.

95. If judges try to please their superiors and fail to be guided by the law while making decisions they would undoubtedly damage the:
A. good name of all other people who are serving in judicial system.
B. integrity of all the officials in the judicial system of the country.
C. high reputation of courts as stable institutions which stand for justice.
D. popular belief that judges are the only officials who are incorruptible.

96. Public officials are expected to have high standard of conduct because all of them are expected to:
A. make no mistake what so ever in performing their duties.
B. provide service to all citizens in an equal and just manner.
C. be morally and intellectually more superior to the people.
D. gain the respect and admiration of the people they lead.

97. The term civil servant refers to:
A. people of a country that earn their living by being employed.
B. all the peoples of a country outside children and pensioners.
C. all employees of a government in the non-military sector.
D. all employees outside those employed by the private sector.

98. An action that is moral for one person may be found immoral for another. This is because:
A. it is impossible to find a common definition for what is moral and immoral in any society.
B. individuals who have different backgrounds may perceive things in different way.
C. Individuals would always tend to disagree on such matters as far as they are benefited from it.
D. laws fail to clearly differentiate between what is moral and what is immoral in any given society.

99. Which of the following is a common cause of the peasant rebellions that took place during the reign of Haile Selassie?
A. Forced incorporation. C. Inefficient administration.
B. Unfair taxation system D. Absence of social services.

100. Which of the following is the consequence of the absence of organized political leadership for a popular struggle?
A. Prevalence of peace and security.
B. Social and economic development.
C. Political stability and social harmony.
D. Political instability and social disorder.

Telegram.https://t.me/ExtremeSeries

Booklet Code: 52

Direction: Each of the following questions is followed by four possible answers. Read each question carefully and **blacken** the letter of your choice at the separate answer sheet provided.

1. Which organ of state has legislative power at state level in FDRE?
A. State Bureau of Legal Affairs. C. State Supreme Court
B. State president. D. State Council.

2. When was the Universal Declaration of Human Rights adopted by the United Nations Organization?
A. 1945 B. 1948 C. 1966 D. 1995

3. All member states of the United Nations Organization have agreed to accept all decisions passed by the
A. General Assembly. C. General Secretary.
B. International Court of Justice. D. Security Council.

4. The Legislative branch of the F.D.R.E. is accountable to the
A. House of Federation. C. Prime minister.
B. peoples of the country. D. President of the Federation

5. The executive members of your kebele pass decisions on matters that affect the residents of the kebele. The residents are subjected to those decisions because the Kebele executive are
A. appointed government officials.
B. Careful to pass decisions that pleases all residents.
C. Capable of punishing those who disagree with them.
D. Elected representatives given power of decision by the residents.

6. The national emblem of the Ethiopian flag reflects the hope of the peoples of the country to
A. build a wealthy country
B. advance in science and technology.
C. live together in equality and unity.
D. live in peace and friendship with their neighbors.

7. Identify the statement that is true of the Ethiopian constitution.
A. It is relative to norms and values.
B. It is absolute and untouchable.
C. It transcends all the other laws that are culture specific
D. It may be invalidated across cultures.

8. As it happens in politics or law, litigation refers to the
A. method of dispute settlement by the parties with the guidance and facilitation of a third party.
B. Discussion between people who are trying to reach an agreement.
C. Process of trying to succeed in ending a dispute between people or groups.
D. Process of making or defending a claim in a court of law

43. One of the following is NOT just enough to qualify a state as a really constitutional one, which?
A. Its having a written constitution.
B. The people enjoying full rights and freedoms as given to them by the constitution.
C. The people feeling as citizens rather than subjects.
D. The people having confidence in the law of the state and keeping trust in their leaders.

44. Among the following, which one is NOT a mark of a totalitarian government?
A. Government is by and for the people.
B. The practice of the free and fair elections.
C. The people are rather subjects, not really citizens.
D. Power of the government is limited.

45. The notion of sovereignty essentially refers to one of the following:
A. The state of being a country with freedom to govern itself.
B. The complete power of a country to interfere in the affairs of a neighboring country.
C. The right of elite persons to assume government offices.
D. The political status of citizens to participate in national elections.

46. In free-market economy the market is governed, theoretically, by the
A. amicable relationship between sellers and buyers.
B. relationship between demand and supply.
C. honesty or dishonesty of merchants in fixing prices.
D. abundance of goods and services in the market.

47. If you seriously believe that no one will stand against you in this county in solving mathematical problems, you could be said to have exaggerated view of
A. self-consciousness.
B. self-importance.
C. self-esteem.
D. self-neglectfulness.

48. A democratic state differs from undemocratic state in that it furnishes its people with
A. equal opportunity for advancement.
B. equal economic and political status.
C. laws that are derived from the constitution.
D. all things necessary for citizens' wellbeing.

49. Contempt to the rule of law by leaders of a state would entail:
A. lawlessness.
B. political advancement.
C. devotion and commitment on the part of the citizens.
D. the culture of tolerance.

50. Among the following, which one is NOT the implication of rule of law?
A. Democratic system.
B. The supremacy of the constitution.
C. Little human rights violations.
D. Rampant corruption

Telegram: https://t.me/ExtremeSeries

4. From the following, which one is NOT a factor to evaluate the goal of good constitutional one, which?
A. Uniformity
B. Acceptability.
C. Partiality
D. Relative stability

5. Separation of power among the agents of the government implies that
A. the agents do not cooperate each other
B. each agent has specialized responsibilities to work with a degree of autonomy.
C. each agent is absolutely independent and there is no area where they may work together
D. specializing on the making of laws, the legislative agent can change the laws without consulting the other agents of the government

6. How is transparency important for building a democratic system is
A. promotes privacy.
B. brings open and accountable bureaucracy
C. eliminates the spirit and determination to ensure justice and respect of human rights.
D. blocks responsibility

7. Which one of the following is NOT a concurrent function and responsibility between the two levels of states in Ethiopia?
A. Levying taxes and collecting revenues.
B. Land and natural resource management.
C. Protecting the federal constitution.
D. Foreign policy execution

8. One of the advantages of having a written constitution is:
A. the existence of a single and accessible document stating the rights of citizens.
B. the uncertainties of political life could prevail.
C. lack of awareness to one's constitutional rights and duties.
D. its changeability without the consent of the people.

9. Which one of the following is true about Terrorism? It:
A. is justifiable unless it targets civilians.
B. enhances the effort to build a democratic society.
C. fosters religious bond and brotherhood.
D. shatters peace and stability.

10. Which one of the following is taken into account by the federal government for provision of social services?
A. Geographical proximity to the center.
B. Racial composition.
C. Low level of development.
D. Territorial extent.

11. Among the following regions of Ethiopia, which one is benefiting from package of affirmative actions?
A. The state of Amhara.
B. The state of Tigray.

C. The state of the peoples of Gambella.
D. The State of Oromia.

59. Identify the correct statement abut the HIV pandemic
A. HIV transmits with physical contact.
B. HIV discriminates between people.
C. HIV knows no racial, linguistic, or religious bounds.
D. HIV transmits with breath.

60. Which one of the following best expresses the equality between and among peoples of Ethiopia?
A. Having approximately equal size of land mass.
B. Having approximately equal size of population.
C. The right to have been allocated equal budget.
D. The equal right of states to full measure of self governance.

61. One of the following shows why people with good moral characters find it easy to appreciate the good work of others. Which?
A. Their feeling that they are better than others.
B. Their little regard for others.
C. Their open mindedness.
D. Their close mindedness.

62. One of the following is NOT among the national responsibilities of a democratic government
A. Adopting a parliamentary system
B. Protecting citizens' wellbeing and property.
C. Provisions of services to citizens.
D. Maintenance of law and order.

63. What would you do as a responsible citizen to prevent your country's historical heritage from destruction?
A. Show no concern about it for it is up to the government to protect the country's heritage.
B. Discuss with others to try to find out ways of preserving it.
C. Selling the heritage to foreign tourists.
D. Taking them for personal uses.

64. Which of the following is a bad practice contributing to environmental degradation?
A. Reforestation.
B. Joining environmental campaigns.
C. Dumping residuals in public areas.
D. Keeping the streets tidy.

65. Which of the following is a way by which a citizen's patriotic feelings can be expressed?
A. Exposing bribery.
B. Respecting minority rights.
C. Collaborating to bring unconstitutional change of government.
D. Paying taxes serving as foreign agent.

66. Violation of the rights of citizens by the state could effectively undermine citizens' patriotic feelings as it
A. promotes the interests of the minority.
B. diminishes the citizens' sense of belongingness to the state.
C. discriminates against non-citizens.
D. fosters the feeling of pride and nationalism.

67. Which of the following was NOT completely true about Ethiopian women?
A. Their representation in politics was very insignificant.
B. Rural women were made to marry at an early age.
C. They were required to carry enormous workload at home.
D. They were deprived of the right to attend higher education.

68. Which of the following acts contradicts the principle that "people in similar situation should be treated similarly"?
A. Establishing gender limitation for exercising voting right.
B. Establishing age limitation foe exercising voting right.
C. Prohibiting people convicted of serious crimes from voting in national elections.
D. Denying some of the basic rights to people imprisoned for serious crimes.

69. Science as a body of knowledge contributes to the development of human society through all of the following EXCEPT?
A. Challenging the fundamental and accepted beliefs of societies
B. Disseminating relevant information
C. Predicting the future
D. Offering solutions for various human problems

70. Unlike political parties, civic societies:
A. can provide alternative political directions.
B. have political agendas.
C. are meant to provide social, economic, and cultural opportunities for individuals or groups.
D. are unlimited by law

71. Civic activities that a student could participate at school level include all of the following EXCEPT?
A. Plotting anti-government activities.
B. Joining environmental campaigns.
C. Joining AIDS campaigns.
D. Joining fund raising activities for charity.

72. How are personal savings important for the growth of investment? Personal savings:
A. are little to contribute for government investments.
B. always end up spent on household consumptions.
C. reduce extravagance and enhance accumulation of capital for investment.
D. undermine the habit of generosity.

Telegram: https://t.me/ExtremeSeries

73. For executing their public functions all the nations, nationalities and peoples of Ethiopia can use the language that
A. is common for all of them.
B. is their mother tongue alone.
C. is the Federal working language.
D. they prefer.

74. Statistics shows the USA emits 20 tons of carbon per person and China emits 6 tons of carbon per person. This information enables us to identify the country that makes the largest carbon emission provided that the
A. population size of both countries is equal.
B. initial year taken for comparison is varied.
C. population size of both countries is known.
D. particular type of gas they release is known.

75. In theory, a global citizen is a person who is ready to
A. claim citizenship and pay taxes to all countries of the world.
B. pay allegiance to the United Nation Organization.
C. refuse to give recognition to national boundaries.
D. take action to make the world a more just and equitable place.

76. Which of the following is one of the principles that is a basis for the present foreign policy of Ethiopia?
A. Promoting national interest at the expense of neighboring countries.
B. Seeking and supporting peaceful solutions to international disputes
C. Supporting the stands of the great powers at international level
D. Forging political and economic union with neighboring nations

77. Adam Smith states that the rich should pay more tax than the poor.
A. they would not have become rich without government protection
B. governments usually help the rich to get advantage of the poor.
C. the poor will not have enough resource to pay as much as the rich
D. government itself is seized by the rich people not by the poor.

78. One of the important activities that the FDRE government is undertaking the develop the pastoralist citizens and done without undermining their life style and pastoralist economy is
A. establishing mobile schools for their children.
B. encouraging them led a settled life in their regions.
C. making them start farming crops instead of raising cattle.
D. setting them on fertile lands where there is enough water and grass.

79. Stereotyping based on sex hurts the female members of society by
A. assigning them roles that are traditionally assigned to men.
B. forcing them to do the easiest and less rewarding types of jobs.
C. limiting their choice of career that best suits their ability or interest.
D. making them perform duties that are despised and hated by society.

80. In inter-personal relationships self-confident individuals are always willing to
A. avoid challenging situations or arguments.

B. give in to arguments of others in order to gain their good will.
C. restrain themselves from telling the truth.
D. accept powerful and logical arguments of their opponents.

81. Ethnocentrism is dangerous for harmonious living of peoples because it
A. considers ones own group inferior compared to other groups.
B. stresses on gaining advantage for ones group at the expense others.
C. accepts that ones group has comparable position with others.
D. refutes that ones group is superior in everything to others.

82. One reason why the Agricultural Development Led Industrialization strategy is chosen over other strategies is because
A. Agriculture is a sector that could be easily developed.
B. Agriculture is the occupation of the majority of the country's people.
C. Industrialization is attained only after agriculture fully developed.
D. Agriculture is a sector that requires no investment and knowledge.

83. In Ethiopia, investment in social services is mainly the responsibility of the government because
A. the government do not want private investors' participation.
B. it enables the government to get political support from the people
C. benefits from social services is regarded as rights of citizens.
D. the peoples prefer to get these services from the government.

84. Procedural justice refers to the ability of a legal system to:
A. gather information and make decisions impartially and openly.
B. equally treat people according to their need, capacity and worth
C. properly and equivalently compensate wrongs and injuries.
D. make someone shoulder responsibility equivalent to benefits gained.

85. Why is it imperative to limit the power of the federal government of Ethiopia can be explained in terms of all EXCEPT?
A. Avoiding a centralized system of administration marked by cumbersome work process.
B. Making corruption impossible to take place.
C. Empowering peoples to set goals and express their creative capacities.
D. Regional states enjoying a degree of autonomy for self determination.

86. What would the peoples of Ethiopia especially miss out if the country currently had a unitary state structure?
A. Unity among the people
B. The right to self governance.
C. Centralization.
D. Citizenship

87. Presidential democracy is a state in which the
A. head of the state usually has little or no executive power.
B. executive is led by the head of the government.
C. executive is led by the head of the state.
D. parliament has highest authority.

88. Which one of the following clearly states the difference between individual rights and group rights?
A. While group rights are political in nature, individual rights are not.

Telegram: https://t.me/exitextremeseries

B. The distinction between the two, is so blurred that it is difficult to understand.
C. While group rights concern promoting common interests of a group, individual rights concern in diminishing them.
D. Individual rights aim at promoting individual human development while group rights aim at promoting common interests of groups.

89. One of the following is NOT a mark of democratic rights as distinct from human rights, which? Democratic rights are:
A. absolute and irrevocable rights.
B. political in type and revocable.
C. relative and can be challenged.
D. artificial and only democratic states do acknowledge them.

90. Which of the following is one of the essential characteristics of qualitative data? It
A. can be assessed and determined exactly.
B. describes facts that can not be expressed numerically.
C. often analyzed using statistics or computer programs.
D. involves the measurement of an amount or quantity.

91. One of the following is NOT a true character of a hard working person. Which?
A. Determination to act.
B. Dependency.
C. He/she is conformist.
D. Too loyal and obedient.

92. Which one of the following NOT among the basic reasons for giving emphasis to speed up technical and vocational education in Ethiopia? To
A. reduce brain drain.
B. promote industriousness and self-run production process.
C. reduce rural-urban migration.
D. increases dependency.

93. Among the following, a factor that could enhance productivity of individuals in Ethiopia is:
A. affirmative action.
B. rapid population growth
C. cumbersome bureaucracy.
D. having a federal structure.

94. Which one of the following is NOT true about time management and its contribution to productivity? Time management
A. enables us to plan our daily activities; thus, achieving the most productive use of time.
B. increases efficiency and productivity.
C. is unproductive by itself; it is only how we conduct our work that really matters.
D. reduces cost productivity

95. The reason for giving great concern and publicity to the HIV pandemic by the international community is because:
A. in most countries the pandemic affects elite people.
B. the pandemic mainly affects women.

C. the pandemic mainly
D. as an incurable disease
whole of human races.

96. Among the following, which has to do with the international responsibility of any government?
A. Adopting a fee market economic system.
B. Respecting the sovereignty of other states.
C. Adopting a multi-party system.
D. Joining regional or international organizations.

97. True about the importance of gathering information for the acquisition of knowledge among the following is? Gathering information:
A. is the only scientifically accepted means to gain knowledge.
B. creates a wider context for further understanding of the issue.
C. has little importance to make inferences.
D. is the only means of arriving at a right conclusion.

98. Which one of the following is NOT among the roles civic societies were required to play in the various political systems of Ethiopia?
A. Providing educational services.
B. Campaigning on social and cultural affairs.
C. Supporting opposition forces.
D. providing cultural opportunities.

99. In order to succeed in his/her education, a student has to:
A. refuse joining extra curricular activities.
B. spend time in idle.
C. spend much time helping his/her parents.
D. properly plan and utilize his/her time as well as resources.

100. Identify the INCORRECT proposition about extravagance practices? Extravagance practices:
A. affect savings at a national level, so do they affect the national economy.
B. are generous acts and should be encouraged.
C. are civilized practices.
D. are real only in rich countries.

Telegram: https://t.me/Extreme Series

UNIVERSITY ENTRANC EXAMINATION (UEE)
CIVIC AND ETHICAL EDUCATION SENE 2003/ JUNE 2011

DIRECTONS: Each of the following questions in followed by four possible alternatives. Read question carefully and blacken the letter of your choice on the separate answer sheet provided.

1. Which Federal body has the power to deal with issues that regard interpretation of the Federal Constitution of Ethiopia?
   A. The House of Federation
   B. The House of people's Representatives
   C. The Federal Judiciary
   D. The prime Minister of the Federation
2. Which of the following could NOT be considered as group right?
   A. The right to self-governance
   B. The right to own property
   C. The right to self determination
   D. The right to develop one's culture
3. When was the first written constitution of Ethiopia revised?
   A.1931 B. 1995 C. 1955 D1965
4. A vote taken on an important issue by all the peoples of a country is referred to as
   A. Referendum C. plebiscites
   B. Popular participation D. direct democracy
5. Which organ of the FDRE government is responsible to enforce laws?
   A. The Judiciary organ
   B. The Legislative organ
   C. The House of people's representatives
   D. The Executive organ
6. The state of being answerable for an obligation is referred to as
   A. Patriotism C. civility
   B. Responsibility D. accountability
7. Treating citizens fairly in the distribution of benefits and burdens implies
   A. Constitutionalism C. good governance
   B. Justice D. the rule of law
8. Which of the following is the most widely practiced indigenous mechanism of saving in Ethiopia?
   A. Agriculture C. Afersata
   B. Industry D. Gudifacha
9. What is the back bone of Ethiopia's economy?
   A. Agriculture C. Commerce
   B. Industry D. Service sector
10. A mechanism by which people spend their money to get profit, expand their business and contribute to the development of national economy is
    A. Planning C. investment
    B. Saving D. cost effectiveness

Telegram: https://t.me/ExtremeSeries

11. Which of the following is one of the basic principles on which the Ethiopian Constitution is founded?
    A. Sovereignty of government
    B. Accountability of government
    C. Supremacy of the Judiciary
    D. Economic equality
12. One important prerequisite for political, social and economic development of a country is
    A. Good governance C. natural resource
    B. Periodic election D. national economy
13. The best instrument by which a country executes its foreign policy is called
    A. Protocol C. Convention
    B. Agreement D. Diplomacy
14. A situation where there is non supreme power or law and where lawlessness reigns in a country is called
    A. Anarchy C. Disillusionment
    B. Revelation D. Disenchantment
15. Ethiopia has played very crucial role in the struggle for the independence of one of the following African countries. Which one?
    A. Tunisia C. Ghana
    B. Nigeria D. Zimbabwe
16. The most recent crime of genocide in Africa was committed against
    A. Hutus in Burundi C. Hutus in Rwanda
    B. Tutsis in Burundi D. Tutsis in Rwanda
17. A person could better enjoy her/ his civil and political rights if she has a minimum level of
    A. sense of humor C. Societal respect
    B. immovable property D. Education
18. Responsible citizens of a country are expected to stand against all of the following EXCEPT
    A. Genocide C. Drug trafficking
    B. Virtuousness D. Terrorism
19. The payment that a person who wants insurance service pays to the insurer is known as
    A. Collateral C. Premium
    B. Indemnity D. Warranty
20. Dr. Aklilu Lema l s known for discovering the medicine to cure
    A. Alzheimer C. Malaria
    B. Bilharzia D. Neumonia
21. What is the difference between civil actions and political participation?
    A. Civil actions are meant for political agenda
    B. Civil actions are activities related to the public or citizens outside politics
    C. Civil actions must be linked to political party activities

D. Civil actions are non-voluntary, profit seeding activities

22. Which of the following is NOT a factor for job satisfaction and effectiveness?
    A. High Interest in the job
    B. Better payment
    C. Disagreement with the boss
    D. positive interpersonal interaction
23. Which of the following is one of the cultural heritages of Ethiopia registered by UNESCO?
    A. The Awash National Park
    B. The 11 Rock- Hewn Churches of Lalibela
    C. The Nech sar National park
    D. The Monastery of Debre Libanos
24. During the Second World War Adolph Hitler's Nazi soldiers atrociously killed millions of the Jewish people. Which of the following can explain this action?
    A. Homicide
    B. Massacre
    C. Genocide
    D. Suicide
25. One of the following is NOT among the basic factors behind unequal status between men and women within the Ethiopian society
    A. Biological difference
    B. Forced early marriage
    C. Little or no access for education
    D. Unfair labor division
26. The major significance of federalism for Ethiopian Nations, Nationalities and peoples is, above all for
    A. Equitable socio- economic growth
    B. Self- administration
    C. Distribution of infrastructures
    D. Self – consciousness
27. Which of the following is true about the FDRE constitution?
    A. It promotes one party system
    B. It clearly puts that ultimate political power comes from the people.
    C. It was promulgated with out active participation of the peoples
    D. It does not need to be amended for the next 20 years.
28. The principle of constitutional law, which protects the life, liberty and property of a person is referred to as
    A. individual right
    B. habeas Corpus
    C. due process of the law
    D. justice
29. One of the following city-states is credited for its contribution of exercising direct democracy
    A. Athens
    B. Rome
    C. Jerusalem
    D. Constantinople

Telegram: https://t.me/EXtremeSeries

Which of the following is true about the current Ethiopian state structure and function?
A. Power is shared among all member states of the federal government
B. Matters of national importance are among major functions of all regional states
C. Regional states are subordinate to the center.
D. Regional states cannot levy taxes.

The Sovereignty of the peoples of Ethiopia is expressed through their
A. direct participation to elect their representatives
B. right to hold government positions on equal basis
C. duty to serve in the armed forces of the country
D. right to enjoy from welfare benefits in the country

The Ethiopian Constitution is the supreme law of the country This means,
A. regional states cannot promulgate their own Constitution
B. all practices including customary ones have to conform to the constitution
C. religious rules have to be drawn out of the articles of the Constitution
D. It is above everything else and cannot be amended

The main reason why many counties of the world are trying to build democratic systems at present is that because it
A. has become fashionable in the modern world
B. enables them to get aid from the developed world.
C. is believed to be a necessary prerequisite for development
D. is one requirement for joining the United Nations Organization

Presidential Democracy is believed to be better than parliamentary Democracy in
A. implementing effective check and balance system
B. respecting human and democratic rights of citizens
C. establishing a fast and fair justice system to citizens
D. accommodating internationally approved conventions.

A government is said to have legitimate authority to govern if it functions within the limits set by a constitution provided that the constitution is
A. prepared and approved by a constituent assembly
B. copied from constitutions of developed countries
C. a written document easily accessible to the people
D. a free expression of the will of the people

One cause of grievance of the Gojjam peasants that drove them to rebels in 1966 was
A. Demand that they contribute money for construction of statue of the emperor
B. Attempt to take Mota and Dega Damot from Gojjam and make it part of Shewa
C. Refusal to accept people appointed from areas outside Gojjam as governors

D. Imperial government attempt to impose religious doctrine of its own on them

37. It is necessary to fight corruption because it
A. enable individuals to gain personal benefits
B. is a practice that could not be mitigated by any measure
C. can be undertaken easily in any circumstances
D. is a threat to the well – being of any society.

38. In order to live together in unity, it is necessary that Ethiopians should
A. be able to speak the languages of each others
B. remove all things that are unique to each other
C. emphasize on matters that make them different
D. respect the cultures of each other

39. Which of the following statement is correct according to the principle of equity of benefits and burdens?
A. Those who work hard should benefit more
B. Those who lead should be given respect
C. Those that carry burdens are always benefited
D. Those that are benefited are those who are burdened

40. While you are enjoying your legal rights, it is mandatory that you
A. refrain from violating the rights of others
B. disregard the benefits and interests of others
C. protect your interest regardless of its effect on society
D. refrain from involvement in political activities

41. How many constitutions could there be in federal state?
A. Only one federal constitution that is the supreme law of the land
B. Two constitutions – one federal and one for the members of the federation
C. One federal plus as many as the number of members of the federation
D. One federal Constitution and nine constitutions of the regional states.

42. Social harmony exists in a country where there is
A. Educated people
B. Strong police force
C. presidential system
D. rule of law

43. What is the concept of progressive taxation system?
A. All people are required to pay similar amount of tax
B. People who earn more money are required to pay more
C. People who earn more pay less tax to encourage them to work more
D. Tax is levied at a flat rate to encourage people to produce more

44. The main benefit that a government gains by performing its duties in a transparent manner is
A. respect from international organizations
B. ability to have control over its sectors
C. trust and respect of the people it leads
D. willing participation of people in government

45. In a democratic system, power is diffused and mutual accountability is instituted in order to
A. Enable a certain group hold state power indefinitely
B. Stop parties from dominating the politics of a country
C. Help smaller parties to mutually compete and take power
D. Prevent domination of a country by an individual or party

46. Most of the harmful traditional practices that affect women in rural Ethiopia can be eliminated by
A. Making the people aware of their ill effects
B. Taking harsh measures against their perpetrators
C. Establishing a task force for their elimination
D. Moving as much women as possible to urban areas

47. In democratic societies, power of a government is believed to be legitimate because it is
A. Derived from the authority given by the people
B. A means of subduing al manifestation of dissent
C. The basis for the authority of the government to rule
D. Used to silence people who refuse to recognize it

48. Saving should be understood as
A. Hording money for the sake of having it
B. Planned and wise use of money
C. Forfeiting necessities to keep money
D. Refraining form spending money

49. A pragmatic person is one who believes that truth is
A. What is universally accepted
B. Something that is unattainable
C. Community defined
D. What works at the time

50. The petition known as Habeas Corpus is in a sense a protest against what the accused believed to be
A. Unlawful detention
B. Public trial
C. trial by a jury
D. legal review

51. If a person lacks self- confidence, he/ she would NOT he expected to
A. Be a member of a certain community
B. Give respect to other people
C. Engage in some type of work
D. Stand up to challenging situations

52. In the execution of his responsibilities as head of government, the prime Minster of FDRE is accountable to the
A. President of the country
B. House of people's Representatives
C. House of Federation
D. Council of Ministers

53. It is possible to reduce the poverty situation in Ethiopia by

Telegram: https://t.me/ExtremeSeries

A. Focusing all attention on agricultural production rather than on industry
B. Providing quality basic education to the uneducated and eradicating illiteracy
C. Working to expand export of unfinished goods in order to import food items
D. Giving financial support to the people who are living in absolute poverty.

54. At present, goods from other countries could be found at a cheaper price than those goods produced in the country. What could be the final result of such practice?
A. Enabling the country to focus on other areas of production and wise use of available resources.
B. Increased dependency on foreign countries and disintegration of the national economy
C. Prosperity to the country as it could import cheaply and use the remaining money for investment
D. Popular support to the policy that enable the people to get needed goods at a cheap price.

55. Which of the following characteristics differentiates an assertive person from an aggressive one?
A. Agreeing with convincing ideas of other people
B. Attempting to gain control over all other groups
C. Assuming one's own stand is always correct.
D. Avoiding hurting others at the expense of one's self

56. One of the shortcomings of the traditional institution Idir for which it is criticized by some people is that it
A. Gives services only to its members, not to non- members
B. Does not give interest on the money saved by the member
C. Is organized by people who have relatively equal position
D. Does not support very needy members while they are alive

57. Which of the following traits is NOT essential for civic participation?
A. Commitment to see things through
B. Dedication to a cause
C. Willingness to work with others
D. Seeding gratitude from others

58. In order to pass sound judgment on anything, it is necessary that the person should
A. Love and respect of the community
B. serve as judge in the legal system
C. Correct information on the issue
D. Have at least university level degree

59. What comparative advantage could Ethiopia offer to potential foreign investors more than anything else now?
A. Modern and up-to date communication system.

B. Highly educated industrial work force.
C. Easy access to maritime transport to facilitate export.
D. Abundant work force that could be easily trained

60. How could banks play important role in the proper use of money in a country?
A. By collecting money that is out of use in the economy
B. By lending money to people who applied for a loan
C. By lending money for most promising productive ventures
D. By issuing new money whenever the previous ones become old.

61. Which of the following personalities defined democracy as " The government of the people, for the people, by the people, by all, for all "?
A. Rousseau
B. John Lock
C. Abraham Lincoln
D. Thomas Jefferson

62. Identify an all- inclusive term from the given alternatives
A. Abuse of power
B. Corruption
C. Embezzlement
D. Nepotism

63. Which of the following is true regarding the fight to equality?
A. It is given by the good will of government
B. It is one of the core elements of democracy
C. It means enjoying similar protection before the law
D. It does not consider equal protection before the law

64. All of the following are qualities of true patriots, EXCEPT
A. Commitment for common good
B. Tolerance and the culture of peace
C. Disclosing national secrets
D. Defending national interests

65. Active community participation implies involvement in
A. The overall socio – economic affairs of a society
B. Religious matters
C. Private business activities
D. Political party activities

66. In order to be more productive worker, one should have the following qualities. EXCEPT
A. Punctuality at work place
B. Good relationship with fellow workers
C. Improper utilization of resources
D. Effective application of skills

67. Which of the following is correct about concurrent powers?
A. Powers shared between regional and federal governments
B. Powers shared between the legislative and executive organs
C. Powers which are solely left to regional states.
D. Powers which are solely left to the Fedeal government

68. A theory that assumes the existence of distinctive physical and psychological characteristics accounting for leadership effectiveness is

Telegram: https://t.me/ExtremeSeries

A. Contingency Leadership Theory
B. Behavioral Leadership Theory
C. Trait Leadership Theory
D. Autocratic system

69. The Ethiopian foreign policy issued in 2003 directs an "inside out" approach as argued by some. What does this generally imply?
A. Ethiopia wants to avoid its relationship with some of its neighbors
B. Ethiopia wants to have wider diplomatic contacts than ever before
C. Ethiopia gives priority to the long established relationship with its neighbors
D. Ethiopia gives priority to national pride, poverty eradication and building a full- fledged democratic system

70. High school students are advised to participate in all of the following extra-curricular activities. EXCEPT
A. Scouts club
B. Anti- HIV club
C. Drama and arts club
D. Chat chewing group

71. Mr.X was accused of serving as an undercover spy for a state, which was an 'enemy' of his own country at the time. What kind of crime did Mr. X commit?
A. Unpatriotic action
B. Treason
C. Revealing state secret
D. Terrorism

72. The period between 1970 and 1980 was known as a period of silence with regard to HIV/AIDS. What does this mean?
A. There were no Clearly noticed symptoms of the disease
B. Unsafe sex was not practiced before the 1970s
C. People were afraid of talking about the disease
D. It was a period of steady expansion of the pandemic

73. Which of the following is true regarding Ethiopian's economic development?
A. .Ethiopia's failure to develop economically is mainly because of the unpatriotic culture of her people.
B. Ethiopia cannot develop economically
C. Ethiopia's failure to develop economically is attributed to absence of good governance in the past, endless wars, famine and drought.
D. Economic development and patriotism do not have any link

74. What is the link between democracy and responsibility?
A. Fulfilling our responsibilities has nothing to do with democracy
B. Fulfilling our responsibilities is one major element in an attempt to promote democratic culture
C. Fulfilling our responsibility is important only at the family level.
D. Responsible citizens can contribute little for the development of democratic culture.

75. What will be the best way of curbing the alarming growth of HIV/ AIDS in Ethiopia?
A. Involving HIV/ AIDS victims in all socio- economic activities
B. Promulgating non- discriminatory laws

C. Promoting the use of condoms in schools
D. Creating better consciousness about the pandemic than ever before

76. What is the link between international economic realities and national economy in the context of Ethiopia?
A. National economy is not affected by international economic realities
B. National economy cannot operate independently and is influenced by international economic realities
C. There is no direct link between international and national economic situations
D. Foreign investment does not support national economy

77. Identify the correct statement from the given alternatives about poverty eradication in Ethiopia?
A. Government institutions alone can succeed in eradicating poverty
B. Humanitarian activity is not associated with poverty eradication at all
C. Government institutions must be supported by civil societies and humanitarian organizations in the effort to eradicate poverty
D. Poverty eradication is not the major agenda of Ethiopian government and peoples

78. To achieve the planned goal of Change and Transformation Ethiopia needs
A. Apply the basic principles of Neo- Liberalism
B. Abolish all indigenous institutions like edir
C. Apply scientific and technological knowledge and promote socio-cultural values simultaneously.
D. endorse closed- door policy to avoid the negative influence of globalization

79. In what way do democratic rights differ from human rights?
A. Democratic rights are naturally given
B. Democratic rights are granted by democratic constitutions
C. Democratic rights are irreversible at any political circumstances
D. Democratic rights, unlike human rights, can sometimes exist even in the absence of the rule of law

80. What were the similar characteristics of the governments of Nazi Germany and Fascist Italy?
A. Both played a big part for the developments of rule of law
B. Both rejected communism
C. Both were good examples of limited government
D. Both possessed and exercised absolute, unlimited power over their respective nations

81. Which of the following statement is NOT necessarily true regarding a parliamentary democracy?
A. The prime Minister should be a member of the parliament
B. All the ministers should be members of parliament
C. The prime Minster is appointed from the majority party

Telegram: https://t.me/ExtremeSeries

D. The prime Minster leads the executive branch of government

82. In the present Ethiopia context the federal system of government is mainly designed to
   A. Strengthen the power of the federation at the expense of the regions
   B. Do away with the grievances of some minority groups
   C. Unite the peoples of the country by respecting their diversity
   D. Enumerate the right and obligations of the Regional States

83. Some peoples argue against the principle of affirmative action because they claim that it
   A. Focuses on the plight of women instead of past injustices
   B. Goes against the basic principle of equality of rights of people
   C. Enables previously disadvantaged groups a chance to compete in life
   D. Serves governments to gain support from the disadvantaged groups

84. When we say there is equality among citizens of a country, what we mean is that the citizens
   A. Enjoy equal political, social and economic status in the country
   B. Are not treated differently because of their social and economic status
   C. are equal in everything and have the right to enjoy the fruits of equality
   D. have equal position in society in using the resources of their country

85. The FDRE Constitution safeguards equality of opportunity between citizens of the country. What does this mean?
   A. Citizens should be made socially, economically and politically equal
   B. Citizens should be provided with equal chance in all aspects of life
   C. All types of inequalities between citizens should be abolished
   D. There should be gender equality in all political and social activities

86. Which of the following is true regarding globalization?
   A. It has increased the interdependence among nations of the world
   B. It has brought about greater prosperity to all countries of the world
   C. It has enabled poor nations to get much greater support from the rich nations
   D. It has effectively stopped civil disturbances from spreading in the world

87. The Universal Declaration of Human Rights was adopted by the United Nations in 1948 with the main aim of codifying human rights in international law so as to
   A. Prevent atrocities such as the holocaust from happening again
   B. Stop political parties that violate human rights from coming to power
   C. Pave the way for parties that respect human rights to come to power
   D. eradicate evil persons who intend to violate human rights of others

88. People would support and respect their government if they believe above everything else that it
   A. manages to bring economic growth irrespective of who is benefited
   B. enables citizens to benefit from national wealth on equitable basis
   C. declares that it resolutely fights all manifestations of corruption
   D. ratifies all important international conventions on human right

Telegram: https://t.me/Extremeseries

One reason why known drug addicts could NOT be expected to be responsible citizens is because there is a very high possibility that they
A. Cannot stop taking drugs even if they are rehabilitated
B. Are afflicted by serious illnesses and could die of them
C. May not have enough money to support themselves
D. Could engage in criminal activities to satisfy their addiction.

If you contribute your share to your community and that contribution is recognized
A. Inspire you to do more for your community.
B. Motivate you to be complacent form then on.
C. Induce you to become leader of your community.
D. Stimulate a desire to continue your education.

Rule of law and rights of citizens are better respected and protected in countries where
A. The people elect members of parliament on regular basis.
B. There are two or more political parties that compete for power.
C. The judicial branch of government is independent and strong.
D. The Legislative branch of government introduces strong laws.

The principal of universal suffrage dictates that
A. legally able citizens of a country have the right to participate in election.
B. Election is a prerogative of all citizens of a country irrespective of race or age.
C. Election right is given only to citizens who permanently reside in the country.
D. The right to elect leaders is a constitutional right and cannot be abridged by law.

Paying tax is an obligation of every citizen who earns income in one way or another. While overcoming this obligation, the individual has a right to
A. Decide and make known the amount of tax that he/she pay.
B. Claim a refund of what he/she paid if he/she stops earning income.
C. Refuse to pay beyond the amount required of him/her by the appropriate law.
D. Say no to a demand by authorities to pay overdue tax payments.

Most of the people who stand and fight for the common good of their society usually do so out of
A. Conviction that societal interest is more important than individual wellbeing.
B. Belief that society would disintegrate if they do not take leadership role.
C. Certainty that they are more educated and enlightened than the rest of the people.
D. Desire to get fame and popularity in their community for being at the forefront.

95. Alex is a self-centered man and his personality is known to all people he knows. Which of the following tasks would be difficult for him to accomplish?
   A. Securing a gainful employment.
   B. Renting a house to live in.
   C. Understanding the needs of others.
   D. Respecting elderly people.
96. Women were unable to contribute for the socio-economic development of our country mainly because they
   A. Were deprived of the opportunity to develop themselves and utilize their potentials.
   B. Are weak to engage in development works compared to their male counterparts.
   C. Fear to take part in activities that are physically and mentally demanding.
   D. Prefer to do household work to involvement in development activities.
97. What is morally acceptable in one culture could be immoral in another. This is because
   A. There is no universally acceptable definition of culture.
   B. Morality is culture specific and could vary form one to another.
   C. Morality has standard definition that is acceptable to all c cultures.
   D. Moral values determine the culture of specific groups of people.
98. Which one of the following is a backward attitude that has hindered Ethiopian women from participating in many income-generating jobs so far?
   A. Belief that women must be shy, disciplined, and meek.
   B. Classification of jobs as "male oriented" versus " female oriented" ones.
   C. The belief that women must be given care and protection.
   D. The refusal of women themselves to participate in income-generating activities.
99. Identify the correct statement regarding Ethiopia's relation with international organization from the given alternatives.
   A. Ethiopia was a founding member of the League of Nations.
   B. Ethiopia hosts the Economic Commission for Africa along with AU and several UN regional bodies.
   C. Ethiopia signed the UN Charter in 1950.
   D. Ethiopia joins OAU two years after its formation.
100. Which of the following is true about the victims of HIV/AIDS?
   A. HIV/AIDS victims do not like social life.
   B. The number of the HIV/AIDS victims is very small in Ethiopia.
   C. HIV/ AIDS is a threat against only the youth.
   D. HIV/AIDS victims have been discriminated due to the poor level of awareness among the society.

Directions: Each of the following questions is followed by four possible alternatives. Read each question carefully and **blacken** the letter of your choice on the separate answer sheet provided.

1. Which of the following is right of an Ethiopian Child?
   A. Right to vote in parliamentary election.
   B. Right to own property in his own name?
   C. Right to have a job.
   D. Right to have shelter.
2. The tendency to consider one's own culture as the **only** right one and believe it to be superior to that of others is called
   A. Stereotyping
   B. Ethnocentrism
   C. Egocentrism
   D. Uniculrurism
3. The transfer of power from the federal to regional governments is known as _____ of power.
   A. Dispersion
   B. Separation
   C. Division
   D. Devolution
4. According to the principle of universal suffrage which group of people is entitled to vote?
   A. All citizens of a country
   B. All people living in the country
   C. All legally able citizens of a country
   D. All people who showed interest in voting.
5. The rotating members of the United Nations Organization Security Council are chosen of the basis of their.
   A. Geographical location
   B. Economic development
   C. Democratization process.
   D. Commitment to fight injustice
6. Which of the following peaceful conflict resolution methods could be undertaken without involving a third party?
   A. Arbitration B. Mediation C. Litigation D. Negotiation
7. The system of rule that was based on rigid raci division in former South Africa was known as
   A. FascismB. ApartheidC. TotalitarianD.Authoritarian
8. Which of the following is accepted as an indicator of poverty?
   A. Small family size
   B. High illiteracy rate
   C. High traffic accident rate.
   D. Low rate of infant mortality
9. Which one of the following is an institution that shaped the modern day era of globalization?
   A. The World Tourist Organization
   B. The International Monetary Fund.
   C. The Global Forum for Social Justice
   D. The Global Alliance Against corruption

Telegram: https://t.me/EXtremeSeries

10. One of the approaches that include bringing corrupt officials to justice is called.
A. Curative approach. C. Preventive approach
B. Remedial approach D Rehabilitation approach
11. Which of the following is generally taken as 'female oriented' job in the Ethiopian tradition?
A. Grazing cattle
B. Collecting firewood
C. Agricultural activity
D. Participation in house constructions
12. Proper recognition of our abilities, strengths and weaknesses implies:
A. Self-esteem B. Self-respect
C. Self-reliance D. Self –awareness
13. According to the provision of current Ethiopian constitution all sovereign political power rests on:
A. The Judiciary
B. The House of Peoples Representatives
C. Nations, Nationalities and Peoples of Ethiopia.
D. The Prime Minister and the Council of Ministers.
14. Which one of the following is **NOT** among the obligations of citizens in a democratic system?
A. Promoting political tolerance
B. Respect for the rights of others
C. Membership in a political party
D. Respect for the national emblem
15. Free access to know the activities of government officials implies:
A. Transparency C. Accountability
B. Responsibility D. Good governance
16. Which of the following hierarchical order is properly entered?
A. Regulations------- specific laws---------constitution
B. constitution --------- specific laws--------regulations
C. Regulations---------- constitution--------specific laws
D. .Constitution --------- regulations-------specific laws
17. One of the following was the motto of the Ethiopian student Movement:
A. "IdgetBehibret" C. "MeretLearashu"
B. "ItiyopiaTiqdem " D. "TimihirtLehulum"
18. Which of the following is **NOT** among the rights of an

accusedperson?
A. The right to appeal
B. The right to access for informant ion.
C. The right to be considered innocent even after he is found guilty.
D. The right to be communicated in the language he/she understands well.
19. What do the green, yellow and red colors of the Ethiopian? National flag represent respectively
A. Scrifices, hope and prosperity
B. Prosperity, hope and sacrifices.
C. Hope, sacrifices and prosperity
D. Hope, prosperity and sacrifices.
20. "I have a dream....." This inspiring quotation is associated with:
A. Rosa Parks B. Mahatma Ghandi
C. Abraham Lincoln D. Martin Luther King
21. The demand for universal suffrage in the past was a demand for
A. All citizens of a country the right to vote
B. Legally able citizens of a country the right to vote
C. People born and living in a country the right to vote
D. People who suffered injustice special privilege in voting
22. One similarity between parliamentary and presidnential democracies are that:
A. The head of state is elected by direct vote of the population
B. Government is formed by the party that won majority of votes
C. State power is transferred from one party to another peacefully
D. The judiciary branch has more effective power than the executive branch .
23. In the present world democracy is becoming more and more a preferred mode of governance because it:
A. is one requirement for becoming full-fledged member of the UNO.
B. insures social, economic and political equality among citizens
C. is believed to be better in creating equal opportunities for citizens
D. guarantees a just and equitable distribution of national wealth in a country.
24. You will be considered to tolerant person if you primarily have the ability to:
A. Always accommodate the wishes & interests of others.
B. Accept and respect differences that exist between you and others.
C. Change your views and beliefs in accordance with interests of others.
D. Give up your own rights for the sake of creating peace with others

Telegram: https://t.me/Extremeselfes

25. Which of the following principles is a basis for the present foreign policy of Ethiopia?
A. Sending peacekeeping forces to countries in conflict or civil war.
B. Forging an promoting economic union with neighboring nations.
C. Seeking and supporting peaceful solutions to international disputes.
D. Respecting international agreements irrespective of their implications.

26. Two communities could resolve disagreement on the use of grazing land through legal means and established procedures provided that.
A. The constitution has provision on usage of that particular grazing land .
B. There is enough grazing land for both communities to feed their cattle on
C. The communities adhere to the values and principles of the constitution.
D. There is strong longstanding social relationship between the communities.

27. Which one of the following could be considered as one of the reasons for the prevalence of corruption in many countries?
A. Existence exemplary ethical leadership.
B. Power monopoly by certain ethnic group
C. Absence of legal system in many countries.
D. Existence of greediness among many leaders.

28. One of the targets of the Millennium Development Goals to be achievd by all staes by the year 2015 is
A. Put an end to mortality of children under the age of five.
B. Ensuring equitable distribution of wealth among citizens
C. Eradicating inequality between boys and girls in access to education
D. Enabling their people to get rid of dire poverty and get modest income.

29. The rule of law above everything else indicates a country has
A. A written constitution
B. A democratic government.
C. Highly organized judicial system
D. Well educated judges and lawyers

30. What does the principle of 'Horizontal Equity' in taxation state?
A. people who have more income should pay more tax.
B. people who earn equal income should pay equal tax.
C. people should be taxed according to their ability to pay
D. people should pay the same tax regardless of their income.

31. NO man is above the law'. This quotation is designed to show the:
A. Existence of powerful law.
B. Preeminence of rule of law
C. Supremacy of peoples' power.
D. incomparability of people to law.

32. Individual right is the right to_____

A. Own property B. self-governance.
C. self determination D. develop one's own culture.

33. Egypt and Sudan closely follow any Ethiopian attempt to use Abay River for development because
A. This river contributes 85% of the water of the Nile River
B. They are afraid that such move would make Nile River dry.
C. They produce 80% of their electricity form dams on this river.
D. They fear that Ethiopia would be developed threat to them

34. In Presidental Democracy such as the UAS, the president is
A. The head of the Judiciary branch of government
B. Member of both Executive and Legislative branches.
C. Elected from among the members of the wining party
D. Elected by the popular vote independently from representatives.

35. Legal limitations could be laid down on the right to freedom of expression if that right is used for
A. Damaging the reputation of individuals
B. Criticizing the government and its actions
C. Publishing documents purely about officials
D. Writing about that are not important to the Society.

36. In democratic system power is diffused and mutual accountability is instituted in order to
A. Enable the society to be free from government officials.
B. Help smaller parties to mutually compete and take power.
C. Stop the society from dominating the politics of a country
D. Prevent domination of a country by an individual or party

37. Nationalism differs from patriotism mainly by the fact that it strongly.
A. Shows dedication for a country's cultural heritage
B. Preaches the superiority on one's country to others
C. Teaches love for ones country and its national symbol.
D. Requires peoples of one country to live in peace and harmony.

38. The principle due process of law refers to
A. petition to free oneself from an illegal arrest
B. Right of the accused to presumed innocent until proven guilty
C. Right to be tried by a court that non-partisan to the disputing parties.
D. Conduct of legal proceedings purely on established rules and principles.

39. Why do whistle blowers expose wrong doings privately and anonymously?
A. They are afraid of reprisal by their bosses or those exposed.
B. They know that what they are doing is unethical and immoral
C. They know that their actions are usually harmful to the people.
D. They know that their action wrong and do not want to be known

40. A constitution of a State will be declared null and void if it is found to be in contradiction with the FDRE constitution because the latter.
A. is written by highly qualified people and is not susceptible to errors.

Telegram: https://t.me/Extremeseries

B. is the expression of the will of nations, nationalities and peoples.
C. has no article that provides for amendment or revision by anyone.
D. has to be copied directly in the former constitutions without change.

41. Which of the following is true about the pre 1991 Ethiopia?
A. There was no written constitution in the country.
B. There was no civil war except the case of Eritrea.
C. The rights of Nations and nationalities were undermined.
D. Government officials were elected through free, fair and periodic election system.

42. Which of the following is NOT true about negotiation?
A) Negotiation requires open mindedness.
B) In negotiation outside parties do not make decisions
C) Negotiation is a conflict resolution mechanism that involves third party.
D) Negotiation involves direct discussion between the parties that are in conflict.

43. What does unity in diversity mean in a multicultural society?
A) Accepting and respecting cultural differences
B) Disregarding importance of cultural differences
C) Striving to create one acceptable uniting culture
D) Subscribing to one among the different cultures.

44. According to the FDRE constitution, who is responsible for signing international conventions and agreements?
A) The Judiciary.
B) The house of Federation
C) The Council of ministers
D) The House of people's representatives.

45. In Ethiopia, any proclamation shall come into force on the date of
A) Ratification by the parliament.
B) Endorsing by the Prime Minister
C) Publication on the NegaritGazeta
D) Signing by the president of FDRE

46. Ethiopian women in general had been economically dependent on their spouses mainly because they were:
A) Forced to marry at an early age
B) Deprived of the right to own property
C) Required to produce and raise children.
D)Prohibited from participating a agriculture

47. Among the following types of rights, Which one is democratic right?
A) The right to live.
B) The right to liberty.
C) The right to assembly
D) The right to security of person.

48. Why is building democratic system very essential in today's world?
A) Because it promotes secularism.
B) Because it promotes a multi-party system.
C) Because it makes all citizens equal in terms of economic benefit
D) Because it upholds values like freedom, equality and human rights.

49. Which of the following is NOT true about parliamentaryDemocracy?
A) The president is the head of state.
B) The prime minister is the chief executive
C) The president will have nominal representation
D) The prime minister is not a member of the legislative organ.

50. Which one of the following is true based on article 61 of the FDRE constitution?
A) TheHouse of Federation is the supreme authority in the country.
B) The house of federation shall have members elected by the state council.
C) The house of federation consists of representatives of the major ethnic groups in the country.
D) The House of federation consist of only 1 representative for each one million members of its population.

51. Article 79 of the FDRE constitution states the :
A) Supremacy of the Rule of law.
B) Independence of courts and judges.
C) Supremacy of the Federal High Court.
D) independence of the legislative organ.

52.One of the following does NOT belong to the so called " Millennium Development Goals"
A) Creating economic equality,
B) Expanding primary education.
C) Reducing infant mortality rate.
D) Eradicating gender inequlality.

53. Accordig to article 19(2) of the FDRE constitution the defendant has
A) no right to appeal.
B) the right to remain silent.
C) no right to prove his innocence.
D) the right to know why is accused.

54. Which one of the following is turue about apartheid?
A) Apartheid was a colonial program thet did not involve racial segregation.
B) Apartheid was the policy of separate treatment of blacks and whites in southe Africa.
C) Apartheid was a political system that helped South Africa to be economically well developed.
D) Apartheid was a racially based segregation system practiced by the white minorities is South Africa.

55. Identify the incorrect statement from the given alternatives:
A) The rule of law plromotes arbitrary judicial practices
B)The rule of law prohibits violation of human rights.
C) The rule of law restricts both the governor and the governed.

Telegram: https://t.me/ExtremeSeries

D) The rule of law obliges citizens to behave in accourdance to the constitutional principles.

56. Which of the following is true about the state and Federal constitutions in Ethiopia?
A) The state constitution is equivalent to specific laws.
B) The Federal constitution has supremacy over the state one.
C)The state constitution can be a substitute for the Federal one.
D) The Federal constitution does not share similar provisions with the state constitution

57. Which of the following is NOT true about the situation of women in Ethiopia?
A) Women are the backbone of Ethiopan society.
B) Positive disctimination is used to compensate women for the past injustice.
C)Ethiopian women were victims of forced, unwanted marriage and abduction.
D) Domestic violence has come to an end following the promulgation of a new family law in Ethiopia

58. What was the immediate cause for the Gojjam peasant rebellion?
A) Religious injustice
B) Corrupt administration
C) Unfair distribution of land among peasants
D) Imposition of forced taxes and contribution

59. Which of the following is true about cultural relativism?
A) It is a comparative study of different cultures
B) It is evaluating a certain culture in a global context.
C) it is evaluating a certain culture in its own particular context
D) it is a tendency of considering ones culture superior to others.

60. Which of the following can be taken as a good example of social injustice juring the imperial regime?
A) Prohibition of interracial marriage.
B) Prohibition of free production of artistic works.
C) Prohibition of involvement in the political affairs of country.
D) Marginalizing some regions and social groups in the distribution of resources.

61. Which of the following is true about ethical relativism?
A) It assumes ethical values to be universal
B) It assumes ethical values to be culture specific
C) It gives emphasis to the consequences of actions rather than motives.
D) It gives emphasis to the motives rather than the consequences.

62. What possible recommendation can be suggested to reduce the negative?impacts of the widely known extravagant practices in Ethiopia?
A) Avoiding traditional religious practices
B) Introducing modern saving mechanisms

C) Avoiding arbitrary, unplanned expenditure
D) Prohibiting all traditional ceremonial practices

63. One of the following is not the behavior of responsible citizens
A) Breaking promises
B) Involvement in public affairs
C) Refusing to pay unreasonable taxes
D) Protecting the culture and heritages of the country.

64. What is the relationship between family responsibility and social responsibility?
A) Family responsibility is a pre-requisite for social responsibility
B) Social responsibility has no connection to family responsibility
C) Social responsibility is pre-requisite for family responsibility
D) Family responsibility plays insignificant role in the social system of a community.

65. Why are the victims of HIV/AIDS discriminated among different societies ? HIV/AIDS victims are discriminated because of :
A) Wrong religious teachings.
B) The fact that the disease is incurable
C) The fear that the victims of the disease may intentionally transmit it
D) The wrongly held view that the disease can be ransmitted through body contacts.

66. Which of the following has become a serious threat to our globe these days?
A) Terrorism
B) Nuclear weapons
C) Environmental degradation
D) Sunami flooding & earth quake

67. Why are handicraftsmen despised in the long established of Ethiopia?
A) Because of their ethnic identity.
B) Because of their poor professional skills.
C) Because they are dangerous to the social will being.
D) Because of backward belief attached to their personality.

68. One of the following is NOT among the factors that can positively affect working environment:
A) Getting reasonable salary
B) Observig professional ethics
C) Cooperation with fellow workers
D) Isolating oneself from fellow workers

69.Identify the statement that does NOT imply assertive behavior:
A) Sharing experiences
B) Criticizing others and hasty generalization
C) Playing a positive role during conversation
D) Questioning meaningless traditions and practices

70. Identify the corerect statement from the given alterenatives:
A) Self-reliance will normally lead to self-confidence
B) Self-confidence does not influence decision making

Telegram: https://t.me/Extremeseries

C) Self-confident individuals are not necessarily self-reliant
D)Decision-making does not have any relationship with self-reliance

71. Litigation differs from other types of peaceful conflict resolution mechanisms in that it is
A) Carried out by direct negotiation between the parties to the dispute.
B) A non-briding mechanism that involves a neutral third party in the process.
C) Resolving constroversy within limits agreed upon by the disputing parties.
D) Formal and binding mechanism that is based on rules, customs or practices.

72. Group rights are different from human rights in that they:
A) Do not apply to all human beings in a similar way
B) Are not ratified by the united Nations Organization
C) More complex and difficult to respect in all countries
D) Have to be respected by all nations in a uniform manner.

73. The principle of the supremacy of the constitution dictates that
A) Elected officials should be accountable to their electorate.
B) Other laws in the country should conform to the constitution.
C)The constitution should have separate provisions for state and religion.
D) The constitution should respect the democratic rights of citizens.

74. Which of the following statements is logically correct?
A)Rule of law prevails in highly literate society irrespective of its political and cultural tradition.
B) If a nation has democratic constitution that is respected by the government in power, rule of law prevails.
C) Rule of law prevails in any country where there is a constitution that has articles for its respect.
D) If a country's constitutions has provisions for human and democratic rights then rule of law prevails.

75. The federal state structure is chosen over the unitary one for Ethiopia because it
A)enables the country to be respected by other nations.
B) guarantees equal economic development for all peoples
C) enables the various peoples to become independent of each other
D) provides equal opportunities for peoples to develop themselves.

76. What is the main difference between the federal and regional constitutions of the FDRE?
A. The former can replace the latter in case of emergency while the latter can be annulled by federal decree.
B. The former is binding in all regional states while the letter will not have full legal force in other regions.

C. The former respects the basic rights of all citizens while the latter respects only the rights of those living in the region.
D. The former provides for rules for transition of power while the latter does not have such provisions of their own.

77. Voluntary organizations that focus on various social, economic and environmental issues are different from purely social groupings because they.
A. are non-profit organizations
B. function independent of government control
C. have some degree of formal organizational setup.
D. depend on their own resources to run their activities

78. It is fact that Ethiopia is one of the poorest countries of the world while having.
A Abundant natural and cultural resources.
B. Peace and social stability in the country.
C. Well educated and hardworking manpower.
D. International economic and political support.

79. The developing countries of the world are highly dependent on rich nations. This practice has brought about
A. speedy economic and social development in the developing nations.
B. Improvement in the ability of the poor nations to stand for their interests.
C. Increased interference of rich nations in the interna affairs of the poor nations.
D. Equitable and mutually beneficial relationship between the poor and rich nations.

80. Check and balance system is more strong in presidential democracy because the:
A. Executive and the Legislative have power of vetoing abill forwarded by the other.
B. Judiciary had to approve all bills introduced by the other two branches before they become law.
C. members of all three branches of government are directly responsible for their electorate.
D. Legislatie has to seek the agreement of the other branches before introducing a bill.

81. The newly launched construction of the Grand Millennium Dam on the Abayriver is an indication of:
A. Ethiopia's long awaited wish to utilize her natural resources for socio-economic development
B. Ethiopia plan to foster her economic advantage at the expense of Egypt and Sudan.
C. Ethiopia's readiness to put into danger the 'acquired right of Egypt'

Telegram: https://t.me/ExtremeSeries

D. Ethiopia's total dependence on the hydro-electric power for her development.

82. What is the major manifestation of the equality of Ethiopian Nations, Nationalities and peoples?
A. The right to participate in local elections
B. The right to receive equal amount of annual budget.
C. The right to self-governance and promotion of their culture and history.
D. The right to participate in various music and dancing shows, particularly in the Day of Nations, Nationalities and peoples of Ethiopia.

83. Which of the following statement is true regarding rights?
A. the FDRE constitution gives priority to individual rights.
B. There is harmony between group rights and individual rights.
C. Group rights facilitate realization of equality among different groups of people
D. There is agreement between affirmative action and equal rights for citizens

84. The right to equality is one of the citizen rights enshrined in the Constitution of the FDRE. What is expected of the government with regard to this rights?
A. Safeguard the interest of the impoverished groups at the expense of the prosperous ones.
B. Take action to assure that economic and social equality of all citizens is achieved in the country.
C. Allocate it's the major portion of its resources for the improvement of the previously disfavored groups .
D. Ensure equal opportunities are available for all citizens to realize their potential and improve themselves.

85. A leader is said to be a motivating one if he/she is
A. Feared and respected by all employees
B. Willing to accept criticism made by employees
C. Capable of doing a job wrongly done by employees
D. Able to give employees a reason to work efficiently

86. For good governance an justice to prevail in a country the fulfillment of which condition is more important?
A. Prevalence of respect for rule of law
B. Existence of many civic organizations.
C. Level ground for competition between parties.
D. Existence of a written democratic constitution

87. Scientific truth differs from other theories of truth in that they are
A. Objective facts and remains of for everyone .
B. Accepted by society so far as they are desirable.
C. Derived from personal views about the universe.

D. Measured by their utility to individuals or groups

88. There has been a mass uprising against governments that occupies political power for many decades in different Arab countries. The uprising costs the lives of many innocent civilians and many other socio-economic problems. What major lesson can be drawn from the situation about democracy?
A. There are no democratic regimes in the Arab world.
B. Democracy is at its infancy in many parts of our world
C. Democracy is perfectly practiced in the western world only
D. Mass uprising is the only way to remove dictatorial regimes and establish democratic ones .

89. It is apparent that Ethiopia contributed peacekeeping forces to Korea, Congo Darfur and now Sudan. What does this imply?
A. The strength of Ethiopia's military power.
B. The fact that Ethiopian solders are good fighters
C. The ability of the UN to mobilize its member states.
D. Ethiopia's readiness to play her part in the peacekeeping process

90. Which of the following is true about objective judgment?
A. Objective judgment is the base for reasoning
B. Objective judgment is all about decision making
C. Objective judgment does not require reasoning ability
D. Objective judgment is the outcome of knowledge and reason

91. which of the following is true about scientific research?
A. Data collection & analysis are the easiest tasks in the process.
B. A theory or hypothesis is the final finding of any research activity.
C. The results of scientific research are irreversible at any circumstances.
D. Collecting and analyzing data involves a very careful testing to arrive at conclusion

92. One of the following Ethiopian intellectuals is the author of a famous book, "MengistinaYehizibAstedader"
A. AfeworkTekle
B. TeklesadikMekuria
C. Bejrond T/Hawariat T/Mariam
D. Negadras G/HiwotBaykedagn

93. Now days, the International community pays huge attention to the threat of the HIV/AIDS pandemic expansion due to the:
A. Fact that all attempts to find medication for the diseases have failed
B. Fact that globalization has created opportunities for interracial marriages.
C. Fact that the disease has became a great obstacle against the socio-economic development of countries
D. Fear that the victims of the disease can no longer contribute to the development endeavors of their nations

94. Which of the following statements is true about the Blue Nile/Abay?

Telegram: https://t.me/Extremeseries

A. The Blue Nile can be an element of co-operation between Ethiopia and Egypt.
B. The Blue Nile is an element of conflict and hatred between Ethiopia and Egypt.
C. The Blue Nile does not have any implication on the Ethio-Sudanese and Egyptian relations.
D. The Blue Nile cannot be used for the economic development of Ethiopia without the approval of Egypt .

95. Which of following is true about job satisfaction?
A. The degree of job satisfaction is high in Ethiopia.
B. Job satisfaction did not have any effect on industriousness
C. Job satisfaction implies the provision of community service.
D. Job satisfaction can help individuals commit themselves to their profession

96. Which of the following is a negative impact of globalization?
A. Competition on markets.
B. Advancement of technology
C. Transfer of global information and knowledge
D. Dependence of economically weak nations on others

97. Chewing Chat, smoking Cigarettes and drinking alcohol are not advised as leisure activates, particularly for young students like you because:
A. They are expensive for students
B. Of the problem of addiction and consequent effects.
C. Such practices would retard the memory of students
D. Such practices are out of the norms and values of Ethiopians.

98. What is the ultimate goal of Agriculture-led Industrialization /ADLI/ policy of the current Ethiopian government?
A. Making Ethiopia self-reliant, developed country.
B. Making Ethiopia leading nation in industrial technology.
C. Making Ethiopia a leading country in the production of cereals.
D. Making Ethiopia the major producer of cash crops like coffee and chat.

99. What is the significance of effective leadership in a democracy?
A. Strengthening multi-party system.
B. Strengthening the power of individuals
C. Enhancing good governance and transparency.
D. Strengthening the authority of the ruling political party.

100. Which of the following is true about demonstrations?
A. Demonstrations are equivalent to violence.
B. Demonstrations are conducted always peacefully.
C. Demonstrations must be organized by government officials.
D. Demonstrations are public displays for or against a given situation.

1. Which of the following which one is among the prerequisites for political, social and economic development of a country?
A. Rule of law C. Periodic election
B. Transfer of power D. Multi pary system

2. A state where religion and state are separate is called
A. Erastian C. Secular
B. Theocratic D. Neutral

3. The members of the House of Federation of RDRE are elected by the
A. Regional state Council
B. Executives the regional states
C. Presidents of the regional states
D. Electrons in the regional states

4. Who has the power to introduce laws at regional state level in FDRE?
A. State administration C. State president
B. State Council D. State Supreme Court

5. Which of the following group of people were considered inferior to others in the traditional Ethiopian society?
A. Priests B. Teachers C. Farmers D. Singers

6. The most vocal opposition to the regime of Haile Sellassie I, particularly during its last stage came from
A. General officers in the army
B. Dissatisfied ordinary peasants
C. Industrial workers and teachers.
D. Students and the educated elite.

How many seats shall be reserved for all minority nationalities and people in the House of People's Representatives as provided by the FDRE constitution?
A) 40 seats B) 20 seats C) 30 seats D) 10 seats

Who is the source of the legitimate power of the FDRE?
A) The Parliament C) The People
B) Supreme Court D) The army

Which of the following is a crime that is mainly committed for the purpose of gaining personal wealth?
A) Homicide B) Corruption C) Treachery D) Torture

Identify a conflict resolution mechanism in which the third party makes decisions.
A) Negotiation B) Arbitration C) Mediation D) Litigation

Which of the following gives a general definition of corruption in a better manner?
A) Corruption is taking bribes for official services.
B) Corruption is the misuse of public property and money.
C) Corruption is the abuse of public office for private gain.

Telegram: https://t.me/ExtremeSeries

D) Corruption is forceful act of obliging people to pay for official services.

12. Which of the following is morally wrong according to the Ethiopian culture?
    A) Respect for elders.
    B) Breaking promises
    C) Helping the needy
    D) Participating in social affairs
13. The Millennium Development Goals include all of the following except?
    A) Providing primary education for all children of the world.
    B) Reducing infant mortality by ninety percent.
    C) Making the poor rich.
    D) Eradicating gender inequality in education
14) The present flag in green, yellow and red probably became the national flag of Ethiopia during the time of :
    A) Lij Iyasu
    B) Emperor Yohannes IV
    C) Empress Zewdittu
    D) Emperor Menelik II
15) What is the major advantage of fulfilling our obligations as Ethiopian citizen?
    A) Socialization
    B) Getting spiritual satisfaction
    C) Enjoying our rights in return
    D) Getting financial reward
16) Which of the following professions(occupations) has long been the most undermined one among many Ethiopian communities?
    A) Teaching
    B) Shoe shining
    C) Pottery
    D) Wood work
17) Which of the following is generally taken as 'male oriented' job in the Ethiopian tradition?
    A) Cooking
    B) Looking after children
    C) Collecting fire wood
    D) Long distance trade.
18) Why do many Ethiopian prefer to use imported products than the local ones?
    A) Because all local products have poor qualities
    B) Because all imported products have good qualities
    C) Because there is a bad habit of giving low value for local products.
19) Which of the following is true about Idir and Equb?
    A) Both are traditional saving institutions in Ethiopia
    B) Both are practiced only in the rural parts of Ethiopia
    C) Idir is not associated with saving
    D) Equb is a backward saving system which is now being totally replaced by modern banking
20) Which of the following is NOT among the components of knowledge?
    A) Truth
    C) Data

Telegram: https://t.me/Extremeseries

D) Myth

B) Evidence

Which of the following is NOT true about democratic constitution?
A) It paves the way for constitutionalism and democracy
B) It limits the power of government
C) It does not allow extra-constitutionalism and democracy.
D) It does not stipulate directly the rights and duties of citizens.

Ethiopia's foreign policy is based. Primarily, on the country's
A) geographical location
B) strong military power
C) benefits and interests only
D) mutual benefits and cordial relation ship

Which of the following is NOT true about direct democracy?
A) It works among communities of small size.
B) It is represented today in the form of referendum or plebiscites.
C) It allows all participants to pass decisions on their own affairs.
D) It was practiced by ancient Greeks where by both men and women were active participants.

What is the major advantage of federalism?
A) Creating regional autonomy
B) Promoting cultures and languages
C) Unifying different peoples of a country
D) Creating interregional economic cooperation

Indentify the country that has combined both parliamentary and presidential systems of democracy together.
A) South Korea
B) England
C) South Africa
D) France

Which of the following is true?
A) Ethiopia's constitution gives priority for individual rights.
B) Ethiopia's constitution gives priority for collective right.
C) Ethiopia's constitution protects collective rights at the expense of individual rights.
D) Ethiopia's constitution protects both collective and individual rights.

The regional state of Afar has full authority to do all of the following, but:
A) Collecting taxes.
B) establishing regional police force.
C) establishing independent diplomatic relations with Djibout.
D) using its natural resources for regional development

Why did Ethiopia remain to be the only African country that was never colonized?
A) Because of her geographical location
B) Because Europeans were not interested in it
C) Because she had successfully defended foreign aggressions.
D) Because she had a very strong foreign policy from the beginning.

What is the major possible motive for corruption in countries like Ethiopia?
A)Poverty
B)Resource scarcity and completion in the economy
C)Absence of the rule of law and constitutionalism
D) Greediness and searching a short-cut for accumulation of wealth.

30. How was the derg overthrown?
A)Through mass revolution
B)Through armed struggle
C)Through political dialogue
D)Through peasent rebellion

31. Which of the following is true about positive discrimination?
A)It is used to make women equal to men
B)It is used to stop domestic violence
C)It is used to restrict women at home.
D)It is used to compensate women for the past injustice.

32. One of the following peasant rebellions was crushed by the joint efforts of the Imperial Government and British forces
A)The Bale Peasant Rebellion
B) The first *woyane* movement
C)The Second *woyane* movement
D)The Gojjam peasant rebellion

33. How and when do multicultural societies like Ethiopia exist in peace?
A)When there is mutual respect for one another.
B)When there is strong military government
C)When there is written constitution.
D)When there is strong. Independent judiciary.

34. According to Ethiopian law the responsibility of signing international conventions and agreements is given to:
A)the executive branch
B)the Judiciary branch
C)the Legislative branch.
D)the House of Federation

35. The period of the derg was characterized by all the following negative experience, EXCEPT
A)red terror
B)literacy Campaign
C)forced military service(Afesa)
D)absence of religious freedom.

36. Identify the WRONG statement from the given alternatives.
A) Payment of taxes should be based on people's capacity to pay.
B) literacy Campaign
C) forced military service (Afesa)
D) absence of religious freedom

37. Which of the following is NOT true about voluntary organizations?
A)They are profit making agencies
B)They are independent of government
C)They usually work towards social well being

D) They cooperate with governments and other organizations to tackle common problems.

38. "I have a dream that my four little children will one day live in a nation where they will not be judged by the color of their skin but by the content of their character. I have a dream today!" This famous quotation is associated to:
A)Martin Luther
B)Rosa Parks
C) Abraham Lincoln
D) Martin Luther King

39. Student Bezawith treats people with respect both in school and outside unlike many of her close friends. Which of the following civic virtues explain her best?
A. Fairness B. Truthfulness C) Civility D) Courage

40. As an Ethiopian student why do you learn your history and culture?
A)In order to build ethnocentric feelings.
B)In order to appreciate past achievements.
C)In order to bring about Cultural Revolution
D) In order to learn from the past and shape your values and beliefs.

41. Which of the following cultural, made heritages of Ethiopia is recently registered by UNESCO as World Heritage?
A)The Wolaita Folk dance.
B)The Evangadi ceremony
C)The Gedeo Megalithic stones
D)The konso Cultural Landscape.

42. One of the following is NOT the behavior of responsible citizens:
A)Treating people as equals.
B)Protecting cultural and natural heritages
C)Working hard to improve living conditions
D)Charging extra payment for official services

43. Which of the following is true about cross- boundary rivers?
A)Ethiopia does not have cross- boundary rivers.
B)Cross- boundary rivers are found only in the Middle East.
C) Cross- boundary rivers create the condition for countries to have a special relationship.
D) Cross- boundary rivers cannot be used for common development projects.

44. Identify the WRONG statement about nuclear weapons.
A) Any war involving nuclear weapons could bring an end to human race.
B) The number of countries that possess nuclear weapons is increasing form time to time.
C) Nuclear weapons do not have long term effects.
D) Hiroshima and Nagasaki are good examples of the danger that can be caused by nuclear weapons.

Telegram: https://t.me/ExtremeSeries

most advisable, safe way of protecting
HIV/AIDS infection as young students?
sing condoms.
B) A voiding boy/girl friends
C) Avoiding pre-martial sexual contact
D)Having only one partner for sexual contact.
Which of the following is NOT true about HIV/AIDS?
A)It is a global problem that needs global intervention
B) it is number one killer disease in the developed world.
C) It conspires with infections such as tuberculosis, malaria, typhoid and hepaties.
D) Improving the living standards of poor nations is one way of combating HIV/AIDS.
Which of the following institutions has no role in shaping globalization?
A)World Bank Group
B) Transparency International
C) World Trade Organization.
D)International Monetary Fund
Because human rights are inalienable and inviolable birth rights government should:
A)Grant them to their peoples.
B)Recognize and respect them.
C)Bestow them on their peoples.
D)Include them in their constitutions
Democratic mode of governance is preferred over other modes of governance because it:
A)insures equality of citizens in everything
B)guarantees equitable distribution of wealth
C)creates equal opportunities for citizens
D)assures economic prosperity of a nation.
Which of the following statements is true regarding Ethiopia in relation to the League of Nations?
A)The League failed Ethiopia when she was invaded by Italy
B)Ethiopia was a founding member of the League of Nations.
C)Ethiopia represented African Nations in the League of Nations.
D)The Ethiopia withdrew from the League when Italy invaded her.
Tolerance to differences among peoples demands primarily one's ability to:
A)identify major differences.
B)do away with differences
C)accept differences as natural
D)learn about national differences.
Whose appointment should be approved by the parliament of FDRE?
A)President of the Federal Supreme Court

B)Chair person of the House of Federation
C)President of a Regional State Council
D)Ethiopian Ambassadors and Envoys
53. Where ever there is a strong and independent judiciary it is safe to expect
A)Complete eradication of corruption
B)Presence of federal system of government
C)Prevalence of respect for rule of law
D)the Executive to be afraid of the courts
54. An unbiased employer should select workers for any type of job by taking into account the applicant's
A)experience and ability
B)family social position
C)race and religious affiliation
D)age and marital status.
55. A government that respects the rule of law in its functions will be able to:
A)stay in power whatever it does to the people
B)win the confidence and acceptance of the people.
C) exercise supreme and absolute power over the people
D)conceal any type of human right abuse of citizens
56. The FDRE government can nationalize privately owned property provided that:
A)the property is owned by a suspected criminal
B)the owner failed to pay taxes or bank loans
C)the owner is amply or adequately compensated.
D)the property is found to be unproductive under the owner.
57. According to Pragmatic theory, truth:
A)worked for a given time and space
B)has universal acceptance at a certain time
C) transcends space but not time
D)never changes from time to time
58. Many people have little understanding that while enjoying the right to equality they are required to
A)ensure that they are capable of enjoying it
B)earn it through their struggle
C)enter into identical obligations with others
D)enjoy similar and identical rights with others
59. Progressive taxation system is a method of
A)levying equal tax on all irrespective of their income.
B)collecting tax a rate proportionate to payer's income
C)encouraging investors by reducing their tax progressively
D) making rich people pay high rate of taxation than poor people
60. Education enables the educated first and foremost to

Telegram: https://t.me/ExtremeSeries

A)become a politician and community leader
B)honestly serve country and people one day
C)become successful and lead comfortable life
D)have better understanding of self and society.

61. Which of the following is one of the major functions of the government of FDRE?
A) Implementing foreign policy
B)Managing primary education
C)Ratifying regional Constitution
D)Appointing regional presidents

62. In theory, markets in free- market economies are regulated by
A)relationship between demand and supply
B)amicable relationship between sellers and buyers
C)freedom given to merchants to fix prices honestly
D)abundance of goods and services in the market

63. What are the two major categories of rights stated in the constitution of FDRE?
A)Group and Individual rights
B)Human and Democratic Rights
C)National and Regional Rights
D)Civil and Social Rights

64. During election campaigning political parties should devote much of their time and energy for
A)Collecting money contributions from their supporters
B)Defaming opposition political parties and their leaders
C)Winning as much support as possible for their policies
D)Promising rewards for their members and supporters

65. Development of a certain country is finally judged by the improvement it brought to the:
A)country's natural and manmade resources
B)nation's health and education facilities
C)lives of peoples of the country
D)communication infrastructure of the country

66. Self- confident individuals are always ready to
A)hold back from telling the truth about others
B)flatter others in order to gain their good will
C) accept logical arguments and learn from them.
D)avoid challenging situations or arguments

67. Many youngsters fall victim to the habit of smoking manly because they:
A)live far from their parents and relatives
B)have income of their own at an early age
C)realize that smoking is mark of modernity
D)lack courage to resist their peers' influence

Telegram: https://t.me/Extremeseries

68. The principles of free market economy include all of the following except:
A)freedom of decision to economic agents
B) less government involvement in the market
C) letting market forces determine the extent of transactions and prices.
D) having prices fixed for commodities by the government.

69. What are two major international monetary institutions?
A)IMF and WHO
B)World Bank and WHO
C) World Bank and IMF
D) IMF and FAO

70. A lawful gathering of people to protest, support or create awareness on a certain issue implies:
A)civic campaign
B)peaceful demonstration
C) popular participation
D) political freedom

71. Which of the following is true about myths?
A) Myths are unscientific stories that people would like to tell and believe
B) Myths are scientific stories based on empirical evidence
C) Myths are found in the traditions and cultures of African peoples only
D) Myths can help us transform our knowledge

72. Which of the following is NOT among the right of persons who are accused of any criminal offense?
A) The right to be considered innocent until the final verdict is given
B) The right to choose judges
C) The right to be tried in the language they understand better.
D) The right to fair justice

73. Which of the following is the major reason that made Addis Ababa one of the diplomatic hubs of the world and the seat of African Union (the former OAU)?
A) Because it is one of the few big, beautiful cities in the world.
B) Because it is the place where cultural diversity and tolerance are well cultivated
C) Because Ethiopia had played commendable role in African and international politics.
D) Because Ethiopia is a good example of wonderful hospitality

74. Which of the following countries has ancient relations with Ethiopia?
A)Italy B) Sudan C) Greece D) Egypt

75. The conduct of legal proceedings according to established rules and principles for the protection of and enforcement of private rights it known as:
A)Procedural justice
B)Due process of law
C) Protection of civil rights
D) Fair justice

76. Which of the following is NOT true?
A)HIV/AIDS is incurable disease

B) HIV/AIDS is transmitted mainly through careless sexual intercourse.
C) HIV/AIDS has nothing to do with the economic situations of a nation
D) HIV/AIDS creates a serious socio-economic challenge in today's world.

77. Which of the following is NOT true according to Article 18 of the FDRE Constituion?
A)No one shall be held in slavery or servitude
B) Trafficking in human beings for whatever purpose is prohibited
C) Everyone has the right to protection against inhuman punishments.
D) Everyone has the right to assemble or demonstrate

78. When do countries usually form closer relationship and create a forum for cooperation?
A)When they have commonly shared objectives.
B)When they have strong, charismatic leaders
C)When they have developed economies
D)When they face social crises.

79. Medical Ethical standards include all of the following EXCEPT:
A)giving loyalty and support to fellow physicians
B)treating all patients equally
C) giving assistance to those who wanted to commit suicide
D)keeping the secretes of patients confidentially

80. Which of the following is NQT true about dependency?
A)It will create unbalanced economic relationship
B)It will create free market accesses to wealthy nations
C) It will make poor countries import manufactured goods obsolete or used.
D) It will result in sustainable economic growth

81. Which of the following is true about Ethiopia's development projects on her rivers?
A) Ethiopia has started to use her rivers primarily for the production of hydro-electric power
B) Ethiopia has fully utilized her rivers for irrigation
C) Ethiopia did not need to used her rivers because she as a lot of mineral deposits.
D) Ethiopia's development policy focuses more on agriculture ignoring the question of utilizing her rivers.

82. Which of the following is true about macro economics?
A)It deals with the behavior of individual economic agents
B)It deals with the working of the overall economy
C)It deals with the activities of industrial sector
D)It deals with traditional saving mechanisms

Telegram: https://t.me/ExtremeSeries

83. Which one of the following is true about modern saving institutions in Ethiopia?
A) Modern saving institutions include Banks and Insurance companies .
B) Modern saving institutions are inaccessible to the majority of the people
C) Modern saving institutions have completely replaced the traditional ones .
D) Modern saving intuitions are owned by the government

84. Which of the following mechanisms can be suggested to improve saving habits in Ethiopia?
A) Avoiding leisure activates at all
B)Avoiding social interactions and commitments
C)Increasing salaries of government workers every year
D)Limiting one's interests to one's income amount

85. Which of the following is true according to Article 29 of the FDRE Constitution?
A) Everyone has the right to freedom of expression without any interference.
B) Everyone has the right to follow any religion of his/her own choice.
C) Anyone has the right to form opposition political party
D) Anyone has the right to be elected.

86. Democratic and non-democratic states widely differ in terms of the use of authority and power, Which of the following statements is correct in this regard?
A) Non-democratic states used their power to enforce their authority.
B) Power is the source of the authority of democratic state
C) Democratic states lack legitimate authority to enforce their power.
D) Non- democratic states use their authority to enforce their power.

87. Societal development is better achieved under a transparent democratic government than otherwise because such a government will have:
A) ability not to make mistakes that seriously hamper development
B) resources useful for economic development at is disposal
C) better chance of winning support of the people for its endeavors
D) no opposition and critiques of its work from among the people.

88. To which of the following activities should the emerging regions of Ethiopia give priority while allocating resources?
A) Construction of big irrigation schemes.
B) Provision of basic health care and basic education
C) Construction of a network of asphalted highways.
D) Expansion of mechanized farms and industries

89. Democratic system of government is the preferred system in many countries of the world because it:

A) eliminates all social, political and economic problems of a country.
B) accepts the principle that majority decisions are always right
C) assures absolute social, economic and political equality in a country.
D) addresses the interests of citizens in a relatively better way.

90. The democratic values in the constitution of FDRE can be part of social structures and practices of the people only if they are
A) approved by the house of peoples' Representatives.
B) part of the universal declaration of human rights
C) able to win the trust and confidence of the society
D) approved by and elected constituent assembly

91. Civic societies can play vital role in building a democratic system in a country by:
A) standing against a government that they believe to be undemocratic
B) encouraging constructive participation of people in public affairs
C) organizing a democratic party and participating in elections.
D) supporting a government disregarding the will of their members.

92. To improve the life of their peoples governments of developing countries should:
A) allocate greater budget for expansion of social services.
B) fix the price of basic necessities like food and clothing
C) give economic subsidies to the poor so that they can compete.
D) provide food and material aid to the needy people.

93. In what language can all the members of the federation execute their public functions according to the FDRE constitution?
A)Language of their own preference
B)Language that is common for all of them
C)Their mother tongue or home language
D)The working language of the Federation.

94. Ethiopian farmers are encouraged to produce and use organic fertilizers, as much as possible, mainly because these fertilizers are:
A)proved to give much greater yield than the inorganic ones.
B) sought after by exporters and could bring foreign exchange.
C) easily produced in large quantities and sell to other farmers
D) environmentally friendly and do not harm the land.

95. According to the modern understanding of freedom a person is not free until he/ she is free from
A)foreign occupation and oppression
B)lack of all basic necessities of life.
C)cruel and oppressive local governments
D) restriction on democratic and human rights.

96. The role assigned to women by traditional societies impeded national development by:
A) depriving the country the chance to fully exploit their potentials.

B) causing irreconcilable difference with their male partners
C) making them mentally weak and emotionally sensitive.
D) encouraging them to demonstrate their skill and ability publicly

97. Many of the communicable disease in Ethiopia could be eradicated by
A)Enabling people to have easy access to modern hospitals
B)Assigning doctors and other professionals in rural areas
C)Making people put simple hygiene rules into practice
D)Distributing medicines for the diseases periodically

98. There is a need for the regional states of FDRE to grow and develop together because the regional states are
A)endowed with identical natural resources.
B)capable of fulfilling their economic needs
C)found at various stage of economic development
D)engaged in developing similar economic capacities

99. To motivate followers and effective leader should
A) be flexible enough to learn from works of others
B) design plan of action and force others to accept it
C) have vision and manage to impose it on followers
D) have commitment for the objective of the task

100. One true measure of a person's greatness could be her/his
A)readiness to criticize the works of others always
B) ability to identify errors made by productive people
C) tangible accomplishment to solve societal problems
D)social position and ability to socialize with others.

Telegram: https://t.me/Extremeseries

ETHIOPIAN UNIVERSITY ENTRANCE EXAMINATION (EUEE) CIVIC AND ETHICAL EDUCATION GINBOT 2006/ JUNNE 2014

BOOKLET CODE:- 52

Directions:- Each of the following questions is followed by four possible alternatives. Read each question carefully and blacken the letter of your choice on the separate answer sheet provided.

1. Which countries benefit most from the new world order of globalization?
   A) Developing countries
   B) Developed countries
   C) Under developed countries
   D) Less developed countries
2. What is the principle of constitutional law that protects the life, liberty and property of a person?
   A) Criminal law
   B) Civil law
   C) Due process of law
   D) Procedural law
3. Which of the following is a name given to human needs for food, Shelter and security?
   A) Maintenance needs
   B) Transformational needs
   C) Societal needs
   D) Actualization needs
4. An amendment to a constitution is
   A) Replacing old constitution by a new one.
   B) Additional article to a constitution.
   C) Decelerating a constitution null and void.
   D) Emphasizing an article of a constitution.
5. A person who refuses to give military service on grounds of religion or other belief is
   A) Conscientious objector
   B) Whistle blower
   C) Unpatriotic citizen
   D) Secret intelligence officer
6. Who was the American Civil Rights Activist sent to prison for refusing to give up her seat in a bus for a white man?
   A) Rosa Luxemburg
   B) Patricia Goldman
   C) Clara Zeitkin
   D) Rosa Parks
7. The ability to hear with un pleasant and annoying situation is known as
   A) Assertiveness
   B) Arrogance
   C) Lenience
   D) Tolerance
8. Which of the following is one of the key institutions that fashioned globalization?
   A) Amnesty International
   B) Transparency International.
   C) International Labor Organization.
   D) International Monetary Fund.
9. Which of the following traits is expected from judges for justice to prevail?

Telegram: https://t.me/EthioExamSeries

A) Compassion B) Sympathy C) Kindness D) Impartiality

10. Which of the following characteristics could be a guarantee for individual freedom?
    A) Rule of law
    B) Right to trial
    C) Trial by jury
    D) Human rights
11. Who presents a nominee for the post of Prime Minister of the FDRE?
    A) The speaker of the House of Peoples' Representatives.
    B) The speaker of the House of Federation.
    C) The Opposition in the House of Peoples' Representatives.
    D) The majority party in the House of People's Representatives.
12. Which of the following is true?
    A) There is no relationship between saving and development.
    B) *Idir* and *Iqub* are backward saving mechanisms that do not help Individuals improve their lives.
    C) Saving plays a crucial role in the development endeavors of countries like Ethiopia.
    D) World Bank is the only global monetary organization.
13. Which of the following was the most negative experience in the period of the derg?
    A) *"Qey Shibir"*
    B) *"Meret Le' arashu!"*
    C) *"Idget Behibret!"*
    D) *"Meserete Timihrt"*
14. In the FDRE, interpreting law is the responsibility of the judiciary organ, whereas interpreting the constitution is the responsibility of:
    A) The Executive Organ
    B) The Legislative Organ
    C) The House of People Representatives
    D) The House of Federation
15. The American Foreign Secretary, John Kerry facilitated a peace talk between Israel and Palestine recently. What was the role of Mr. Kerry?
    A) Arbitration B) Negotiation C) Mediation D) Resolution
16. Which of the following is the correct order that should be followed while formulating a democratic constitution?
    A) Drafting ---Ratification ---Discussion
    B) Drafting ---Discussion --- Ratification
    C) Ratification --- Discussion---Drafting.
    D) Discussion --- Drafting --- Ratification.
17. Which of the following is an inclusive term?
    A) Good governance
    B) Transparency
    C) Accountability
    D) Responsibility
18. Which of the following countries combine both presidential and parliamentary government systems?
    A) USA and England
    C) Russia and South Africa

B) Ethiopia and USA D) South Africa and USA

19. Which one of the following government system differs from others?
A) Unlimited government C) Unitary government
B) Totalitarian government D) Federal government

20. Which of the following is correct according to article 10 of the FDRE constitution?
A) Human rights are inviolable and inalienable.
B) Human rights do not emanate from the nature of mankind.
C) Human rights emanate from the good will of governments.
D) Human rights do not have any relation with democratic rights.

21. Which one of the following is correct?
A) Constitution and specific laws have equal weight.
B) Constitution and specific laws do not any relation.
C) Constitution is a mother law from which specific law can be derived.
D) Constitution is a mother law that emanates from specific laws.

22. What was the name of the corruption in which president Nixon had been involved during his presidency?
A) 'The Wastergate Scandal' C) 'The Great Depression'
B) 'The Communist Scandal' D) 'The American Scandal'

23. Which of the following is correct about budget allocation for different regions of Ethiopia?
A) The federal government allocates and distributes money reasonably based on the specific situations of different regions.
B) The federal government allocates equal amount of money to all Nations, Nationalities and peoples of Ethiopia.
C) The federal government allocates less amount of money to less developed regions.
D) The federal government allocates large amount of money to better developed regions.

24. Which one of the following personalities is different from others?
a) Nyerere B) Mengsistu C) Mobutu D) Idi Amin

25. Which one of the following women was famous for her political career in the modern history of Ethiopia?
A) Empress Zewditu C) Empress Taitu
B) Empress Menene D) Yodit Gudit

26. One of the following Ethiopian rivers is different from others:
A) Abay B) Awash C) Tekeze D) Baro

27. Which one of the following is true based on article 61 of the FDRE constitution?
A) The House of Federation is the supreme authority in the country.

B) The House of Federation consist of only one representative for each one people of the contrary.
C) The house of federation shall have members elected by the state council.
D) The house of federation consists of representatives of the major ethnic groups in the country.

28. Which one of the following is **INCORRECT** about conflict resolution in Ethiopia?
A) All Ethiopians peoples have indigenous conflict resolution mechanisms.
B) Elders/ shimagles are highly respected in the indigenous conflict resolution process.
C) Elders/shimagles cannot resolve conflicts that involve homicide.
D) Arbitration and mediation are employed to resolve conflicts.

29. Which of the following gives ultimate decisions according to the Ethiopian law?
A) The Federal Supreme Court
B) The Federal High court
C) The State Supreme Court
D) The State High Court

30. Which of the following countries is a unique example in the world that does not have a written constitution?
A) China B) France C) Nethrlands D) England

31. Which of the following is the mandate of the federal government of Ethiopia under the current constitutional provision?
A) Establishing the regional police force.
B) Establishing the foreign relations.
C) Writing constitutions for all regional governments.
D) Organizing cultural events in the regions.

32. Which of the following groups were the most undermined ones in the long history of Ethiopia?
A) Potters B) Peasants C) Merchants D) Women

33. Which one of the following is true about illegal human migration in the current Ethiopian context?
A) All illegal migrants are from Southern Ethiopia
B) The major destination of women migrants is South Africa.
C) Illegal migration has become the major socio-cultural problem.
D) Illegal migration is assisted by government agencies.

34. Which of the following is true about benefits and burdens?
A) Benefits and burdens must be shared equally among all peoples in a society.
B) Benefits and Burdens must shared on the education level of individuals.

Telegram: https://t.me/Extremeseries

C) Individuals who carry more burden should get less benefits.
D) Individuals who carry less burden must not enjoy their political rights.

35. Which of the following is a negative impact of globalization?
A) Transfer of global information and knowledge.
B) Dependence of economically weak nations on the developed ones.
C) Market competition between economically weak and strong nations.
D) Advancement of technology.

36. In what way did Ethiopia protect her independence from foreign aggression in the past few centuries?
A) Through diplomatic negotiations.
B) Through patriotic resistance struggle.
C) Through peaceful submission
D) Through the support of Western Great powers.

37. One of the following problems was the most serious one that aggravated socio-economic problems and consequent oppositions against the imperial regime of Ethiopia and brought about its downfall:
A) The war with Eritrean rebel groups
B) Land problem
C) The question of Nationalities
D) Marginalization of Occupational groups

38. In what possible ways can student like you protect themselves from peer influence?
A) By avoiding relations with peer groups.
B) By studying hard always.
C) By making planned activities
D) By staying at home after school.

39. Which one of the following is **NOT** among the peoples of Southern Ethiopia?
A) Ari B) Hammer C) Nuer D) Surma

40. Which one of the following social groups is the most disadvantaged group in Ethiopia?
A) Artisans C) Children
B) Women D) Persons with Disabilities.

41. In democratic countries obligations of citizens arise from their
A) Rights that they enjoy C) Gender identity
B) Duties and responsibilities D) Position in society

42. Why are elected officials accountable to their electorate? It is because
A) They should listen to the grievance of the electorate.

B) The electorate is the source of their political power.
C) They need to have constant contact with their electors.
D) The electorate can replace them with another if anytime.

43. The table below shows a comparison of two different systems of government.

Federal System	Unitary System
?	?

Which information completes the above table?
A) Role of government limited/ Role of government expanded.
B) Rule by a democracy/ Rule by a monarchy.
C) Power divided among the branches/ Power held by one branch.
D) Power shared / Power delegated.

44. One of the following Declarations was promulgated following the humanitarian crisis of Second World War:
A) Universal Declaration of Human Rights
B) World Health Day
C) Declaration of Environment Protection
D) March Against HIV/AIDS.

45. What was the greatest gain of the Women's Suffrage Movement?
A) Holding citizenship rights.
B) Holding public office.
C) Vote in national elections.
D) Gin employment rights.

46. Identify the Incorrect Statement from the given alternatives:
A) Industriousness leads to Self-reliance.
B) Self-Reliance leads to Self – confidence
C) Addictions could affect the working habits of the youth.
D) Male-oriented jobs make individuals more industrious.

47. Assertive behavior includes all of the following , except:
A) Questioning C) Critical arguments
B) Positive Thinking D) Hasty generalization

48. Who is the Author of famous Ethiopian Novel, "Oromay"?
A) Kebede Mikael C) Haddis Alemayehu
B) Be'alu Girma D) Mamo Wudneh

49. What do an absolute monarchy and an autocracy have in common?
A) Supremacy of a single ruler.
B) Absence of written constitution.
C) Strong national court system.
D) Single legislative house.

50. Which of the following is true about foreign aid in Ethiopia?
A) Ethiopia is fully dependent on foreign aid.
B) Foreign aid is the base of Ethiopian economy.
C) Ethiopia's goal is to be self – reliant in the future.

Telegram: https://t.me/ExtremeSeries

D) Ethiopia is the major beneficiary foreign aid in Africa

51. One similarity between parliamentary and presidential democracies is that in both cases:
    A) The head of state elected by the members of parliament.
    B) The head of state is elected directly by popular vote.
    C) There is a smooth and peaceful transition of state power.
    D) Government is formed most of the time by coalition of parties.
52. Which of the following statements is true about progressive taxation?
    A) Varying tax rates are levied on people who earn equal income.
    B) People are expected to pay tax at a rate proportionate to their income.
    C) People who earn more pay equal tax with those who earn less.
    D) Reich people are taxed more while the poor are exempted from it.
53. Civic minded citizens serve their community wholeheartedly mainly to get
    A) Fame and recognition
    B) Monetary reward
    C) Appointment to public office
    D) Personal satisfaction.
54. What was the objective of the non aligned movement?
    A) To become an alternative force for the East and West in the Cold War.
    B) To promote economic cooperation between developing countries.
    C) To advocate solutions to global social and political problems.
    D) To be neutral in the East and West confrontation of the Cold War.
55. Which of the following is a characteristic feature of the demography of developing nations?
    A) High birth rate and High death rate.
    B) High birth rate and low death rate.
    C) Low birth rate and High death rate.
    D) Low birth rate and low death rate.
56. What is ethics? It is a
    A) Collection of moral standards accepted by society.
    B) Part of philosophy that studies moral standards.
    C) Set of rules and regulations issued by democratic states.
    D) Reference material on moral standards issued by UNO.
57. An altruist is identified by his/her readiness to
    A) Initiates action for constitutional amendment.
    B) Considered to be revolutionary.
    C) Automatically made null and void.
    D) Replaces some articles in the constitution.

Telegram: https://t.me/ExtremeSeries

58. An altruist is identified by his/her readiness to
    A) Lead people in protests against injustice.
    B) Oppose any form of dictatorship and tyranny.
    C) Sacrifice ones happiness and welfare for others.
    D) Use any means to gain power and respect.
59. Concerning Human Rights what governments are required to do is
    A) Insure that they are not violated by anyone.
    B) Introduce new types of rights for their people.
    C) Coordinate their use with social development.
    D) Use any means to gain power and respect.
60. In democratic countries officials are expected to be non-partisan while performing their duties. This means?
    A) They should not be prejudiced in any way.
    B) They should not be party members.
    C) They should not belong to an ethnic group.
    D) They should not belong to a religious group.
61. Which of the following is an attribute of a person who is self-confident?
    A) Readiness to accept criticism.
    B) Determination to help other people.
    C) Reluctance to take on challenges.
    D) Having personal role models.
62. Ten countries are elected for a period of two years to serve in the Security Council taking into consideration their
    A) Acceptance by other members.
    B) Economic development.
    C) Geographical location.
    D) Willingness to serve.
63. For citizens to understand their rights and duties properly laws and rules should be
    A) Willingly agreed to by citizens.
    B) Attractive enough in design.
    C) Amended periodically.
    D) Clear, easy and well designed.
64. In Ethiopia it possible to get rid of many off the communicable diseases by:
    A) Teaching the people to keep personal hygiene.
    B) Distributing medicines in the diseases prone areas.
    C) Increasing the number of health professionals.
    D) Building health centers and make them accessible.
65. What does domesticating of international laws and norms mean?
    A) Fashioning domestic laws and norms with the international ones.

B) Changing the international ones to suit the conditions of the land.
C) Including some domestic provisions to the international ones.
D) Making the international laws and norms part of the law of the land.

66. Which of the following functions is exclusively given to the Federal Government of Ethiopia by the Constitution?
A) Administration of secondary education.
B) Administration of taxes.
C) Establishing a police force
D) Formulation of foreign policy.

67. Any place where the Ethiopian flag is flown, be it in embassy or on ship, is considered as:
A) Part of Ethiopian sovereign territory.
B) Territory free from any international obligations.
C) Free transit area for Ethiopian trade with others.
D) An integral part of the host country or area.

68. During state of emergency the Ethiopian government could take action that could
A) Dismantle all articles of the constitution.
B) Limit the democratic rights of citizens.
C) Breakup the federal arrangement.
D) Reinforce commitment for human rights.

69. What does wise saving mean?
A) Saving money even before satisfying your basic needs.
B) Saving money after satisfying all your personal wants.
C) Saving money after satisfying your basic needs.
D) Saving money by reducing from your basic needs.

70. Accountability of state requires a government to:
A) Consult elected representatives before making any decision.
B) Settle its accounts and make them for auditing on time.
C) Take responsibility for all its decisions and functions.
D) Build modern and trustworthy financial system.

71. What could be a basic possible reason that leads people to corruption?
A) Greediness and selfishness.
B) Poverty
C) Lack of self - confidence
D) Lack of civic education

72. Which of the following was the common feature of the Ethiopian peasant rebellions that took place during the imperial regime at different places?
A) Participation of women as solders.
B) The fact that all rebellions were attached to land questions.

C) The fact that all rebellions were suppressed by military action supported by air force.
D) The fact that all rebellions took place in northern Ethiopia.

73. How do we ensure equality between men and women in a democratic system?
A) By assigning women to all key positions in the government.
B) By creating equal chance and opportunities for both men and women.
C) By increasing the number of women politicians.
D) By providing free chance of education for women.

74. Article 20/2 of the FDRE constitution provides for accused persons to be given the charges against them in order to
A) Help them remember how and where the crime is done.
B) Prove that the legal system works in transparent manner.
C) Enable them Gather evidence to prepare their defense.
D) Reinforce popular confidence in legal system of the nation.

75. Which of the following African country is associated to a fairly recent phenomenon of genocide?
A) Rwanda
B) Burundi
C) South Africa
D) Democratic Republic of Congo

76. Which of the following statements about the African continent is not correct;
A) It is the most utilized river in Ethiopia.
B) It is the bond that links Ethiopia, Egypt and Sudan.
C) It is the only cross boundary river in Ethiopia.
D) It is the only river that has never attracted the attention of the Europeans.

77. One of the following personalities is associated to the 1st written constitution of Ethiopia:
A) Bejrond Teklehawariat Tekeleamriam.
B) Dejazzmach Kebede Tesema
C) Ras Desta Damtew
D) Ras Abebe Aragay.

78. Which of the following National Holidays is associated to the identities and cultures of Ethiopian people?
A) National Flag Day
B) Nations and Nationalities Day
C) National Day of Unity.
D) National Day of Ethiopian Workers.

79. Ethiopian Students like you come from different social, economic, cultural and linguistic backgrounds. Which of the following is the most essential virtue that you need to live in harmony in such a situation?

Telegram: https://t.me/ExtremeSeries

A) Civility
B) Tolerance
C) Open mindedness
D) Civic mindedness

80. Who is the Commander in Chief of the FDRE armed forces?
A) The President
B) The Minister of Defense
C) The Prime Minister
D) The Chief of Staff of the army

81. What makes the governments of Nazi party of Germany, Fascist party of Italy and National Party of South Africa similar?
A) Rule by minority racial group.
B) Segregation of colored group.
C) Absence of rule of law.
D) Gender inequality and oppression.

82. The fight against corruption could bear fruit if the government in power
A) Pays high salaries for its officials to deter them from taking bribe.
B) Manages to make all its officials honest and truthful.
C) Is fully committed to root out corrupt practices in the country.
D) Works with international anti-corruption organizations.

83. Read the following data carefully and answer the question that follows.

Country	Literacy rate	Infant mortality rate	Life expectancy rate	GDP/person (USD)
A	44.6%	94.4%	45 years	146.8
B	98%	5.6%	80 years	4,00,000

From the table above it can be safely concluded that
A) Literacy rate has positive relation with infant mortality.
B) Literacy rate has positive relation with life expectancy.
C) Infant mortality has positive relation with life expectancy.
D) Infant mortality rate has positive relation with GDP growth.

84. Identify the correct statement concerning HIV/AIDS from the given alternatives
A) HIV/AIDS has become a great obstacle against socio-economic development among societies.
B) The victims of HIV/AIDS can no longer contribute to the development endeavors of their respective nations.
C) Globalization and human mobility played their part in the expansion of HIV/AIDS.
D) HIV/AIDS is the only incurable disease.

85. Which of the following statements is correct regarding the prevalence or rule of law?
A) It forms a state dominated and ruled by judges.

B) It hinders the smooth functioning of courts.
C) It establishes rule by law courts and judges.
D) It guarantees the safety and security of citizens.

86. Which of the following is NOT among the moral values of Ethiopian societies?
A) Supporting the poor
B) Respect for elderly people.
C) Conflict resolution through indigenous mechanisms
D) Ethnic based violence

87. Which of the following factors essentially necessitated the revision of 1931 constitution in 1955?
A) The promulgation of Universal Declaration of Human Rights.
B) The Federation of Eritrea with Ethiopia.
C) The outbreak of peasant rebellions.
D) The expansion of modern education.

88. What could be the essential base for Ethiopia's development endeavors?
A) Better education and fostering socio-cultural values.
B) The Grand Millennium Dam which is under constructions.
C) The expansion of global information centers.
D) Opening new legal migration opportunities to citizens.

89. Which of the following is INCORRECT?
A) Ethiopian is the seat of many international organizations.
B) Ethiopia is a member of many international organizations.
C) Ethiopia is the seat of ECOWAS
D) Ethiopia is a founding member of OAU.

90. What could be one of the long-term results of the constitutional protection of free speech?
A) Expansion of public radio.
B) More informed society
C) Growth of unbiased media.
D) Reliable election results.

91. Great Scientists and thinkers of all times contributed great things driven primarily by their desire to
A) Attain self-fulfillment through knowledge.
B) Eradicate poverty and injustice.
C) Transform their country and world.
D) Earn respect for their creations and thoughts.

92. A self - reliant person is one who is
A) Self sufficient in everything and requires no help from others.
B) Dependable to find solutions for serious social problems.
C) Refuses to accept the opinion of others at any time.
D) Aware of her/his own strengths and weaknesses.

93. Ethnocentrism is a belief that one's own ethnic group is superior in everything to others. Therefore it

Telegram: https://t.me/ExtremeSeries

A) Insures respect by other ethnic groups.
B) Demands the loyalty of other ethnic groups.
C) Deserves recognition by other ethnic groups.
D) Hinders interaction with other ethnic groups.

94. If a government truly respects rule of law in any country then
A) There will be a democratic constitution.
B) There is no need to hold periodic elections.
C) It is not necessary to have courts and prisons.
D) Most certainly justice will prevail.

95. League of Nations has indirectly encouraged the 1935 Italian invasion of Ethiopian by:
A) Forbidding Russia to come to the support of Ethiopia.
B) Impeding British and French support to Ethiopia
C) Passing arms embargo on both Ethiopia and Italy.
D) Refusing to send peace keeping troops to Ethiopia.

96. Article 20/7 of the FDRE constitution stipulates that an accused person has a right to:
A) Be tried in a court that uses language o f the accused.
B) Request for assistance of an interpreter at government expense.
C) Keep silent and refuse to give testimony against oneself.
D) Not to appear in court and not be accused of contempt of court.

97. The Universal Declaration of Human Rights is incorporated in the FDRE constitution. This is significant because it:
A) Guarantees full respect of the stated right for the peoples of Ethiopia.
B) Compel future governments to respect them for the people of Ethiopia.
C) Will serve as a standard for respect of human rights in the country.
D) Enables the UN to take action against a government that abuses them.

98. Unlike previous times the Nations, Nationalities, and Peoples of Ethiopia have the right to:
A) Promote and the develop their language and culture.
B) Move and settle anywhere in the country.
C) Join the various ranks of the country's civil service.
D) Marry in to and live together with other ethnic groups.

99. "In theory, markets in free market economies are regulated by the relationship between demand and supply". What does this statement imply?

A) There could be other practical factors that regulate free market economy.
B) Demand and supply exist only in the markets of fee market economy.
C) Others types of economies do not have a theory that regulates them.
D) Free market economy is healthy because Demand and Supply grow equally.

100. Identifying the commonly shared valued, interests and goals of the Ethiopian Nations, Nationalities and Peoples and respecting them is important to:
A) Facilitate national assimilation.
B) Bring national consensus.
C) Celebrate festivals together.
D) Appreciate the disparity among them.

Telegram: https://t.me/ExtremeSeries

CIVIC AND ETHICAL EDUCATION ANSWERS WITH EXPLANATION 2001/2009

1. A. The principle of democracy which asserts that the power of government derives from the freely given consent of the people is known as democratic election.
2. D. The FDRE foreign policy is based on the principles of self interests including promoting development, developing democratic system and national pride and globalization not territorial expansion.
3. B. According to Article 46 of FDRE constitution the regional states shall be delimited on the basis of the settlement patterns, language identity and consent of the peoples concerned.
4. A. Negotiation is the mechanism of conflict resolution which involves the coming together of those who are in conflict to make discussion
5. B. In Ethiopia the affirmative action policy is important for bridging the gender gap in unfair division of labour at the house hold level.
6. A. Citizen's devotion and commitment for the protection of democratic values and ideas is known as patriotism.
7. D. In Ethiopia's constitution history four constitution were promulgated the 1st was in 1931, the 2nd the 1955, the third was in 1987 and the 1993 constitutions.
8. D. Responsibility is the state of being answerable for an obligation.
9. D. Active involvement of citizens in socio-economic and political sphere is known as community participation.
10. B. The state of processing truths, facts, principles or other objects of perception through instruction, study and research is known as knowledge.
11. B. All citizens of democratic country are bound to be abided by the constitution of their country. They should do so even though some of them disagreed with some of its provision.
12. B. In democratic system citizens give state official authority so that the official rule the country on behalf of the electorate.
13. A. The House of Peoples' Representatives is the highest authority of the Federal government that empower the executive organ to enforce rules and regulations.
14. A. The principle of constitutional law which protect the life, liberty and property of a person is known as criminal law.
15. C. In Ethiopia true patriotism had for along period meant a person's readiness to sacrifice oneself for the country.
16. C. Mediation is the most formal methods of peaceful conflict resolution.
17. A. According to the constitution the exectutive branch of government given the power to ratify international treaties.
18. C. Civil disobedience is utilizes mainly illegal method of protest.
19. B. The functions of court includes.
    - Enforcing judicial decisions

Telegram: https://t.me/ExtremeSeries

    - Deciding over controversies
    - Imprisoning convicted criminals.
20. C. Air pollution is one of an environmental hazard that is caused by a country's unwise utilization environment of other states.
21. A. Accountability of states is explain the system in which elected public officials are made responsible to the people that elected them.
22. D. The similarity between parliamentary and presidential democracy is that in both case the judiciary is independent from legislative and executive branches.
23. D. The source of authority of democratic government is the consent of the governed (people)
24. C. The necessity of tolerance in a diversities society it fosters a variety of view points, new ideas and ways of looking at and solving problems.
25. C. The exclusive power and functions of the Federal government of Ethiopia, regulating foreign exchange, establish and administer national defense, administer national bank deploy armed forces, declare and lift national state of emergency and soon...
26. D. Accountability and Transparency used to develop performance.
27. D. Rules and Laws can be evaluated by using fairness or free from prejudice, favoritism and self interest is easy to understand and well – designed to achieve its purpose and clear as to what is expected to be achieved.
28. The reciprocal relationship between rule of law and democratic system in a country prevalence of good governance.
29. B. The difference between limited government and unlimited government is that the limited government is characterized by the constitution is the highest law.
30. A. The consequences of the absence of the rule of law is anarchy and arbitrary rule, absence of predictability and absence of independent judiciary.
31. D. The consequence of corruption includes discouraging the prevalence of the rule of law, dangerous to national security and order, under development the rise of the coast of goods and services, antithesis to socio – economic development.
32. B. The nations, nationalities and peoples of Ethiopia have equal right to share the results of the country's development.
33. A. Affirmative Action is important to build the ability of the nation and nationalizes of Ethiopia to self administration.
34. A. The impact of terrorism on the effort to build a democratic society is disrupts peace and security.
35. B. The reason that the Judiciary is setup as an independent organ of FDRE is administer justice without the interference of the executive and legislative organs.
36. C. The patriotic actions includes – maintaining domestic peace defending

national interests and standing for the truth.

37. B. The present Ethiopian Agricultural Development Led Industrialization (ADLI) economic strategy primary aimd at developing peasant farming system.
38. B. Actions should be taken to present our historical heritages from destraction includes proper protections registration and documentation of our historical heritages.
39. B. The National responsibilities of democratic government includes protecting the nation's cultural and historical heritages.
40. D. Gender sensitive work places explains women's disadvantaged position in work place.
41. A. Efficient use of time in work places improve productivity indicated the relation ship between time and productivity.
42. A. Self respect does not prevent a person from appreciating the good work of others because he/she knows that individuals can acquire different skills and specializations.
43. D. Ethiopia's continued dependency on of foreign aids can be accounted for absente of shortage of human resources or skilled human power.
44. A. One of the impact of extravagant practices on the national economy is reducing the capacity for investment.
45. D. A person is not expected to pay tax only because he/she is a citizen of a certain national state.
46. C. The major obstacle of implementation of the various protocol to reduce global warning is refusal of the USA to agree to the protocols.
47. B. The cultural and historical heritages of Ethiopia have invaluable importance for the peoples of the country in a sense that they enable to earn money from tourists that came to visit them.
48. C. The difference of HIV/AIDS pandemic from other pandemics is the HIV/AIDS poses threat to the development of the entire human race.
49. C. Disciplinary measure may not be taken on the worker who fails to establish good relationships with other workers.
50. D. Leisure refreshes the mental and physical capacity of worker. As a result it increase their efficiency and productivity then it is possible to conclude that it is in the best interest of employers should arrange sufficient leisure time only for workers who are productive.
51. A. League of Nations was the first International Community to cooperate for peace between the two world wars.
52. B. Nationalism is different from patriotism in that by its very nature it discourages cultural superiority of a group over other.
53. C. Workers job satisfaction is based on their attitude and the value the give to the job.
54. B. Ethical rules of certain profession enable workers to minimize misunderstanding between them by enabling them to realize the similarity and difference between various professions.

55. C. Any type of assistance should be executed properly to produce the desired result. Otherwise it may back fire and may end up in making the beneficiaries loathe the donors.
56. A. Citizens are advised to buy domestically produced goods as much as possible this is because these goods enable the country to save foreign currency and accumulate capital.
57. D. An International monetary fund is one of the Brittan woods institutions.
58. C. The Agricultural Development Led Industrialization Policy of Ethiopia emphasizes the need to encourage private investors to participate in agricultural development.
59. C. Saving is using resources in a planned manner.
60. B. Microfinance institutions are extended credit in cash or kind to economically active poor people.
61. A. Banks get the money they lend to various investors from personal saving of individuals process from their own investment.
62. D. Voting in elections is the most common mode of political participation in which all legal citizens can engage in and that requires nothing more than willingness on the part of participant.
63. B. Money is considered as one of the most important factors that could increase the chance of party to win election because it would enable the party to disseminate its policy and agenda among the electorate easily.
64. A. Group can be best defined as two or more people who interact to achieve a defined objective.
65. C. Eventhough you are refused to vote for election so you have respect the decision of the majority and respect her/his leadership.
66. B. An effective leader gets others to perform a job willingly if above all other trait he/she has persuasiveness ability.
67. C. Modern psychologists termed human needs for food, shelter, security and so on as the societal needs.
68. B. Stereotyping a group of people as having a certain good or bad trait in common it is usually wrong because is related to only to bad traits that people have in common.
69. A. All living things need air to live is false promise.
70. A. Both individuals and group rights overlap in the area of equality.
71. D. One of the advantage of Federal state structure for Ethiopia is securing peoples right to self governance.
72. A. Providing equal opportunities for all peoples of Ethiopia not leads to conflict of interests leading to violence.
73. B. One of the way by which Ethiopia protest her national interest is reducing a foreign threat or danger.
74. C. Valid reason about the necessity of fighting corruption to promote good governance.
75. D. The need for limiting the power of the Federal Government of Ethiopia

Telegram: https://t.me/Extremeseries

to secure peoples right to self government, it provides peoples with opportunities to work togther for their common good and it limited the authority of the centeral government.

76. D. The common good is the sum total of personal benefits
77. A. Irresponsible use of natural resources practices contribute to environmental degradation.
78. D. The International Community gives a great concern to HIV/AIDS pandemic because it is a global issue affecting the human race as a whole.
79. A. One of the indicator of people with lack of confidence is absence of fair assessment of one's potentials and limitations.
80. C. Political organizations different from civic societies political organization monitoring and influencing public policy.
81. B. One of the role played by civic societies in Ethiopia is enacting public policy.
82. B. Participating in a civic society enable student to improve your academic performance.
83. D. The process of gathering relevant information for acquiring knowledge is recording, transcribing and analyzing information.
84. B. The constitution of the FDRE gives the Federal Government the right to intervene in the internal affairs of regional states of the Federation when the violation of human rights of citizens is reported in that state.
85. C. The basic ground for International Human Rights Organizations persistently campaign for the abolition of capital punishment is that a right to life is respected by international conventions.
86. B. A country is accepted as new member of the UNO only if the General Assembly accepts its application by 2/3 majority.
87. A. Gender roles and expectations it is the possible to see variations in them between and within cultures.
88. C. The direct effect of corruption on a country's economic development because it weakens a government's ability to invest in vital social services.
89. D. Article 20(2) of the FDRE constitution states "Accused person have the right to be informed with sufficient particularly of the charge against them and to be given the charge against them" This provision is important because it enable the accused to gather evidence and prepare his/her defense.
90. A. Registering the income and wealth high level government officials are required before they assume office because it could be relatively easy to identify officials who are living beyond their means and take appropriate measures in time.
91. D. Any act is considered a crime only if it is declared as porohibited by an international or national law.

Telegram: https://t.me/ExtremeSeries

92. B. Terrerists are willing to use anything to make their targeted society live in fear.
93. B. The 2008 financial crisis that began in the USA has triggered economic crush in many countries. This situation could not have happened if the economies of the countries of the world had been independent of each other and been self sufficient by themselves.
94. A. Western culture in developing countries as harmful because they believed that it could cause the target people to repudiate their local identities.
95. C. If judges try to please their superiors and fail to guided by the law while making decisions they would undoubtely damage high reputation of court as stable institutions which stand for justice.
96. B. Public officials are expected to have high standared of conduct because all of them are expected to provide service to all citizens in an equal and just manner.
97. C. Civil servant refers to all employees of government in non – military sector.
98. B. An action is moral for one person may be found immoral for another. This is because individual who have different backgrounds may perceive things in different way.
99. B. Common cause of peasant rebellion that took place during the relign of Haile Selassie I was unfair taxation system
100. D. Political instability and social disorder is the consequence of the absence of organized political leadership for a popular struggle.

CIVICS AND ETHICAL EDUCATION ANSWERS WITH EXPLANATIONS 2002/ 2010

1. A. According to the Federal Constitution of Ethiopia both the Federal and Regional government have their own branch of government. The legislative power at state level resided on the state council.
2. B. The universal declaration of Human Rights adopted by thes United Nation Organization in 1948 to ensure the democratic and human rights.
3. A. All member states of the united Nation Organization have agreed to accept all decision passed by the General Assembly
4. A. The Legislative branch of the FDRE is accountable to House of people Representative and House of Federation.
5. C. The executive members of the kelebe pass decision on the matter that affect the residents of the keble. The residents are subjected to those decisions because the kebele executive capable of pusing those who disagreed with them.
6. C. The National emblem the state of the Ethiopian flag reflects the hope of the peoples of the country to live together in equality and unity.
7. C. The Federal Democratic Republic of Ethiopian constitution is above all the other laws that are culture specifics and supreme law of the land.
8. A. Litigation refers to the method of dispute settlement by the parties with the guidance and facilitation on the third party.
9. B. To serve the public fairly and equally government officials should necessary by respectable.
10. A. The necessary precondition that people who want to form an association should have objectives or goals in common.
11. D. Negotation is the most formal and binding conflict resolution when you are trying to reason with someone about something that is negotiation.
12. B. Ethiopia is the home of the head quarters of
    - the united Nation Economic Commission for Africa (UN-ECA)
    - African Union (AU) parliament
    - the African Union (AU)
13. A. Costly marriage ceremonies and 'Teskar' are among the widely known extravagant practices in Ethiopia.
14. B. Extravagancy is the habit of spending more money than one can afford.
15. D. HIV/Aids positive persons can still innovate despite their health problem.
16. C. The current constitution of FDRE was drafted in 1991 and promulgated by Nations, nationalities and peoples of Ethiopia in 1995.

17. A. The Judiciary and courts at state leve are specialized organ of the government is concerned with interpreting the law of the state.
18. D. The accused person has the right to use his/her language and even to examine witness against him/her is a minimum standard to make affair trial.
19. C. Any idea is labeled as scientific when it is experimented in laboratory.
20. D. The traditional methods of saving includes Iqub and saving in tangible asset where as Banking is the modern institution in Ethiopia.
21. C. According to article 52 of the FDRE constitution the power given to the regional states includes enacting and executing state constitution.
22. B. In a democratic system protection for the right of minorities are guaranteed by the law and institutions of the country.
23. B. Affirmative Action is the positive measure that the governments take to guarantee equal chance of opportunities for citizens that have deprived of opportunities in the previous regimes such as women and ethnic groups.
24. C. One mark of the USS Revised constitution of Ethiopia is that it was presented as a grant given by the emperor to his subject.
25. D. Article 89/2 provides that 'Government' has the duty to ensure that all Ethiopians get equal opportunities to improve their economic condition and promote equitable distribution of wealth among them this use the wealth of the nation among the citizen on equal basis.
26. B. Many parents of disabled children used to hide their children rather than sending them to school primarly because they fear social stigma caused by ignorance about disability.
27. D. Government policies that promote rights to equality among citizens may in fringe on liberty of some individuals. Nevertheless such policies are introduced because government are first and foremost desirous of realizing the common good of society rather than individuals.
28. B. The Ethiopia penal law, civil law and property law are primarly desined to keep social peace and do away with anarchy.
29. C. During the initial year of the dergue regime binding decision on the life, liberty and property of Ethiopia was mostly taken by committees at various levels.
30. C. An agreement or Treaty that signed by more than two countries is called the multilateral treaty.
31. B. Globalization has resulted in greater interdependence among nation of the world
32. D. According to Article 20/1/ FDRE, constitution, a court may hear a case in closed session such as:

Telegram: https://t.me/ExtremeSeries

to secure peoples right to self government, it provides peoples with opportunities to work togther for their common good and it limited the authority of the centeral government.

76. D. The common good is the sum total of personal benefits
77. A. Irresponsible use of natural resources practices contribute to environmental degradation.
78. D. The International Community gives a great concern to HIV/AIDS pandemic because it is a global issue affecting the human race as a whole.
79. A. One of the indicator of people with lack of confidence is absence of fair assessment of one's potentials and limitations.
80. C. Political organizations different from civic societies political organization monitoring and influencing public policy.
81. B. One of the role played by civic societies in Ethiopia is enacting public policy.
82. B. Participating in a civic society enable student to improve your academic performance.
83. D. The process of gathering relevant information for acquiring knowledge is recording, transcribing and analyzing information.
84. B. The constitution of the FDRE gives the Federal Government the right to intervene in the internal affairs of regional states of the Federation when the violation of human rights of citizens is reported in that state.
85. C. The basic ground for International Human Rights Organizations persistently campaign for the abolition of capital punishment is that a right to life is respected by international conventions.
86. B. A country is accepted as new member of the UNO only if the General Assembly accepts its application by 2/3 majority.
87. A. Gender roles and expectations it is the possible to see variations in them between and within cultures.
88. C. The direct effect of corruption on a country's economic development because it weakens a government's ability to invest in vital social services.
89. D. Article 20(2) of the FDRE constitution states "Accused person have the right to be informed with sufficient particularly of the charge against them and to be given the charge against them" This provision is important because it enable the accused to gather evidence and prepare his/her defense.
90. A. Registering the income and wealth high level government officials are required before they assume office because it could be relatively easy to identify officials who are living beyond their means and take appropriate measures in time.
91. D. Any act is considered a crime only if it is declared as porohibited by an international or national law.

Telegram: https://t.me/ExtremeSeries

92. B. Terrerists are willing to use anything to make their targeted society live in fear.
93. B. The 2008 financial crisis that began in the USA has triggered economic crush in many countries. This situation could not have happened if the economies of the countries of the world had been independent of each other and been self sufficient by themselves.
94. A. Western culture in developing countries as harmful because they believed that it could cause the target people to repudiate their local identities.
95. C. If judges try to please their superiors and fail to guided by the law while making decisions they would undoubtely damage high reputation of court as stable institutions which stand for justice.
96. B. Public officials are expected to have high standared of conduct because all of them are expected to provide service to all citizens in an equal and just manner.
97. C. Civil servant refers to all employees of government in non – military sector.
98. B. An action is moral for one person may be found immoral for another. This is because individual who have different backgrounds may perceive things in different way.
99. B. Common cause of peasant rebellion that took place during the relign of Haile Selassie I was unfair taxation system
100. D. Political instability and social disorder is the consequence of the absence of organized political leadership for a popular struggle.

CIVICS AND ETHICAL EDUCATION ANSWERS WITH EXPLANATIONS 2002/ 2010

1. A. According to the Federal Constitution of Ethiopia both the Federal and Regional government have their own branch of government. The legislative power at state level resided on the state council.
2. B. The universal declaration of Human Rights adopted by thes United Nation Organization in 1948 to ensure the democratic and human rights.
3. A. All member states of the united Nation Organization have agreed to accept all decision passed by the General Assembly
4. A. The Legislative branch of the FDRE is accountable to House of people Representative and House of Federation.
5. C. The executive members of the kelebe pass decision on the matter that affect the residents of the keble. The residents are subjected to those decisions because the kebele executive capable of pusing those who disagreed with them.
6. C. The National emblem the state of the Ethiopian flag reflects the hope of the peoples of the country to live together in equality and unity.
7. C. The Federal Democratic Republic of Ethiopian constitution is above all the other laws that are culture specifics and supreme law of the land.
8. A. Litigation refers to the method of dispute settlement by the parties with the guidance and facilitation on the third party.
9. B. To serve the public fairly and equally government officials should necessary by respectable.
10. A. The necessary precondition that people who want to form an association should have objectives or goals in common.
11. D. Negotation is the most formal and binding conflict resolution when you are trying to reason with someone about something that is negotiation.
12. B. Ethiopia is the home of the head quarters of
    - the united Nation Economic Commission for Africa (UN-ECA)
    - African Union (AU) parliament
    - the African Union (AU)
13. A. Costly marriage ceremonies and 'Teskar' are among the widely known extravagant practices in Ethiopia.
14. B. Extravagancy is the habit of spending more money than one can afford.
15. D. HIV/Aids positive persons can still innovate despite their health problem.
16. C. The current constitution of FDRE was drafted in 1991 and promulgated by Nations, nationalities and peoples of Ethiopia in 1995.

17. A. The Judiciary and courts at state leve are specialized organ of the government is concerned with interpreting the law of the state.
18. D. The accused person has the right to use his/her language and even to examine witness against him/her is a minimum standard to make affair trial.
19. C. Any idea is labeled as scientific when it is experimented in laboratory.
20. D. The traditional methods of saving includes Iqub and saving in tangible asset where as Banking is the modern institution in Ethiopia.
21. C. According to article 52 of the FDRE constitution the power given to the regional states includes enacting and executing state constitution.
22. B. In a democratic system protection for the right of minorities are guaranteed by the law and institutions of the country.
23. B. Affirmative Action is the positive measure that the governments take to guarantee equal chance of opportunities for citizens that have deprived of opportunities in the previous regimes such as women and ethnic groups.
24. C. One mark of the USS Revised constitution of Ethiopia is that it was presented as a grant given by the emperor to his subject.
25. D. Article 89/2 provides that 'Government' has the duty to ensure that all Ethiopians get equal opportunities to improve their economic condition and promote equitable distribution of wealth among them this use the wealth of the nation among the citizen on equal basis.
26. B. Many parents of disabled children used to hide their children rather than sending them to school primarly because they fear social stigma caused by ignorance about disability.
27. D. Government policies that promote rights to equality among citizens may in fringe on liberty of some individuals. Nevertheless such policies are introduced because government are first and foremost desirous of realizing the common good of society rather than individuals.
28. B. The Ethiopia penal law, civil law and property law are primarly desined to keep social peace and do away with anarchy.
29. C. During the initial year of the dergue regime binding decision on the life, liberty and property of Ethiopia was mostly taken by committees at various levels.
30. C. An agreement or Treaty that signed by more than two countries is called the multilateral treaty.
31. B. Globalization has resulted in greater interdependence among nation of the world
32. D. According to Article 20/1/ FDRE, constitution, a court may hear a case in closed session such as:

Telegram: https://t.me/ExtremeSeries

- protecting the privacy of the parties concerned
- when the case is considered to be harmful to public moral
- when the case is believed to Jeopardiee national security

33. A. The Ethiopian Federal Ethics and Anti corruption commission is trying to combat corruption is by raising the awarness of the citizen about the evilness of corruption.
34. D. According to Article 521 the Ethiopian penal code, a person who deliberately transmits deadly disease to a healthy person could be reprimanded for being negligent
35. B. At present countries of the world collaborate to fight terrorism because they considered it as an act that is a threat to global peace and security.
36. C. When the House of peoples' Representatives of the FDRE ratifies an international agreement signed by the executive, that government becomes on integral part of the law of the land.
37. C. Our national environment could be protected effectively only if every body stires for its protection actively are willingly.
38. C. Cultural and ethic similarities are the factors that considered for demarcating the boundary of the regional states of FDRE.
39. D. Historical and cultural heritages of Ethiopia is found scattered throughout the country.
40. --- According to the pragmatism theory truth is
41. B. Women's Associations are established with the main aim of struggling for attaining social, political and economic equality of the sexes.
42. C. Use of terror by the Jacobins as means of political repression and social control was the negative legacies of the French Revolution.
43. A. It is Just enough to quality a state as realy constitutional one if the people enjoying full rights and freedom as given to them by the constitution, peoples feeling as citizens and having confidence in the law of state and keeping trust in their leaders.
44. C. Totalitarianian rule is the political system where the state usually under the control of single political person. The people are rather subjects not realy citizens.
45. A. The notion of sovereignty essentially refers to the state of being a country with freedom to govern itself.
46. C. In free market economy the market is governed, theoretically by the relationship between demand and supply.
47. C. If you seriously believe that no one will stand against you in this country in solving mathematical problem you could be said to have exaggerated view of self-esteem
48. A. A democratic state differs from undemocratic state in that it furnishes its people with equal opportunities for advancement.

49. B. Contempt to the rule of law by leaders of the state could entail political advancement
50. C. The implication of the rule of law includes
    - democratic system
    - the supremacy of the constitution
51. D. Factors to evaluate the good of law are uniformity, acceptability and partiality
52. B. Separation of power among agents of the government implies that each agents has specialized responsibilities to work with a degree of autonomy.
53. B. Transparency important for building democratic system because it bring open and accountable bureaucracy.
54. D. The concurrent functions and responsibilities betweens the two levels of state in Ethiopia includes – levying taxes and collecting revenue.
    - Land and natural resource management
    - protecting the Federal constitution
55. C. One of the advantage of having a written constitution is the existence of a single and accessible document stating the rights of citizens.
56. D. Terrorism is the systematic use of terror especially as means of coercion. It shatters peace and stability.
57. C. Low level of development is taken into account by the federal government for provision of social services.
58. A. Less developed states such as Gambella, Beni-shangul-gumuz, Afar and Somali are benefiting from package of affirmative actions.
59. C. HIV/AIDS has no racial, linguistic or religious bonds.
60. D. The equal right of states to full measure of self governance best express the quality between and among peoples of Ethiopia
61. C. Open-mindedness people with good moral character final it easy to appreciate the good work of other.
62. A. The Responsibilities of democratic government includes
    - protecting citizens well being and property
    - provision of service of citizens
    - maintenance of law and order
63. C. Responsible citizen to prevent your country historical heritage from destruction discuss with other to try to find out ways of preserving it.
64. C. Joining environmental campaigns is bad practice contributing to environmental degradation.
65. B. Among the civic responsibilities of patriotic citizens are tolerance of all culture civility, loyality to the constitution respect other human and democratic rights and laws of the country.
66. B. Violation of the rights of citizens by the state could effectively undermine citizens patriotic feeling as it diminishing the citizens sense of belong ness to the state.

Telegram: https://t.me/ExtremeSeries

67. D. In the past and even at the present day the women in Ethiopia, their representation in politics was very insignificant, they made to marry at early age, and required to carry enormous workload at home.
68. A. Establishing gener limitation for exercising voting right acts contradicts the principle that "people in similar situation should be treated similarly because it disseminated on the base of gender.
69. A. Science as a body of knowledge contributed to the development of the human sóciety through disseminating relevant information, predicting the future and offering solutions for various human problems.
70. C. Unlike political parties, civic societies are meant to provide social economic, and cultural opportunities for individuals or groups.
71. A. Civil activities that students could participate at school level include Joining environmental campaigns joining HIV?AIDS campaigns and joining fund raising activities.
72. C. Personal saving important for the growth of investment personal savings reduce extravagance and enhance accumulation of capital for investment.
73. B. For executing their public functions all nations, nationalities and peoples of Ethiopia can use the language that is their mother toung alone.
74. B. Statistics show that USA emits 20 tons of carbon per person and chine emits 6 tons of carbon per person. This information enables us to identify the country that makes that the largest carbon emission provides that the initial year taken for comparisons is varied.
75. A. In theory, a global citizen is a person who is readly to take action to make the world a more just and equitable peace.
76. B. The major principle that is the basis for the present foreign policy of Ethiopia is seeking and supporting peace for solution to international disputes.
77. C. Adam smith states the rich should pay more tax than the poor. The argument he present in support of this theory was that the poor with not have enough resources to pay as much as the rich.
78. A. Establishing different facilities such as mobile schools for their children's, health center and other activities that the FDRE government is undertaking to develop the pastoralist citizens and done with out underming their life style and pastor wist economy.
79. B. Stereoty ping based on sex harts the frneale members of society by ing or forcing them to do the easies and less rewarding types of jobs.
80. D. In inter-personal relationships – self – confident individuals are always willing to accept powerful and logical arguments of their opponents.

81. B. Ethnocentrism is dangerous for harmonious living of peoples because it stresses gaining advantage for one group at the expense of other
82. B. One reason why agricultural development led industrialization strategy in chosen over other strategies is because Agriculture is the occupation of the majority of the countries people (85%)
83. C. In Ethiopia, investment in social services is mainly the responsibilities of the government because benefit from social services is regarded as right of citizens.
84. A. Procedural Justices refer to the ability at legal system to gather information's and make decision impartially and openly.
85. B. Limiting the power of the Federal government of Ethiopia can be explained interms of ..... avoiding a centralized system of administration marked by cumbersome work and express their creative capacities and regional states enjoying a degree of autonomy for self determination.
86. B. If the country currently had a unitary state structure the peoples of Ethiopia would mis the right of self governance.
87. C. Presidential democracy is a state in which the presidents as the nation's head of state and active chief executive authority.
88. D. The difference between individual right and group right is individual rights aim at promoting individual human development while group rights aim at promoting common interests of group.
89. B. Democratic Rights are generally the right to participate in political activities and the right to a democratic form of government. It is political in type and revocable in its nature.
90. B. One of the characteristics of the qualitative dat is describes facts that can not be expressed numerically.
91. B. The character of a hard working person includes determination to act, he/she is conformists and to loyal and obedient.
92. C. The basic reasons for giving emphasis to speed up technical and vocational education in Ethiopia is to reduce brain drain, promote industriousness and self production process and reduce rural urban migration.
93. C. cumbersome bureaucracy is one of the factor that could enhance productivity of the individual in Ethiopia.
94. C. Time management enables us to plan our daily activities, thus achieving the most productive use of time, increases efficiency and productivity and reduces cost of productivity.
95. D. The reason for giving great concern and publicity to the HIV pandemic by the international community is because as an incurable disease the spread of the pandemic is threat against the whole of human races.

Telegram: https://t.me/ExtremeSeries

96. B. Respecting the sovereignty of other states is one has to do with the international responsibility of any government whether it is democratic or undemocratic government.
97. B. Gathering information creates a wide extent for further understanding of the issue.
98. C. The Role of civic societies required to play the provision of educational services campaign ob social and cultural affairs and providing cultural opportunities.
99. D. In order to succeeded in his/her educational, a student has ti properly plan and utilize his/her time as well as his/her resources.
100. A. Extravagance practices are affect saving at a national level, so do they affect the national econ

2003 E.C GRADE 12TH CIVIC AND ETHICAL EDUCATION EXAM ANSWERS WITH EXPLANATIONS.

1. **A** - The major duty of the House of Federation in Ethiopia is to interpretive the Federal constitution of Ethiopia.
2. **B** - The group right of the Nations, nationalities and peoples of the Federal democratic Republics of Ethiopia includes the rights of self, governance, the right of self determination and the right to develop one's own culture.
3. **C** - The first written constitution of Ethiopia was drafted in 1931 and revised in 1955 during the region of Emperor Hailesulassie I.
4. **A**- Referendum is a vote taken on an important issue by all peoples of the country.
5. **A** - The judicial branch of the government is responsible for the enforcement of the laws of the country.
6. **B** Responsibility is the state of being answerable for an obligation.
7. **B**- Justice (fairness) is treating citizens fairly in the distribution of benefit and burdens in the country.
8. ***Iqub*** is the most widely practiced indigenous mechanism of saving in Ethiopia.
9. **A**. Agriculture is the major or the back bone of the Ethiopian economy
10. **C**- Investment is a mechanism by which people spend their money to get profit expand their business and contributed to the development of the national economy.
11. **B**- The principle of the any democratic constitutions include popular sovereignty, Accountability of the government, limited government power, separation of powers check and balance, Judicial reviews and federalism.
12. **A**- Good governance is the most important prerequisite for the political social and economic development of a country.
13. **D**- The best instrument by which a country executes its foreign policy is called Diplomacy.
14. **A**- Anarchy is a situation where there is no supreme power or law and where law less ness reigns in the country.
15. **D**- Ethiopia has played very crucial role is the struggle for the independence of Zimbabwe in 1970s.
16. **D**- Most of the dead in recent crime of genocide in Rwanda in 1994 were Tutsis and most of those who perpetuated the violence were the Hutus in Rwanda.
17. **C**- A person could better enjoy his/ her civil and political rights it he/ she has a minimum level of societal respect.
18. **B**- Responsible citizens of a country are expected to stand against genocide, terrorism, and corruption, drug trafficking.
19. **D**- Warranty is the payment that a person who wants insurance service pays to the insurance.

Telegram: https://t.me/Extremseries

20. **B** - Doctor Aklilu Lemma is known for discovering the medicine to cure bilharzias.
21. **B**- The difference between civil action and political participation is that civil actions are activities related to public or citizens outside politics. Where as political participation linked to the political activities.
22. **C** - Factors for job satisfaction and effectiveness included higher interest in job, better payment, and positive interpersonal interaction.
23. **B** - The Eleven Rock-Hewn Churches of Lalibella are one of the cultural heritages of Ethiopia registered by UNESCO.
24. **C**- During the WWII Adolph Hitler's Nazi soldiers atrociously killed millions of Jewish people. This action called Genocide.
25. **A** - The basic factors behind unequal status between men and women with in the Ethiopia society were forced early marriage little or no access for education and unfair labor division.
26. **B**-The major significance of federalism for Ethiopia Nations Nationalities and peoples is above all self administration.
27. **B**- The FDRE constitution it clearly puts that ultimate political powers came from the people.

A- The principle of constitutional law which protects the life, liberty and property of a person is referred to as Individual Rights.

A - Athens was one of the Greek cities State is credited for its contribution of exercising direct democracy by about 1000 B.C.

30. **A**- Power is sharing among all member states of the Federal government of Ethiopia.
31. **A**- The sovereignty of the peoples of Ethiopia is expressed through their direct participation to elect their representatives.
32. **B**- The Ethiopia constitution is the supreme law if the country means all practice including customary ones have to conform to the constitution.
33. **C**- The main reason why many countries of the world are trying to build democratic systems at present is that because it is believed to be a necessary prerequisite for development.
34. **C**- presidential democracy is believed to be better than parliamentary democracy in establishing fast and fair justice system to citizens .
35. **D**- A government is said to have legitimate authority to govern if it functions with in the limited set by a constitution provided that the constitution a free expression of the will of the people.
36. **C**. One causes of grievance of the Gojjam peasants that drove the to rebel in 1966 was refusal to accept people appointed from areas outside Gojjam as governors.
37. **D**- it is necessary to fought corruption because it is threat to the well-being of any society.
38. **D**- In order to live together in unity it is necessary that Ethiopians should respect the cultures of each other.

39. **A**- According to the principle of equality of benefits and burden those who work hard should benefit more.
40. **A** -While you are enjoying your legal right it is mandatory that you refrain from violating the rights of others.
41. **C**- The Federal state have one Federal plus as many as the numbers of the federation.
42. **D**- Social harmony exists in a country where there is rule of law.
43. **B**- Progressive taxation system people who earn more money are required to pay more.
44. **C**- The main benefits that a government gains by performing its duties in a transparent manner is trust and respect of the people it leads.
45. **B**- In a democratic system power is diffused and mutual accountability instituted in order to stop parties from dominating the politics of a country.
46. **A**- most of the harmful traditional practice that affect women in rural Ethiopia can be eliminated by making the people a ware of their ill effects.
47. **A**- In democratic societies. Power of a government is believed to be Legitimate because it is derived from the authority given by the people.
48. **B**- Saving is (should) be understood as planned and wise use of money.
49. **A**- A pragmatic person is one who believes that truth is what universally accepted.
50. **A** The petition known as Habeas corpus is in a sense a protest against what the accused believed to be unlawful detention or imprisons.
51. **D**- If a person lacks self- confidence, she/ he would be expected to be a member of certain community, give respect to other people and engage in some type of work.
52. **B**- In the execution of his responsibilities as head of government the prime minister of FDRE is accountable to the House of Peoples Representatives.
53. **B**- It is possible to reduce the poverty situation in Ethiopia by providing quality basic education to the uneducated and eradicating illiteracy.
54. **B**- At present, goods from other countries should be found at cheaper price than those good produced in a country increased dependency on foreign countries and disintegration of the national economy.
55. **A**- An assertive person from an aggressive person is that a greeting with convincing idea of other people.
56. **D**- The short comings of the traditional **Idir** for which it is criticized by some people is that it does not support very needy members while they are alive.
57. **D**- The traits that is essential for civic participation includes, commitment to see things through, dedication to a cause and willingness to work with others.

Telegram: https://t.me/Exitnseries

58. **C** - In order to pass sound judgment on anything, it is necessary that the person should correct information on the issue.
59. **D-** The comparative advantage could Ethiopia offer to potential foreign investors more than any things else now is abundant work force that could be easily trained.
60. **C-** Banks play important role in the proper use of money in a country by lending money for most promising productive ventures.
61. **C-** "Democracy is the government of people, for the people, by all for all "Abraham Lincoln.
62. **B-**Corruption includes abuse of power, Embezzlement and Nepotism.
63. **B-**The right of equality is the one of the core element of democracy.
64. **C-**The qualities of true patriot's commitment for common good tolerance and culture of peace and defending national interests.
65. **A-**Active community participation implies involvement in the socio-economic affairs of a society.
66. **C-**In order to be more productive worker one should have the following qualities.
    - Punctuality at work place.
    - Good relationship with follow workers
    - Effective application of skills
67. **A-** Concurrent powers are the power that is shared by the state and the Federal government.
68. **A.** Contingency Leader Theory of leadership focus on particular variables such as the existence distinctive physical and psychological characteristics accounting for leadership effectiveness.
69. **D-** The Ethiopian foreign policy issued in 2003 directs an "inside out" approach as argued by some that Ethiopia gives priority to national pride poverty eradication and building a full fledge democratic system.
70. **D-** High school students are advised to participate in all of the following extra curricular activities such as scout club, Drama and art clubs…
71. **C-** Mr'X' was accused of serving as under lover spy for a state which was an enemy of his won country at a time Mr 'x' committed a crime of Revealing state secret.
72. **D** -The period between 1970 and 1980 was Known as a period of "silence" with regard to HIV/ AIDS implies it was a period of steady expansion of the pandemic.
73. **C** -The Ethiopian economic development is attributed to absence of good governance in the past.
74. **B-** The link between democracy and responsibility is that fulfilling our responsibility is one major element in attempt to promote democratic culture.
75. **D-** The best way curbing the alarming growth of HIV/ AIDS in Ethiopia is creating better consciousness about the pandemic than ever before.

76. **B-** The link between international economic realities and national economy in the context of Ethiopia is that the natural economy cannot operate independently and is influenced by international economic realizes.
77. **C-** Government institutions must be supported by civil societies and humanitarian organizations in the effort to eradication of poverty.
78. **C-** To achieve the planned goal of change and Transformation Ethiopia needs to apply scientific and technological knowledge and promote socio-economic values simultaneously.
79. **B-** Democratic right differs from human rights that democratic rights are granted by democratic institutions.
80. **D-** Both Nazi Germany and Fascist Italy possessed and exercised about unlimited power over their respective nations.
81. **B-** In Parliamentary democracy the prime ministers should be a number of parliament appointed from the majority party and leads the executive branch of government.
82. **C-** In The present Ethiopia context, the Federal system of government is mainly designed to unite the peoples of the country by respecting their diversity.
83. **B-** Some people argue against the principle of affirmative action because they claim that it goes against the basic principle of quality of rights of people.
84. **B-** When we say there is equality among citizens of a country what we mean is that the citizens are not treated differently because of their social and economic status.
85. **B-** The FDRE constitution safeguards equality of opportunity between citizens of the country means citizens should be provided with equal chance in all aspects of life.
86. **A-** Globalization has increased the interdependence among nations of the world.
87. **A-** The UDHR was adopted by the UN in 1948 with main aim of codifying human rights in international law so as to prevent atrocities such as the holocaust from happening again.
88. **A-** People would support and respect their government if they believe above everything else that it manages to bring economic growth irrespective of who is benefited.
89. **D-** Drug addicts could not be expected to responsible citizens is because there is a very high possibility that they could engage in criminal activities to satisfy their addiction.
90. **A-** if you contribute your share to your community and that contribution is recognized the very act of acknowledgement given to your contribution is bound to inspire you to do more for your community.

Telegram: https://t.me/ExtremeSeries

91. C- The rule of law and rights of citizens are better respected and protected in countries where the judiciary branch of government is independent and strong.
92. A- The principle of universal suffrage dictates that all legally able citizens of a country have the right to participate in election.
93. A- Paying tax is an obligation of every citizen who earn income in one way or other while overcoming this obligation the individual has the right to decide and make known the amount of taxes that he/ she shall pay.
94. A- Most of the people who stand and fight for the common good of their society usually do so out of conviction that societal interest is more important than individual wellbeing.
95. C -Alex is self- centered man and his personality is self-centered man and his personality is known to all people he known understanding the need of others.
96. A- Women were unable to contribute for the social and economic development of our country mainly because they were deprived of the opportunity to develop themselves and utilize their potentials
97. D- One morally acceptable in one culture could be immoral in another culture because moral values determine the culture of specific group of people.
98. C- The back ward attitude that has hindered Ethiopia women from participating in much income generating job so far the belief that women must be given care.
99. B- Ethiopia hosts the Economic commission for Africa along with Au and several UN regional bodies.
100. D- HIV / AIDS victims have been discriminated due to the poor level of awareness among the society.

Telegram: https://t.me/Extremeseries

2004 E.C UNIVERSITYENTRANCE EXAMINATIONSENE
2004/JUNE 2012 ANSWERS AND EXPLANATIONS

1. D. The right of any democratic country according the UNICEF convention signed in 1989 in Newyork childe means under age of 18 Childs have some creation basic rights including the right to life, his or her own name and identity to be raised by his or her parents, have the right to express their own opinion and right to have shelter.
2. B. The tendency to consider one's own culture as the only right one and believed to be superior to that of others is called **Ethnocentrism**.
3. D The transfer of power from Federal to the reginal government is known as **Devolution** of power.
4. C. According to principle of universal suffrage all legally able citizens of country are entitled to vote.
5. D The rotating member of the UNO Security council are chosen on the basis of their commitment to fight in justice.
6. D **Negotiation** is peaceful conflict resolution methods could be undertaken without involving a third party.
7. **B** The System of rule that was based on rigid racial division in the former South Africa was known as **Apartheid**
8. B. The three indicators of the human poverty are
    1. Short life expectancy
    2. Lack of basic education/illiteracy/
    3. Lack of access to public resources
9. B The International Monetary Fund is an institution that shaped the modern era of globalization
10. A. **Curative Approach** is one of the approaches that includes bringing corrupt officinal to Justice
11. B. Collection fire woods and fire woids water is generally taken as female oriented Job in Ethiopia tradition .
12. D. **Self awareness** is the proper recognition of our abilities, strengths and weakness.
13. C. According to provision of the current Ethiopian constitution all the sovereign political power rest on the nations, nationalities and peoples of Ethiopia.
14. C. **Obligations** of citizens in democratic system are includes:-
    - Promoting political tolerance.
    - Respecting the rights of others flag
    - Respect for the national
15. A. **Transparency** is free access to known the activates of government officials
16. B. The correct hierarchical order is constitution specific laws regulations.

17. C. The main motto /slogan/ of Ethiopian student movement in 1970s was "MeretLearashu" or "Land to the tillar"got wide support among the mass or the peasants of Ethiopia
18. C. The Rights of an accused person includes:
    - ✓ The right to appeal
    - ✓ The right to be considered innocent before the is found guilty
    - ✓ The right to access for information
    - ✓ The right to communicated in a language he/she understand
19. B. The colours of Ethiopian national flag represent
    - ✓ The Green represent prosperity
    - ✓ The Yellow represent hope and
    - ✓ The Red color represent scarifies respectively
20. D. "I have a dream ....." This inspiring quotation is associate with the Black Afro-American Human Right movement leader DR.Martin Luther King in 1960s
21. B. The demand fro Universal Suffrage in the past was a demand for legally able citizens of a country the right to vote.
22. C. One similarity between parliamentary and presidential democracies is that in both case state power is transferred form one party to another through peaceful way.
23. B. In the present world democracy is becoming more and more a preferred mode of governance became it is ensures social, economic and political equality among citizens.
24. B You will be considered a tolerant person if you primarily have the ability to accept and respect differences that exist between you and others.
25. C. One of basic principle for the present foreign policy of Ethiopia is seeking and supporting peaceful solutions to international disputes.
26. C. Two communities could solve disagreement on the use of grazing land through legal means and established procedures provide that the communities adhere to the values and principles of the constitution.
27. D one of the reasons for the prevalence of corrupt in may countries is the existence of greediness among many leaders .
28. D. The Targets of the Millennium Development Goals /MDG/ achieved by all states by the year 2015.

    Are:- 1. Eradicate extreme hunger and poverty
    2. Achieve universal primary education
    3. Promote gender equality and empower women
    4. Reduce child mortality
    5. Improve mathernal health
    6. Combat HIV/AIDs, malaria and other diseases
    7. Ensue environmental sustainability
    8. Develop a Global partnership for development

29. B. The rule of law above everything else indicates a country has a democratic government .
30. B **Horizontal equity** is an economic theory that sates that individuals with similar income and assts should pay the same amount of taxes.
31. B. The rule of law above everything so "No man is above the law" This quotation is designed to show the preeminence of the rule of law.
32. A. Individual rights is a legal term referring to what one is allowed to do and what can be done to an individual . It is an inheretpart of a human being. They are your birth given to you by nature and permit you to act in yourself interest with total control over your own life and property.
33. D. Egypt and Sudan closely follow any Ethiopian attempt to use Abayriver for development because they fear that Ethiopia could be developed and a threat to them.
34. D. In presidential democracy such as the USA, the president is elected by the popular vote independently from the representatives.
35. A Legal limitation could be laid down on the right to freedom of expression if that right is used for damaging the reputation to individuals
36. D. In a democratic system power diffused and mutual accountability is instituted in order to prevent domination of a country by an individual or party .
37. D **Nationalism** differsfrom**patriotism** mainly by the fact that it requires peoples of one country to live in peace and harmony.
38. D. The principle due process of law refers to the legal requirement that the state must respect all the legal right that are owned to a person. Due process of law balance the power of law of the land and protect individual persons from it.
39. B **The Whistle blower**is person who tells the public or some one in authority about alleged dishonest or illegal activities or misconduct occurring in government department, a public or private organization. Because they know what they are doing is unethical and immoral.
40. B. A constitution of regionalstate will be declared Null an Void if it is found to be in contradiction with the FDRE constitution because the FDRE constitution is the expression of the will of nations and nationalities and people of Ethiopia.
41. C In the pre 1991 Ethiopia the rights of nations nationalities were undermined
42. C. Negotation is a dialoged between two or more parties, intended to reach and understanding, resolve point of difference or gain advantage in out come of dialogue. It intended to aim at compromise.
43. A unity in diversity means in a multicultural Society is accepting and respecting cultural difference.
44. C. According to the FDRE constitution, who is responsible for signing international conventions and agreements is the council of ministers

Telegram: https://t.me/ExtremeSeries

45. C. In Ethiopia, any proclamation shall come in to forces on the date of publication of the NegaritGazea
46. B. Ethiopian Women in general had been economically dependent on their spouses mainly because they were derived of the right to own property .
47. C. The democratic rights includes the right to assembly expressing your opinion member of political party....
48. D. Building democratic system very essential into days world because it upholds vales like freedom, equality and Human rights.
49. D. A parliamentary system, is a system of government in which the ministers of executive branch get their legitimacy from legislature and are accountable to that parliament body Accordingly the president is the head of state, the prime minister is the chief executive.
50. B Base on Article 61 of the FDRE constitution The House of Federation is the representative of all federal states nations, nationalities and peoples. At least eachnations, nationalities and peoples represented by one representative additional representative for one million people members of House of Federation shall have elected by state council.
51. B. Article 79 of the FDRE constitution state that the Judge and Courts independence form any interference form government officials.
52. A. The Millennium Development Goals /MDGs/ includes expanding primary education, reducing infant mortality rate and eradicating genders inequality.
53. D According to Article 19/2/ of the FDRE constitution the defendant has the right to known why he/she accused
54. Apartheid was a radically based segregation system practiced by the white minorities in South Africa.
55. D The rule of law obliges citizens to behave in accordance to the constitution.
56. B The Federal constitution is the supreme law of the country has supremacy over the state one.
57. D. The situation of women in Ethiopia includes They are the backbone of Ethiopia Society positive discrimination is used to compensate women for the past injustice called Affirmative Action and they were victim of forced, unwanted marriage and abdication.
58. D. The immediate cause for Gojjam peasant rebellion on was the imposition of forced taxes and contribution by the imperial rule.
59. C **Cultural Relativism** is the view that all beliefs, customs and ethics are relative to individual with in his social context .
60. D Social injustice is the concept relation to the claimed unfairness or injustice of society in its divisions of rewards and burdens and other incidental in equalities'
61. B. Ethical relativism is the theory that holds that morality is relative to the norms of one's culture.

62. C The recommendation can be suggested to reduce the negative impact of widely known extravagant practices in Ethiopia is avoiding arbitrary, unplanned expenditure.
63. A. The behavior of responsible citizens includes involvement in public affairs, refusing to pay unreasonable taxes and protection the culture and heritages of the country.
64. A.The relationship between family responsibility and social responsibility is that family responsibility is a pre requisite for social responsibility.
65. A. The victim of HIV/AIDS discriminated among different societies. HIV/AIDS victims are discriminated because of wrong religious teachings.
66. A Terrorism has become a serious threat to our globe these day.
67. D. Handicraft men despised in the long established tradition of Ethiopia because of backward beliefs attached to their personality.
68. D Factors that can positively affect working environment includes is getting reasonable salary, observing professional ethic and cooperation with fellow workers.
69. B Assertive behaviors includes:- Sharing experiences playing a positive role during conversation and questioning meaningless traditions and practices.
70. A. Self-reliance will normally leads to self –confidence.
71. D Litigation is differs from other types of peaceful conflict resolution mechanism in that it is a formal and biding mechanism that is based on rules, customs or practices.
72. A Group rights are rights held by a group rather than by its member separately or a rights held only by individuals within the specified group in contrast.
73. B. The principle of the supremacy of the constitution that other laws in the country be accountable to their electorate.
74. B If a nation has democratic constitution that is respected by the government in power, rule of law prevails.
75. D. The federal state structure is chosen over the unitary one for Ethiopica because it provides equal opportumtics for peoples to develop themselves.
76. B. The difference between the federal and regional constitutions of the FDRE is that the Federal constitution is the binding in all reginal state while the reginal constitution will not have full legal forces in other regions.
77. C. Voluntary organization is a group of individuals who enter into an agreement as volunteer to from a body or organization to accomplish its purpose.
78. A. It is fact that Ethiopia is one of the poorest countries of the world while having abundant natural and cultural resources .

Telegram: https://t.me/ExtremeSeries

79. C. The developing countries of the world are highly dependent on rich nations. This practice has brought about increased in dereference of rich nations in the internal affairs the poor nations.
80. C. Check and balance power system in more strong in presidential democracy because the members all three branches of government are directly responsible for their electorate.
81. A. The newly launched construction of the Grand Millennium Dam on the Abayriver is an indication of Ethiopia's long waited wish to utilize hre natural resources for socio-economic development.
82. C The major manifestation of equality of Ethiopian Nations and Nationalities and Peoples are the right to self governance and promotion of their culture and history.
83. B. There is harmony between group rights and individual rights.
84. D The right to equality is one of the citizen rights enshrined in constitution of the FDRE to ansure equality opportunities are available for all citizens realize their potential and improve themselves.
85. D. A leader is said to be a motivating one if he/she is able to give employees a reasons to work efficiently.
86. A. For good governance and Justice to prevail in a country the fulfillment of which condition is more important for prevalence of respect for rule of law.
87. A. **Scientific truth** differs from other theories of truth in that they are based on objective facts and remains so for everyone.
88. A. There has been a mass a uprising against governments that occupies political power for many decades in different Arab countries. This indicate mass uprising is the only way to remove dictatorial reigns and establish democratic ones.
89. D. It is apparent that Ethiopia contributed peace keeping forces to Korea, Congo, Darfur and Sudan imply that Ethiopia's readiness to play her part in the peace keeping process.
90. D **Objective Judgment** is the outcome of knowledge and reasons.
91. D **Scientific Research** is a body of techniques for investigating phenomena acquiring new knowledge or collecting and integrating previous knowledge. It includes collecting and analyzing data involves a very careful testing to arrive at conclusion.
92. D NegadrasGebreHiwotBaykedagn was one of Ethiopian intellectualand the author of the famous book MengestinaYehizibAstedader.
93. C Now days the international community pays huge attention to the threat of the HIV/AIDS pandemic expansion due to the fact that the disease has become a great obstacle against the socio-economic development of countries.
94. B. The Blue Nile is an element of conflict and hatred between Ethiopian and Egypt.

95. D Job satisfaction can help individuals commit themselves to their profession.
96. D One of the negative impact of globalization is dependence of economically weak nation on others developed countries.
97. B Chewing, Chat, Smoking cigarettes and drinking alcohol are not advised as leisure activities, particularly for student because of the problem of addiction and consequent effect.
98. A. the ultimate goal of Agriculture-ledDevelopmentIndustrialization /ADLI/ policy of the current Ethiopia government is to make Ethiopia self-reliant, develop country.
99. C. The significance of Effective leadership in democracy is enhancing good governance and transparency
100. D. Demonstrations are public display for or against a given situation.

Telegram: https://t.me/ExtremeSeries
Extreme Series
t.me/ExtremeSeries

2005 CIVIC AND ETHICAL EDUCATION ANSWER WITH EXPLANATION

1. A Rule of law or good governance is one the prerequisite of the poltical, social and economic development of a country.
2. C A state where religion and state are segment is called secular.
3. A According to Article 61of FDRE constitution the members of the house of Federation are elected by regional council.
4. B State council has the power to introduce laws at regional state level in the FDRE.
5. D Singers(Azmars) were group of people who were considered inferiors to others in the traditional Ethiopian society.
6. D The Ethiopian revolution of 1974 was the result of combined effects of various in 1960 and 1970 uprisings and students movement which strongly challenged the regime. They hoding slogan such as ''Land to Tiller''
7. B According to Article 54/3 the total seats for the house of Peoples Representative not more than 550 and not more than 20 seats shall be reserved for all minority nationalities and peoples in the House of People's Representatives as provided in the constitution.
8. C The nations, nationalities and peoples of are the source of the legitimate power of the EFDR
9. C **Treachery** is trickery, cheating and deceit, it is a crime that mainly committed for the purpose of gaining personal wealth
10. C **Mediation** is a method of non-binding dispute resolution involving a neutral third party. Who tried to help the disputing parties
11. C Corruption is the about abuse of public office for private gain
12. B Breaking promise is morally wrong according to the Ethiopian culture. There is a saying known as ተናገሩት ከማፉ የወለዱት ይፉ!
13. C The Millennium Development Goals includes
    1. Eradicate extreme poverty and hunger
    2. Achieve universal primary education
    3. promote gender equality and empower women
    4. Reduce child mortality
    5. Improve maternal health
    6. combat HIV/AIDS, malaria and other disease
    7. Ensure environmental sustainability
    8. Global partnership for development
14. D Emperor Minilik II was believed to have used a flag in Green Red and Yellow and the present flag of Ethiopia become the national flag (green, yellow and Red) during empless Zewditu 1918
15. C The major advantage of fulfilling over obligation as Ethiopian citizen is enjoying our rights in return.
16. C Pottery has long been the most undermined occupation among many Ethiopian communities

17. D The long distance trade was generally taken as male oriented job in the Ethiopia history because the nature of the trade
18. C many Ethiopians prefer to use imported products than local one because there is a bad habit of giving low value for local products.
19. A The traditional saving institutions are those established by the community in order to save for example equb and Iddir
20. D Knowledges gain from information, data, evidence or truth.
21. D Features of democratic constitution includes it limits the power of the government it paves the way for constitutionalism and democracy and democracy and does not allow extra- constitutional authorities.
22. D Ethiopian foreign policy is based, primarily on the country's mutual benefits and cordial relation ship
23. D Direct democracy is the ancient form which still working among communities of small size like ancient Greek bout only adult male citizens can participate. It is represented today in the form of referendum or plebiscites. It also allow all participates to pass decisions on their own affairs
24. A The major advantages of federalism is creating regional autonomy.
25. D France exemplified the hybrid system of democracy includes parliamentary and presidential democracy
26. D FDRE constitution protects both collective and individual rights together.
27. C The regional states including Afar has full authority to do the following
    - Collecting taxes
    - Establishing regional police forces
    - Using natural resources and the like
28. C The main reason why Ethiopia remain to be the only African country that was never colonize was because she had successfully defended foreign aggressors.
29. D The major possible moture for corruption in searching a short cut for accumulation of wealth.
30. D The dictatorship of the derg regime over throw by the armed stratggle by the Ethiopian peoples in 1991.
31. D Positive discrimination(Affirmation Action is used to compensate women for the past injustice
32. B In 1943 the first Woyane Rebellion which led by Dej- Hailemariam Redda is Tigrai crushed by both Imperial and British military forces.
33. A Multicultural societies like Ethiopia exist is peace when there is mutual respects in cultural, language race, and like for one another.
34. A According to Ethiopian law the responsibility of signing international convention and agreement is given to the executive branch of the government but it approved by house of people's representative but it approved by house of people's representatives

Telegram: https://t.me/Extremeseries

35. B The period of the Derge 9 (1974-1991) characterized by negative experience like red terror, forced military services, absence of religious freedom but there was some positive experiences like the literacy campaign.

36. D Payment tax should be based on people capacity to pay and tax payers have the right to get services from the government and have the right to be informed about government budget

37.A Voluntary organization are independent from government they work toward social well being and they cooperate with governments and other organization to tackle common problems and they did not work for profit.

38. D '' I have a dream that my four little children will one day live in a nation where them will not be judged in a nation where them will not be judged by their colour of their skin but by the content of their character I have a dream today'' this famous quotation is associated to Dr. Martin Luther King

40.D As an student we learn our history and culture inorder to learn from the past and shape your values and belief.

41.D Ethiopia world cultural heritages recently registered by UNESCO as world Heritage is the Konso cultural land scape in 2011
- Acsum- 1980, Fasil Gimb 1979, Jugole Gimb 2006. Konso cultural Land scape 2011, Lalibela 1978, Tiya, 1980…..

42. D The behavior of responsible citizens includes
- Treating people as equals
- Protecting cultural and Natural heritages
- Working hard to improve living conditions

43. C Cross- boundary rivers create the condition for countries to have special relationship

44. C Nuclear weapons have hong term effect and could bring and end to human race if any war involveing it. Today a number of countries possess it. Hirshimo and Nagaski are good example of the danger that can caused by nuclear weapons

45. C Avoiding pre-marital sexual contact is the most advisable, safe way of protecting yourself from HIV/AIDS infection as a young students.

46. B HIV/AIDS is the global problem that needs global intervention. It conspires with infections such as tuberclulosis malaria, typhoids and hepatitis and improving the living standards of poor nation is one way of combating HIV/ AIDS.

47. B Institution has the role in shaping globalization includes
- world bank group (WB)
- world trade organization(WTO)
- Imf- International monetary fund(IMF)

48. D Because human rights are inalienable and inviolable birth rights government include them in their constitution.

49. C Democratic mode of government is preferred over other modes of governance because it creates equal opportunities for all citizens

50. A In 1924 Ethiopia become member of LN, however the LN, failed when it unable to stop the invasion of Italy in 1935.

51. C Tolerance to difference among peoples demands primarily one's ability to accept differences

52. A The appointment of president of the federal supreme court should be approved by the parliament of FDRE according to article 81 of the constitution

53. C Where ever there is a strong and independent judiciary it is safe to expect prevalence of the respect for rule of law.

54. A An unbiased employer should select workers for any type of job by taking into account the applicants' experience and ability

55. B A government that respects the rule of law in its functions will be able to win the confidence and acceptance of the people

56. D The FDRE government can nationalize privately owned provided that the property is found to be unproductive under the owner

57. B The pragmatic theory say truth measured by its usefulness or utility. It has universal acceptance at a certain time

58.C Many people have little understanding that while enjoying the right to equality they, are required to enter into identical obligations with others.

59.B Progressive taxation system is the method of tax rate increases as the taxable base a mount increase

60.D Education enables the educated first and for most to have better understanding of self and society

61.A One of the major functions of the government of FDRE is implementing foreign policy

62. A In theory, markets is free market economies are regulated by relationship between demand and supply

63. B Human and Democratic rights are the two major categories of right stated in the constitution of FDRE

64. C During election campaing political parties should devote much of their time and energy for winning as much support as possible for their policy

65. C Development of a certain country is finally judged by the improvement it brought to the lives of the peoples of the country.

66. C Self confident individuals are always ready to accept logically arguments and learn from them.

67. D Many youngsters fall victim to the habit of smoking mainly because of they lack courage to resist their peers influences

68. D The three economic principles of the free market economy are
- Freedom of decision to economic agents
- Lower government involvement in market
- Letting market forces determine price and transactions

Telegram: https://t.me/ExtremeSmes

69. C The two major international monetary institutions are IMF(International Monetary Fund) and WB (the World Bank

70. B Peace full demonstration is a law full gathering of people to protest, support or create awareness on certain issues

71. A Beliefs that are not true may be called myths. They are not prove to be true through scientific methods of investigations

72. B The right of person who are acussed of any criminal offense includes
- The rights to be considered innocent until the final verdict is given
- The right to fair justices
- The right to be tried in the language they understand better.

73.C The major reason why Addis Ababa made one of the diplomatic hubs of the world and seats of AU is that Ethiopia had played commendable rule in African and international politics

74. D Egypt has ancient relation with Ethiopia in commerce, religion and polities

75. B The conduct of legal proceeding according to established rules and principles for the protection of enforcement of private rights is known as due process of law.

76. C HIV/AID is incurable disease, it transmitted mainly through carless sexual intercourse and creates a serious socio-economic challenge in the world.

77. D According to FDRE constitution article
- No one shall be held slavery or servitude
- Trafficking in human beings is prohibited
- Everyone has the right to protection against in human punishment

78. A Countries usually form closer relationship and create a forum of cooperation when they have common shared objective

79. C Medical ethical standards includes
- giving equal treatment to all patients
- giving loyalty and support to fellow physicians
- Keeping the secretes of patients confidentially

80. D Dependency has a relative economic impact on individuals and families that support them because the dependent one make no contribution and may develop a dependency syndrome

81. A Ethiopia had started to use her rivers primarily for the production of hydro-electric power

82. B Microeconomics studies the behavior of consumers and producers where as the macro economics studies the inclusive so macro economics deals with the working of over all economy.

83. A Modern institutions of saving in Ethiopia includes banks and insurances, where as the traditions institutions includes Ikub and Idir.

84. D Limiting one's interests to one's income amount is the mechanism can be suggested to improve saving habits in Ethiopia

85. A According to Article 29 of FDRE constitution everyone has the right to freedom of expression with out any interference

86. D Democratic and non-democratic state, widely differ in terms of use of Authority and power is that non-democratic state use their power to enforce their power.

87. C Social development in better achieved under a transparent democratic government than other wise because such government will have better chance of winning support of the people for its endarovrs

88. B Currently the government of Ethiopia giving priority for provision of basic health care and basic education while allocating resources.

89. D Democratic system of government is the preferred system in many countries of the world because it adres the interests of citizens in relatively better way.

90. C The democratic values in the constitution of FDRE can be part of social structures and practices of the people only if they are able to win the trust and confidence of the society

91. B Civic societies can play vital role in building a democratic system in a country by encouraging constructer participation of people in public affairs

92. A To improve the life of their peoples in government of developing countries should allocate greater budget for expansion of social services

93. D The member of the Federation execute their public functions according to the FDRE constitution by working language of the federation

94. D Ethiopian farmers are encouraged to produce and use organic fertilizer as much as possible maily because these fertilizers are environmentally friendly and do not harm the land

95. C According to modern understanding of freedom a person is not free until he/she is free from cruel and oppressive local government

96. A The role assigned to women by traditional societies impeded national development by deriving the country the chance to fully exploit their potential

97. C Many of the communicable diseases in Ethiopia could be eradicated by making people put simple nggien rules in to practices

98. C There is a need for the regional states of FDRE to grow and develop together because regional states are found at various stage of economic development

99. D To motivate followers an effective leader should have commitment for the objective of the task

100.C one true measure of a person's greatness could be her/his tangible accomplishment to solve societies problems

Telegram: https://t.me/ExtremeSeries

2014 EUEE CIVIC AND ETHICAL EDUCATION ANSWER AND EXPLANATIONS

1. **B** Developed Countries of the world are most benefited from the new world order of **globalization.**
2. **C** Due process of Law is the principle of constitution which provides "No person shall be deprived of life, Liberty or property without **due process of the law.**
3. **C** Societal needs are name given to human needs for shelter and security
4. **B** An amendment to a constitution is additional article to a constitution
5. **A** "A Conscientious Objector is an individual who has claimed the right to refuse to perform military service on the ground of freedom of thought, conscience and religious objector states only considered in the context of military conscription and is not applicable to volunteer military forces
6. **D** **Rosa Louis McCauley Parks** was Afro- American civil right activist, whom the USA Congress called the first lady of civil rights and mother of freedom movement. She refused to obey bus drivers order that she giving her seat to the white passenger.
7. **B** The ability to bear unpleasant and annoying situation is known as **Arrogance**
8. **D** International Monetary Fund (IMF) and world Bank (WB) are the key institution that fashioned Globalization
9. **D** Impartiality is one of the traits expected force judge for justice to prevail
10. **D** Human right could be a guarantee for the individual freedom.
11. **D** The majority party in the House of People Representatives are presents a nominee for the post of Prime Minister of EDRE.
12. **C** Saving play a crucial role in the development endvours of the countries like Ethiopia.
13. **A** Qeyshibir **(Red Terror)** was the most negative experience during the period of Derg.
14. **D** In the Federal Democratic Republic of Ethiopia, interpreting law is the responsibility of the judiciary organ, where as the House of Federation interpreting the Constitution of the FDRE.

15. **C** The term **medition** broadly refers to any instance in which a third party help other to reach agreement of foreign secretary John Kerry mediate a peace take between Israel and Palestine recently.
16. **D** The correct order of formulating a democratic constitution is first discussion then drafting and finally ratification (discussion → drafting → ratification)
17. **A** Good governance is inclusive terms because it includes the rule of law, transparency, accountability, Responsibility, effectiveness, Participatory and others.
18. **C** Russia and South Africa are the countries that who combine both **Presidential** and **Parliamentary** government system in the world.
19. **C** The unlimited type of government system includes to talitarian and unitary governments.
20. **A** According to Article 10 of the FDRE constitution human rights are inviolable and inalienalde.
21. **C** Constitution is a mother law from which specific laws can be derived.
22. **A** The Watergate Scandal was the major political scandal that occurred in the US in 1970s a result in the break in the democrat National Committee
23. **A** The Federal Government allocates and distributes money reasonably based on the specific situation of different regions.
24. **A** Mengistu Hailemariam of Ethiopia Mabutu seseko of Kongo and Idim Amin of Uganda are all military dictatorship in Africa, where as J. Nyerere of Tanzania was democratic leader.
25. **C** Empress Taitu (wife Emperor Menilik II) was very famous woman for her political career in the modern history of Ethiopia. who made series of political **Shum shir** during the last Period of Menilik II (1906-1913).
26. **B** Awash river is the only inland drainage river in Ethiopia which started and end in Ethiopia where as the other are class boundary rives or cross an international boundary.
27. **C** According to Article 61 of the FDRE Constitution the House of Federation shall have members elected by the state councils.

Telegram: https://t.me/ExtremeSeries

28. **C.** All Ethiopian Peoples have indigenous conflict resolution mechanism. For example Elders/Shimagles are highly respected in the society and arbitration and mediation are employed to solve conflicts.
29. **A** The Federal Supreme Court gives ultimate decision according to the Ethiopian law.
30. **D** England is a unique example in the world that does not have written constitution.
31. **B** The mandate of the Federal government of Ethiopia under the current Constitutional provisions are
    - ⇒ Protecting the Constitution
    - ⇒ Prepare the overall economic, social and development plan
    - ⇒ Establishing foreign relation
    - ⇒ Control the national Banks and issue currency
32. **A** Potters were the most undermined ones in the long history of Ethiopia".
33. **C** Illegal immigration has become the major socio cultural problem in the current Ethiopia context.
34. **A** **Benefits and burdens** must be shared equally among all people in the society.
35. **B** The negative impact of globalization is that dependence of economically weak nations on the developed ones.
36. **B** Ethiopia is the only African country in Africa that protect her independence from foreign aggressions in the 19th C against Egyptian and Italian.
37. **C** The Question of nationalities was the most serious problem that aggravated the Socio-economic problems and consequently opposition against the imperial region of Ethiopia and brought about its dawn fall
38. **C** By making planned activities is one way of protecting our self from peer influences
39. **C** The peoples in SNNP includes mainly the Omotic speaking and some Kushitic language Speakers includes Ari, Hamar, Surma were as the Nuer and the Speakers in the Nilo-sahara Language found in the Gambella National States
40. **D** Even today and in the past persons with **disabilities** are the most disadvantaged group in Ethiopia.

41. **B** In democratic countries obligation of the citizens arise from their duties and responsibilities.
42. **B** The elected of officials are accountable to their electorate, because the electorate is the source of their political power.
43. **D** In unitary system of government, the central government gives power to sub-national government (Provinces) or power is delegated where as in Federal system of government the local of regional government share the power
44. **A** The universal Declaration of Human was promulgated on Dec.1948 following the Humanitarian Crisis during WWII
45. **C** The greatest gain of the women's suffrage movement was Vot in National Election and the stand for election office. Voting right gain by women in Sweden, Filand in Some USA states in the Late 19th C
46. **D** Addiction could effect the working habit of youth self reliance had to self confidence and industriousness leads to self reliance.
47. **D** Assertiveness behavior is a key communication skill. It allows you to pursue your own objectives and stands up for your own right, values and belief all while respecting the right, values and beliefs of other. It is the ultimate win- win behavior. It allow you to seek best possible result It also help to maintain a positive and friendly relationship. It also included questioning mind, positive thinking and critical arguments.
48. **B** Be'alu Girma, was the author of the Famous Ethiopian Novel "Oromay".
49. **A** The common characteristics of both an absolute monarchy and an autocracy is that the supremacy of a single power.
50. **C** Ethiopia's goal is to be self reliant in the near future.
51. **D** One similarity between Parliamentary and Presidential democracies is that in both cases there is a smooth and peaceful transition of state power.
52. **B** Progressive taxation is the taxing mechanism in which the taxing authority charge more taxes as the income of the tax payer increases
53. **D** Civic minded citizens serve their community whole heartedly mainly to get **personal Satisfaction**

Telegram: https://t.me/ExtremeSeries

54. **D** The aim of **Non-Aligned** movement is to stabilization of world peace respect the Human Rights, peaceful coexistence between nations opposition of military alliance preservation of independence, Condemnation of racial discrimination.

55. **A** The demographic characteristics of developing countries high birth rate and high death rate, law life expectancy, high number of youth and low number older age.

56. **A** Ethics is the collection of moral standard accepted by the society.

57. **C** Any law that contradicts a constitution of a country is automatically made null and void.

58. **C** An **altruist** is who set up a charity to help marginalized, communities, while battling to earn a living herself has won our Bounquet of the week.

59. **A** Concerning human right the governments are required to insure that they are not violated by anyone.

60. **A** In democratic countries officials expected to be non- partisans, while performing their duties. This means they should not be prejudiced in any way.

61. **A** One of the attribute of a person who is self confident is readiness to accept critism.

62. **C** The Security Council shall consist of 15 member of the UN 5 countries, USA, Russia, China, Britain and France had permanent member or Veto power where as the other 10 members on non-permanent shall elected by the General Assembly of the UNO on the bases of contribution of member of UN to maintain international Peace and Security based equitable geographical distribution.

63. **D** For citizens to understand their rights and duties properly laws and rules should be clear, easy and well designed.

64. **A** In Ethiopia it is possible to get rid off many of the communicable diseases by teaching the people to deep personal **hygine.**

65. **B** Domesticating international laws and norms means changing the international ones to suit the condition of the land.

66. **D** The functions that exclusively given to the Federal government Ethiopia by the constitution includes:-

- Formulating foreign policy

- Administrating air, railway, sea transport, postal and telecommunication service
- Administrate the national Bank and prepare the National economic, social and development policy, Strategies
- Organize the army and federal police

67. **D** Any places where the Ethiopian flag in flown, be it in embassy or ship is considered as an integral part of the host country or area.

68. **B** During the state of emergency the Ethiopian government could take action that could limit the democratic rights of citizens.

69. **C** Wise saving is saving money after satisfying your basic needs.

70. **C** Accountability of state requires a government to take responsibility for all its decisions and functions.

71. **A** The basic possible reason that lead people to corruption is greediness and selfishness.

72. **B** One of the common features of the Ethiopian peasant rebellions that took place during the imperial regime in 1960s and 1970s at different places were the fact that all rebellion in the North and South were attached to the land questions.

73. **B** We ensure equality between men and women in democratic system by creating equal chances and opportunities for both men and women.

74. **C** Article 2012 of the FDRE constitution provides for accused person to be given the charge against them in order to enable them gather evidences to prepare their defense.

75. **A** Beginning from April 1994 Hutus began slaughtering the Tutsis in the Rwanda. As the brutal killing continued the world stood idly by and Just watched the slaughter lasting 100 day the Rwanda Genocide left approximately 800,000 Tutsis and Hutu sympathizer dead.

76. **B** Abay River, is the bond that link Ethiopia, Egypt and Sudan.

77. **A** The first written constitution of Ethiopia was drafted by Bejround Tekelehawwarat Teklemariam in 1931 and later revised in 1955.

78. **B** Nations and Nationalities day which celebrate in Hider 29 is associated to the identities and cultures of Ethiopian peoples.

Telegram: https://t.me/ExtremeSeries

79. **B** Ethiopian students like you come from different social, economic, cultural and linguistic background, essential to have tolerance to live in harmony in such a situation.

80. **C** The Prime Minister in the commander in chief of the FDRE armed force, according to the constitution.

81. **C** The governments of Nazi party of Germany, Fascist party of Italy and national part of South Africa were similar in that there were absence of the rule of law.

82. **B** The fight against corruption could bear fruit if the government in power manage to make all its official honest and truthful.

83. **B** A country with high literacy rate has high life expectancy. So they have positive relationship.

84. **A** Especially in developing country HIV/AIDS has become a great obstacle against socio-economic development among societies.

85. **D** The prevalence of the **rule of law** in a country result in
   - ➢ Security of private property right and it
   - ➢ Guarantees the safety and security of citizens

86. **D** The moral value of Ethiopian societies includes
   - ➢ Supporting the poor and disabled person
   - ➢ Respecting the elderly people
   - ➢ Conflict resolution through indigenous mechanism

87. **B** The Federation of Eritrea with Ethiopia was one of the factor that necessitated the revision of the first written Constitution of Ethiopia in 1955.

88. **A** Better education and fostering socio cultural values are essential bases for Ethiopians development endeavors.

89. **C** Ethiopia is a seat of many international organization and member of many international organization and it also the founding member of OAU in 1963.

90. **B** One of the long term result of the constitutional protection of free speech having well **informed society.**

91. **A** Great scientists and thinkers of all time contributed great things driver primarily by their desire to attain self fulfillment through knowledge.

92. **D** A self reliant person is one who is aware of his/her own strength and weakness.

93. **D** **Ethnocentrism** is a belief that one's own ethnic group is superior in everything to other. Therefore it hinders interactions with other ethnic group.

94. **D** If a government truly respects rule of law in any country than most certainly justice will prevail.

95. **C** LNs has indirectly engaged the 1935 Italian invasion of Ethiopia by passing arms embargo on both Ethiopia and Italy.

96. **B** Article 20/7 of the FDRE constitution stipulate that an accused person has a right to request for assistance of interpreter at government expense.

97. **B** The Universal Declaration of Human Rights is incorporated in the FDRE, because it will serve as the standard for respect of Human Rights in the country.

98. **A** Unlike previous times the Nations, Nationalizes and Peoples of Ethiopia have the right to promote and develop their language and culture.

99. In theory, market is free market economies are regulated by the relationship between supply and demand. This implies that free market economy is health because demand and supply grow equally.

100. **B** Commonly shared values interests and goals of the Ethiopian Nations, Nationalities and Peoples and respecting them is important to bring national consensus.

Telegram: https://t.me/ExtremeSeries